INFORMATION TO USERS

This manuscript has been reproduced from the microfilm master. UMI films the text directly from the original or copy submitted. Thus, some thesis and dissertation copies are in typewriter face, while others may be from any type of computer printer.

The quality of this reproduction is dependent upon the quality of the copy submitted. Broken or indistinct print, colored or poor quality illustrations and photographs, print bleedthrough, substandard margins, and improper alignment can adversely affect reproduction.

In the unlikely event that the author did not send UMI a complete manuscript and there are missing pages, these will be noted. Also, if unauthorized copyright material had to be removed, a note will indicate the deletion.

Oversize materials (e.g., maps, drawings, charts) are reproduced by sectioning the original, beginning at the upper left-hand corner and continuing from left to right in equal sections with small overlaps. Each original is also photographed in one exposure and is included in reduced form at the back of the book.

Photographs included in the original manuscript have been reproduced xerographically in this copy. Higher quality 6" x 9" black and white photographic prints are available for any photographs or illustrations appearing in this copy for an additional charge. Contact UMI directly to order.

UMI

A Bell & Howell Information Company
300 North Zeeb Road, Ann Arbor MI 48106-1346 USA
313/761-4700 800/521-0600

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

The Greek Reduplicated Aorist

A Dissertation
Presented to the Faculty of the Graduate School
of
Yale University
in Candidacy for the Degree of
Doctor of Philosophy

by

Miles Christopher Beckwith

Dissertation Director: Professor Brent Vine

November 1996

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

UMI Number: 9712751

Copyright 1996 by
Beckwith, Miles Christopher

All rights reserved.

UMI Microform 9712751
Copyright 1997, by UMI Company. All rights reserved.

This microform edition is protected against unauthorized
copying under Title 17, United States Code.

UMI
300 North Zeeb Road
Ann Arbor, MI 48103

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

©1996 by Miles Christopher Beckwith
All rights reserved

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

Abstract
The Greek Reduplicated Aorist
Miles Christopher Beckwith
1996

The Greek verbal system is tripartite, having three tense-aspects: present, aorist and perfect. Perfect formations in the Greek typically show initial reduplication, as does the occasional present. In addition, a small number of forms exist which can only be classed as reduplicated aorists, but these forms typically do not occur in prose and have a very limited distribution in the attested Greek texts. The following dissertation examines the use of these forms and attempts to unravel their history.

Etymological analysis shows that a small number of these forms are archaisms preserved from Indo-European while an equally small number are innovations of the Greek era. However, the majority of the attested reduplicated aorists resist such analysis, and are more difficult to place within their individual verbal paradigms. Fortunately, a careful analysis of the metrical distribution shows that many of the remaining forms occur only in specialized metrical environments and suggests that these forms are artificial creations of poetic diction.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

TABLE OF CONTENTS

Acknowledgments....iv
Bibliographical Abbreviations....v
Other Abbreviations....xvi

INTRODUCTION....1

CHAPTER I: *Ce-CC-e/o-*....7

CHAPTER II: *Ce-CRH-e/o-*....25

CHAPTER III: *Ce-CRC*-e/o-....48

CHAPTER IV: *Ce-CaC-e/o-*....89

CHAPTER V: *HRe-HRC-e/o-*....130

CHAPTER VI: *aC-aC-e/o-*....147

CHAPTER VII: CONCLUSIONS....173

INDEX of Principal Greek Forms....178

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

For Elinor

Grau, teurer Freund, ist alle Theorie
Und grün des Lebens goldner Baum
-Goethe

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

Acknowledgments

Many people have contributed to this work in one way or another and to each of them I owe a debt of gratitude. Far and away the greatest debt is to my advisor, Professor Brent Vine, now at UCLA, whose patience and help have been boundless. Words cannot express how greatly I appreciate his help and encouragement. I would also like to thank my teachers: Jay Jasanoff, Craig Melchert, Stephanie Jamison, Calvert Watkins, and special thanks to Michael Weiss and Stanley Insler, both of whom offered many fine suggestions on earlier versions of this work. Each in their own way has taught me a great deal about Linguistics, about Indo-European, and also what it means to be a scholar. Special thanks are also due to George Prager who read the entire manuscript and saved me from more than a few embarrassing mistakes, although any remaining errors are of course my own responsibility. Finally I would like to thank my wife Mary who has endured countless drafts, late night typing, and my occasional ill-humor throughout the writing of this document.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

Bibliographical Abbreviations

AArmL	=	*Annual of Armenian Linguistics* (Cleveland).
AJP	=	*American Journal of Philology* (Baltimore).
Adams, *THPM*	=	D.Q. Adams, *Tocharian Historical Phonology and Morphology* (New Haven 1988).
Arnold, *Vedic Metre*	=	E.V. Arnold, *Vedic Metre in its Historical Development* (Cambridge 1905).
Aura Jorro, *DMic*	=	F. Aura Jorro, *Diccionario Micénico* (Madrid 1985).
Autenrieth, *Hom.Dic.*	=	G. Autenrieth, *Homeric Dictionary for use in Schools and Colleges* (New York 1876).
Bartholomae, *Verbum*	=	Bartholomae, *Das altiranische Verbum* (Munich 1878).
BB	=	*Beiträge zur Kunde der indogermanischen Sprachen*, hrsg. von A. Bezzenberger (Göttingen).
Beekes, *Development*	=	R.S.P. Beekes, *The Development of the Proto-Indo-European Laryngeals in Greek* (The Hague 1969).
Benveniste, *Origines*	=	E. Benveniste, *Origines de la formation des noms en indo-européen* (Paris 1935).
Benveniste, *Vocab.*	=	E. Benveniste, *Le vocabulaire des institutions indo-européenes* (Paris 1969).
Birwé, *Gr.-Ar.Sprach.*	=	R. Birwé, *Griechisch-Arische Sprach-beziehungen im Verbalsystem* (Hessen 1956).
Brugmann, *Kurze*	=	K. Brugmann, *Kurze vergleichende Grammatik der indogermanischen Sprachen* (Leipzig 1903).
Buck, *Greek Dialects*	=	C. D. Buck, *Greek Dialects: Grammar, Selected Inscriptions, Glossary* (Chicago 1955).
Buttmann, *Aus.Gr.Spr.*	=	P. Buttmann, *Ausführliche Griechische Sprachlehre*, Bd. II^2 (Berlin 1839).
Cardona, *Them.Aor.*	=	G. Cardona, *The Indo-European Thematic Aorists* (Yale Diss. 1960).
Chantraine, *DELG*	=	P. Chantraine, *Dictionnaire étymologique de la langue grecque* (Paris 1968-1980).
Chantraine, *Form.*	=	P. Chantraine, *La formation des noms en grec ancien* (Paris 1933).
Chantraine, *Gr.Hom.*	=	P. Chantraine, *Grammaire homérique* (Paris 1958-1963) —citations below refer to vol. one unless otherwise noted.
Chantraine, *Morph.hist.*	=	P. Chantraine, *Morphologie historique du grec* (Paris 1991).
Chantraine, *Parfait*	=	P. Chantraine, *Histoire du parfait grec* (Paris 1927).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

CHD	=	*The Hittite Dictionary of the Oriental Institute of the University of Chicago*, ed. by H.G. Gütterbock and H.A. Hoffner (Chicago 1989-).
Clackson, *Arm.-Grk.*	=	J. Clackson, *The Linguistic Relationship between Armenian and Greek* (Oxford 1994).
CLI	=	*Compendium Linguarum Iranicarum*, ed. by Rüdiger Schmitt (Wiesbaden 1989).
Comm.Il.	=	*The Iliad: A Commentary*, ed. by G.S. Kirk (Cambridge 1985-1993).
Comm.Od.	=	*A Commentary on Homer's Odyssey*, ed. by A. Heubeck (Oxford 1988-1992).
Cowgill, *Diss.*	=	W. Cowgill, *The Indo-European Long-Vowel Preterits* (Yale Diss. 1957).
Curtius, *Grk.Verb*	=	G. Curtius, *The Greek Verb*^{2} (London 1883), trans. of *Das Verbum der griechischen Sprache* (Leipzig 1876).
Diakonoff-Neroznak, *Phrygian*	=	I.M. Diakonoff and V.P. Neroznak, *Phrygian* (1985).
Dybo, *Slavjanskaja*	=	V.A. Dybo, *Slavjanskaja akcentologija* (Moskva 1981).
Emmerick, *Saka Gr.St.*	=	R.E. Emmerick, *Saka Grammatical Studies* (London 1968).
Encyclopaedia Iranica	=	*Encyclopaedia Iranica* (London 1982-).
Ernout-Meillet, *DELL*	=	A. Ernout and A. Meillet, *Dictionnaire étymologique de la langue latine*^{4} (Paris 1957).
Etudes Hom.	=	*Etudes homériques*, [ed. by] M. Casevitz (Lyon 1989).
Evidence	=	*Evidence for Laryngeals*, ed. by W. Winter (The Hague 1965).
Explanation	=	*Explanation in Historical Linguistics*, ed. by G. Davis and G. Iverson (Amsterdam/ Philadelphia 1992).
Fachtagung VI	=	*Lautgeschichte und Etymologie, Akten der VI Fachtagung der Indogermanischen Gesellschaft*, hrsg. von M. Mayrhofer [et.al.] (Wiesbaden 1980).
Fachtagung VIII	=	*Rekonstruktion und relative Chronologie, Akten der VIII Fachtagung der Indo-germanischen Gesellschaft*, hrsg. von R.S.P. Beekes [et.al.] (Innsbruck 1992).
Fachtagung IX	=	*Früh-, Mittel-, Spätindogermanisch, Akten der IX Fachtagung der Indogermanischen Gesellschaft*, hrsg. von G. Dunkel [et.al.] (Wiesbaden 1994).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

Flobert, *Déponents*	=	P. Flobert, *Les verbes déponents latins des origines à Charlemagne* (Paris 1975).
Forssman, *Unters.*	=	B. Forssman, *Untersuchungen zur Sprache Pindars* (Wiesbaden 1966).
Fournier, *Verbes "dire"*	=	H. Fournier, *Les verbes "dire" en grec ancien* (Paris 1946).
Fraenkel, *LEW*	=	E. Fraenkel, *Litauisches etymologisches Wörterbuch* (Heidelberg 1962-65).
Frame, *Myth of Return*	=	D. Frame, *The Myth of Return in Early Greek Epic* (New Haven 1978).
Francis, *Disyl.Roots*	=	E.D. Francis, *The Greek Disyllabic Roots: the Aorist Formations* (Yale Diss. 1970).
Frisk, *GEW*	=	H. Frisk, *Griechisches etymologisches Wörterbuch* (Heidelberg 1960-1972).
FsBrown	=	*Indological Studies in Honor of W. Norman Brown*, ed. by E. Bender (New Haven 1962).
FsChantraine	=	*Mélanges de linguistique et de philologie grecques offerts à Pierre Chantraine*, [ed. by] F. Bader (Paris 1972).
FsKuiper	=	*Pratidānam. Indian, Iranian and Indo-European Studies Presented to Franciscus Bernardus Jacobus Kuiper on his Sixtieth Birthday*, ed. by J.C. Heesterman [et.al.] (The Hague 1968).
FsLane	=	*Studies in Honor of George Sherman Lane*, ed. by Walter W. Arndt [et.al.], (Chapel Hill 1967).
FsLejeune	=	*Etrennes de Septantaine. Travaux de linguistique et de grammaire comparée offerts à Michel Lejeune...* (Paris 1978).
FsNeumann	=	*Serta Indogermanica, Festschrift für Günter Neumann zum 60 Geburtstag*, [ed. by] J. Tischler (Innsbruck 1982).
FsPolomé	=	*Languages and Cultures: Studies in honor of Edgar C. Polomé*, ed. by M.A. Jazayery & W. Winter (Berlin-New York 1988).
FsRisch	=	*o-o-pe-ro-si Festschrift für Ernst Risch zum 75.Geburtstag*, hrsg. von A. Etter (Berlin 1986)
FsRix	=	*Indogermanica et Italica*, [ed. by] J. Bendahman [et.al.] (Innsbruck 1992).
FsSchmidt	=	*Indogermanica et caucasica: Festschrift für Karl Horst Schmidt zum 65. Geburtstag* (Berlin 1994).
FsStrunk	=	*Verba et structurae: Festschrift für Klaus Strunk zum 65. Geburtstag* (Innsbruck 1995).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

FsSzemerényi	=	*Studies in Diachronic, Synchronic, and Typological Linguistics: Festschrift for Oswald Szemerényi...*, ed. by B. Brogyanyi (Amsterdam 1979).
FsWackernagel	=	*Ἀντίδωρον. Festschrift für Jacob Wackernagel* (Göttingen 1923).
Gershevitch, *Hymn*	=	I. Gershevitch, *The Avestan Hymn to Mithra* (Cambridge 1959).
GsCowgill	=	*Studies in Memory of Warren Cowgill (1929-1985)*, ed. by C. Watkins (Berlin 1987).
GsKretschmer	=	ΜΝΗΜΗΣ ΧΑΡΙΝ. *Gedenkschrift Paul Kretschmer*, [ed. by] H. Kronasser (Wien 1956-57).
Godel, *Introduction*	=	R. Godel, *An Introduction to the Study of Classical Armenian* (Wiesbaden 1975).
Goto, *I Prs.*	=	T. Goto, *Die "I. Präsensklasse" im Vedischen* (Wien 1987).
Grundriss	=	K. Brugmann and B. Delbrück, *Grundriss der vergleichenden Grammatik der indogermanischen Sprachen*^{2} (Strassburg 1897-1916).
Harðarson, *Studien*	=	J.A. Harðarson, *Studien zum urindogermanischen Wurzelaorist* (Innsbruck 1993).
Haas, *Phrygische*	=	O. Haas, *Phrygische Spachdenkmäler* (Sofia 1966).
Hirt, *Hb.Gr.*	=	H. Hirt, *Handbuch der Griechischen Laut- und Formlehre*^{2} (Heidelberg 1912).
Hirt, *Hb.Urg.*	=	H. Hirt, *Handbuch des Urgermansichen* (Heidelberg 1932).
Hoffmann, *Injunktiv*	=	K. Hoffmann, *Der Injunktiv im Veda* (Heidelberg 1967).
Hoffmann, *Aufsätze*	=	K. Hoffmann, *Aufsätze zur Indoiranistik* (Weisbaden 1975-1976).
Hollifield, *System*	=	P. Hollifield, *On the System of Conjugation in Proto-Indo-European* (Harvard Diss. 1977).
HS	=	*Historische Sprachforschung = Historical Linguistics* (Göttingen).
Humbach, *GZ*	=	H. Humbach, *Gāthās of Zarathushtra*, in collaboration with J. Elfenbeim and P.O. Skjaervo (Heidelberg 1991).
IIJ	=	*Indo-Iranian Journal* (Dordrecht).
IF	=	*Indogermanische Forschungen. Zeitschrift für Indogermanistik und allgemeine Sprachwissenschaft* (Berlin).
Insler, *GZ*	=	S. Insler, *The Gāthās of Zarathustra* (Leiden 1975).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

Jacquinod, *Double acc.*	=	B. Jacquinod, *Le double accusatif en grec d'Homère à la fin du Ve Siècle avant J.-C.* (Louvain 1989).
Jamison, *áya-Form.*	=	S. W. Jamison, *Function and Form in the -áya-Formations of the Rig Veda and Atharva Veda* (Göttingen 1983).
Jasanoff, *SMIE*	=	J. Jasanoff, *Stative and Middle in Indo-European* (Innsbruck 1978).
Joachim, *Mehrfachpr.*	=	U. Joachim, *Mehrfachpräsentien im R.gveda* (Frankfurt a.M. 1978).
Joseph, *Problems*	=	L.S. Joseph, *Problems in the Development of the Indo-European Laryngeals in Celtic* (Harvard Diss. 1980).
Kellens, *Verbe avestique*	=	J. Kellens, *Le verbe avestique* (Wiesbaden 1984).
Keller, *Verbes en-sc-*	=	M. Keller, *Les verbes latins à infectum en -sc-* (Bruxelles 1992).
Klingenschmitt, *Verbum*	=	G. Klinengschimitt, *Das altarmenische Verbum* (Wiesbaden 1982).
Krause-Thomas, *Toch.El.*	=	W. Krause and W. Thomas *Tocharisches Elementarbuch*, Bd. I (Heidelberg 1960).
Kuiper, *Nasalpräsentia*	=	F.B. Kuiper, *Die indogermanischen Nasalpräsentia* (Amsterdam 1937).
Kurylowicz, *Apophonie*	=	J. Kurylowicz, *L'apophonie en indo-européen* (Wroclaw 1956).
KZ	=	*(Kuhns) Zeitschrift für vergleichende Sprachforschung auf dem Gebiete der idg. Sprachen* (Göttingen).
Lardinois, *Diss.*	=	A. Lardinois, *Wisdom in Context: the Use of Gnomic Expressions in Early Greek Poety* (Princeton Diss. 1994).
Laryngaltheorie	=	*Die Laryngaltheorie*, hrsg. von Alfred Bammesberger (Heidelberg 1988).
Latacz, *Zum Wortfeld*	=	J. Latacz, *Zum Wortfeld "Freude" in der Sprache Homers* (Heidelberg 1966).
Lejune, *Phonét. hist.*	=	M. Lejune, *Phonétique historique du mycénien et du grec ancien* (Paris 1972).
Lehmann-Feist, *GEW*	=	W.P. Lehmann, *A Gothic Etymological Dictionary Based on the third edition of Vergleichendes Wörterbuch der Gotischen Sprache by Sigmund Feist* (Leiden 1986).
Leumann, *Kl.Schr.*	=	M. Leumann, *Kleine Schriften* (Zürich 1959).
Leumann, *HW*	=	M. Leumann, *Homerische Wörter* (Bale 1950).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

Leumann, *LLuF*	=	M. Leumann, *Lateinische Laut- und Formenlehre* (München 1977).
LfrgE	=	*Lexikon des frühgriechischen Epos*, begründet von B. Snell (Göttingen 1955-).
Lindeman *Introduction*	=	F.O. Lindeman, *Introduction to the Laryngeal Theory* (Oslo 1987).
Lowenstam, *Patroklos*	=	S. Lowenstam, *Death of Patroklos* (Königstein 1981).
Macdonell, *Vedic Gr.*	=	A.A. Macdonell, *Vedic Grammar* (Strassburg 1910).
Mackie, *Talking Trojan*	=	H. Mackie, *Talking Trojan* (Princeton Diss. 1993).
Mayrhofer, *EWAia*	=	M. Mayrhofer, *Etymologisches Wörterbuch des Altindoarischen* (Heidelberg 1986-).
Mayrhofer, *Idg.Gr. I/2*	=	M. Mayrhofer, *Indogermanische Grammatik*, Bd. I/2, *Lautlehre* (Heidelberg 1986).
Mayrhofer, *KEWA*	=	M. Mayrhofer, *Kurzgefasstes etymologisches Wörterbuch des Altindischen* (Heidelberg 1956-1982).
McCone, *IE Origins*	=	K. McCone, *Indo-European Origins of the Old Irish Nasal Presents, Subjunctives and Futures* (Innsbruck 1991).
Meier-Brügger, *Gr.Spr.*	=	M. Meier-Brügger, *Griechische Sprachwissenschaft* (Berlin 1992).
Meillet, *Esquisse*	=	A. Meillet, *Esquisse d'une grammaire comparée de l'arménien classique* (Vienna 1936).
Meiser, *Lautgeschichte*	=	G. Meiser, *Lautgeschichte der umbrischen Sprache* (Innsbruck 1986).
Meister, *Kunstspr.*	=	K. Meister, *Die homerische Kunstsprache* (Leipzig 1921).
Mélanges Boisacq	=	*Mélanges Emile Boisacq* (Bruxelles 1937).
Melchert, *AHP*	=	H.C. Melchert, *Anatolian Historical Phonology* (Amsterdam 1994).
Melchert, *SHHP*	=	H.C. Melchert, *Studies in Hittite Historical Phonology* (Göttingen 1984).
Monro, *Gr.Hom.Dia.*	=	D.B. Monro, *Grammar of the Homeric Dialect* (Oxford 1891).
Most, *Measures*	=	G.W. Most, *The Measures of Praise* (Göttingen 1985).
MSL	=	*Mémoires de la Société de Linguistique de Paris* (Paris).
MSS	=	*Münchener Studien zur Sprach-wissenschaft* (München).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

Narten, *Sigm.Aor.*	=	J. Narten, *Die sigmatischen Aoriste im Veda* (Wiesbaden 1964).
Neu, *Anitta*	=	E. Neu, *Der Anitta-Text* (Weisbaden 1974).
NTS	=	*Norsk Tidsskrift for Sprogvidenskap* (Oslo).
Nussbaum, *Head and Horn*	=	A. Nussbaum, *Head and Horn in Indo-European* (Berlin 1986).
Oettinger, *Stammbildung*	=	N. Oettinger, *Die Stammbildung des hethitischen Verbums* (Nürnberg 1979).
Onians, *Origins*	=	R.B. Onians, *Origins of European Thought* (Cambridge 1951).
Parry, *MHV*	=	M. Parry, *The Making of Homeric Verse*, ed. by A. Parry (Oxford 1971).
Pedersen, *Tocharisch*	=	H. Pedersen, *Tocharisch vom Gesichtspunkt der indoeuropäischen Sprachvergleichung* (Copenhagen 1941).
Pedersen, *Sprachgesch.*	=	H. Pedersen, *Zur tocharischen Sprachgeschichte* (Copenhagen 1944).
Pedersen, *Vgl.Gr.*	=	H. Pedersen, *Vergleichende Grammatik des keltischen Sprachen* (Göttingen 1909-13).
Peters, *Untersuchungen*	=	M. Peters, *Untersuchungen zur Vertretung der indogermanischen Laryngale im Griechischen* (Wien 1980).
Pinault, *Introduction*	=	G.-J. Pinault, *Introduction au tokharien*, *Lalies* 7 (Paris 1989).
Poetae Comici Graeci	=	*Poetae Comici Graeci*, [ed. by] R. Kassel and C. Austin (Berlin 1983-).
Pokorny, *IEW*	=	J. Pokorny, *Indogermanisches etymologisches Wörterbuch* (Bern 1959).
Poultney, *Tables*	=	J.W. Poultney, *The Bronze Tables of Iguvium* (Baltimore 1959).
Prévot, *Aoriste* - θην	=	A. Prévot, *L'aoriste grec en* - θην (Paris 1935).
Puhvel, *HED*	=	J. Puhvel, *Hittite Etymological Dictionary* (1984-).
Reichelt, *Av.Elem.*	=	H. Reichelt, *Avestisches Elementarbuch* (Heidelberg 1909).
RPh	=	*Revue de Philologie, de littérature et d'histoire ancienne* (Paris).
Risch, *Wortbildung*	=	E. Risch, *Wortbildung der homerischen Sprache*[2] (Berlin 1974).
Rix, *Hist.Gr.Gr.*	=	H. Rix, *Historische Grammatik des Griechischen: Laut- u. Formenlehre* (Darmstadt 1976).
Ruijgh, *Script.Min.*	=	C.J. Ruijgh, *Scripta minora ad linguam graecam pertinentia* (Amsterdam 1991).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

Schmitt, *Gr.Kl.Arm.*	=	R. Schmitt, *Grammatik des Klassisch-Armenischen mit sprachvergleichenden Erläuterungen* (Innsbruck 1981).
Schrijver, *Reflexes*	=	P. Schrijver, *The Reflexes of the Proto-Indo-European Laryngeals in Latin* (Amersterdam 1991).
Schulz, *Wurzel* Πειθ-	=	S. Schulz, *Die Wurzel* Πειθ- (πιθ-) *im älteren Griechischen* (Freiburg 1952).
Schulze, *Kl.Schr.*	=	W. Schulze, *Kleine Schriften*^{2} (Göttingen 1966).
Schulze, *Quaest. epicae*	=	W. Schulze, *Quaestiones epicae* (Gütersloh 1892).
Schwyzer, *Gr.Gr.*	=	E. Schwyzer, *Griechische Grammatik* I (Munich 1934).
Seebold, *EtWb*	=	E. Seebold, *Vergleichendes und etymologisches Wörterbuch der germanischen starken Verben* (The Hague 1970).
Shipp, *Studies*	=	G.P. Shipp, *Studies in the Language of Homer*^{2} (Cambridge 1972).
Sihler, *NCGGL*	=	A. Sihler, *A New Comparative Grammar of Greek and Latin* (Oxford 1995 [1994]).
Sprache	=	*Die Sprache. Zeitschrift für Sprachwissenschaft* (Wiesbaden/Wein).
Strunk, *Nasalpräsentien*	=	K. Strunk, *Nasalpräsentien und Aoriste* (Heidelberg 1967).
StIdgW	=	*Studien zum indogermanischen Wortschatz* (Innsbruck 1987).
Studia-Slavico-Byz.	=	*Studia Slavico-Byzantina*, vol. I *Studies on the Slavo-Byzantine and West-European Middle Ages* ([Sofia] 1988).
Szemerényi, *Syncope*	=	O. Szemerényi, *Syncope in Greek and Indo-European and the Nature of Indo-European Accent* (Naples 1964).
Solmsen, *Untersuch.*	=	F. Solmsen, *Untersuchungen z. griech. Laut- u. Verslehre* (Strassburg 1901).
Thieme, *Plusquamperfectum*	=	P. Thieme, *Das Plusquamperfectum im Veda* (Göttingen 1929).
Thurneysen, *GOI*	=	R. Thurneysen, *A grammar of Old Irish* (Dublin 1946).
Tichy, *Onomat.*	=	E. Tichy, *Onomatopoetische Verbalbildungen des Griechischen* (Wien 1983).
TIES	=	*Tocharian and Indo-European Studies* (Reykjavik).
Tischler, *HEG*	=	J. Tischler, *Hethitisches etymologisches Glossar* (Innsbruck 1983-).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

Tucker, *Creation*	=	E.F. Tucker, *The Creation of Morphological Regularity: Early Greek Verbs in -éō, -áō, -óō, -úō and -íō* (Göttingen 1990).
Universals of Hum.Lg.	=	*Universals of Human Language*, ed. by J. Greenberg (Stanford 1978).
VBGS	=	*Vermischte Beiträge zur griechischen Sprachkunde* (Basil 1897).
van der Valk, *Researches*	=	M. van der Valk, *Researches on the Text and Scholia of the Iliad* (Leiden 1964).
Van Nooten-Holland, *RV*	=	*Rig Veda: a Metrically Restored Text with an Introduction and Notes*, ed. by B.A. van Nooten and G.B. Holland (Cambridge 1994).
Van Windekens, *Tocharien*	=	A. J. Van Windekens, *Le tocharien confronté avec les autres langues indo-européennes* (Louvain 1982).
Vasmer, *REW*	=	M. Vasmer, *Russisches Etymologisches Wörterbuch* (Heidelberg 1955).
Veitch, *GV*	=	W. Veitch, *Greek Verbs* (Oxford 1871).
Vendryes, *LEIA*	=	Vendryes, *Lexique étymologique de l'irlandais ancien* (Dublin-Paris 1959-).
Ventris-Chadwick, *Doc.*	=	M.Ventris and J.Chadwick, *Documents in Mycenaean Greek*^{2} (Cambridge 1973).
Vine, *Diss.*	=	B. Vine, *Indo-European Verbal Formations in *-d-* (Harvard Diss. 1981)
Wackernagel, *Kl.Schr.*	=	J. Wackernagel, *Kleine Schriften* (Göttingen 1953-).
Wackernagel, *Spr.Unt.*	=	J. Wackernagel, *Sprachliche Untersuchungen zur Homer* (Göttingen 1916).
Wackernagel-Debrunner, *Ai.Gr.*	=	J. Wackernagel and A. Debrunner, *Altindische Grammatik* (Göttingen, 1896-).
Walde-Hofmann, *LEW*	=	A. Walde and J.B.H. Hofmann, *Lateinisches etymologisches Wörterbuch* (Heidelberg 1938-54).
Walde-Pokorny, *VWIS*	=	A. Walde and J. Pokorny, *Vergleichendes Wörterbuch der indogermanischen Sprachen* (Leipzig 1930).
Watkins, *Celtic Verb*	=	C. Watkins, *The Indo-European Origins of the Celtic Verb*, Pt.I *The Sigmatic Aorist* (Dublin 1962).
Watkins, *HKD*	=	C. Watkins, *How to Kill a Dragon: Aspects of Indo-European Poetics* (Oxford 1995).
Watkins, *Idg.Gr. III/1*	=	C. Watkins, *Indogermanische Grammatik III. 1, Geschichte der indogermanischen Verbalflexion* (Heidelberg 1969).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

Watkins, *Selected Wr.*	=	C. Watkins, *Selected Writings*, ed. by L. Oliver (Innsbruck 1994).
Weiss, *Studies*	=	M. Weiss, *Studies in Italic Nominal Morphology* (Cornell Diss. 1993).
West, *Theogony*	=	Hesiod, *Theogony*, ed. by M. West (Oxford 1978).
Whitney, *Skt.Gr.*	=	W.D. Whitney, *Sanskrit Grammar* (Cambridge 1960 = 1889).
Whitney, *Roots*	=	W.D. Whitney, *Roots, Verb-Forms, and Primary Derivatives of the Sanskrit Language* (New Haven 1945 = Leipzig 1885).
ZPE	=	*Zeitschrift für Papyrologie und Epigraphik* (Bonn).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

Ancient Texts

Ancient sources are noted in square brackets. The text of Homer used for this study is the standard *OCT* text *Homeri Opera*, ed. by T. W. Allen, in four volumes. Commentaries (e.g., *Comm.Il.*, *Comm.Od.*) and other specialized studies (e.g., van der Valk, *Researches*) have been consulted for those passages where the textual tradition is uncertain. Greek and Latin texts are cited by standard abbreviations. Indo-Iranian text abbreviations and sources are:

AV	=	*Atharva Veda*
AVP	=	*Atharva Veda (Paippalāda)*
RV	=	*Rig Veda*. from the edition T. Aufrecht, *Die Hymnen des Rigveda* (Berlin 1861-1863). The new Van Nooten-Holland edition was also consulted. Translations, when not my own, are taken from K.F. Geldner, *Der Rig-Veda* (Cambridge 1938).
ŚB	=	*Śatapatha Brāhmaṇa*
TS	=	*Taittirīya-Samhitā*
Vd	=	*Vendidad*
Y	=	*Yasna*. Gathic references are noted to Insler and Humbach above.
Yt	=	*Yasht*. No specific text was used for the Younger Avestan citations. Transcriptions and translations have been taken from the standard handbooks, e.g., Kellens, *Verbe avestique*, and are so noted.

A Note on Citation

Homeric forms are quoted when possible. In general, against the common practice (e.g., Chantraine, *Gr.Hom.*; Risch, *Wortbildung*) I have not quoted verbs in the first person singular unless that form is attested in the Homeric texts. First person sg., third person sg., or infinitive forms are used as citation forms when possible. Forms which do not occur in the attested tradition but which are secure from paradigmatic patterns are noted with an asterisk following the form. A double asterisk marks an unattested and improbable form.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

Other Abbreviations

Languages

Aeol.	=	Aeolic (Greek)	OCS	=	Old Church Slavic
Arm.	=	Armenian	OE	=	Old English
Av.	=	Avestan	OH	=	Old Hittite
Cret	=	Cretan (Greek)	OHG	=	Old High German
CT*	=	Common Tocharian	OL	=	Old Latin
Dor.	=	Doric (Greek)	ON	=	Old Norse
Ion.	=	Ionic (Greek)	OP	=	Old Persian
Khot	=	Khotanese	TA	=	Tocharian A
Lat.	=	Latin	TB	=	Tocharian B
Lith.	=	Lithuanian	Toch.	=	Tocharian
Myc.	=	Mycenaean (Greek)	Y.Av	=	Younger Avestan
M.Ir.	=	Middle Irish			
O.Ir.	=	Old Irish			

Grammatical Abbreviations

act.	=	active	pl.	=	plural
aor.	=	aorist	plpft.	=	pluperfect
imp.	=	imperative	prs.	=	present
ind.	=	indicative	prt.	=	preterite
intrns.	=	intransitive	sg.	=	singular
mid.	=	middle	subj.	=	subjunctive
pft.	=	perfect	trns.	=	transitive

Other abbreviations used below are self-explanatory: e.g., frag[ment], bib[liography], esp[ecially], etc..

---

*For reconstructions of Tocharian I follow Hilmarsson in noting the CT front and back low vowels as *æ* and *ȧ*. The source of the "Fremdvokal" is noted as *ə*, palatalized consonants are marked with an apostrophe.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

INTRODUCTION

§0.1 *Background*

The reduplicated aorist is at best a marginal formation in Greek, attested to some thirty roots in Homer, and a few scattered forms in later Greek.[1] Synchronically, these aorists sometimes stand in relation to non-reduplicated forms, often with a variation in semantics, e.g., δέδαε 'taught' : (ἐ)δάην 'I learned'. Such a contrast is not without parallels. Thus in Sanskrit, where intransitive verbs often build derived transitives in *-aya-*, the aorists to such derived forms are typically reduplicated, e.g., *pṛ* 'cross over' > *pārāyati* 'makes cross over' : (aor.) *á-pī-parat*.[2] There is also evidence for a similar formation in Tocharian (dialect A only), which forms a reduplicated transitive preterite, e.g., *wik-* 'vanish': *wiks-* (prs.) : *wawik-* (prt.) 'destroy'.[3]

---

[1] Curtius, *Grk.Verb* 292 f., lists 41 forms which serves as the base list used here (see Index for locations). Schwyzer, *Gr.Gr.* 748 f., adds only ἔγρετο improbably from *γεγρ- ε/ο- and ἔκιχον. The prothesis of ἔγρετο is probably best explained by a laryngeal, cf., the perfect ἐγρήγορα < **h₁g(r)e-h₁gor-* ≡ Skt. *jāgára* (Beekes, *Development* 117). The latter form, although synchronically an aorist in Homer, is certainly a relic of an older present-imperfect formation *κίχημι (Schwyzer, *op.cit.* 688; Risch, *Wortbildung* 243) and need not concern us here. In addition to the Curtius' forms I have added a discussion of ἀείδω, which J. Wackernagel, *KZ* 29 (1888) 151 f., derives from an unattested reduplicated aorist (see §5.3). Other reduplicated aorists have been posited but most such analyses are not valid. Compare, e.g., E. Tichy's explanation of δειδέχατο, δειδίσκομαι, *Glotta* 54 (1976) 71 f., from an unattested *δεδεχθαι. This analysis is unlikely both because of the unexpected form (full grade root, athematic endings, see §1.1) and the semantics (see esp. §4.1.4, §7.2). For an alternative analysis see B. Forssman, *Sprache* 24 (1974) 3 f.

[2] Double object causative formations are also attested in Vedic, but apparently belong to a later strata of diction. For a discussion of this, as well as the entire history of the 'causative' paradigm in Indic, see Jamison, *aya-Form*. 178 f. and passim.

[3] For a discussion of the Tocharian class II preterite and its possible relationship to both the Greek and Sanskrit forms see Birwé, *Gr.-Ar.Sprach.* 29 f. (also J. Jasanoff, *GdCowgill*

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

From a purely typological point of view the correspondence in form and function between these three languages is striking. Further, a small number of forms are certainly archaic: εἶπον = Skt. *ávocat* (Av. *vaocaṭ*) 'spoke' < **u̯e-u̯k̑*-e/o-* (§1.1); εὗρον 'I found' = O.Ir. *·fuar* < **u̯e-u̯r-e/o-* (§2.1); ἐνεγκεῖν 'to bear' ≅ Lat. *nanc-* ≅ Av. *nąsaṭ* (but see below §5.1) and perhaps also ἦραρε 'fit, attached' = Arm. *arari* 'I made' (§6.1). Nevertheless, there are certain considerations which make us pause. First, of these relic forms only ἦραρε (< **ar-* 'fit' [intrs.]) shows evidence for a transitive or causative formation.[4] Second, apart from εἶπον (Cret. Fειπ-; Ionic εἶπα-) and ἐνεγκεῖν (and its by-form ἐνεικεῖν), there is essentially no inscriptional or dialectal evidence for these forms or any other reduplicated aorist outside of epic poetry. Finally, the Greek formation showing *e*-reduplication with a zero-grade root differs formally both from the Indic (*ī*-reduplication)[5] and Tocharian formations.[6]

---

93 n. 4). For general discussion of the Tocharian preterite system and its relation to the Indo-European aorist see most recently Pinault, *Introduction* 147 f.

[4] While the Sanskrit formation is best described in terms of transitivity, such a syntactic distinction is less useful in describing the Greek forms, where the difference is primarily semantic encompassing a variety of syntactic formations (cf. below §4.1.4, §6.1, §6.2).

[5] The Indic formation also shows zero-grade, although the treatment of some root structures is distinct from Greek (see §2.0 f. and §2.4.1).

[6] The reconstruction of the TA class II preterite is difficult. For a form such as *cacäl* 'raised' (root *täl-*) a mechanical projection backwards might argue for **cacäl-ā* < **t'at'el-* < **t'æt'el-* < **tētel-*. F.O. Lindeman, *NTS* 23 (1969) 15 f., in fact proposes such a reconstruction, but this is unlikely. First, evidence for long vowel reduplication in Indo-European is all but non-existent (Beekes, *Development* 113 f.). Second, other reduplicated forms in Tocharian have gone through considerable restructuring, e.g., TA *kakmu* TB *kekamu* 'gone' < **kæ-kəm-u̯o-* < **ǵo-ǵm̥-u̯os-*, remade from older **ǵe-ǵm̥-u̯os-* (Pinault, *Introduction* 107) or the TB subj. *tättā-* 'will set, place' apparently from Indo-European **dhi-dh(e)h₁-* (or **sti-sth₂-t* see Pinault, *Introduction* 133). In both these examples the expected palatalization has been blocked. Thus for *cacäl*, an analogical source for the palatalization is more likely (W. Winter, *Fachtagung VI* 559 f.). Hence, it is not clear whether *cacäl* descends from **tetel-* or **totel-* (**te-tl̥hₓ-* or **te-tol-*?). Further, if this formation is related to the analogous class II prt. in TB (the *cāla*-type) analogical restructuring must have taken place in both languages.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

Regarding the first point, the existence of word equations which do not show causative force is not in principle insurmountable. Indo-European presents in **-ske/o-* were iterative (iterative-causative?) in the parent language (Sihler, *NCGGL* 505 f.), while individual relic forms have largely been bleached of their original semantics, e.g., Latin *poscō* 'I ask' = Sanskrit *pṛcchāmi* 'id.' < **pṛḱ-ske/o-*. As to the second point, it is true that poetry often preserves archaisms lost in the spoken language. And it is at least possible that the reduplicated aorist of Homer was preserved from some ancient era. Nevertheless, it is also true that the metrical constraints of Homeric diction have forced the creation of newer forms which are peculiar to the Kunstsprache (Parry, *MHV* 5). In what follows I will show that a large number of reduplicated aorists are in fact late analogical creations. This is especially true of the 'causative forms', most of which can be safely dated to the era of Homeric composition. Hence, the typological equation between the Greek reduplicated aorist and the Indic and Tocharian data is an illusion.

In a verbal system in which two of the three tense-aspect systems either allowed reduplication (in the case of the present[7]) or required it (in the case of the perfect), it is not surprising that the third tense-aspect, the aorist, would also acquire reduplication. Analogical expansion seems to have happened from both directions. That is, at least one reduplicated aorist can be explained only as a late backformation from the perfect (τετυκεῖν see §3.1.3), while another is best explained as built from the older present (δέδαε §3.4.1). Nevertheless, these are more or less isolated cases. The vast majority of reduplicated aorists would seem to have been built in analogy to a core of inherited reduplicated forms, e.g., *ἀράρε : ἤραρε :: ὄρωρε : X, where X = ὤρορε (§7.2). The exact analogical process cannot always be recovered, but some possible sources will be described below.

---

[7] In the following study, for simplicity's sake I use the term 'present' to refer to the present system, including the imperfect.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

§0.2 *Methodology*

The most important tool for a study of this kind is comparative analysis. Each reduplicated aorist below is examined both within its Greek paradigm and in its etymological background. In many cases it can be shown that not only is there no comparative evidence for a reduplicated aorist, but that such a formation would be unlikely in the parent language. In addition to a purely etymological analysis, however, a careful study of the philological problems of the text before us can often be just as illuminating. In metrical texts such as the Homeric poems, this philological study should pay special attention to metrical problems. As it turns out, metrical distribution is key to understanding the spread of the reduplicated aorist in epic poetry.

The metrical distribution of the reduplicated aorist is best understood in relation to the augment. In Greek poetic texts the augment was often dropped for metrical reasons. Thus in Homer we find both augmented and unaugmented forms side by side without any distinction in meaning, e.g., ἐτέτυκτο : τέτυκτο 'was made' (see Chantraine, *Gr.Hom.* 479 f.; Schwyzer, *Gr.Gr.* 650).[8] If we look at the aorist forms of πευθ- in Homer, we find both the augmented ἐπύθοντο and the unaugmented πύθοντο sometimes occurring in the same position, except that the augmented form is preceded by a formular construc-

---

[8] Attempts to find 'injunctive' uses of the non-augmented aorist and imperfect in Homer have been largely unproductive (but see M.L. West, *Glotta* 67 [1989] 135 f., for possible relic injunctive uses in Hesiod and the Homeric Hymns). Others have tried to see some specific force to the augment, not necessarily linked to the Indo-European injunctive. See most recently L. Basset, *Etudes Hom.* 9 f., who sees, *op.cit.* 14, in the augment a marker of "un événement antérieur". Alternatively see H.J. Blumenthal, *IF* 79 (1974) 76: "[the augment is] linked with the age of the various forms affected...". For possible sociolinguistic factors see R. Janko, *Il.Comm.* IV 11 n. 15 (with bib.) "[the augment is] typical of the Mycenaean lower class". All in all these approaches, typically relying on statistical analysis, are not probative. Further, the lack of common ground between the different analyses does not inspire confidence. The older metrical approach of Schwyzer and Chantraine is preferable.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

tion ending in a long syllable, while the unaugmented form is preceded by a formula ending in an trochee. Outside of the indicative where the use of the augment is prohibited, no such distribution is possible. However, a reduplicated form can fill the same metrical slot that the augmented form fills in the indicative, giving the following distribution:

Metrical environment	Indicative	Non-Indicative
after – ∪	πύθ-οντο	πυθ-οίατο
after —	ἐ-πύθ-οντο	πε-πύθ-οιτο

In this way reduplicated forms can function as needed metrical by-forms in non-indicative modes. Significantly, several reduplicated aorists never occur in the indicative, but are only attested in modal or non-finite forms alongside a non-reduplicated thematic aorist.

Such a distribution does not in itself explain the reduplicated aorist, since some forms must predate this system to provide an analogical model for the creation of the newer πεπύθοιτο. Nevertheless, in what follows I will show that the only meaningful distinction between πυθοίατο and πεπύθοιτο is metrical. Further, I will show that none of the extravagant claims concerning the semantic force of the reduplicated aorist are valid.[9] Although metrical analysis is useful in only a handful of forms,[10] these examples in

---

[9] Compare especially Curtius' claims, *Grk.Verb* 295: "[a]n intensive force may be detected most clearly in the Homeric use of ἐκέκλετο by the side of κέλετο, in κέκλυθι as compared with κλῦθι, and in the two 'aorists of rebuke' ἐνένιπε and ἠνίπαπε. The isolated aorist τεταγών too has certainly derived its forcible meaning of 'seizing' as compared with Lat. *tangere* by the help of reduplication. We might conjecture the same for ἀμπεπαλών. In the case of λελάκοντο, as we have already intimated, the meaning 'howl' as compared with λακεῖν 'to sound' depends upon the same. The force of ἀκαχεῖν, ἀραρεῖν, δεδαεῖν, κεκαδεῖν is decidedly causative, and the same is especially clear in ὀρορεῖν, in λέλαχον and λέλαθον. ...if we compare the causative force of this device of language in Sanskrit aorists, and in the presents like ἵστη-μι = *sisto* its use can hardly be a matter of chance".

[10] For metrical origins see especially: ἔσπετο §1.3.2; πεπύθοιτο §3.1.1; πεπιθεῖν §3.2.1; τετάρπετο §3.3.1; λελαβέσθαι §4.1.2; λέλαθον §4.1.4.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

themselves are sufficient to show the 'artificial' quality associated with this formation in epic poetry.

§0.3 *Organization*

Although the focus of this dissertation is both formal and functional, it has seemed best to arrange the material on purely formal grounds. Hence, Chapter One describes zero grade formations of the type *Ce-CC-e/o-*, Chapter Two *Ce-CRH-e/o-*, etc. Because metrical expansion primarily effected forms of the shape *Ce-CRC-e/o-* and *Ce-CaC-e/o-*, the reader will find most of this discussion in Chapters Three and Four. Chapters Five and Six describe the vowel initial forms. Chapter Seven briefly addresses the status of the reduplicated aorist in Indo-European.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

CHAPTER I

§1.0 *Ce-CC-e/o-*

In a short paper, H. Ebel, *KZ* 2 (1853) 46-8, noted the equation between the Sanskrit reduplicated aorist *ávocat* 'spoke' and Greek εἶπε. Ebel reconstructed the Greek preform as a full grade formation Fε- Fεπ-, although it is now universally assumed that εἶπε represents a zero-grade formation with dissimilation of the second *u̯*, εἶπε < **u̯eu̯ipe-* < **u̯e-u̯kʷ-e/o-* (Rix, *Hist.Gr.Gr.* 62 & 216; Chantraine, *Gr.Hom.* 131 f.; Frisk, *GEW* I 464). Three qualities mark this formation: 1) *Ce*-reduplication; 2) a zero-grade *C(e)C* root; and 3) thematic endings. Three other aorists, ἔπεφνον 'slew', ἕσπετο 'followed', and ἕζετο 'sat', are formally equivalent (or at least can be so interpreted) and will be described below (see §1.2, §1.3.2 and §1.4 respectively). A fourth form sometimes cited in this context, ἔσπετε 'speak!', is best taken as the relic of a compound formation (see §1.3.1).[1]

§1.1 εἶπον 'I spoke'

The reduplicated aorist εἶπον, epic augmented ἔειπον, Cret. Fειπ- < **u̯e-u̯kʷ-e/o-*, is one of the few reduplicated aorists to survive into Classical Greek, no doubt because it had ceased to be recognized as a reduplicated formation. The form is ubiquitous in Homer, e.g., εἰπὲ θεοπρόπιον ὅ τι οἶσθα# 'speak the prophesy that which you know' [A 85, cf. A 68 +], as well as in later Greek. From an historical point of view, the comparison with Skt. *ávocat*, Av. *vaocaṭ* guarantees its antiquity (Pokorny, *IEW* 1135 f.; Frisk, *loc.cit.*; Chantraine, *DELG* 362). Hence, we seem to have clear evidence for at least one reduplicated aorist in Indo-European. However, before committing ourselves to this interpretations there are several possible objections which require comment.

---

[1] For *C(e)RH* roots see Chapter II.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

First, the comparison with the Indo-Iranian forms requires the dissimilation *u̯-u̯-kʷ* > *u̯-i̯-kʷ-*, a process all but unique to this lexeme. But this is only an apparent problem. Similar dissimilations exist in the history of Greek: εἴρημαι < *FέFρημαι (cf. Lejeune, *Phonét. hist.* 228; Schwyzer, *Gr.Gr.* 257; Fournier, *Verbes "dire"* 4); perhaps (F)ἑαμαι < **u̯i-i̯i̯-e/o-* < **u̯i-u̯i̯-e/o-* (as per Harðarson, *FsRix* 163) and ἀείδω (< *αFε-Fδ-ε/ο- ? cf. §5.3).[2] More importantly, Fειπ-ε/ο- is otherwise inexplicable. Full grade formations are rare in Greek thematic aorists.[3] Even if we were to assume a full grade here, the -ι- would still have no explanation. Hirt assumes, without comment, *Hb.Gr.* 552, that εἶπον is an *s*-aorist, presumably *Fεπ-σ-m̥ > *Fειπ-α > (F)ειπ-ε/ο-. But if this were the case why should *-s-* be lost after an obstruent? Further, as the inscriptional evidence

---

[2] The environment before a liquid is, of course, not entirely parallel. Further, we might expect the de-labialization of the sequence *-u̯kʷ-* as in βουκόλος < **gʷou̯-kʷol-os* (Weiss, *Studies* 153 f.). It is therefore necessary to assume two levels of change. First, the de-labialization was blocked so as to keep the form in agreement with ἕπος < **u̯ekʷ-os* etc. Next the vowel sequence *-eu-* became *-ei-* through dissimilation as above. For an important study of the chronology see A. Morpurgo Davies, *Classical Review* n.s. 20 (1970) 206 f.

[3] Not counting those forms with α-vocalism, the following aorists show apparent full grade in Homer: (*e*-grade) ἀγερ-ε/ο- 'gathered', γεν-ε/ο- 'born', ἑζ-ε/ο- 'sat', ἑλ-ε/ο- 'took', ἐνεικ-ε/ο- 'bore', ἑσπ-ε/ο- 'followed' and ἐσπ-ε/ο- 'speak'; (*o*-grade) θορ-ε/ο- 'leapt', μολ-ε/ο- 'went', πορ-ε/ο- 'gave', τορ-ε/ο- 'pierced', ὀλ-ε/ο- 'perished', γο-ε/ο- 'moaned' (see Cardona, *Them.Aor.* 55 f.; Risch, *Wortbildung* 242 f.). For ἑσπ-ε/ο-, ἑζ-ε/ο- and ἐνεικ-ε/ο- see below (§1.3.1, §1.3.2, §1.4 and §5.1, respectively). ᾽Αγέροντο is apparently formed from ἀγρόμενος for metrical reasons (Chantraine, *Gr.Hom.* 395). Γένετο is certainly an older root aorist **genh₁-to* (Risch, *loc.cit.*). The *Cor*-forms are also likely to represent laryngeal final roots (see Beekes, *Development* 233 f.; Harðarson, *Studien* 158). The etymology for ἑλ-ε/ο- is unclear but a root **selh₁-*, aorist **sl̥h₁-e/o-* > ἑλεῖν is possible (Klingen-schmitt, *Verbum* 268 n.2). A number of other *e*-grade thematic aorists in Greek may have developed in the same way (see Harðarson, *Studien* 155 f). Only τεκ-ε/ο- 'born' [*Hymn to Aphr.*] and θεν-ε/ο- 'killed' [Attic only] cannot be explained from older *set* bases. But the last named form has probably been backformed to the present θείνω (see below n. 13), and τεκ-ε/ο- is also best taken as an analogical formations. For a general discussion see P. Monteil, *FsLejeune* 139 f.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

makes clear, the ει here is not a spurious diphthong (Solmsen, *Untersuch.* 237).[4] No solution via compensatory lengthening, reinterpreted augment or long vowel is possible. Hence, even without comparative data εἶπον is best taken as a reduplicated formation.

Second, only two branches attest this formation. But again this complaint does not invalidate the comparison. Meillet's insistence on three branches for comparison arose because of the need to distinguish true archaisms from parallel developments. But the likelihood of both Indo-Iranian and Greek independently developing a reduplicated aorist for the same root is vanishingly small. A common preform **u̯e-u̯kʷ-e/o-* is the simplest solution.

Finally, the possibility exists that **u̯e-u̯kʷ-e/o-* was not an aorist in the parent language, but a reduplicated imperfect or pluperfect. Here the agreement between Greek εἶπον, clearly an aorist, and Sanskrit *ávocat*, also an aorist, seems probative. But it is worth noting that Reichelt, *Av.Elem.* 124, takes Avestan *vaocaṯ* as a pluperfect, although later, quoting this form and others, he notes: "...[d]as Plusquamperfektum hat die Bedeutung eines Imperfekets oder eines Aorists..." (*op.cit.* 312).[5] This identification can be traced to the works of Reichelt's teacher. Bartholomae, *Verbum* 83, states that the only certain reduplicated aorist in Avestan is *vac-*, but he goes on to note that *vaocā* [*Y* 45.3] is "sicher perfektisch". But *vaoca=cā* in *Yasna* 45.3 is certainly a 1sg. aorist subjunctive (cf. Insler, *GZ* 75 '...as I shall...speak of it'; Humbach, *GZ* I 164 '...I shall ...pronounce it...'). Despite Reichelt and Bartholomae, most scholars today take *vaocaṯ* as an aorist (Kellens, *Verbe avestique* 374 & 408, *CLI* 50; K. Hoffmann, *Encyclopaedia Iranica* A 61). But

---

[4] Compare especially Old Attic ειπέν = εἰπεῖν [*IG* I² 57 18] with the digraph in the first syllable against the spurious diphthong in the final syllable (where the infinitive ending *ē* descends from **-e(h)en*).

[5] With the loss of the augment and the elimination of the modal system of the aorist, Avestan has largely merged the aorist and imperfect; hence, any synchronic difference would be surprising (cf. Kellens, *Verbe avestique* 431 f.; and Reichelt himself at *Av.Elem.* 95 f.).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

even if Avestan *vaocaṭ* is synchronically not an aorist, it is unlikely that **u̯e-u̯kʷ-e/o-* was ever a pluperfect in the parent tongue.

The Greek aorist εἶπον is isolated, with a suppletive present (Homer φημί, Attic λέγω). A backformed present εἴπουσι is late and of no historical significance (Fournier, *op.cit.* 99). Only the 'alphathematic' εἰπα- has a chance of being old (cf. the Homeric 2sg. εἶπας [A 106, 108 γ 227 δ 204 π 243 χ 46], 2pl. (imperative) εἴπατε [γ 427 φ 198], and Ionic εἰπα-). Opinion concerning εἰπα- has shifted through the years. Thus, J. Wackernagel, *VBGS* 48 = *Kl.Schr.* I 809, states: "...εἶπα mit εἶπον wenigstens gleich alt...". However, later, *Spr.Unt.* 112 f. n. 2, he takes this form as an innovation. This is the standard view today (cf. Chantraine, *Gr.Hom.* 384 f.). Recently, K. Strunk, *Laryngaltheorie* 570 f. and M. Meier-Brügger, *KZ* 100 (1987) 314 n. 5, have returned to this problem, noting the existence of the Sanskrit *vócati* (*RV* I[2]; V[1]; X[2]), which seems to be a short vowel subjunctive arguing for an underlying root formation. An athematic **u̯e-u̯kʷ-m̥* could give the Greek 1sg. εἶπα 'I spoke' from which the remaining forms would then be built (cf. also Schwyzer, *Gr.Gr.* 745).

However, as to the Vedic forms, while Book Five is generally considered to be archaic, Books One and Ten contain some of the youngest hymns,[6] and, hence, do not constitute the best evidence for archaic inflection. More importantly, the Greek 1sg. εἶπα is not attested until Herodotus. Without a 1sg. ending **-m̥*, there is no reason to assume that εἶπας is a relic of an old athematic form, as opposed to a late analogical creation. That is, a purely phonological analysis requires that the first person singular or third plural

---

[6] For dating of the hymns see Arnold, *Vedic Metre* 250 f. For I 105 Arnold notes, *op.cit.* 45, some trace of epic rhythm. For late features in I 123, X 11 and X 16 see *op.cit.* 272, 45 & 285 respectively. Note, however, that Van Nooten-Holland, *RV* xii f., argue that many of the metrical criteria used by Arnold and others are illusory and based on elisions introduced in later redactions. Nevertheless, viewing books II-VIII (the so-called 'Family Books') as older has proved useful in a number of analyses of the hymns and will be followed here.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

predate the second person singular. Further, the distribution of ἐιπα- in Homer, only at A 106, 108 with all other occurrences in the *Odyssey*, in itself points to a late formation (cf. again Wackernagel, *Spr.Unt.* 112 f. n. 2). Since the thematic aorist typically shows zero grade, while full grade *s*-aorist forms take alphathematic inflection, it is likely that ειπ- adopted alphathematism once it had ceased to be recognized as a zero-grade formation.[7] The alphathematism in εἶπας, whatever its ultimate origin, is best taken as a development of Greek, and not an Indo-European archaism.

Although the nominal forms of the root **u̯ek$^{u̯}$-* are represented in several Indo-European branches, e.g., the *s*-stem in Greek ἔπος, Skt. *vácas-*, or the root noun, Lat. *vōx*, Skt. *vāk*, Av. *vāxš-* (Pokorny, *IEW* 1135 f.), the verbal forms of this root are more difficult to trace. Outside of Greek, the Latin *vocāre* is clearly derived from the noun *vōx* (Ernout-Meillet, *DELL* 753 f.). Similarly, Armenian *goč'em* 'ich rufe, schreie' appears also to be denominal in origin < **u̯ok$^{u̯}$-i̯e/o-* (Schmitt, *Gr.Kl.Arm.* 64). Hence, only Indo-Iranian and Greek (and perhaps Tocharian) can contribute to our understanding of the proto-paradigm.

In this respect, the Indo-Iranian languages offer more information as to the state of this verb in the parent language. Both Sanskrit and Iranian have more or less complete paradigms, but even here there is evidence for suppletion. Thus, Sanskrit has, in addition to *ávocat*, a passive aorist *avāci*, a present *vívakti* (epic *vakti*), passive *ucyáte*, perfect *uvā́ca*, and other derived formations (Whitney, *Roots* 151; Mayrhofer, *EWAia* II 489 f.). But some of this paradigm represents an Indic innovation.[8] Iranian lacks the reduplicated present and instead supplies the present through suppletion, Av. *mraomi* agreeing with Sanskrit *brávīti* (cf. Fournier, *op.cit.* 4). Considering that both Greek and Indo-Iranian

---

[7] Alphathematism is also found in the reduplicated πεπάλασθε (§4.2.4) and ἤνεγκον, 2sg. ἤνεγκας (§6.1).

[8] Mayrhofer, *loc.cit*, considers it possible that the present *vivakti* is also an inherited formation, although the comparative evidence is all but non-existent (see §5.4).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

offer evidence of a suppletive paradigm (cf. also Tocharian B *we-sk-* 'sagen' < **u̯ok$^{u̯}$-sḱ-*, where the present in A is suppletive), this was most likely the original situation in Indo-European, implying that both *vakti* and *vivakti* are purely Indic developments. The Indic perfect *uvâca* is also an innovation while the Avestan *vaoxəma* (< **u̯au̯kma*) [*Y* 34.5] may be a relic formation (Hoffmann, *Injunktive* 64). Finally, the aorist passive *avāci* has a counterpart in Avestan *vāci*,[9] while the present passive reflected in Avestan *vašyetē* 'is taught' may match the Sanskrit *ucyáte*, which has similar semantics (cf. Insler, *GZ* 247).[10]

Hence, the comparative evidence adduced above suggests that the root **u̯ek$^{u̯}$-* built a reduplicated aorist active and perhaps also a middle aorist **u̯ok$^{u̯}$-e*, a relic of which is attested in the Indo-Iranian passive aorist. It is unlikely that any other forms of this verb existed in the parent language, as there is considerable evidence to suggest various suppletive relationships. Indo-European, therefore, had at least one reduplicated aorist. However, we should not immediately assume that the existence of **u̯e-u̯k$^{u̯}$-e/o-* requires us to postulate a whole class of reduplicated aorists in Indo-European. Such an hypothesis cannot be maintained without the support from other demonstrably archaic forms. A number of possible candidates are presented below.

§1.2 ἔπεφνε 'slew'

The aorist ἔπεφνεν 'slew' is limited to epic diction (Veitch *GV* 590). In Homer, ἔπεφν- ε/ο- occurs in different positions in the line, although the majority of forms appear

---

[9] Although the formation of the present passive is Indo-European, the usage is largely a functional development of Indo-Iranian (cf. Brugmann, *Kurze* 527 f.). The status of the aorist passive is far from certain. Thus, S. Insler, *IF* 73 (1968) 312 f., explains the formation as an Indo-Iranian development. Alternatively, J. Jasanoff, *TIES* 2 (1988) 52 f., explains the passive aorist as a relic of an old *o*-grade aorist (see also S. Insler, *FsStrunk* 91, for a recent analysis substantially agreeing with Jasanoff, but with important critism).

[10] Note however, Mayrhofer, *KEWA* III 221: "...Av. *vašyete* wohl nicht hierher, sondern zu *vañc-*..." (also Humbach, *GZ* II.155).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

either at the beginning of the third foot (ἐ-)|πεφν- (28/61), or in line final position (20/61). Further, the form shows a degree of formulaic consistency in usage, occurring primarily in the third person singular and being all but exclusively limited to the slaying of men, e.g., πάντας ἔπεφν(ε) 'he slew them all' [Δ 397, cf. E 69, Z 12, 29, 180 +]. There are only three exceptions: 1) the Chimaera Χίμαιραν ἀμαιμακέτην ἐκέλευσε / πεφνέμεν 'he bade him kill the raging (?) Chimaera' [Z 179-80]; 2) the killing of a bull in the simile ἠύτε ταῦρον ἔπεφνε λέων [Π 487], and 3) the single instance in the third plural ἠδ' ὡς Ἡελίοιο Βόας κατέπεφνον ἑταῖροι 'and how the companions slew the cows of the Sun' [ψ 329].

Like εἶπον, ἔπεφνε is more or less isolated in Greek, having been completely separated from the etymologically related θε(ι)ν- (see below). In addition to the aorist, one finds a middle perfect πέφαται δ' ὥριστος Ἀχαιῶν 'the best of the Achaeans is slain' [P 689, cf. T 27 +], and a future πεφήσεται [O 140 +]. But this latter form is certainly secondary, perhaps having been created on the model of πεφήσεται : φαίνομαι (Buttmann, *Aus.Gr.Spr.* 273; K. Brugmann, *KZ* 24 [1879] 279; Chantraine, *Morph.hist.* 254; Watkins, *HKD* 480). Perhaps a root aorist is attested in the Hesychian gloss ἀπέφατο· ἀπέθανεν. But Frisk, *GEW* I 657 f., takes this as an error for ἐ-πέφατο, i.e., a middle corresponding to ἔπεφνε, but with a vocalized zero grade. He is certainly correct. A root aorist would be extremely unlikely where unambiguous evidence of a root present exists: Skt. *hánti* = Av. *jainti* = Hittite *kuen-zi* (Pokorny, *IEW* 491; Mayrhofer, *KEWA* III 575 f.).[11]

Etymologically ἔπεφνε and θείνω are both traced to an Indo-European root **g^{u}hen-* (Pokorny, *loc.cit.*; Frisk, *loc.cit.*; Chantraine, *DELG* 425 f.). In the parent language **g^{u}hen-* apparently meant 'strike, kill'. In Greek, however, the two stems have

[11] In addition to the forms above note also the Hesychian παφών· κτείνας which is certainly a late analogical form made on the pattern of such pairs as τέμνω : ταμών (Schwyzer, *Gr.Gr.* 749 with bib.).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

completely divided the semantic notion (P. Chantraine, *Sprache* 1 [1949] 143 f.). Thus, as seen above, ἔπεφνε means 'killed, slew' while the present θείνω means 'strike' [cf. A 588, Z 135 +]. Although the two verbs are not related in the synchronic system of Greek, they were once part of the same verbal system, presumably a virtual **g^{u}h(e)n-i̯e/o-* : **g^{u}he-g^{u}hn-e/o-*. But the status of both forms in Indo-European is far from certain. Only the root present is assured to date from the common era, although the *i̯e/o*-present of θείνω may also be old, cf. Lith. *gimi* (prt. *gýniau*) 'wehren, verteidigen', OCS *žinjǫ* 'ernten, scheiden' < **g^{u}hn̥-i̯e/o-* (Fraenkel, *LEW* 152). Other relics of an old *i̯e/o*-stem may lie in the Armenian *jnjem* 'abwischen, reinigen, abschaffen usw.' **jinj-* < **g^{u}hen-i̯e/o-* (Pokorny, *loc.cit.*; Schmitt, *Gr.Kl.Arm.* 65; some doubt in Klingenschmitt, *Verbum* 154) and perhaps Latin *-fendō* if we accept Sihler's etymology, *NCGGL* 206, with *-nd-* < **-ni̯-*, hence = θείνω.[12,13] The status of the aorist is less clear.

Since both Grassmann's Law and the changes in the inherited labiovelars were relatively late developments (cf. Lejeune, *Phonét. hist.* 43, 57; R. Janko, *Glotta* 55 [1977] 1 f.), a preform **g^{u}he-g^{u}hn-e/o-* could be the product of very early Greek, and need not be traced all the way back to Indo-European. However, the divergence in semantics strongly argues against this possibility, and a purely Greek analogical solution is very difficult to motivate. Alternatively, one could try to derive ἔπεφνε from φόνος 'murder', but there is no evidence for reduplication as a denominal formation. Hence, the internal evidence

---

[12] For an earlier discussion of this sound law see H.M. Hoenigswald, *Archivio Glottologico Italiano* 60 (1975) 55 f. (with bib.).

[13] Ernout-Meillet, *DELL* 224 f., argue that the present θείνω is recent, formed from the aorist θεν- ε/ο- , itself a relic of the old athematic present, but this is highly unlikely. The distribution of forms would seem to indicate that the aorist is the more recent form. It is true that the alternation between full and zero-grade **-i̯e/o-*presents is unusual, but this distribution is not unparalleled. Compare Indo-European **ǵher-i̯e/o-* / **ǵhr̥-i̯e/o-* 'delight' which A. Nussbaum, *Glotta* 54 (1976) 248, argues must have replaced an ablauting root present in late Indo-European (see below §4.1.3). Since we have clear evidence for a root formation here the same solution may well apply.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

suggests that ἔπεφνε is an archaism.

Again, as with εἶπον, the comparative evidence for an Indo-European **g^{h}he-g^{h}hn-e/o-* rests on Indo-Iranian. Unlike εἶπον, however, the equation here is not with a well established form, but instead with a rare form in Younger Avestan, and a participle in epic Sanskrit. The comparison of ἔπεφνε = Y.Av. *ava-jaγmaṭ* 'er schlug' can be found in the standard handbooks (Frisk, *GEW* II 657; Chantraine, *DELG* 426; Schwyzer *Gr.Gr.* 748; Rix, *Hist.Gr.Gr.* 216): *yō fraēštəm usaγanąm ... ava.jaγmaṭ* 'welcher am meisten von den sehr bösen erschlug' [*Yt* 13, 105] (trans. from Reichelt, *Av.Elem.* 312-313). Again, as with *vaocaṭ*, Reichelt takes this form as a pluperfect, but at the same time notes, *Av.Elem.* 312, that it has the usage of an imperfect or aorist. It is also possible that *-jaγmaṭ* is neither an aorist nor a pluperfect, but instead an intensive imperfect, comparable to Sanskrit *jánghanti* (Kellens, *Verbe avestique* 194). But this assumes either a rule of dissimilation (Kellens, *Verbe avestique* 195 n.4) or some other novel formation. Intensives in Avestan show either full root reduplication, e.g., *carkarə-*, or overt lengthening of the first syllable, e.g., *pāpərət* < *part-* 'fight'.

Finally, there is the participle in epic Sanskrit *jaghnant-*. Some scholars (e.g., Schwyzer, *Gr.Gr.* 748; Chantraine, *DELG* 426; Beekes, *Development* 225) have accepted the equation of *ja-ghn-ant-* = πε-φν-όντα [Π 827]. Unfortunately the lateness of both this form and the Avestan *-jaγmaṭ* seriously compromises any potential word equation to Greek ἔπεφνον. Nevertheless, I am inclined to accept this aorist as archaic, not because of the strength of word equations, but because I see no other way of generating ἔπεφνον within the early history of Greek.[14]

---

[14] The status of this form in Indo-European remains a matter of guesswork (cf. §7.3). For extensive discussion of this root and its formular usage see Watkins, *HKD* 297 f. Watkins, *op. cit.* 301 n. 9, 336 n. 3, considers ἔπεφνον a relic of an older intensive.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

§1.3.1 ἔσπετε 'speak!'

Ebel, *loc.cit.*, refers to two nearly isomorphic aorists ἔσπετε 'speak!' and ἕσπετο 'followed' (as if from *σε- σπ- ε/ο- < **se-sk$^{\mu}$-e/o-*). For the first form there is all but universal agreement that it is not a reduplicated aorist (Chantraine, *DELG* 349 f.; Frisk, *GEW* I 520; E. Risch, *ZPE* 60 [1985] 1 f.). The imperative ἔσπετε [B 484, Λ 218, Ξ 508, Π 112] occurs alongside the more common aorist ἐνισπε [B 80 +], imperative ἐνίσπες [Λ 186, Ξ 470], ἐνίσπε (10X *Odyssey*). The verb also shows a present ἔννεπε 'speak!' [α 1, B 761, cf. Θ 412], ἐνεπ- ε/ο- (1X *Iliad*, 5X *Odyssey*) and two future formations ἐνίψω [β 137 +] and ἐνισπήσω [ε 98].

Even without comparative evidence, the contrast between the sg. ἐνίσπε[ς] : pl. ἔσπετε, as well as the present forms in ἐν(ν)-, implies that ἔσπετε descends from an earlier *ἐν(ι)- σπ- ε/ο- (Chantraine, *loc.cit.*; Frisk, *loc.cit.*). The Latin *inseque* 'speak' (also from **en-sek$^{\mu}$-e/o-*) confirms this (Walde-Hofmann, *LEW* I 702 f.; Ernout-Meillet, *DELL* 318). A *tertium comparationis* may lie in the Old Irish *insce* 'speech', also from the compound < **en-sek$^{\mu}$-* (Pokorny, *IEW* 897). Since the preverb is used in all other attested forms of this verb in Greek (as well as the Latin cognate and Irish noun), it is certain that ἔσπετε is simply a reduction of *ἐν- σπ- and the plural of ἐνίσπες (again cf. Chantraine, *loc.cit.*; Frisk, *loc.cit.*.).[15]

§1.3.2 ἕσπετο 'followed'[16]

Many of the older handbooks follow Ebel, *loc.cit.*, in his designation of the rare ἕσπετο as a reduplicated form, as if from *σε- σπ- ε/ο- < **se-sk$^{\mu}$-e/o-* (Curtius, *Grk.*

---

[15] The Hesychian σπέτε· εἴπατε may indicate an uncompounded relic, although more likely this is a backformation.

[16] Note that the older equation of active ἕπω, middle ἕπομαι is a conflation of two separate verbs, the former of which is best traced to a root **sep-* (see most recently, B. Vine, *IF* 93 [1988] 52 f.).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

Verb 294; Schwyzer, *Gr.Gr.* 748; Chantraine, *Gr.Hom.* 395; Pokorny, *IEW* 896). This aorist is found in Homer (16X *Iliad*, 7X *Odyssey*) and occasionally in later poetry (Veitch, *GV* 231 f.; B.K. Braswell, *Glotta* 58 [1980] 203 f. for the forms in Pindar). Significantly, the aorist ἔσπετο also occurs alongside a zero-grade thematic aorist σπ- ε/ο- (4X *Iliad*, 9X *Odyssey*). The simple thematic is generally compounded, e.g., μετα- σπόμενος [N 567], although the uncompounded imperative occurs once in Homer, σπεῖο [K 285], and there are three instances of the simple infinitive σπέσθαι [E 423, δ 38, χ 324].[17] The relationship between these two forms has been a constant source of controversy.

Before Ebel's analysis, the standard approach took σπεῖο as primary with ἔσπετο as a zero-grade thematic aorist with aspirated augment (e.g., Buttmann, *Aus. Gr.Spr.* 173). More recently, this older view was revived by Debrunner, *GsKretschmer* I 81 f., who shows that the evidence for a reduplicated form is slender indeed. In Homer there are no augmented forms of ἑσπ- ε/ο-, i.e., *ἐἰσπόμην < *ε- σε- σπ-*e/o*-. Further, all examples of ἔσπετο in Homer are either in the indicative or after elided vowel, e.g., ὁ δ' ἑσπόμενος περὶ δουρὶ / ἤσπαιρ' 'he, following after the spear tug, expired' [N 570]. Given these considerations, Debrunner argues that ἑσπ- ε/ο- is a late creation, a modification of the thematic σπ- ε/ο-, augmented and with secondary aspiration so as to keep the aorist in agreement with the present ἕπομαι. Frisk, *GEW* I 544 f., accepts this analysis as does Chantraine, *DELG* 361 f., although the latter scholar still admits the possibility of a reduplicated aorist to this root (cf. especially the wording in his *Morph.hist.* 175).

A number of scholars go further and defend the reduplicated analysis (Schwyzer,

---

[17] The numbers here are subject to some variation. Thus, I. Bekker removed all non-indicative forms of ἑσπ- ε/ο- from his 1858 edition of Homer, e.g., ἐπὶ δὲ σπώνται for (Allen's) ἐπὶ δ' ἑσπωνται [μ 349] (see B.K. Braswell, *op.cit.* 205 n. 1; also Veitch, *GV* 231 f.; *LfrgE* [II] 653 f. for editorial practice). For the Hellenistic examples see the discussion in Veitch, *loc.cit.*

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

Gr.Gr. 748; Pokorny, *loc.cit.*; Forssman, *Unters.* 3 n. 1; J.B. Hainsworth, *Comm.Il.* III 183; R. Janko, *Comm.Il.* IV 116). Although I do not find this view convincing, there is some circumstantial evidence in its favor. First, the distribution of forms in Homer, with ἐσπ- ε/ο- on the decline and σπ- ε/ο- on the rise would seem to indicate that ἐσπ- ε/ο- is the more archaic (Chantraine, *Gr.Hom.* 388: "...la forme à redoublement semblant plus ancienne").[18] More importantly, a number of instances of ἐσπ- ε/ο- in Pindar cannot be explained in the manner envisioned by Debrunner (B. Forssman, *loc. cit.*; B.K. Braswell, *loc.cit.*). Finally, in addition to the evidence of distribution, some scholars have seen a word equation between ἐσπ- ε/ο- (**se-sp-e/o-*) and Sanskrit *saśc-* (Schwyzer, *loc.cit.*; Pokorny, *loc.cit.*). It is this last point which appears to offer the strongest evidence for a reduplicated aorist, although a quick survey of the comparative evidence will show that the word equation between ἐσπ- ε/ο- and Sanskrit *saśc-* is an illusion.

In the present this verb was apparently a medium tantum in the parent speech: ἕπομαι = Latin *sequor* = Skt. *sácate* = Av. *hacaite* = Irish *sechur* = Lith. *sekù* all from **sekᵘ-e/o-* with their respective endings (Pokorny, *IEW* 896 f.; Frisk, *loc.cit.*, Chantraine, *loc.cit.*; Walde-Hofmann, *LEW* II 519 f.; Ernout-Meillet, *DELL* 616; Mayrhofer, *KEWA* III 417 f.; Goto, *I Prs.* 319; Fraenkel, *LEW* 773). An active thematic present is also attested in Gothic *saihvun* 'see' if this form is cognate (cf. Sihler, *NCGGL* 449, "[i]n some forms the middle developed a special sense, for example **sekʷetor* orig. *'keeps in sight' (root **sekʷ-* 'see, view') but already in PIE with the sense of 'follows'").[19]

---

[18] Schwyzer, *loc.cit.*, notes that σπ- ε/ο- is found mostly in compound verb formations, e.g., ἐσπ- ε/ο-, but ἐπι- σπ- ε/ο-. It is tempting to compare the Latin treatment of compound verbs such as *ce-cīd-ī* but compounded *oc-cīd-ī*. Needless to say, however, no such explanation is possible. The loss of reduplication in compounds in Latin has its starting point in vowel syncope, e.g., *occīdī* < **ob-ke-kaid-ai* (Sihler, *NCGGL* 580) and has no parallel in Greek.

[19] For further discussion of the possible connections, as well as the relationship of these forms to the homonym **sekᵘ-* 'say', see also P. Baldi, *JIES* 2 (1977) 77 f.; Risch, *op.cit.* 7

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

In Indic, a perfect *saściré*, a root aorist *sacāna-*, an *s*-aorist *asakṣata* and the active present *sí-ṣakti* (3sg.) : *sáścati* (3pl.) are all attested in addition to the original present reflected in *sácate* (Mayrhofer, *loc.cit.*). Of these forms at least the *s*-aorist and reduplicated present are datable to the Indo-Iranian era. Compare Avestan *haxšāi* 'to follow' [*Y* 46.10] and (athematic) *ā.hiš.haxti* [*Vd* 5.34] : (thematic) *hišcamaidē* [*Y* 40.4] 'we follow'.[20] But the reduplicated present probably represents an Indo-Iranian innovation since one would not expect to find any active forms to this root (assuming the bifurcation of active and middle **sekᵘ-* 'see/ follow' is correct). Even aside from such *a priori* assumptions, there are problems with a direct equation between Greek ἑσπ- ε/ο- and Sanskrit *sašc-*. To begin with, matters are complicated because the Vedic *sac-* is a conflation of two separate forms, [1]*sac-* 'follow' and [2]*sac-* 'stocken, versiegen' (Joachim, *Mehrfachpr.* 163; Mayrhofer, *KEWA* III 417 f. with bib.). Of the reduplicated forms of [1]*sac-* most are reduplicated in *-i-*, e.g., *sí-ṣakti* = *-hiš.haxti*.[21] The majority of forms which do show *a*-reduplication can generally be traced to [2]*sac-*, and have no bearing on an Indo-European **sekᵘ-* 'follow'. A few instances of *sašc-* 'follow' certainly do occur: *yásyéndrasya síndhavaḥ sáścati vratám* 'the rivers obey the command of Indra' [*RV* I 101.3]. But, the athematic nature of the form makes any comparison with Greek ἕσπετο unlikely. Most likely this form has been built to the older perfect stem (Debrunner, *op.cit.* 83; Mayrhofer, *loc.cit.*).

---

n. 22 (with bib.).

[20] Kellens, *Verbe avestique* 185, takes this latter form as a subjunctive.

[21] The status of the *i*-reduplication in *siṣakti* is subject to a number of possible interpretations. Harðarson, *Studien* 36 & 28 n. 14, assumes that the *i*-reduplication has replaced the older forms in *-a-* in both Sanskrit and Iranian (cf. also Joachim, *loc.cit.*). Goto, *I Prs.* 320 n.771, considers it possible that *síṣak-ti* : *sáścati* descend from the same archaic paradigm with *-i-* before the "konsonantisch anlautend Endung" and *-a-* before the consonant group *-śc-*, originally with the thematic ending *-e-*. Finally, one additional approach takes the reduplicated forms (*siṣakti* and *ā.hiš.haxti*) instead with the root *sajati* (cf. Mayrhofer, *KEWA* III 418, with reference to Burrow).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

More importantly, while the evidence for a reduplicated **se-sk$^{\mu}$-e/o-* is all but non-existent, Indo-Iranian offers solid evidence for a root aorist. Thus, the Avestan shows the 1pl. *haxma* (Cardona, *Them.Aor.* 54; Harðarson, *Studien* 120), perhaps a 1sg. subj. *hācā* [*Y* 46.1],[22] and a 3pl. *scantū* 'let them follow' [*Y* 53.2] (Kellens, *Verbe avestique* 318 "peut-être d'un thème d'aoriste"). Sanskrit also shows evidence for a root aorist, although only in the single middle participle *sacāná-*. For our purposes the most significant form is the zero-grade middle *scantū*, reflecting a zero-grade **sk$^{\mu}$-éntu*, which matches the Greek zero-grade thematic middle σπεῖο.[23] In view of the above it is likely that ἔσπετο is a late creation of Greek built to an augmented *ε- σπ- ε/ο- just as Debrunner explains it.

A final question remains: when did the replacement of the older ἔ- σπετο with ἔσπετο occur? Certainly, both the ambiguity of the Homeric attestations and the absence of this form in prose might imply that ἔσπετο is a purely redactional change, perhaps a product of the Alexandrian era (or even later), and, hence, a 'ghost form'. But this analysis is contradicted by the irrefutable existence of ἔσπετο in Pindar [*Ol.* 8.11, +]. Hence, ἔσπετο and σπεῖο exist side by side, but only in epic or stylized poetry. The inescapable conclusion is that ἔσπετο is the newer form, but this form did not replace the simple thematic entirely, because the older σπεῖο was still useful in the Kunstsprache as a metrical variant (cf. again Braswell, *op.cit.* 213).

---

[22] The status of this form is unclear. Insler, *GZ* 80 and 263, noting that the verse lacks a syllable, takes this as error for the perfect [*hā-*]*hācā* (=**ha-haca*). Kellens, *Verbe avestique* 105 f. n. 30 and 354, gives the root aorist interpretation, but with some hesitation.

[23] For the origin of the zero-grade thematic aorist from the weak stem of older athematic forms see Cardona, *Them.Aor.* 71 f. (with bib.). For the possible relationship between the root aorist and the full grade present see below §3.1.1 f.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

§1.4 ἕζετο 'sat'

The aorist ἕζετο occurs throughout Homer, e.g., ὣς εἰπὼν κατ' ἄρ' ἕζετο 'having spoken thus he sat down' [A 68, 101, 246, +]. Schwyzer, *Gr.Gr.* 652 n. 5, gives three possibilities for the origin ἕζετο: 1) an original imperfect from **sed-i̯e/o-*; 2) an augmented thematic aorist **e-sd-e/o-* with aspiration introduced from the present ἵζει; 3) a reduplicated aorist **se-sd-e/o-*. The first possibility can be immediately discounted, since the present ἕζεαι is demonstrably late, with a single occurrence in the *Odyssey*: τίφθ' οὕτως,Ὀδυσεῦ, κατ' ἄρ' ἕζεαι ἶσος ἀναύδῳ 'why do you, Odysseus, sit thus like to one mute...' [κ 378]. Further, the standard present in Homer is the reduplicated present ἵζει 'sets' [N 281 +], intransitive 'sat' [cf. B 53, +] < **si-sd-e/o-*, which is certainly old, cf. Lat. *sīdō* = Skt. *sídati* < **sīḍati* < **si-zd-e/o-*, cf. Pokorny, *IEW* 884 f.; Frisk, *GEW* I 445; Chantraine, *DELG* 313 f.; Walde-Hofmann, *LEW* II 509; Ernout-Meillet, *DELL* 611; Mayrhofer, *KEWA* III 472 f.). Most likely ἕζεαι is simply a back-formation from the aorist (E. Risch, *Gnomon* 37 [1965] 3).[24]

Cardona, *Them.Aor.* 85. f., argues for the second possibility, an augmented thematic aorist **e-sd-e/o-* with secondary aspiration analogous to ἕσπετο < **e-sp-e-to* (see above). But in his analysis of ἕσπετο, Debrunner showed that the distribution of forms in Homer implied that ἕσπετο was secondary to the unreduplicated thematic σπεῖο. No such simple distributional factors govern the use of ἕζετο. Further, unlike ἕσπετο, ἕζετο has a much wider use in Greek literature, including occasional uses in prose (e.g., Herodotus IV 168, cf. Veitch, *GV* 188).

The third possibility, that ἕζετο is an archaic reduplicated aorist, may be considered the standard theory (cf. Risch, *Wortbildung* 243; G. Meiser, *FsRix* 284 n. 24; note, however, that both Frisk, *GEW* I 446, and Chantraine, *DELG* 314, admit either a redupli-

---

[24] Klingenschmitt, *Verbum* 196 (with bib.), takes Armenian *hecani-* from **sed-i̯e/o-* = Greek ἕζεαι. But the Greek form can hardly be old and the Armenian form could be explained in a number of ways. See below for an alternative analysis.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

cated or zero-grade preform with secondary aspiration). There are certain arguments which support an interpretation as a reduplicated formation, although none is probative.

First of all, a word equation between ἕζετο the Avestan *hazdyāṭ* is often cited in the comparative literature (Schwyzer, *loc.cit.*; Chantraine, *DELG* 314). The form occurs only once in a Younger Avestan text [*Yt* 13 13]

> *nī antarə ẓam asmanəmca druj manivå hazdyāṭ*
> *nī antarə ẓam asmanəmca druj manivå vaonyāṭ*
> le trompeur d'entre les deux esprits se serait installé entre la terre et le ciel,
> le trompeur d'entre les deux esprits aurait été vainqueur entre la terre et le ciel
> (trans. from Kellens, *Verbe avestique* 422).

Clearly *hazdyāṭ* is in a parallel construction with the optative *vaonyāṭ*; hence, it is likely that either both forms are aorists or both forms perfects. But since each form is unique, neither designation has any overriding authority. All things being equal, it is far more likely that these forms are perfect optatives. Reduplicated aorists are, at best, rare in Iranian. Further, the reduction of the modal system of the aorist in Avestan (Kellens, *Verbe avestique* 431) would require that both *hazdyāṭ* and *vaonyāṭ* were relics of the oldest sort. Finally, both *hazdyāṭ* and *vaonyāṭ* are athematic, while Avestan shows clear evidence for a thematic reduplicated optative in *vaocōimā=cā* [*Y* 35.3],[25] so that we might expect a reduplicated ***hazdōiṭ*. Hence, if we knew that ἕζετο was a reduplicated aorist preserved from Indo-European we could perhaps use this to explain the Avestan form. However, the equation is too weak to be used the other way around, in support of the antiquity of ἕζετο.

G. Klingenschmitt, *KZ* 92 (1978) 1 f., gives a rather involved argument in favor of a reduplicated aorist. He begins by stating that Narten (i.e. acrostatic) presents should

---

[25] Cf. also *vāurōimaidī* (1pl. opt.) [*Y* 28.5c] which is likely to be another Avestan reduplicated aorist (Insler, *GZ* 126 and below §2.1.1).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

have acrostatic iterative-causatives of the shape *CōRC-i̯e/o-*, not *CoRC-e̯i̯e/o-*, cf. Latin *sōpiō* 'einschläfern'. However, O.Ir. *sáidid* 'sit' seems to come from an Indo-European **sōd-e̯i̯e/o-*. In order to avoid positing a second variety of 'Narten' causative-iteratives, Klingenschmitt postulates that the lengthened grade was backformed from a reduplicated aorist of the shape **sēd-e/o-* < **se-sd-e/o-*.[26] However, this line of reasoning is not probative. Even if we were dead set against positing an Indo-European *CōRC-e̯i̯e/o-*, the length could be analogized from elsewhere.[27]

In his monograph, Klingenschmitt, *Verbum* 130 f., carries the argument further, suggesting that not only the lengthened grade forms of this root, Lat. *sēdāre*, O.Ir. *sáidid*, Lith. *sēdmi*, are descended from a reduplicated formation, but perhaps the short vowel forms as well. Thus he explains Sanskrit *a-sád-a-t* as "eine Umbildung des wegen seiner vermeintlichen Langstufe abnormen **sáda-*..." (cf. also Pokorny, *loc. cit.*, who sees a possible reduplicated form behind the intransitive *e*-grade present: Lat. *sedeō*, Umbrian **zeřef** 'sedens').

Finally, Klingenschmitt, *loc.cit.*, notes that the relationship between reduplicated present and aorist **si-sd-e/o-* : **se-sd-e/o-* has a parallel in Greek πι-πτ-ε/ο- : Skt. *pá-pt-a-*. But the relationship of reduplicated present to reduplicated aorist need not be old. The existence of any reduplicated aorists, say **u̯e-u̯kʷ-e/o-*, would make possible an analogical **se-sd-e/o-* built to **si-sd-e/o-* but with *Ce-* for *Ci-*. Klingschmitt's arguments are interesting but do not constitute proof.

More importantly, Klingenschmitt's attempt to explain away the root aorist **sed-t(o)* may be misdirected. Thus, C. Barton, *IF* 84 (1989) 147, convincingly shows that

---

[26] Tracing the lengthened grade forms of *sĕd-* to an older reduplicated formation is, of course, not a new idea and goes back to A. Holtzmann in his 1844 treatment of ablaut (for a complete discussion of the history of this idea, with bib., see Cowgill, *Diss.* 21 f.).

[27] Compare, e.g., Jasanoff, *SMIE* 114 f., who explains the lengthened grade of the Balto-Slavic forms (e.g., Lith. *sēdėti*) from an Indo-European reduplicated perfect middle.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

Armenian *hecay* is best explained as a neo-sigmatic aorist **sed-s-* < **sed-* comparable to *etu* < **e-dō-so-m* (> OCS *daxŭ*).[28] The existence of a root aorist in pre-Armenian gives a word equation to the Sanskrit **a-sad-t(o)* presumed on the basis of the subjunctive *sádathas* [*RV* V 67.2] (cf. G. Cardona, *Language* 39 [1963] 14 f.). Hence, the existence of an Indo-European root aorist **e-sed-t(o)* renders our **se-sd-e/o-* a redundant formative in the parent language. Given these considerations it may be best to consider the Greek aorist as an innovation of epic diction, built in analogy to the reduplicated present.[29]

The development of **se-sd-e/o-* built to **si-sd-e/o-* is all but identical to Cardona's ἐζ- ε/ο- < *ζ- ε/ο- after ἵζει with two slight variations. First, the formation is posited at an earlier period. Second, the formation need not involve the augment. These distinctions are necessary precisely because ἐζ- ε/ο- does not show the ambiguous metrical positioning of ἐσπ- ε/ο- where the change from ἐ- σπ- ε/ο- to ἐσπ- ε/ο- could even be post Homeric.

§1.5 Conclusions on *Ce-CC-e/o-* aorists

Of the five forms above, εἶπον, ἔπεφνεν, ἔσπετε, ἔσπετο, and ἔζετο, a reduplicated aorist is reconstructible for εἶπον < **u̯e-u̯k̑-e/o-* and perhaps ἔπεφνεν < **gʷhe-gʷhn-e/o-*, although the comparative evidence is significantly weaker for this second form. Of the remaining forms only ἔζετο < **se-sd-e/o-* can be even synchronically classed as a reduplicated formation. But here the formation could just as easily represent a late analogical creation after **si-sd-e/o-* as opposed to a relic from some older epoch.

---

[28] A key point here is that Klingenshmitt's proposed **sed-i̯e/o-* (above n. 20) should have yielded an Armenian present **hecem* (again cf. C. Barton, *op.cit.* 148 n. 43; also Harðarson, *Studien* 103 n.44) See also F. Kortlandt, *AAmL* 15 (1994) 27 f., for arguments against deriving a newer nasal present from an older *i̯e/o*-formation.

[29] Although we lack the forms for a real 'four part' analogy, this may still be out best solution.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

CHAPTER II

§2.0 *Ce-CRH-e/o-*

A number of reduplicated aorists are built to *CRC* roots. The majority of these forms are described below in Chapter III. There is some evidence to suggest, however, that roots in a final laryngeal were treated differently. Three such roots are dealt with here (§2.1), as are a number of forms sometimes described as roots in final $*h_3$ (§2.2).

§2.1.1 εὗρον 'I found'[1]

Like εἶπον, εὗρον is frequent in Homer and later poetry (*Iliad* 48X, *Odyssey* 41X).[2] And like εἶπον it is one of the few reduplicated aorists to exist in prose as well as poetry. Finally, and not unrelated, is the fact that like εἶπον, εὗρον is not recognizable as a reduplicated formation and its identification as such has been the subject of considerable and skeptical analysis. In addition to this aorist, only a present εὑρίσκω [τ 158] occurs in Homer, although the future εὑρήσω [*Hymns*, Hesiod +] and perfect εὕρηκα [Herodotus +] forms are also likely to be old.[3]

The standard handbooks list εὗρον as a reduplicated formation from **(u̯)e-ur-e/o-*, citing as a possible comparandum the Old Irish *·fuar* 'id.' < **u̯our-* < **u̯e-u̯r-e/o-* (Schwyzer, *Gr.Gr.* 709 note 2.; Frisk, *GEW* I 592; Chantraine, *DELG* 387).[4] However, in

---

[1] An earlier version of the section appeared in *Glotta* 72 (1994[95]) 24 f. I have not altered my central conclusions; nevertheless the text here contains a number of small changes and supersedes the published version.

[2] For a discussion of the metrical distribution of forms see below §2.1.3.

[3] The future is attested in both Ionic and Doric and is therefore probably datable to common Greek. I assume the perfect also dates from that era. The present εὑρίσκω, however, is certainly a late addition to the paradigm, cf. B. Vine, *HS* 106 (1993) 53 n. 22. The present forms in -ίσκω are discussed in detail in §6.6.

[4] See also Walde-Pokorny, *VWIS* I 280. Pokorny, *IEW* 1160, takes *·fuar* as descended

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

his 1980 book, M. Peters, *Untersuchungen* 22 f., discards the traditional equation of Greek εὗρον 'I found' = Old Irish *·fuar*. While accepting Old Irish *·fuar* from **u̯e-u̯r-e/o-*, Peters offers several analyses of the aorist εὗρον, but to my mind none are as persuasive as the older explanation. To be sure there are problems with any analysis of this form. Phonologically, the Greek εὗρον would seem to imply a pre-form **(h)eur-e/o-* or a reduplicated **(u̯)e-ur-e/o-*. But, as seen above, the evidence for a reduplicated aorist in Indo-European is scanty at best. Further, the Greek pre-form does not seem to be quite what we expect based on the extra-Greek cognates. Before going on to discuss the reduplicated formation, let us examine the possible cognates in question as well as the competing theories.

The Old Irish *fríth* 'found' is generally considered to descend from an Indo-European **u̯reh$_1$-tó-* (Thurneysen, *GOI* 428; Pedersen, *Vgl.Gr.* II 368; Pokorny, *IEW* 1160). An Indo-European root **u̯reh$_1$-* with a dental enlargement also underlies Old Church Slavic *obrětŭ* 'I found' < **u̯reh$_1$-t-* and the Lithuanian *randù* 'I find' < **u̯rh$_1$-n-d(h)-* (Pokorny, *loc.cit.*; Klingenschmitt, *Verbum* 125.).[5] However, the Greek aorist εὗρον and future and perfect stem εὑρη- cannot be directly equated. First, the Greek forms show no trace of the initial *u̯-*, but instead show rough breathing and an apparent full grade vowel followed by *u̯*. Second, although the future and perfect forms, εὑρήσω and εὕρηκα, may indicate a *set* root, the aorist shows no trace of the root final laryngeal. No obvious preform will work here. The zero grade **u̯rh$_1$-e/o-* would have given Greek *Fαρε : *Fαρον, equivalent to, say, ἔβαλον, or perhaps *Fερε : *Fερον.[6] Nor would

---

from a reduplicated formation, but is unsure of εὗρον "...mit unklaren ευ...". For the Irish forms as well as an extensive discussion of this root see Francis, *Disyl.Roots* 168 f.

[5] Vasmer, *REW* II 244, prefers to connect the Slavic forms with Gothic *wraton** 'travel', but the semantic mismatch makes such an approach unlikely (cf. also Fraenkel, *LEW* 700).

[6] For this view of the development of *CRh$_1$-V-* see Beekes, *Development* 221 f.; Harðarson, *Studien* 156 f. & 213.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

a reduplicated formation seem to fare much better (see below).

Alternatively, it is possible that εὗρε is in origin a root aorist which was later reshaped into the attested form. Thus, Cardona, *Them.Aor.* 72 f., believes an archaic root aorist **e-u̯rē* (3 s.) : **e-uren(t)* (3 p.) formally looked too much like a passive aorist, but because this form was transitive, a newer **(h)e-ur-e/o-* replaced **(h)e-urē*. However, as Francis, *Disyl.Roots* 168 f., points out "...**éurē* could be more effectively characterized as transitive by a sigmatic **eu̯rēsē*". Further, Cardona's paradigm **(h)eurē* (3 s.) : **(h)euren* does not stand up to close scrutiny. Although there is some evidence that the augment could be resegmented in this way (as in ἐσπ- ε/- above §1.3.2), there is no evidence for such a resegmentation here. More importantly, however, since **(h)eurē* is a *se̯t* formation, we expect a third plural (as above) **(h)eu̯aren(t)* < **-u̯rh$_1$-ént*.

Given these problems, Peters, *op.cit.* 52, is not sure that εὗρον belongs with the root of the Irish forms,[7] but notes that the root etymology could be saved if one accepts one of two metathesis rules. Thus, he considers it at least possible that εὗρε descends from **u̯érh$_1$-t* with "eine optionale Metathese **u̯eC-* > **eu̯C-*" (*op.cit.* 23). He seems more inclined, however, to accept a slightly different metathesis rule: one originally proposed by H. Rix to explain the nearly homophonous εὐρύς 'wide'. H. Rix, *FsSzemerényi* 737, derives εὐρύς from **u̯rh$_1$-u-*, through the application of a rule such that the Indo-European sequence **u̯rh$_1$V* becomes pre-Greek **h$_1$urV* (also Peters, *op.cit.* 31 after Rix; G. Klingenschmitt, *KZ* 92 [1978] 6). This same metathesis rule would also explain our εὗρον from a thematic aorist **u̯rh$_1$-e/o-*.

However, this analysis has several failings. First, neither Rix nor Peters offer any

---

[7] See also J. Taillardat, *Revue de philologie* 34 (1960) 231 f., who argues that εὗρον descends not from **u̯reh$_1$-*, but from an otherwise unattested root **su̯er-*. Such an approach cannot really be argued with, but it is strongly counter-intuitive. Given a well attested Indo-European root **u̯reh$_1$-* with the proper meaning, it is difficult to ignore the comparison with Greek εὗρον.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

other example of this supposed law. Second, the metathesis rule of ABC -> CAB is unlikely both from a cross-linguistic view,[8] and from what we expect in Indo-European.[9] Third, there is no evidence for a zero-grade aorist here. Although Cardona's attempt, *op.cit.*, to explain away all thematic aorists except **h₁ludh-e/o-* and **u̯id-e/o-* may have been too restrictive, it is clear that this was a rare formation in Indo-European. It will not do simply to posit a new thematic aorist for which there is no evidence. Fourth, a possible counterexample may exist in Greek (F)αρήν 'sheep' < **u̯rh₂ēn* (for the final laryngeal see Nussbaum, *Head and Horn* 286).

This brings us to the older analysis, attributed to Windisch,[10] which sees εὗρον as a reduplicated aorist. But this analysis has its own problems. Thus, the pre-form **u̯e-u̯rh₁-e/o-* might have been expected to give **u̯e-i̯rhe/o-*, with the labial dissimilation of εἶπον, which should in turn have yielded **u̯e̯i̯ar-e/o-* > Greek **ἔαρον*. But the labial dissimilation may have worked differently before *-r-*.[11] Moreover, as noted above, there is one additional piece of evidence: the Old Irish preterite *·fuar* 'I found' *·fuair* (3sg.) < **u̯ou̯r-* < **u̯e-u̯r-e-t*.

Now if we assume that the initial **u̯-* of Greek εὗρον was dissimilated in a manner

---

[8] A metathesis rule X...Y -> Y...X is possible, but such changes typically require like sounds (e.g., Spanish *milagro* < Old Spanish *miraglo*) which is hardly the case for **u̯* and **h₁*. For a general theoretical discussion see R. Ultan, *Universals of Hum.Lg.* II 376 f. & esp. 381.

[9] Consider the pre-consonantal metathesis rule sometimes proposed for this same sequence *(C)u̯rC* > *(C)ruC* as in **kʷetu̯r-C* > *kʷetru-* (Mayrhofer, *Idg.Gr. I/2* 161).

[10] For a discussion of the Irish forms see E. Windisch, *KZ* 23 (1877) 218. His analysis of the Greek forms was never published, but the attribution can be found in older handbooks.

[11] However, there is also the evidence of εἴρημαι < *FέFρημαι. Frisk, *GEW* I 592, assumes that the aspiration is analogical, and given such forms as ἁλίσκομαι (Arcadian F-), with aspiration after αἱρεῖν, this may be the best solution. Note that the competing theories, other than Taillardat's, are also forced to explain the initial aspiration by analogy and do not, in this respect, offer a better alternative.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

distinct from εἶπον we are in better shape, although the missing final laryngeal still needs to be explained. Here, however, rather than a problem, our missing laryngeal may help confirm our etymology. The significant point here is that the Irish preterite is also atypical in that it cannot descend from a preform **u̯reh₁-* which would presumably have given **·foair*, but like the Greek εὗρον must come from a reduplicated *aniṭ* root **u̯e-u̯r-e/o-*. Hence, both Greek and Irish show the same pattern: a reduplicated *aniṭ* aorist contrasting with an otherwise *seṭ* root as in the perfect forms εὕρηκα : *fríth*.

It is this very contrast which assures the etymological connection between the Greek and Irish forms, despite the unexplained initial of Greek εὗρον. It is extremely unlikely that both languages would have independently developed a reduplicated formation which differed from the other forms in their respective paradigms. Why should the reduplicated aorist have developed without the laryngeal of the perfect forms? As it turns out, this form is entirely predictable from what we now know of Indo-European phonology, although it depends on a rule which has left few traces in the western Indo-European languages.

A zero grade formation **Ce-CRh₁-e/o-* would first have given **Ce-CR-e/o-* directly with loss of the laryngeal after sonant in a reduplicated formation or compound (for recent formulations see Mayrhofer, *op.cit.* 129; R. Beekes, *Laryngaltheorie* 60). The existence of such a rule has been known for over a hundred years (J. Schmidt, *KZ* 25 [1881] 50 f.). However, the exact status of this rule in Indo-European and its specific environment is still a subject of dispute. There is ample evidence in Indo-Iranian for the loss of root final laryngeals in compounds. One can cite such forms as Sanskrit *devá-ttas* 'von Gott gegeben'; *ā-ttas* 'genommen' with *-tt-* < **-dt-* not **ditas*, cf. aorist *adita* = ἔδοτο, or *āhutis* 'Anrufung' from the *seṭ* root *hū* (see Wackernagel-Debrunner, *Ai.Gr.* II 82 f. & 93 f.). Apparently, in Sanskrit any root final laryngeal was lost in a compound, regardless of the environment, although many have been restored analogically.

F.B.J. Kuiper, *Sprache* 7 (1961) 14 f., has analyzed the data in detail and con-

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

cluded that this is a purely Indo-Iranian phenomenon. In one sense he is probably correct, as it is likely that the original rule has been extended in Indic. Nevertheless, some forms clearly can be dated to Indo-European. Greek offers at least ἴγκρος 'brain' < **en-kr-* < **en-krh₂-*[12] and νεογνός apparently cognate with Gothic *niuklahs* < **-knahs* < *-*ǵn(h₁)-e/o-*, also the similar Phrygian ουεγνω < **su̯e-gno-*.[13] Note also the Latin deity *Cōnsus* < **com-dh-to-* < **com-d*h₁-to-* (Schrijver, *Reflexes* 328 f.).

Matters are slightly more complex for reduplicated formations. The standard example is γίγνομαι, which had long puzzled scholars (cf. Cowgill's attempt, *Evidence* 148, to argue that this Greek present did not descend from Indo-European **ǵenh₁-*!). This example and perhaps also πίμπλαμεν 'we fill' = Sanskrit *ápiprata* 'füllte' < **-pl̥(h₁)-* point to **h₁* (cf. Mayrhofer, *op.cit.* 129; Beekes, *op.cit.* 60 f.; Weiss, *Studies* 151 after Peters), while the status of root final **h₂*, **h₃* is not yet clear. But note Jasanoff's explanation, *FsPolomé* 299 f., for O.Ir. *boí* < **bewe* < **bebwe* < **bhebhuh₍e*. He offers, (p.c.), the same explanation for Sanskrit *bhavati* 'is' < *bhe-bhu̯e-(ti)* < **bhebhuh₍-e* 'has become'.

Using this 'Law of Laryngeal Loss'[14], our present understanding of Indo-European

---

[12] Nussbaum, *Head and Horn* 72 ff., has discussed this first form at length and offers an alternative solution. Noting the existence of φαλακρός 'bald' < *φαλα- κ- ρος, he argues this may have been reinterpreted as *φαλα- κρ- ος 'bald head' yielding a new morpheme - κρ- . In the end, however, after extensive discussion, Nussbaum himself accepts the - κρ- as a reduced form of **-krh₂-*. Further, no such analogical solution will work for νεογνός.

[13] A possible counterexample may exist in the comparison of Sanskrit *abhvà-* 'monster' < 'Unwesen' < **n̥-bhu̯u-o-* < *-bhuh₍-o-* (cf. F.B.J. Kuiper, *Lingua* 11 [1962] 225 f.) and Greek ἀφύη 'Bratfischchen' < **n̥-bhuh₍-o-* (for a recent discussion see M. Meier-Brügger, *MSS* 52 [1991] 123 f.).

[14] One problem with this rule is that it may be 'too powerful'. Thus, we might expect the plural cases of the perfect to show laryngeal loss. In fact there is some evidence for this in Sanskrit *dadhmási* (1pl. prs. with laryngeal loss) versus *dadhima* (1pl. pft. without loss) (cf. Wackernagel-Debrunner, *Ai.Gr.* II 82), although an analogical explanation for these

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

phonology would predict *aniṭ* εὗρον : Old Irish *·fuar* contrasting with *seṭ* εὕρηκα : *fríth*. But since the rule was productive only in Indic, the existence of this contrast in both Greek and Celtic strongly suggests that εὗρε : *·fuair* reflects an original **u̯e-u̯r(h₁)-e-t*. It is also possible that a *tertium comparationis* exists in the Avestan *vāuroimaidī* [*Y* 28.5c], *vāurayā* [*Y* 31.3c], *vāurāitē* [*Y* 47.6d]. Insler, *GZ* 126 f., takes this as a reduplicated aorist to the root *vṛ* 'turn' and glosses the forms as 'we would turn', 'I might convert'. Humbach, *GZ* II 61, translates the verb as 'we would receive', etc. This latter gloss would make a comparison possible between *vāur-* < **u̯au̯r-* and εὗρε = *·fuair* (cf. C.H. Werba, *Sprache* 32 [1986(88)] 354, for an explicit equation of *vāur-* and *·fuair*). The difficulty in interpretation does not permit us to have absolute confidence in this analysis. Nevertheless, even without the Avestan form, the comparison between the Greek and Irish is beyond reproach, since neither form could be easily generated analogically in the post Indo-European era.

Thus, not only does this form provide evidence for a poorly understood phonological rule, but also gives us another example of a reduplicated aorist in Indo-European. Since the comparative evidence for the reduplicated aorist is weak, the methodology above offers us an additional means to prove the antiquity of the formation by using internal reconstruction. Assuming that the rule of laryngeal loss functioned only in Indo-European times, the paradigmatic mismatch between *aniṭ* reduplicated aorists and *seṭ* roots implies that the aorist form must be ancient. In addition to εὗρον, two other Greek aorists, ἐκέκλετο and τέτμε, may show this morphological mismatch. After an analysis of each of these forms a short section follows describing other possible laryngeal final roots and their aorist formations.

---

forms is also possible. In Indo-European the rule may only have worked in specific environments (before the thematic vowel?), but analogical changes have obscured the original distribution. In any case the full grade suffix form of εὕρηκα would not be subject to loss under any formulation.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

§2.1.2 ἐκέκλετο 'called to'

The older handbooks, both school editions and academic works, give (ἐ)κέκλετο as the aorist to κέλομαι and gloss it as 'bid, order, call to' (cf. Autenrieth, *Hom.Dict.* 160; Pokorny, *IEW* 549; Risch, *Wortbildung* 245). This bifurcation in meaning between 'order' (with an infinitive) and 'call to' (with a dative or accusative) has led to two radically different etymological approaches. Some scholars see the root meaning as 'call to', derived from the root of καλέω (e.g., Wackernagel, *Spr.Unt.* 175). An alternative approach takes the sense of 'order' as primary and instead equates κέλομαι with forms of κελεύω 'bid, command' (e.g., Walde-Pokorny, *VWIS* I 442 f.) as well as the formally similar (ὀ)κέλλω : κέλσαι 'drag, draw'[15] (Frisk, *GEW* I 816 f.; Chantraine, *DELG* 513). Matters are further complicated by attempts to equate κελεύω with κλύω 'hear' (either with or without κέλομαι). Thus, Fraenkel, *Mélanges Boisacq* 367 f., argues that we have a contamination of two verbs: an original κέλομαι and *ἔκλευσα 'made hear' (the transitive *s*-aorist of ἔκλυον) to yield the new κέλευσα -> κελεύω (for an earlier derivation of κέλευω from **kleu̯-* see M. Bloomfield, *BB* 23 [1897] 113).[16]

Thus, even before we begin to view the extra-Greek cognates we have an amalgam of possible connections. No amount of etymological analysis will clarify this problem unless we first examine the philological details. In point of fact there is a simple solution

---

[15] The present κέλλω is attested only in grammatical writings, while the prefixed ὀκέλλω occurs in standard Attic (Frisk, *GEW* I 817). In Homer only the aorist occurs (ἐ)κελσ- 'beach (a ship)': κελσάσῃσι ... νηυσὶ 'when the ships were beached' [ι 149]. The aorist occurs eight times, all in the *Odyssey*, all but one in the first person narrative of Odysseus.

[16] Pokorny, *IEW* 548 f., apparently accepts this analysis, having a *seṭ* root [6]**kel-* 'rufen, schreien, lärmen, klingen' for καλέω and κέλομαι : ἐκέκλετο, with a cross reference, *op.cit.* 549, to **kleu-* for κελεύω (although I find no mention of κελεύω in that entry).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

which becomes clear after a review of the use of κέλομαι and ἐκέκλετο: ἐκέκλετο is the original aorist to καλέω, which during the Homeric era became increasingly associated with the formally similar κέλομαι. Thus, our two different usages, 'call to' and 'order', do in fact represent different etymological backgrounds. Both etymological sources are valid because we have, in origin, two separate verbs. A quick review of the Homeric material will make this clear.

Ἐκέκλετο 'called to' usually takes the dative case: Ἀπόλλων / ...Τρώεσσι δὲ κέκλετ' ἀΰσας 'Apollo, shouting, called to the Trojans' [Δ 507-508, cf. Z 66, 110, 287, Θ 172, 184, +]. However, at least three other syntagmas occur. On two occasions we find the same semantics but with the accusative, e.g., κέκλετο δ' Ἥφαιστον 'she called to Hephaestus' [Σ 391, cf. I 454]. This aorist is also used intransitively with a prepositional phrase as its complement, although this usage is restricted to the participle: κεκλόμενος καθ' ὅμιλον 'calling (each other) through the throng' [Λ 460 and N 332]. Finally, in two instances we find (ἐ)κέκλετο with an accusative-infinitive construction: κέκλετο δ' ἄλλους / Τρῶας φευγέμεναι 'he bid the other Trojans flee' [Π 657-658] and κέκλετο δ' ἄλλους / ...σπέσθαι 'he bid the others follow' [δ 37-38].

The present κέλομαι, on the other hand, consistently means 'bid, order', usually with the accusative-infinitive construction: ἀλλά σ' ἔγωγε / παύεσθαι κέλομαι [Γ 433-434, cf. A 385 (without accusative), E 810, I 517, +].[17] Over sixty years ago F. Specht, *KZ* 59 (1932) 85 f., commented on the mismatch between the semantics of κέλομαι : (ἐ)κέκλετο, pointing out that the diction supports κέλευσα as the aorist of

---

[17] At least one instance of κέλομαι should perhaps be assigned to a different verb ^{2}**kel-*, cognate with Latin *calāre* 'heat', Lith. *šìlti* 'warm werden': αἶψα δ' ἰαίνετο κηρός, ἐπεὶ κέλετο μεγαλήϊς /'Ηελίου τ' αὐγὴ 'Υπεριονίδαο ἄνακτος· 'and the wax suddenly grew warm, when the great power and light of the lord Helios Hyperion heated (?) it' [μ 175], (F. Specht, *KZ* 59 [1932] 85 f.; Fraenkel, *LEW* 984).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

κέλομαι (already noted in Buttmann, *Aus.Gr.Spr.* 214 n.). Compare 'Αργείους ἐκέλευσα ἀλήμεναι 'I bid the Argives gather...' [E 823, cf. Z 179 +]. To be sure, there are the two instances of (ἐ)κέκλετο with the accusative-infinitive construction, which may indicate the translation 'bid, order' as opposed to 'call'.[18] In addition, κελεύω is also found with the dative υἱάσι δὲ Πριάμοιο . . . κέλευεν 'he called to the sons of Priam' [E 463]. Nevertheless, these forms seem to indicate little more than a growing confusion between κέλομαι, κελεύω and ἐκέκλετο. The diction of Homer points to the conclusion that κελευσ- was originally an aorist to κέλομαι 'order, command', while ἐκέκλετο is best traced to καλέω. Only in the period of Homeric composition did the reduplicated form begin to be associated with the formally similar κέλομαι.

In Wackernagel's analysis, *loc.cit.*, καλέω was backformed from the root aorist of κέλομαι, a relic of which was preserved in the form κέντο 'κέλετο' [Alcman frag. 139] < *κέλτο < *$k\bar{l}$*to* (cf. also Schwyzer, *Gr.Gr.* 678).[19] A connection between καλέω and κελομαι is likely, but not in the manner envisioned by Wackernagel whose solution is phonologically impossible. A root aorist * $k\bar{l}$*to* (i.e. *klh_1*-tó*) should give **κλητο, analogous to ἔβλητο < *-$g^{\mu}lh_1$*-to* or πλήτο < *plh_1*-to* (e.g., Beekes, *Development* 226 f.). As will be seen below, a better analysis, both phonologically and semantically, takes κέκλετο as an aorist to καλέω. Only after the conflation of κέκλετο and κέλομαι does the sense of 'call to' gradually become associated with the latter present.

The semantic match between καλέω and κέκλετο is unproblematic. The Greek present καλέω 'call' has, as in English, two distinct semantic uses. It is used either in the

---

[18] Another passage which might support such a reading is δμώεσσιν ἐκέκλετο, τοι ἐπίθοντο 'he called to the servants and they obeyed' [ζ 71]. Here the translation 'call to' works but the addition of the phrase τοι ἐπίθοντο may imply a stronger semantic usage then just 'calls'.

[19] Szemerényi, *Syncope* 188 f., takes κέντο < *κέλτο as a syncopated form from κέλετο. This may be our best option since this form is glossed as κέλετο.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

sense of 'to call to someone' καλέει σε 'she (Penelope) calls you' [ρ 553; cf. N 740, +; also with object deletion in equi Δ 204, Ω 88, +]; or in the sense of 'to name' ὃν Βριάρεων καλέουσι θεοί 'whom the gods call Briareus' [A 403; cf. E 306, Y 74 +]. The middle voice and perfect forms give the passive ἀθάνατοι καλέονται 'they are called immortals ' [E 342; cf. A 294, Ξ 210, +]; σὴ παράκοιτις / κέκλημαι 'I am called your wife' [Δ 60-61, +]. [20] Although ἐκέκλετο lacks the latter sense of 'name', the 'call to' usage forms a better match for this aorist than the sense of 'order, command' of κέλομαι.

The phonological development is more complex, but also poses no real problem. The present καλε- is variously explained as an **-e̯ie/o-* formation **kal(h₁)-e̯ie/o-* (Harðarson, *Studien* 82 n.98) or perhaps as a 'Lindeman' variant **ḱll-eh₁-e/o-* (Harðarson, *loc. cit.*). Both options are unlikely. Outside of Greek, the root **kalĕ-* (**kalh₁-*, **kelh₁-*?) is found in Hittite *kallešš-mi* 'zum (Opfer)mahl herbeirufen, einladen' (Oettinger, *Stammbildung* 197 with bib.), and Italic: Latin *calēre* 'to call'; Umbrian **kaŕetu** 'it shall be called to' < **kalētōd* (Walde-Hofmann, *LEW* I 141 f.; Ernout-Meillet, *DELL* 87 f.; Meiser, *Lautgeschichte* 207). The antiquity of the formant **kalĕ-* immediately refutes Wackernagel's attempt to generate this form from some older root aorist. Still, its origins are peculiar. The alternation between the full grade **kalĕ-* and the zero-grade forms such as κλήδην 'by name', κικλήσκεις, and the Latin *clāmāre* clearly point to a root final laryngeal (Ernout-Meillet, *DELL* 88; Beekes, *Development* 186 f.; Chantraine, *Form.* 361). Even positing a *se̯t* root, the quality of the root vocalism remains puzzling.

The obvious parallel is a root like **kore* : **krē* of κορέσαι 'sate' : Lat. *crēscō*

---

[20] The same variation between 'call to/name' exists in the aorist with an important distinction: in the aorist the middle voice is not distinct semantically τοῦ σ' εἵνεκα δεῦρο κάλεσσα 'on account of which I called you here' [Ω 106, +] and ' Ελένην ἐκαλέσσατο 'he (Priam) called Helen' [Γ 161; cf. A 54, +]. The middle forms are best treated as metrical creations as explained by Meister, *Kunstspr.* 19 f. For additional discussion see §3.2.1.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

'grow', Lith. *šérti* 'nourish' < **ke/orh$_1$-* (C.J. Ruijgh, *FsRix* 376 f.). A similar preform **kalh$_1$-*, ablauting with a zero grade **k̥lh$_1$-* > κλη-, would seemed to be implied here (see Oettinger, *loc. cit.*; Mayrhofer, *Idg.Gr. I/2* 128). This approach assumes an **a/∅* ablaut datable to the common era, although most of the evidence cited by the Neogrammarians for such patterns are now explained with laryngeals. There is, however, some evidence for an intrinsic relationship between the Indo-European velar series and Indo-European **a*-vocalism (see most recently Schrijver, *Reflexes* 420 f.). Hence, perhaps our root **kalh$_1$-* is simply from an older **kelh$_1$-* (Schrijver's **(s)kelh$_1$-*) with allophonic */a/* for */e/*.

Given a root **k(a)lh$_1$-* 'call', we expect the *set* base **klē-* to reduce to **-kl-* in reduplication just as εὗρον above.[21] Thus, ἐκέκλετο is the expected outcome of **ke-klh$_1$-e/o-*, and its derivation from the root of καλέω is entirely lautgesetzlich. Unlike the attempts to somehow connect κέλομαι and καλέω, no *ad hoc* phonological manipulations are necessary here. Without strong analogical support, which εὗρον could hardly provide, the formal correspondence between the stems of καλέω and ἐκέκλετο was soon forgotten, and the form became increasingly associated with κέλομαι. Hence, ἐκέκλετο was originally the aorist to καλέω, but created at a time when the rule of laryngeal loss was still functioning.

Unfortunately, without comparative evidence the archaic status of ἐκέκλετο remains somewhat uncertain. Further, the existence of the *s*-aorist (ἐ)κάλεσσα might argue against taking ἐκέκλετο as the original aorist to καλέω. However, this *s*-aorist need not be old. Note C. Barton's comment, *Laryngaltheorie* 51 f., that "...PIE set root aorists active of the shape 3sg. **CéRH-t* were commonly replaced in pre-Greek by neosigmatic formations". Moreover, even if κάλεσσα is old, this does not preclude the ex-

---

[21] For a similar analysis of κέκλετο see Francis, *Disyl.Roots* 160 f. Although he considers this aorist built to κέλομαι, he also notes the possibility of an etymological connection with καλέω. However, he is forced to explain the aorist through a variety of analogic changes.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

istence of a polymorphic reduplicated variant.[22] Although in that latter case we would expect to find some difference in usage.

Before closing, let us say a few words about the presents κελεύω and κέλομαι. Fraenkel's analysis of κελεύω, from *κ(ε)λευσ- 'make hear', requires that the sense 'order' predates the use 'drive'. This chronology, however, is unlikely, since the noun κέλευθος 'way, path' must derive from 'a driving place' just as in the pair ἄγυια : ἄγειν. Further, the ablaut variant ἀκόλουθος 'following, follower' may well indicate that κέλευθος is an archaism, since a newer creation would probably not be built with an *o*-grade in this way (A. Pârvulescu, *IF* 91 [1986] 191 f. with bib.).[23]

It is likely that ***kel-*** is somehow a by-form of the older ***keleu-***. There is no exact parallel, but similarly enlarged and unenlarged forms are not unknown in Indo-European morphology (cf. Latin *gaudeō* < ***geh₂-u̯-d(h)-***, perfect *gāuīsus sum* < ***geh₂-u̯-i-***, but Greek γηθέω < ***geh₂-dh-***). Both forms are likely to descend from the root ***kel-*** 'set in motion', cf. Latin *celer*, Greek κέλης 'race horse' < ***kel-es-***, as well as Sanskrit *kalayati* 'drives' and the Tocharian *käl-* 'lead' Pokorny, *IEW* 548. Whether or not this root is further connected with (ὁ)κέλλω is not significant for this study, but the evidence would seem to argue against it.[24]

---

[22] The Hittite *kalleššˇ-* might be taken as evidence for an archaic aorist; however, the relationship of the Hittite *s*-presents to the Indo-European aorist is a matter of some controversy (see J. Jassanof, *Laryngaltheorie* 227 f. for one analysis). Further, this equation is largely accidental since Greek ἐκάλεσσα is apparently from ***kalh₁-s(s)-***, while the Hittite requires ***kal(h₁)e-s-*** or even ***klh₁-es-*** (cf. Oettinger, *loc.cit.*; Melchert, *AHP* 80). Any relationship between the two forms would require some restructuring.

[23] Further evidence for the antiquity of these forms lies in the Mycenaean *ke-ro-u-te-u*. The exact interpretation of this name is not clear, but *κέλουθευς 'follower' is probable (Ruijgh, *Sc.Min.* 20; Aura-Jorro, *DMic.* I 352).

[24] Note R.S.P. Beekes' comment, *HS* 105 (1992) 173 n. 1, that the gloss 'to drive' is "...pure phantasy, for the sake of the etymology". Beekes, *loc.cit.*, goes on to suggest that (ὁ)κέλλω : (ἐ)κελσ- is most likely non-Indo-European.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

§2.1.3 ἔτετμε 'found'

The aorist ἔτετμε(ν), τέτμε(ν), is rare, occurring exclusively in epic (3X *Iliad*, 5X *Odyssey*, 1X Hesiod, also Theoc. & A.R.). Metrically, in the *Odyssey* ἔτετμε(ν) usually occurs in the last place in the line (4/5X), while in the *Iliad* we find two instances in the third foot ἔ|τετμε(ν), one in the sixth. This tendency for line final position contrasts with εὑρ- ε/ο-, which shows a preference for placement in the first foot: 41/81 (line initial) + 13/81 (end of the first foot e.g., τὸν δ' εὗρ-|) = 54/81. Aside from the metrical difference, the contrast between these two verbs is slight. Thus, consider the passage in the sixth book of the *Iliad* when Hector seeks his wife: #οὐδ' εὗρ' Ἀνδρομάχην... 'he did not find Andromache...' [Z 371]. Two lines later the thought is repeated: Ἕκτωρ δ' ὡς οὐκ ἔνδον ἀμύμονα τέτμεν ἄκοιτιν 'but as Hector did not find (his) blameless wife within...' [Z 374].

The usage of ἔτετμε(ν), however, is narrower than that of εὗρε. Like εὗρε, in Homer ἔτετμε(ν) is transitive (ineffectual) and takes an accusative complement: Ἕκτορα δῖον ἔτετμεν ἀδελφεόν 'he found his brother, shining Hector' [Z 515, cf. also Δ 293, Z 274, α 218, γ 256 ε 58, 81, ο 15]; however, in each instance the accusative complement is a person. No such restriction applies to εὗρε, which can describe finding things as well as people [cf. Γ 24, I 49, M 121 +].

A single instance of τέτμῃ occurs in Hesiod; however, its usage, syntactic, semantic, and stylistic, is distinct from the Homeric examples. It takes a genitive, occurs in mid-line, and has a rather different sense: ὅς δέ κε τέτμῃ ἀταρτηροῖο γενέθλης 'whoever happens to have a mischievous sort (of wife)' [*Th.* 610] (cf. West, *Theogony* 355). The Hellenistic examples are best considered Homeric imitations.

The paradigmatic connections for ἔτετμε(ν) are less than clear. In his treatment, Ebel, *op.cit.* 48, takes this form as a reduplicated aorist to τέμνω 'cut' (cf. also Francis, *Disyl.Roots* 161 f.; considerable skepticism in both Frisk, *GEW* II 884 f., and Chantraine,

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

DELG 1109). Formally, τέμνω 'cut' descends from a *set* root as is clear from the alternation of τεμ- : τμη- (e.g., εὐτμητῳ [H 304 +] 'well cut') as well as the full grade thematic aorist, ἔτεμε (Ion.-Dor. ἔταμε) < ***témh$_1$-t*** (3sg.) : ***tm̥h$_1$-ént*** (3pl.) (see most recently, Harðarson, *Studien* 214).[25] As the evidence of εὗρον and κέκλετο above has shown, ἔτετμε(ν) would be the expected phonological outcome of ****(e-)te-tmh$_1$-e/o-***, and the derivation is formally possible.

It is, however, in the realm of usage that this derivation fails. A change of 'cut (for oneself)' to 'get a portion' to 'find' is of course possible. But in this case we would expect a sense of 'find' to refer primarily to material possession, and only secondarily (if at all) to people. Since ἔτετμε(ν) has only the latter usage in Homer, a connection to ****temh$_1$-*** is unlikely. Further, with solid evidence for a root aorist, the reduplicated ****te-tm(h$_1$)-e/o-*** becomes an unnecessary polyform.

Unfortunately, Greek shows no other roots of the shape ****tem(h$_x$)-*** with one possible exception. At N 706 f. we find the following passage:

> τὼ μέν τε ζυγὸν οἷον ἐϋξοον ἀμφὶς ἐέργει
> ἰεμένω κατὰ ὦλκα, τέμει δέ τε τέλσον ἀρούρης
> 'the polished yoke alone separates the two (oxen) labouring through the furrow until it attains (?) the limit of the field'

Chantraine, *Gr.Hom.* 309, takes this as a present to ἔτετμε: "qui ne doit pas être rapproché de τέμνω mais signifie 'atteindre' et constitue le présent attendu en face de l'aoriste ἔτετμον". Nevertheless, as noted above, ἔτετμον always refers to meeting people, a sense completely out of place here. Further, I see no problem with a translation of 'cuts to

---

[25] For the alternation between τέμνω and τάμνω see G. Cardona, *Language* 36 (1960) 502 f.; also B. Forssman, *Glotta* 44 (1967) 5 f. For the Lesbian τόμοντες see most recently Harðarson, *Studien* 157 f. (with bib.). O.Ir. *tamnaid* 'cuts' is certainly cognate with the above but may descend from an enlarged proto-form (see §4.2.0).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

the limit' (cf. R. Janko, *Comm.Il.* 136).[26, 27]

One last etymological *comparandum* may lie in the Tocharian verb *täm-* 'geboren werden' (Frisk, *GEW* II 885 with reference to Van Windekens, although he himself apparently abandoned this approach, cf. *Tocharien* I 500 f.).[28] This comparison is strengthened, however, by Hesiod's τέτμῃ ἀταρτηροῖο γενέθλης, which, coupled with the Homeric restriction to persons, may well imply a derivation from 'engender'. No absolute conclusions are possible here. Ἔτετμον lacks an etymology, but both the formular consistency and the isolation of the form implies an archaism.

§2.2 Possible outcome of *Ce-CRh₃-e/o-*

Peters, *Untersuchungen* 30, discusses *Ce-CoR-e/o-* as a possible outcome of *Ce-CRh₃-e/o-* (although he himself notes that analogic solutions are equally possible). Nevertheless, the evidence for this development is far from persuasive. Hesychius gives two reduplicated aorists of similar form: 1) πεπορεῖν· δοῦναι; 2) πεπαρεῖν· ἐνδεῖξαι, σημῆναι. Hence, we have two aorists: πεπορεῖν 'give' and πεπαρεῖν 'show'. The

---

[26] Janko, *loc.cit.*, also accepts the connection with the reduplicated ἔτετμον.

[27] K. Strunk, *Glotta* 68 (1990) 50 f., offers a different analysis of τέμει. Citing the Hesychian gloss τέμνοντα· ἀμέλγοντα and noting that ἀμέλγειν has the meaning 'trinken' in Hellenistic texts, Strunk equates τέμει with Sanskrit *-camati* 'schürft', both from a preform **k̑ʷem-e/o-* 'schürfen, schlucken'. He translates the τέμει δέ-phrase as 'und der Rand des Ackers (ihn) aufsaugt', that is 'absorbs the sweat' (ἰδρώς l. 705). I do not find Strunk's analysis convincing. The distance between ἰδρώς in line 705 and the verb makes this reading improbable.

[28] Other etymologies for Tocharian *täm-* are largely unsatisfactory. Thus, attempts to derive Tocharian *täm-* from the root of Greek δέμω (e.g., Pedersen, *Sprachgesch.* 21 n. 1) face the obstacle of explaining the initial *t-* as opposed to *ts-* (W. Winter, *IF* 67 [1962] 27 f.). Emmerick, *Saka Gr.St.* 46 & 69, argues for an Indo-European **dhem-* as a by-form of **dheh₁-*, parallel to **gʷem-* : *gʷeh₂-*. But aside from the fact that such a by-form is otherwise unattested, both this etymology and that of Pedersen require an odd semantic shift from 'make' or 'set' to 'engender'.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

first of these is simply a reduplicated form of the familiar Homeric πόρε 'gave' [A 72 +], while πεπαρεῖν is built to a root found only in reduplicated forms in Greek. There are only three attestations: this entry; an adjective also known only from Hesychius πεπαρεύσιμον· εὔφραστον, σαφές; and finally the same infinitive found in Pindar's Second Pythian Ode: τὺ δὲ σάφα νιν ἔχεις ἐλευθέρᾳ φρενὶ πεπαρεῖν (57). The sense seems clear enough 'you clearly have this to show forth with a free spirit'.

However, E.D. Floyd, *AJP* 92 (1971) 676 f., offers a different reading of the Pindaric form. He notes that the scholia to this ode indicate that πεπαρεῖν means 'give, furnish',[29] so that the verb is therefore equivalent to Hesychius' πεπορεῖν. Floyd, *op.cit.* 678, translates the phrase '*[y]ou* clearly have this to offer with liberal spirit'. Significantly, however, he does not argue for an emendation of πεπαρεῖν to πεπορεῖν, but instead, he states that πεπαρεῖν is the correct reading and that πεπαρεῖν is etymologically built to πόρε.

Because such a conclusion would have wide repercussions for our understanding of Indo-European and Greek historical phonology, it will be necessary to examine this problem carefully. Now, we know that the outcome of *Ce-CRh$_1$-e/o-* is *Ce-CR-e/o-* with laryngeal loss. However, the status of both the sequences *Ce-CRh$_2$-e/o-* and *Ce-CRh$_3$-e/o-* has yet to be determined. For Floyd, a preform **pe-prh$_3$-e/o-* would have yielded πεπορεῖν in most dialects but πεπαρεῖν in Doric, hence the forms in Hesychius and in Pindar.[30]

The exact preform of πόρε, and the other *o*-grade thematic aorists, is still a sub-

---

[29] δύνασαι καὶ ἑτέροις πεπαρεῖν, ἤτοι πορίζειν καὶ παρασκευάζειν — quoted by Floyd, *op.cit.* 676 f.

[30] Floyd bases this interpretation on a discussion in Beekes, *Development* 260 f. Beekes notes such forms as ἰαρός : ἱερός; however, his central conclusion, *loc.cit.* 259, is instructive: "[t]his examination... has not yielded any clear indications of dialectal differences [in the treatment of laryngeals]".

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

ject of dispute, but we apparently have a root with final *-h_3, cf. the perfect πέπρωται 'it is fated < is given' < *pepr̥h_3-toi; Skt. *pūrdhi* 'gib!' < *pr̥h_3-dhi (cf. Mayrhofer, *Idg.Gr. I/2* 129 with bib.; Beekes, *Development* 216 f.).[31] The status of the sequence *peprh$_3$-e/o- is, however, more controversial. Three possibilities exist:

1) *peprh$_3$-e/o- > *pepr-e/o- (with laryngeal loss as in εὗρον)
2) *peprh$_3$-e/o- > *pepr̥h$_3$-e/o- > *peparh$_3$-e/o- > πεπαρ-
3) *peprh$_3$-e/o- > *peporh$_3$-e/o- > *pepor-e/o- > πεπορ-

Despite the scholia, however, it is not clear that Floyd's reading of πεπαρεῖν as 'give, furnish' is "...in some ways easier" (Floyd, *op.cit.* 676). Purely on a philological basis, Floyd's interpretation has been challenged by B.C. Dietrich, *Grazer Beiträge* 3 (1975) 69 f., who notes that the diction itself may argue against such a reading: "it seems unlikely that πεπαρεῖν ... can mean 'to give'... Hiero would not have had the power to dispose of such wealth even had he been inclined to do so". Compare also Most, *Measures* 92 f. n. 87: "the fact that πεπαρεῖν is by far the best attested reading... and that Hesychius glosses πεπαρεῖν as ἐνδεῖξαι, σημῆναι should decide the question".

Finally, it is not clear how we are to understand this relationship between πεπορεῖν and πεπαρεῖν. Although Hesychius is often misleading in his glosses we cannot simply ignore the existence of a Greek root -παρ- meaning 'show'. The agreement between the Hesychian forms and Latin *pārēre* 'to appear' can hardly be coincidental (cf. Schrijver *Reflexes* 144). Floyd, *op.cit.* 697, seems too quick to dismiss this distinction

---

[31] Since thematic aorists in Greek typically show zero grade of the root, it is possible that forms such as πόρε 'gave' represent the outcome of *pr̥h_3-e/o- (for a discussion of this older view see Beekes, *Development* 210 f. with bib.). However, it is also possible that these forms were originally athematic formations *perh$_3$-t- > *pero-t > *pore- with metathesis (M.S. Ruipérez, *Emerita* 18 [1950] 386 f.). For *CeRo* > *CoRe*, compare Myc. *re-wo-to-ko-wo* = λεFo-τροχοFων 'bath-attendants' beside alphabetic λο(F)ετροχόος [σ 346] (Ventris-Chadwick, *Documents* 160; Beekes, *Development* 232 f.; Peters, *Untersuchungen* 30 n. 19, 321; Harðarson, *Studien* 158 f.).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

when he says that "[a]ctually, a meaning 'give' is not so distant from 'show'". Although it is true that one can imagine both notions coming from the same root, Latin *pariō* 'to bring forth' is a better match for Greek πόρε (cf. Schrijver, *Reflexes* 195 f. and 211). Hence, we require two separate roots: 1) **păr-* (**peh₂r-*?) reflecting Greek (πε-)παρ- and Latin *pārēre*; 2) **perh₃-* for Greek πόρε, Latin *pariō*. In this case, however, it is unlikely that (πε)παρ- descends from a *set* formation, so we have not made any real progress in our analysis of *Ce-CRh₂-e/o-* and *Ce-CRh₃-e/o-*.

In addition to πεπορεῖν, Hesychius gives two other o- grade reduplicated aorists:

1) the three forms: τετορεῖν· ἔτρωσε; ἐτέτορεν· ἐτρύπησεν; τετόρῃ· τρώσῃ
2) τεθορεῖν· διαπηδῆσαι.

Of these, at least the first is probably *set* based on the evidence of the *s*-aorist ἔτρωσε, presumably from **e-trh₃-s-*. The existence of these forms would seem to argue for the third outcome above (cf. Beekes, *Development* 291). However, it may be significant that no such *o*-grade reduplicated aorist is attested in Homer. If these forms were later Greek developments it is easy enough to imagine an analogical creation of a newer πε- πορεῖν from πορεῖν.

As to the outcome of *Ce-CRh₂-e/o-* and *Ce-CRh₃-e/o-*, if the *o*-grade forms are analogical, there is no evidence to suggest any outcome other than *Ce-CR-e/o-* with the laryngeal loss demonstrable for **h₁*. Of forms discussed above only πεπαρεῖν (and the Hesychian forms) is likely to be old, but here there is no evidence for a *set* root here.[32]

---

[32] Note that some versions of the law of laryngeal loss (above n. 14) would also effect a formation **pe-ph₂r-e/o-*. However, this offers little proof here both because the exact specification governing the rule of laryngeal loss are unknown (above n. 14) and because this root need not have had an internal laryngeal.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

§2.3 Conclusions on *Ce-CRH-e/o-* Aorists

In addition to **u̯e-u̯kᵘ-e/o-*, a second reduplicated aorist is also reconstructible for Indo-European based on the comparison of εὗρον = O.Ir. *·fuar*. Further, this comparison offers additional proof for a poorly understood law of Indo-European phonology. The existence of this law suggests that ἐκέκλετο and perhaps also τέτμε may be archaic formations. For ἐκέκλετο, since this form is built to a *set* root, a reduplicated aorist offers the easiest phonological solution. The evidence is considerably weaker for τέτμε since the etymology and structure of this root are unknown. Finally, there is no evidence that the reduplicated aorists of the shape *Ce-CoR-e/o-* are old, and these probably represent inner-Greek developments.

* * *

§2.4.0

The existence of a sound law *Ce-CRH-e/o-* > *Ce-CR-e/o-* has certain repercussions for our understanding of the reduplicated aorist in Indo-European, both in regards to the Sanskrit *ajījanat* aorist, and the Tocharian class II preterite.

§2.4.1 *Excursus*: The Sanskrit *ajījanat* Type

As noted above in the Introduction, Sanskrit has a class of reduplicated aorists, the so-called *ajījanat* type, which is synchronically paired with the transitive *aya*-present.[33] Some scholars have seen a relationship between this formation and the reduplicated aorist in Greek (e.g., Birwé, *Gr.-Ar. Sprach.* 29 f.; Adams, *THPM* 87). The *ajījanat* type, however, differs from the Greek formation in both the quality and quantity of the reduplicated vowel (*ī* or *ū-* in roots with a rounded vowel). Moreover, although both the Indic and

[33] For a fairly complete synchronic analysis of the Sanskrit *Cī*-reduplicated aorist, see Jamison's *aya-Form.* (*passim* and 216 f.) and Thieme's *Plusquamperfectum*.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

Greek formations show zero-grade, *CRH* roots are treated quite differently in the Indic formation.

The length of the reduplicating vowel is not such a problem. Jamison, *aya-Form.* 214 f., argues that the change in the length of the reduplicating syllable was motivated to give these aorists the same metrical template as their corresponding presents, e.g., *pārayat-i* : *a-pīparat* 'makes cross over', where, ignoring the augment and primary ending, both forms show the metrical shape – ∪∪. Hence, it is likely that these long-vowel reduplicated forms are replacements for older **a-jĭ-janat, *a-pĭ-parat.* The quality of the reduplicating vowel is not so easily solved.

Since *i*-reduplication is typically associated with the present system, M. Leumann, *FsBrown* 152 f., argues that these aorists should be seen as transfers to the aorist from older present forms. The proof for this transfer is the aorist *ajījanat* itself, which is matched in Avestan not by an aorist, but by the present participle *zīzanənti*, which in turn makes a passable word equation with the Greek γίγνομαι and Latin *gignit.*[34] Leumann sees the motivation for the transfer in the development of the newer intransitive *jā́yate* and transitive *janáyati*. However, the older **jījanati* was not eliminated, but simply forced into a new position in the paradigm.

Although this scenario is possible, there is a serious problem in Leumann's word equation. While the Greek γίγνομαι and Latin *gignit* show the expected outcome of **ǵi-ǵn(h$_1$)-e/o-*, the Indic and Iranian forms descend a reduplicated form without laryngeal loss, i.e., **ǵi-ǵn̥h$_1$-e/o-*. Further, the Avestan form is not quite as telling as Leumann implies, since without finite forms its designation as an imperfect is not historically significant. Avestan has largely merged the old imperfect and aorist outside of the indicative (Kellens, *Verbe avestique* 431 and above §1.1 n. 5). From an Indo-Iranian point of view it

---

[34] An alternative approach is set forth by Hirt, *Hb.Gr.* 520 f., who takes γίγνομαι as a relic of an earlier aorist.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

is equally likely to view *zīzanənti* as a transfer from the aorist as opposed to viewing *ajījanat* as the innovation. Hence, *ajījanat* is not a word equation with either Latin *gignit* or Greek γίγνομαι, at least not without assuming some degree of restructuring in the Indo-Iranian formation.

Further, not only can *ajījanat* not be a direct relic of the Indo-European **ǵi-ǵn̥h$_1$-e/o-* but it cannot be equated with the reduplicated aorists above. Thus, an alternative approach to the origin of the *ajījanat* formation assumes that Indo-Iranian inherited a formation similar to that reconstructed for Greek (*Ce-CC[H]- e/o-*), but for some reason this formation was largely altered to *ī*-reduplication.[35] Despite the obvious lack of motivation, this hypothesis finds some supporting data. First, Sanskrit has a small number[36] of reduplicated aorists with *Ca-* (Indo-European *Ce-*) reduplication. Second, there is occasional evidence for substitution of *Ca-* with *Cī-*, e.g., *jajastám* 'entkräftet' (2 dual) [*RV*] > *ajījasata* [*ŚB*] (cf. Mayrhofer, *EWAia* I 581). But even assuming such a change in the reduplicating vowel, a preform **ǵe-ǵnh$_1$-e/o-* should have given ***ajījnat*.[37] Hence, the "Rule of Laryngeal Loss" compromises this approach as well.

---

[35] This approach is implicit in the accounts of Birwé, *loc.cit.* and Adams, *loc.cit.*, although neither scholar gives any details as to why this should have occurred.

[36] The clearest example of a *Ca*-reduplicated aorist in Indo-Iranian is Sanskrit *ávocat* = Avestan *vaocaṯ* = Greek εἶπον, < **u̯e-u̯kʷ-e/o-* (see §1.1 above). Whitney, *Roots* 224, lists ten aorists with *a*-reduplication for the oldest language, although several of these are probably pluperfects. For the problem of distinguishing pluperfects from reduplicated aorists see Thieme, *Plusquamperfectum* 4 and *passim*. The following Sanskrit *Ca*-reduplicated forms are certainly aoristic: 1) *jajastám* (see above); 2) *paptat* 'fell' (Mayrhofer, *op.cit.* II 71); 3) *tatanat* 'thundered' (see below); 4) *vavr̥t* 'made turn' (Hoffmann, *Aufsätze* II 588 f.). 5) *neśat* 'destroyed (trans.)', (analogically reformed from older **naṁśat*) ≅ Av. *nąsat* 'disappeared' [*Y* 53.6, 7], < **na-nś-a-t* 'destroyed (?)' (Mayrhofer, *op.cit.* II 28; K. Strunk, *Laryngaltheorie* 580 f. —a *tertium comparationis* may also lie behind the Umbrian *ninctu* 'destroy! (?)' [*Tables* Vba 60 etc.] < **ne-nḱ-e/o-*). For the Avestan *vāuroimaidī* [*Y* 28.5c], see above §2.1.

[37] Sanskrit *tatanat* 'thundered' clearly shows this problem. The form is synchronically a reduplicated aorist (Hoffmann, *Aufsätze* II 568; J. Narten, *FsRix* 324), but Narten's recon-

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

Ultimately I suspect that both proposed origins for the Sanskrit reduplicated aorist are true in part: 1) the Indic *Ca*-aorists are related to the proto-type reconstructed for Greek and; 2) that the *ajījanat* type aorists have been built from older presents (or at least have some present influence). Nevertheless, no simple explanation is possible for these forms, and the relations between Greek and Indo-Iranian reduplicated aorists can be proposed only with the greatest of care.

§2.4.2 *Excursus*: The Tocharian Class II Preterite

The apparent *seṭ* quality of the Sanskrit *ajījanat* type aorist complicates any direct comparison with the Greek reduplicated aorist. A similar problem also arises in an analysis of the Tocharian class II preterite, which shows the standard *-*ā*-endings of the preterite. It is generally agreed that these endings descend from older *seṭ* roots and later spread analogically to the other preterite formations (Pinault, *Introduction* 147). Here, as with the problems associated with the quality of the reduplicating vowel (above §0.1 n. 6), this fact does not prove or disprove any association between the Class II Preterite and Indo-European reduplicated aorist, since we know that analogy has spread these endings to older *aniṭ* forms. What it does show is that the Tocharian formation has been so thoroughly restructured that any equation between this formation and the Greek remains highly speculative.

---

struction, **e-te-tn̥h₂-et*, requires a formation without laryngeal loss. The Hittite reduplicated *tetha* (Old Hittite *titha*) 'es donnert' also belongs to the same root, but this present formation cannot be directly equated (see H. Eichner, *Sprache* 24 [1978] 161 n. 73)

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

CHAPTER III

§3.0 *Ce-CR̥C-e/o-*

The preceding discussion has shown that the aorists εἶπον, εὗρον, and perhaps also ἔπεφνον and ἐκέκλετο are datable to the parent language. In addition ἐνεγκεῖν (§5.1) and ἄραρε, (§6.1) must predate the Greek era (see below). A few other reduplicated aorists may also be archaisms, e.g., πέφραδον (§3.4.2), for which no analogical model is readily available. Nevertheless, the vast majority of reduplicated aorists described in the next four chapters represent innovations, analogical forms created in the Kunstsprache. Many of these aorists are functionally equivalent to older non-reduplicated forms. In such cases, it can usually be shown that the reduplicated aorist has been created for metrical reasons. Other reduplicated aorists function as active causatives contrasting with older intransitive middles. Despite their special status in the paradigm, however, in most cases it can be shown that these causative reduplicated aorists are also late analogical creations.

The order of presentation will again follow purely formal divisions, with semantic discussions added as needed. The *CuC* roots, πεπύθοιτο, κεκύθωσι, τετυκεῖν and τετυπόντες are described first, followed by the *CiC* forms, πεπιθεῖν, πεφιδέσθαι, τέθιγεν, next the *CṛC* root (ἐ)τετάρπετο, and then the *C(R)n̥C* roots δέδαε and πέφραδε. One form which may be a reduplicated aorist in *Ce-CḷHC-e/o-* is described at the conclusion of this chapter.

§3.1 *Ce-CuC-e/o-*

§3.1.1 πεπύθοιτο 'should learn'

The aorist πεπύθοιτο occurs only three times: εἴ κεν ἐμὲ ζωὸν πεπύθοιτ' ἐπὶ νηυσὶν Ἀχαιῶν 'if he might learn of me, yet living, among the Achaian ships...' [Z 50 = K 381, almost identical at Λ 135]. As with κεκύθωσι below, the form could be a pluperfect, but the zero-grade vocalism and the transitive usage both argue for an aoris-

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

tic interpretation (see below for discussion). A number of different tense forms are attested to this *medium tantum* root in Homer:

	Middle
prs.	πεύθομαι, πυνθάνομαι
aor.	πυθόμην, πεπύθοιτ[ο]
pft.	πέπυσμαι (plpft. πέπυστο)
fut.	πεύσομαι

Before going on to analyze the Greek system, it may be worthwhile to say a few words about the Sanskrit cognate *bodh-* 'wake/awaken'. We will concern ourselves only with the prs. forms *bódhati, búdhyate, bodháyati* and the aorists *abodhi, budhánta, abhutsi* and the reduplicated *abūbudhat* (Mayrhofer, *EWAia* II 233 f.). A number of comparisons could be drawn here between the Greek and Sanskrit forms. In both languages the root **bheudh-* builds a thematic present, a thematic middle aorist, and a reduplicated aorist. But no simple comparison between the these two paradigms is possible.

The correspondence of Greek πεύθομαι; Skt. *bódhati* (Epic *bodhate*); Goth. *ana-biudan* 'proclaim' would seem to assure a thematic present (Pokorny, *IEW* 148 f.; Frisk, *GEW* II 625 f.; Chantraine, *DELG* 954 f.; Mayrhofer, *loc.cit.*; Lehmann-Feist, *GED* 30 f.).[1] But the antiquity of this Sanskrit thematic present is subject to debate. Thus, Hoffmann, *Injunktive* 231 f., takes at least one instance of *bodhat* [*RV* IV 15.7] as an aorist subjunctive. S. Insler, *Language* 48 (1972) 559 f., expands on this list considerably. Hollifield, *System* 179, noting the occurrence of full grade forms in the aorist subjunctive, speculates that the present thematic *bodhati* was probably built to an older aorist subjunctive: "Vedic *bodhati*, etc., does not occur in Saṃhitās in augmented forms or in the participle, and is therefore probably subjunctive in origin" (*op.cit.* 181).[2] In a recent paper C.

---

[1] The TA *pot-* TB *pautotär* 'flatter' may reflect the thematic present or perhaps a relative of the Skt. passive *bodhi* : *abudhran*.

[2] For additional discussion of the aoristic nature of *bodh-a-* see Joachim, *Mehrfachpr.* 113

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

Watkins, *StIdgW* 309 f. = *Selected Wr.* II 722 f., has similarly argued against an archaic thematic present "Greek and Germanic both show... an innovated ***bheudhe(ti)***, quite possibly independently".

The comparison of the thematic aorists ἐπύθοντο with the Sanskrit *budhanta* looks especially promising, but this is also an illusion. The so-called aorist passive in *-i* (3sg.) : *-ran* (3pl.) regularly builds a 3pl. injunctive in *-anta* (cf. Hoffmann, *Injunktive* 227 n. 225; Cardona, *Them.Aor.* 66; also Insler, *FsStrunk* 92 for a recent discussion with bib.). Hence, this isolated form offers no evidence for a thematic aorist to this verb. In this same way the *s*-aorist *ábhutsata* has probably replaced an older athematic ***ábudhata*** by substituting *-sata* for *-ata* (cf. Insler, *op.cit.* 558; Narten, *Sigm.Aor.* 178 with bib.).

Finally, concerning the causative paradigm, neither the Sanskrit *aya*-present or the reduplicated aorist of Sanskrit and Greek is likely to be old. Watkins, *op.cit.* 310, notes that, in addition to the thematic presents of Greek and Sanskrit, a nasal present is also well represented, as in Greek πυνθάνομαι, O.Ir. *ad·boind* 'refuse' and Lith. *bundù* 'erwachen', and even the sporadic existence of nasal presents to this root in modern Indian languages (cf. also Pokorny, *loc.cit.*; McCone, *IE Origins* 43; Fraenkel, *LEW* 62). From this Watkins concludes that the Indo-Iranian *aya*-present is a replacement of an older nasal formation.

Synchronically the aorist to *bodháyati* is the reduplicated *abūbudhat*, as in *ká idáṃ no abūbudhat* 'Wer hat uns da geweckt' [*RV* I 161.13]. Ignoring the Indic reduplication in *ū*, *abūbudhat* is a theoretical formal match to Greek πεπυθ- ε/ο- (see above, §2.4.1, for the length of the reduplicating syllable). Functionally, however, things are more complex. Although both forms are transitive, the two forms play fundamentally different roles

---

(with bib.). The process of creating thematic presents from old aorist subjunctives is well documented in Indo-European. For the canonical example cf. the root aor. ***h$_1$eu̯gh-to-i*** (Av. *aoxte*), > the thematic subj. ***h$_1$eu̯gh-e-to-i*** (Skt. subj. *ohate* [*RV* VII 16.11]) > thematic present ***h$_1$éu̯gh-e-*** as in Greek εὔχομαι (see J. Narten, *FsKuiper* 10 f.).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

in their respective languages. While *būbudhat* 'wake' (transitive) contrasts functionally with the 'passive' aorist *bodhi* : *budhran -anta*, the transitive usage of πεπύθοιτο is in no way extraordinary, since the thematic aorist πυθ- ε/ο- takes both a genitive complement [P 427, O 224 +] and an accusative: τάδε πάντε πυθοίατο '(if) they should learn all' [A 257, cf. E 702, O 379 +]. The lack of a strong functional match considerably weakens our possible equation. It now becomes necessary to ask whether or not some other explanation exists for the origin of our reduplicated πεπύθοιτο.

The reduplicated πεπύθοιτο is a polymorphic variant which is functionally redundant. However, its metrical use is different from that of the non-reduplicated forms. It is at least possible that this distinction played a part either in the creation of our form, or at least in its distribution in the Homeric texts. As noted above, the restriction to non-indicative modes is crucial here and ultimately related to the use of the augment in the Kunstsprache. In the indicative the alternation of augmented and unaugmented forms allows for a variety of metrical shapes as in the contrast between

#οἱ δ' ὡς οὖν ἐπύθοντο –|^{4}– ∪∪|^{5}–∪∪|^{6}–× [Σ 530]
#εἰ γὰρ ἐγὼ πυθόμην ∪|^{4}– ∪∪|^{5}– –|^{6}–× [δ 732][3]

Outside of the indicative reduplication can offer the same contrast as in the following

|^{3}– ∪ πυθοίατο – ∪∪|^{6}– × [A 257]
|^{3}– πεπύθοιτ' ∪∪|^{5}– ∪∪|^{6}–× [Z 50 = K 381, Λ 135]

If the reduplicated πεπύθοιτο were created to produce a form with one additional syllable, it would be redundant in the indicative where such variation is already possible. Conversely, a reduplicated aorist which exists only in non-indicative forms may indicative a metrical innovation. This distribution is not peculiar to this verb. Below are other exam-

[3] In this example both the augmented and non-augmented begin in the second foot. Note also the infinitive with the same placement #ἐλθεῖν ἠδε πυθέσθαι ∪|^{4}–∪∪|^{5}–∪∪|^{6}–× [κ 152].

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

ples of reduplicated aorists attested only in non-indicative forms. Although not probative in itself such a distribution gives circumstantial evidence that these forms were created for metrical reasons.

§3.1.2 κεκύθωσι 'they will hide (trans.)'

The standard handbooks list the reduplicated κεκύθωσι as an aorist (cf. Schwyzer, *Gr.Gr.* 748; Frisk, *GEW* I 834; Chantraine, *Gr.Hom.* 396; *DELG* 521-22). However, Risch, although he lists the form as a reduplicated aorist, *Wortbildung* 245, adds the marginal "Perf.?". In fact, there is room for doubt. The form occurs only once in Greek: ἀλλ' ὁπότ' ἄν σε δόμοι κεκύθωσι[4] καὶ αὐλή# 'but when the halls and court shall enclose thee...' [ζ 303].

Both the semantics (the form is transitive) and the form (zero-grade against the usual perfect κέκευθε) might argue for an interpretation as an aorist subjunctive. However, as for the function, although the form is transitive it is also ineffectual, and this is exactly the standard usage of the perfect forms of this verb: ὅσα τε πτόλις ἥδε κέκευθε# 'whatever so much the city holds' [X 118, cf. also the transitive at γ 18, ι 348]. Hence, there is nothing in the usage which requires an aoristic as opposed to a perfective reading. The status of the vocalism, however, is more complex.

The root vocalism of the perfect subjunctive in Homer shows considerable variation. All three main vowel gradations occur: 1) *o*-grade, πεποίθῃς [A 524], πεποίθομεν [κ 335]; 2) *e*-grade, εἰδῶ, εἴδομεν, εἴδετε; and 3) Ø-grade πεφύκῃ [Δ 483 cod. πεφύκει]. However, an *o*- grade formation has no bearing here since such subjunctives as πεποίθομεν are built to *o*-grade perfect indicatives. Nor is the evidence of πεφύκῃ helpful, since there is no evidence for a full grade for φυ- anywhere in

[4] Note the variant κεύθωσι in the app. crit. See below for discussion.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

Greek.[5] Hence, the forms above essentially show two patterns: 1) an archaic *e*-grade subjunctive[6] built to the *o*-grade perfect; 2) perfect subjunctives which are built to the indicative stem without vocalic alternation. Since κεκύθωσι fits neither pattern it is likely that this form is an aorist. However, since ευ-verbs in Greek seem to resist the standard o-grade perfect, cf. δέδυκε, βέβρυχε, μέμυκε all with zero-grade (εἰλήλουθα, δεδουπότος and (προ-)βέβουλα are exceptional, cf. Chantraine, *Morph.hist.* 191), it is at least possible that κεκύθωσι reflects an archaic perfect form, while the full grade has been innovated on the model of the present (cf. the discussion of τέτυκτο : τετεύχατο below). But although such variation is possible, a simpler solution to the origin of κεκύθωσι is proposed below.

Further discussion should begin with an examination of the forms of this root in Homer.

	Active	*Middle*
prs.	κεύθω, ἐκευθάνον [Γ 459]	κεύθωμαι [Ψ 244]
aor.	κύθε, κεκύθωσι, -κεύσης	
pft.	κέκευθε, ἐκεκεύθει	
fut.	κεύσω	

The middle voice is passive: εἰς ὅ κεν αὐτὸς ἐγὼν Ἄιδι κεύθωμαι 'until I am hidden in Hades' [Ψ 244]. The active forms are both transitive [I 313, τ 212 +] and intransitive with an implied object: οὐδ' ἐπικεύσω# '...and I will not hide (anything)' [E 816, cf. A 363, Π 19 +]. Three aorists are attested, but each only once in Homer (or elsewhere): 1) κεκύθωσι (assuming this is an aorist); 2) κύθε in ὅπου κύθε γαῖα

---

[5] All evidence suggests that this verb never participated in paradigmatic ablaut, cf. Latin *fuī*, Sanskrit *babhūva*. Even the supposed full grade *bhavati* may be a zero-grade formation < **bhe-bhuh$_x$-et*, as Jasanoff explains it (cf. above §2.1.1).

[6] For the ε-grade perfect forms see Schwyzer, *Gr.Gr.* 748. J. Jasanoff has recently argued for *e*-grade vocalism in the weak grades of the perfect noting such forms as Av. *cikōitərəš* 'appear(ed)' (1993 ECIEC paper).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

'where the earth hid (him)' [γ 16]; and 3) the *s*-aorist in εἰπέ μοι εἰρομένῳ νημερτέα μηδ' ἐπικεύσῃς# 'speak truly to me asking and do not hide (anything)' [ο 263]. This last form, however, is clearly an innovation built on the same formula as the future, οὐδ' ἐπικεύσω#, or the present, μηδ' ἐπικεύθε# [σ 171]. Here an aorist has been built so as to keep the tense in sequence with εἰπέ. For the two remaining forms, κεκύθωσι : κύθε, etymological analysis is required.

Two competing etymologies exist. The standard handbooks, Pokorny, *IEW* 950 f.; Chantraine, *DELG* 521-22; Frisk, *GEW* I 834, equate κεύθω with Old English *hȳdan* 'hide' < **(s)keudh-*. H. Nowicki, *KZ* 92 (1978) 184 f., defends the older etymology (with considerable discussion and bib.) in his derivation of Latin *custōs* 'guard' < **kudh-to-sd-s* with **k(e)udh-* as the root of Greek κεύθω. The Latin etymology has much to recommend it,[7] and **(s)keudh-* is a possible source for Greek κεύθω.[8,9]

A second etymology connects κεύθω with Sanskrit *gūhati* 'hides' (aor. *guh-a-*) < **gheudh-* (Hollifield, *System* 68; O. Szemerényi, *Studia Iranica* 9 [1980] 45 f.; Klingenschmitt, *Verbum* 226; Mayrhofer, *EWAia* I 502).[10] Although this approach offers evi-

---

[7] Nowicki, *loc.cit.* 188, notes that the oblique cases of *custōs*, Gen. *custōdis* pose problems which are best explained by assuming an underlying compound formation. Here the compound is made of **kudh-to* plus the zero grade of the root **sed-* giving **kudh-to-sd-* > **kusto-sd-* > **kustōss* 'der beim Verborgenen sitzt'.

[8] Armenian *sowzanem* 'submerge, hide', **ḱeu-?-an-* (?) may also belong here (cf. Frisk, *loc.cit.*) although the final *-z-* is difficult to motivate. It is also possible that *-VdhV-* gives Armenian *-VzV-* directly, a position ascribed to Klingenschmitt by J. Greppin, *JIES* 17 (1989) 167, although to my reading Klingenschmitt, *Verbum* 226, is considerably more circumspect. A better analysis takes *-z-* from **-dhs-* so that our nasal present is built to an older aorist stem **ḱeudhs-* (F. Kortlandt, *AArmL* 15 [1994] 29).

[9] One additional form may also point to a root **ḱeudh-*. D.Q. Adams, *HS* 107 (1994) 310 f., has recently derived TB *tuk-* be hidden', OE *dēog* 'he concealed himself' from **ḱeudh-* through metathesis as in such forms as Greek σκέπτομαι : Lat. *speciō*.

[10] An etymological connection between Greek κεύθω and Skt. *gūhati* is not a particularly new idea (cf. H. Hübschmann, *KZ* 23 [1877] 393; J. Schmidt, *KZ* 25 [1881] 166 f.).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

dence for a thematic aorist, it also suffers from serious phonological difficulties concerning both the vocalism and the reconstruction of the consonants. The lengthened *ū* of the Indic present is a small problem, which can be explained as secondary after the past participle *gūḍha* (cf. Cardona, *Them.Aor.* 50; cf. also Goto, *I Prs.* 296 f. n. 704). More difficult is the status of the consonants. The initial aspirate is assumed only on the basis of the Greek comparison (with κεύθω parallel to πείθω < **bheidh-*), but the outcome of the final **dh* as Indic *h* is more suspect. Although Indo-European **dh* does yield *h* in some circumstances (e.g., *hita-* : *-dhita-* 'set'), and Szemerényi notes *gudhera-* 'beschützend' (lexicographical sources only), we would expect stronger traces of the final dental in the verbal forms. Because of this, Mayrhofer, *loc.cit*, gives a second etymology with *gūhati* from **gheugʰ-*. The Iranian forms in **gaud-* are taken as dialectal (presumably from Old Persian where **ǵh* > **ź* > *d*) while Greek is still considered under this root but with a consonant dissimilation κύθε < **(é-)kʰutʰe* < **(é)-kʰukʰe*.

The aorist *guhas* is rare: *támobhir indra tám guhas* '...den versenktest du in Finsternis, o Indra' [*RV* VIII 6.17]. However, its existence is key to the attempt to equate *gūhati* and κεύθω. Since the phonological evidence is weak, the morphological comparison is needed to maintain the etymology (again cf. Hollifield, *loc.cit.*). But Cardona, *loc.cit.*, takes this as an "original" sixth class imperfect, and not an aorist at all. In defense of the Szermerényi etymology a sixth class present is unlikely here, where clear evidence exists that this verb once had a nasal present. Thus, several Middle Iranian forms show a nasal present (e.g., Khot. *uysgun-* < **uz-gu-n-d-* < **ghu-n-d-*, Klingenschmitt, *Verbum* 226; see also Szemerényi, *loc.cit.*). If the Greek root is cognate then the Hesychian κυνθάνει· κρύπτει might also suggest that κεύθω replaced an older nasal present. In this scenario both Greek κεύθω and Sanskrit *gūhati* are innovated from an earlier **ghundh-* (**ghungʰ-*?) contrasting with a thematic aorist.

But even if this etymology holds and Sanskrit *guhas* = κύθε, it might still be possible to take **ke-kudh-e/o-* as a polymorphic derived transitive since κεκύθωσι is gram-

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

matically transitive, while κύθε is not. But this line of reasoning is also suspect. The active forms of κεύθω can also be transitive, hence it is difficult to see why a specifically marked transitive would be necessary.

Hence, neither etymological approach offers any evidence for viewing κεκυθ-ε/ο- as an archaism, while the latter analysis (with *guhas* = κύθε) might support κύθε as a relic form. Both etymological approaches have problems and I see no way to decide the issue here. However, since neither approach offers any evidence in favor of viewing κεκυθ- ε/ο- as an archaic reduplicated aorist, the immediate question is can this reduplicated aorist be explained in some other way. In fact, a number of possible analogic solutions present themselves.

The most straightforward approach is to apply the same explanation proposed above for πεπύθοιτο. That is, since the subjunctive prohibits the augment, while the meter requires one syllable between δόμοι and the root κυθ- , perhaps the meter forced the creation of a newer reduplicated aorist.[11] Alternatively, κεκύθωσι could be a misreading for an older collocation *κε κύθωσι either with the so-called deictic **ke*,[12] since Nausikaa is pointing towards the house, or perhaps the modal particle κε(ν).[13] Finally, given that the Oxyrhynchus Papyrus offers the simple present κεύθωσι here, the change to the reduplicated form may be the result of later redactors, possibly even an Alexandrian

---

[11] There are a number of ways we can imagine this to have happened. Perhaps the phrase σε... δομοι κεκύθωσι καὶ αὐλή was reworked from an older narrative δόμοι ἔκυθον [με?] καὶ αὐλή or the like.

[12] For the deictic **ke* see Chantraine, *DELG* 329 under ἐκεῖ. A similar explanation is sometimes given for κέκλυτε perhaps from *κε κλυ- with a deictic particle **ke* (cf. Schwyzer, *Gr.Gr.* 804; and below §4.1.8 n. 15).

[13] Although in the wrong place in the line, the use of κε(ν) would not be entirely unexpected since κε(ν) can be used with ὁπ(π)ότε and the subjunctive to express a particular future event (cf. Chantraine, *Gr.Hom.* II 257 and Monro, *Gr.Hom.Dia.* 264). The position could be explained by assuming that this line was reworked from a construction with slightly divergent formula.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

creation. The exact formular process which gave rise to κεκύθωσι cannot be stated with any certainty. Each of the options above is possible; but given that the use of the reduplicated aorist in non-indicative modes is systemic and applies to a number of additional forms, a metrical creation from an older[14] thematic aorist is most likely.

§3.1.3 τετυκεῖν 'to prepare'

The reduplicated aorist τετυκεῖν 'to prepare' is used exclusively to describe the preparation of food. The active form occurs twice: δεῖπνον ἐνὶ μεγάροις τετυκεῖν ἅλις ἔνδον ἐόντων '(I bid them) make a meal in the halls of the abundant store of what is within' [ο 77 = 94]. More common is the middle which occurs largely in a sacrificial context with the slaughter of an animal: αὐτὰρ ἐπεὶ παύσαντο πόνου τετύκοντό τε δαῖτα 'and when they had ceased from their labor they prepared a meal' [A 467 = B 430 = H 319 = π 478, cf. θ 61 μ 283, 307, + all with δαῖτα, δεῖπνον, or δόρπον].

The present τεύχει 'prepares, makes' occurs almost exclusively in the active voice in Homer. A single instance in the middle present occurs in the *Odyssey*: χεῖρας νιψάμενοι τεύχοντ' ἐρικυδέα δαῖτα 'having washed their hands they prepared a glorious feast' [κ 182]. Outside the present system one finds: 1) a future τεύξω (always active in Homer!); 2) an isomorphic *s*-aorist ἔτευξα; 3) the reduplicated aorist τετυκεῖν : τετύκοντο; 4) a passive aorist ἐτύχθη; and 4) a number of middle perfect and pluperfect forms such as τετεύχαται 'they were fashioned', and the zero-grade pluperfect forms τετύγμην 'I was made...', or ἀρητὴρ ἐτέτυκτο 'he was made priest' [E 78]. In some instances the pluperfect τέτυκτο is used as a simple predicate 'is'. We can correlate the forms and usage as follows

---

[14] Note that such an explanation does not require that κύθε be an Indo-European archaism; merely that κύθε is older than the reduplicated form at ζ 303. This thematic aorist may itself be an analogical creation (e.g., πευθ- ε/ο- : πυθ- ε/ο- :: κευθ- ε/ο- : X, X = κύθε).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

	Active	*Active*	*Middle*	*Middle/Passive*
	'make'	'prepare'	'prepare'	'was made'
prs.	τεύχω		τεύχοντ[ο]	
aor.	ἔτευξα	τετυκεῖν	τετύκοντο	ἐτύχθη
pft.				τετύγμην
fut.	τεύξω			τετεύξεται

The usage 'prepare (a meal)' is essentially restricted to the reduplicated aorist and the rare middle present τεύχοντ[ο]. But this latter form is certainly a late addition to the paradigm. This usage is quite narrow only referring to food, while other sorts of preparation may take the active forms, cf. τεῦχε κυκειῶ 'she prepared a potion' [Λ 624].

The restriction in usage may well indicate an archaism.[15] Nor is there evidence here for a metrical creation where indicative forms are common (8X ind., 3X inf., 2X opt.). But the restriction in usage need not be old. Further, simply stating that a form has not been created for metrical reasons does not guarantee that it is an archaism. For that matter the very notion of metrically created reduplicated forms requires *a priori* the existence of a model on which to base the creation of such forms.

The key problem for τετυκεῖν, however, is etymological: the aorist root in final -κ- does not match the final aspirate of τεύχω. In principle, it is possible to view the aorist as the more archaic formation, with the present having aspiration from the perfect τετεύχαται. Consider the Homeric δέχομαι (Cret. δεκομαι) with the aspirated final from the perfect δεδέχαται (Chantraine, *DELG* 267 f.). However, no such explanation will work here. There is good evidence that the final -χ- is original. The closest etymological cousins point in this direction, e.g., τυχή 'fortune', τεῦχος 'tool' : pl. τεύχεα 'armor', as does the slightly less certain comparison with the formally similar τυγχάνω :

---

[15] Note that this is a general feature of some reduplicated aorists. Of the forms above, ἔπεφνον and πεπύθοιτο are also limited to a singular formular context.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

τυχεῖν 'chance upon' (Pokorny, *IEW* 271; Frisk, *GEW* II 888 & 940 f.; Chantraine, *DELG* 1111 f.). The attested forms for this second verb in Homer are:

	Active	*Middle*
prs.	τύγχανε	—
aor.	τύχε, τύχησας, *τετύχῃσι[16]	—
pft.	τετύχηκε	—
fut.	---	τεύξεσθαι

If we take the enlarged forms in *-ē-* as secondary, the overall paradigm is much simplified: a nasal present contrasting with a future and aorist τύχε. Before going on to discuss the semantic differences between these two Greek verbs, we should try to explicate the formal differences. Comparing this with the paradigm for τεύχω, given in brackets, allows us to postulate a purely formal paradigm

	Active	*Middle*
prs.	τύγχανε, [τευχω]	—
aor.	τύχε, *τετύχῃσι [ἔτευξα, τετυκεῖν]	[τετύκοντο] [ἐτύχθη]
pft.		[τετύγμην]
fut.	[τεύξω]	τεύξεσθαι, [τετεύξεται]

We see that the polymorphy of the proto-paradigm is small, and that these forms could descend from a single paradigm. The contrast between nasal present 'come upon' and thematic present 'make' is reminiscent of both κεύθω : κυνθάνει and πεύθομαι : πυνθάνομαι. Here, however, the morphological contrast is matched by a variation in semantics. The broader etymological connections help clarify this contrast.

---

[16] The reduplicated aorist *τετυχῃσι is given as a reading by Aristarchus to Λ 116 "τετύχῃσι ὡς λελάχωσι". The line in question reads ἡ δ' εἰ πέρ τε τύχῃσι μάλα σχεδόν, οὐ δύναταί σφι χραισμεῖν· 'but she [a doe], although she chances to be near, is unable to be of use to them'. Here, however, even if we were to accept this reading such a form would be likely candidate for a 'late form in a simile' as per Shipp, *Studies*.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

The standard handbooks compare the Greek forms with Goth. *daug* 'συμφέρει', Skt. *dogdhi* 'milks' (Pokorny, *loc.cit*; Frisk, *loc.cit*; Chantraine, *loc.cit*; Lehmann-Feist, *GED* 96 f.; some doubt in Mayrhofer, *EWAia* I 748) where an original **dheugh-* 'berühre' gives either 'chance upon' or 'make' which in turn gives 'derive benefit from -> extract, milk'.[17] This Sanskrit root has a thematic sixth class imperfect *áduhat* which is almost certainly an archaic middle form *áduha-t* as J. Wackernagel, *KZ* 41 (1907) 309 f. = *Kl.Schr.* 498 f., explains it. Jasanoff, *SMIE* 118 f., goes so far as to suggest that *áduha-t* forms a word equation with Greek τύχε.[18]

Even without the extra Greek cognates, the connection between τεύχω and τυγχάνω implies a proto-form **dheugh-*, which in turn argues that τ(ε)υκ- is an innovation from the older τευχ-. Thus, τετυκ- ε/ο- cannot be an archaic formation. The origins of this aorist cannot be stated with absolute certainty. Most likely this form derives from the older pluperfect.[19] Although it is not clear what verb served as the paradigmatic example, τετυκεῖν must have been built as an active transitive in contrast to the older intransitive pluperfect τέτυκτο.[20]

---

[17] Jasanoff, *loc.cit.*, also accepts the connection but gives the original sense 'gives as a benefit' and derives the other uses from this.

[18] According to this scenario the process of creating thematic aorist from root aorist dates from Indo-European (see also Hollifield, *System* 57 f.).

[19] The pluperfect singular τέτυκτο is matched by a third plural (ἐ)τετεύχατο 'were fashioned' [λ 808, Σ 574]. It is unclear whether or not the full grade plural is an archaism preserved from Indo-European (above n. 6) or is late, having been metrically restored from the past participle τετεύχαται < *τετυχFώς (cf. Lejeune, *Phonét. hist.* 81 n. 6). The latter is probably to be preferred based on the Myc. *te-tu-ko-wo-a* = *τετυχFοhα (Aura Jorro, *DMic* II 625).

[20] For possible paradigmatic example see πέποιθα (intrns.) : πεπιθεῖν (trns.) (§3.2.1) or perhaps λέλασται (intrns.) : -λέλαθον (trns.) (§4.1.4). Although none of these transitive forms is particularly old, they may pre-date the Homeric era by a sufficient margin for the development of such tertiary forms as τετυκεῖν.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

§3.1.4 τετύποντες 'striking'

This hapax in Callimachus [H. 3.16] may be a reduplicated aorist to the root τύπτω 'strike, beat' < **tup-ie/o-* : aor. τύψε, τυπεῖν (Frisk, *GEW* II 945 f.; Chantraine, *DELG* 1145 f.). Even the synchronic designation of this form as an aorist is not clear since the poet may also have been thinking of Aeolic participle inflection of πεπλήγ-οντες (see §3.5.1)or κεκλήγοντες (see Curtius, *Grk.Verb* 294 for accent). Under either analysis the form is a late artificial creation (Schwyzer, *Gr.Gr.* 749) and need not concern us here.

§3.1.5 Conclusion on the *Ce-CuC-e/o-* aorist

It is unlikely that any of the four forms above predate the Greek era. This is clearest for τετύκοντο where the final velar stop precludes an archaic formation. Only πεπύθοιτο has a chance of being old, but for this form, as stated above, a metrical solution is likely. The three *Ce-CiC-e/o-* aorists present more interesting and problematic cases.

§3.2 *Ce-CiC-e/o-*

§3.2.1 πεπιθεῖν 'persuade'

Although the reduplicated aorist πεπιθεῖν occurs frequently in Homer, there are no indicative forms. Thus we have the subjunctive twice: ὥς κέν μιν ἀρεσσάμενοι πεπίθωμεν# 'that we, making amends, may persuade him' [I 112]; μή σ' ἐπέεσσι παραι-πεπίθῃσιν Ὀδυσσεὺς# 'lest Odysseus persuade you with words' [χ 213]. In the optative, in addition to the active forms πεπίθοιμεν 'that we might persuade' [A 100] and πεπίθοιεν 'they might persuade' [I 181, Ψ 40], a single middle form is attested: πεπίθοιτο '(a man) might trust' [cod. πεπίθοιθ' K 204]. Finally, a number of non-finite forms are also attested: 1) the infinitive πεπιθεῖν [I 184]; 2) the participles

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

παραι-πεπιθοῦσα [Ξ 208]; πεπιθοῦσα [O 26]; παρ-πεπιθόντες [Ψ 37]; and παρ-πεπιθὼν [ξ 290 & ω 119]; all active meaning 'persuade'.

Two other aorist formations occur in Homer: 1) a thematic aorist πιθέσθαι 'trust' [O 156, Γ 260 +] which is always middle and 2) an *s*-aorist (παρ-)έπεισεν [H 120 = N 788, Ψ 606, ξ 123] which is always active and transitive. Clearly related to the aorist formations, the texts present us with both an *s*-future πείσομαι [H 286, +] and a single instance of a reduplicated future πεπιθήσω [X 223].

Outside the aorist system, the verb is noted for having both an active and middle present/imperfect πείθω 'persuade' : πείθομαι 'obey, trust'. The perfect πέποιθα and pluperfect πεποίθεα forms are always active in Homer, but with the semantics of the middle. Hence, the overall verbal system is as follows.

	Active	*Middle/Passive/Perfect*
	'persuade' (trns.)	'trust, obey' (intrns.)
prs.	πείθω	πείθομαι
aor.	πεπιθεῖν, -πεισεν	πιθέσθαι, πεπίθοιτο
pft.		πέποιθα
fut.	πεπιθήσω	πείσομαι

The contrast between present πείθω : πείθομαι and aorists πεπιθεῖν/-πεισεν : πιθέσθαι is striking. Since the opposition of transitive and intransitive forms is marked in the present by a simple change of voice, we might have expected a thematic aorist ἔπιθον opposed to the middle πιθέσθαι. In fact such a form does exist in Greek, but it is not attested until Pindar. Yet even aside from its late attestation there are reasons for thinking that the Pindaric ἔπιθον is an innovation. The diathetic system here is reminiscent of other verbs where in the present the contrast of active to middle marks a contrast of transitive to intransitive, while in the aorist voice in itself is not sufficient to mark this opposition. Compare also ἐγείρω -ομαι 'wake/be awoken' but with the corresponding

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

aorists ἤγειρα : ἠγρόμην or φύω : φύομαι 'engender/grow', aorists ἔφυσα : ἔφυν.[21]

Apparently, in Indo-European both present and aorist system marked transitive and intransitive pairs through a combination of stem variation and voice endings. Thus, in both Indo-Iranian and Germanic the relics of the Indo-European *o*-grade *e̯ie/o*-presents are used to mark especially transitive verbs contrasting with intransitive root or thematic forms. Further, there is evidence that in the parent language the nasal infix[22] and perhaps also reduplication[23] could also be used to mark transitive forms in the present system. However most languages have only scattered traces of such contrasts.[24]

The present contrast of πείθω - ομαι, then, represents an innovation within the history of Greek. Latin *fīdō* (OL *feidō*) 'trust' may argue that the active form is the older, which would require that the creation of the middle paradigm in Greek caused the older πείθω '*trust' to become 'make trust -> persuade'. However, even in Latin the present *fīdō* is linked to a deponent perfect *fīsus sum* which may indicate that the active inflection of *fīdō* is an innovation in that language (Walde-Hofmann, *LEW* I 493 f. ; Ernout-Meillet, *DELL* 233 f.; Flobert, *Déponents* 498 f.). Further, within Greek, the distribution of forms points to the conclusion that the active present πείθω was backformed to the older middle present, since active forms are significantly less common in Homer (middle 182X : active 47X) (Schulz, *Wurzel* Πειθ- esp. 24 f.).[25]

---

[21] Compare also καλέω - μαι 'call to/be called' : κάλεσσα - ατο 'call to' above.

[22] See Jamison, *aya-Form*. 186 n. 12; Sihler, *NCGGL* 499; G. Meiser, *FsRix* 280 f., (with bib.).

[23] See B. Delbrück, *Grundriss* IV 26 f.; G. Klingenschmitt, *MSS* 28 (1970) 84 n. 5

[24] In Greek not only have the reduplicated and nasal presents been largely bleached of their semantic content, but there is also almost no evidence for an especially transitive class of *o*-grade *e̯ie/o*-presents —see Tucker, *Creation* 134 f. Greek does, however, preserve some traces of transitive formations, e.g., the reduplicated *-ske/o*-presents. This formation is briefly discussed in Chapter VI.

[25] For this chronology compare also Benveniste's concise statement, *Vocab.* I 115, "...le

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

The points above argue that Indo-European **bheidh-e/o-* originally inflected only in the middle voice. However, two branches of Indo-European also show evidence for a derived transitive to this *medium tantum* root. Thus, Germanic **baid-jan*, Goth. *baidjan* 'compel', OE *bædan* and the OCS *bežd̨o* 'zwingen' each point to a so-called causative formation **bhoidh-e̯ie/o-* (Pokorny, *IEW* 117; Frisk, *GEW* I 487 f.; Chantraine, *DELG* 868 f.; Lehmann-Feist, *GED* 56 for the semantic interpretation). Now the scenario proposed above assumes that this transitive form vanished without a trace in Greek, and the newer transitive πείθω was created in opposition to the middle πείθομαι. However, it is at least possible that our Indo-European **bhoidh-e̯ie/o-* survived into Greek, but was altered on the model of the middle present. Thus, F. Specht, *KZ* 66 (1939) 205 f., assumes that the active transitive πείθω is a neologism for an older *ποιθέω = Goth. *baidjan* (also Walde-Hofmann, *LEW* I 495). The existence of this older *ποιθέω is, for Specht, confirmed by the reduplicated aorist πεπιθεῖν. A second point in favor of assuming a pre-Greek *ποιθέω might lie in the existence of the form πιθήσεις [ϕ 369], since a future in -ησε/ο- might imply a contract-present *πιθεω which would get us closer to a preform *ποιθέω.

Neither of these points, however, carries much weight. Specht's argument requires that the reduplicated aorist was paired with the *e̯ie/o*-present in Indo-European, but such productive pairing is unlikely. As to the *ē*-future, this in no way requires a present *πιθέω. Most likely this future has been built to the corresponding negative ἀπιθήσω 'I do not believe' [Ω 300] (for a recent discussion see Tucker, *Creation* 68 n. 76; earlier Schulz, *op.cit.* 48, saw the underlying pattern ἀπιθής > *ἀπιθέω > ἀπίθησα > πιθήσας). The reduplicated future πεπιθήσω [X 223] is presumably a late addition to

---

présent actif *peíthō* "persuader" est secondaire; il a été bâté assez tardivement sur *peíthomai* "obéir". Conformément à une alternance de caractère archaique, *peíthomai* a pour parfait *pépoitha* comme *gígnomai* : *gégona*".

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

the paradigm.[26]

In the aorist, however, the situation is even more complex. In addition to a distinction of transitivity, there are redundant forms in the aorist system. Unfortunately, the comparative evidence for any aorist to this root is lacking (Pokorny, *loc.cit.*). Any attempt to unravel the forms must rely only on the Greek forms before us. If we assume that the middle aorist πιθέσθαι is old (perhaps replacing an earlier athematic form), the oldest Greek forms would then be the middle present πείθομαι, aorist πιθέσθαι and perfect πέποιθα, all with usage 'trust'.

It is likely, then, that all active forms of π(ε)ιθ- are innovations within the history of Greek (again cf. Schulz, *op.cit.* 62). The analogical models for the present and πείθω and active *s*-aorist (3X *Iliad* 1X *Odyssey*) pose no problem. The origin of the reduplicated aorist are less clear. Certainly a backformation of the perfect πέποιθα, or better the zero-grade pluperfect ἐπέπιθμεν [Δ 159 +] is possible. The proportion would easily follow from a form like λελαθ-

λέλασται (intrns.) : -λέλαθον (trns.) :: *πέπιστο : X, X = πεπιθ- ε/ο-

But, as noted below (§4.1.4) -λέλαθον is itself a recent formation. Further, although such analogies may have played some in the creation of this form, they are reasons for assuming a metrical variant here.

As with πεπύθοιτο and κεκύθωσι above, πεπιθεῖν - οιτο is attested only in non-indicative forms. However, there is no active πίθον in Homer (although the form is attested in Pindar) which could serve as a pivot form in the same way that ἐπύθοντο

---

[26] Four reduplicated futures are built to reduplicated aorists in Homer: κεκαδήσει < κεκαδών; κεχαρήσεται < κεχάροντο; πεφιδήσεται to πεφιδέσθαι; and πεπιθήσω under discussion. The model for the creation of these forms was probably the reduplicated futures such as ἐιρήσεται [Ψ 795] to perfect ἔιρηται, which are themselves innovated from older desiderative formations (cf. Schwyzer, *Gr.Gr.* 783 f.; Risch, *Wortbildung* 350 f.; Chantraine, *Morph.hist.* 254 f.). See §3.5 for additional discussion.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

served for πεπύθοιτο. Nevertheless a metrical solution is possible here if we assume that the middle πεπίθοιτο[27] is the oldest form, created as a metrical variant of (ἐ)πίθοντο. The analogical creation of πεπίθοιτο from the older metrically equivalent thematic middle ἐπίθοντο [Γ 260 +] would follow in the same manner as πεπύθοιτο from the older ἐπύθοντο where all four forms have the same metrical shape ∪∪ – ∪. The external analogical source for our new reduplicated form would probably be πεπύθοιτο itself, as there is considerable formal similarity between the two verbs.

With the creation of the active πείθω 'persuade' there would now be two possible choices to fill the corresponding active aorist slot. A transitive aorist could be built either to the older thematic aorist or the innovated reduplicated πεπίθοιτο. But with the preference in Greek for differentiated aorist stems the creation of the transitive πεπιθεῖν would not be unexpected.

§3.2.2 πεφιδέσθαι 'to spare'

Φειδόμαι 'spare' (usually with the genitive) attests the single *s*-aorist φείσατο: οὐδέ νυ τοῦ περ / φείσατ' ἐνὶ μεγάροις ὁ γέρων 'not even this did the old man spare in the halls' [Ω 235-236]. Slightly more common (3x) is the reduplicated πεφιδ-ε/ο- : εἴ πώς εὐ πεφίδοιτο 'if he might spare him...' [Υ 464]; οὐδ' ἂν ... πεφιδοίμην / οὔτε σεῦ οὔθ' ἑτάρων 'I would not spare either you or your comrades...' [ι 277-78]; τόφρα τί μοι πεφιδέσθαι ἐνὶ φρεσὶ φίλτερον ἦεν / Τρώων 'till then it was more pleasing to me to spare any of the Trojans' [Φ 101]. A reduplicated future is also attested [Ο 215, Ω 158, 187].

---

[27] The analysis has the disadvantage of tracing all reduplicated forms of πεπιθ- ε/ο- to the single occurrence of πεπίθοιτο found in the Doloneia: #ὦ φίλοι οὐκ ἂν δή τις ἀνὴρ πεπίθοιθ' ἑῷ αὐτοῦ / θυμῷ τολμήεντι 'O friends, is there no man who might trust his own enduring thumos...' [K 204-5]. For apparently "late" features in this passage, as well as all of book K, see most recently J.B. Hainsworth, *Comm.Il.* III 173. Nevertheless, taking πεπίθοιτο as the pivot form is the easiest solution.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

	Middle
prs.	φείδεο
aor.	φείσατο, πεφιδοίμην
pft.	----
fut.	πεφιδήσεται

Despite the semantic variation the handbooks relate φειδ- to Lat. *findere* 'split', Goth. *beitan* 'bite' and Skt. *bhinátti* 'splits' (Pokorny, *IEW* 116 f.; Frisk, *GEW* II 999; Chantraine, *DELG* 1185; Walde-Hofmann, *LEW* I 500 f.; Ernout- Meillet, *DELL* 235; Seebold, *EtWb* 96 f.; Lehmann-Feist, *GEW* 66; Mayrhofer, *EWAia* II 273 f.). Chantraine, *loc.cit.*, notes that the semantic change is from 'separate' to 'je me sépare de qqch'. More explicitly O. Panagl, *Fachtagung VI* 317 f., shows that the diction in Homer may support this reading. Panagl notes several passages, the most telling of which is when Pandarus laments his leaving behind his horses when coming to Troy ἵππων φειδόμενος 'sparing his horses'. Panagl, *loc.cit.* 326, gives a reconstructed translation as 'sich von den pferden trennend/abwendend auf sie verzichtend'.

Now, accepting this etymology, a nasal present is confirmed by Skt. *bhinátti* = Lat. *findere*. Hence, we also expect a root aorist as in Sanskrit *abhet* [*RV* I 33.13], subjunctive *bhedati* [*RV* V 86.1 +] (L. Renou, *BSL* 33 [1932] 24; J. Narten, *Sprache* 14 [1968] 125 f.). The Latin perfect *fidī* may also represent an old aorist (Sihler, *NCGGL* 581). As with **bheudh-* above the full grade aorist subjunctive probably served as the starting point for the full grade present, which then has the same origin as the present πεύθομαι discussed above, although here the move to present tense is not yet complete in the Sanskrit of the *Rig Veda*.

Now with the transfer of the older aorist to the present, it is not surprising that Greek would engender a new aorist, and hence, we both expect and find an *s*-aorist, φείσατο. A sigmatic aorist may also be attested in Sanskrit, although the evidence is somewhat indirect. Thus, Macdonell, *Vedic Gr.* 381, takes *bhitthās* [*TS* +] as an *s*-aorist,

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

but this is hardly a solid basis for comparison. Slightly better is the form *abhītsam* [*AVP*] which when corrected to **abhaitsam* makes a passable aorist formation (see Narten, *Sigm.Aor.* 180 for bib. and discussion). The lateness of these forms only confirms that φείσατο is a Greek innovation.

There is no evidence that the reduplicated aorist is old. Although Sanskrit also shows a reduplicated aorist *abībidhat*, the form is listed only in the grammatical writings. Further, the formation is irregular, as if from a root **bheidh-* with the root final consonant aspirated. Most likely the form was built on the model of *būbudhat*. Πεφιδοίμην is therefore likely to be a late analogical creation, although the source for this form is difficult to find. Analogical expansion from the perfect, as in τετυκεῖν, is unlikely, since there is no evidence that Greek inherited a perfect for this verb.[28]

The metrical analysis would then seem to be our only option, but this explanation, which works well for πεπύθοιτο and πεπίθοιτο, is somewhat more suspect here. The problem is that there is no archaic evidence for a thematic aorist φίδοντο which would serve as a building block for our reduplicated formation (although such a form is attested in the late Greek Anthology, see Veitch *GV* 590). However, once other reduplicated aorists have been built to older simple thematic aorists, an analogical source becomes possible, e.g., πείθομαι : πιθέσθαι :: φείδεο : X, X= πεφιδέσθαι. The problem remains, however, why would this form have been created over the more obvious φίδοντο. Once again the metrics of the passages above require ∪∪ – ×, but augmented thematic forms such as **ἐφίδοιτο **εφιδοίμην and **εφιδέσθαι are prohibited.

§3.2.3 τέθιγε 'to touch'

The form is known only from a Hesychian gloss, and an emended one at that:

---

[28] A perfect is found in the *Rig Veda*, e.g., *bibheda* [*RV* II 14.6 +]. It is also possible that the Irish noun *bibdu* 'coupable' is, in origin, an old perfect participle **bhe-bhid-u̯ot-* as Vendryes, *LEIA* B 49, explains it (cf. also Walde-Hofmann, *LEW* I 500).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

τέθιγεν· ἥψατο [cod. τέθειγεν] (Curtius, *Grk.Verb* 293). Even if this reading is accepted, τέθιγε is unlikely to represent an archaism. Although the nasal present θιγγάνω 'touch' (: aor. θίγον) does not occur in Homer, this present is probably old judging from the cognates, Latin *fingō* 'id.', O.Ir. *comtuinc* 'er baut' < ***kon-es-uxs-dinget*** and perhaps Armenian *dizane-* 'heap up' < ***dhingh-*** (Frisk, *GEW* I 675; Chantraine, *DELG* 437; Klingenschmitt, *Verbum* 183).[29] With a nasal present predating the Greek era, we also expect to find evidence of a root (thematic or athematic) aorist. Hence, if we accept θίγον*[30] as the original aorist, a reduplicated τέθιγε* (if it really ever existed) would be redundant in the system. As with τετύποντες above (§3.1.4), Schwyzer, *Gr.Gr.* 748, accepts the existence of the form but adds the comment "spät (künstlich)".

§3.2.4 Conclusion on the *Ce-CiC-e/o-* aorist

There is no comparative evidence suggesting that any of our three *Ce-CiC-e/o-* aorist forms predate the Greek era. Moreover, in each case an analogical or metrical source is likely.

§3.3 *Ce-C̥C-e/o-*

§3.3.1 τετάρπετο 'delighted'

The reduplicated aorist (ἐ)τετάρπετο occurs: 1) with the accusative τεταρπόμενος τε φίλον κῆρ# 'delighting at heart' [α 310; cf. I 705]; 2) with the dative αὐτὰρ ἐπεὶ φρεσὶν ἧσι τετάρπετο δαίδαλα λεύσσων# 'but when he delighted at heart

---

[29] Instead of nasal present, Sanskrit shows a root formation *degdhi*. However, as Mayrhofer notes, *EWAia* I 746 f., the athematic present may have been made in analogy to the formally similar *dohá-* : *dugdhá-*.

[30] It is necessary to assume that the expected *τιχεῖν was remade to θιγεῖν on the model of the present (Chantraine, *loc.cit.*). Still this aorist may well be old judging from the Sanskrit *adihan* ≅ θίγον (Frisk, *loc.cit.*; Chantraine, *loc.cit*).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

seeing the wonders' [T 19; cf. Ξ 244]; 3) with the genitive ὀλοοῖο τεταρπώμεσθα γόοιο# '(when) we delighted in wretched wailing [Ψ 10, 98 cf. λ 212, Ω 513]. Metrically, the reduplicated aorist always occurs in the same slot in the line, after the feminine caesura. The two accusative constructions, α 310 and I 705, are best taken as accusatives of respect.[31]

The present τέρπει attests both active and middle forms in Homer. The active voice is usually transitive, τὸν ἔτερπε λόγοις 'he delighted him glad with words' [O 393], contrasting with the middle present, e.g., τέρπονται Κύπρις τε ...'Απόλλων 'Cypris and Apollo rejoice' [E 760]. It is likely that the active present is, like πείθω above, a late addition to the paradigm since it is relatively rare: 4X in the *Iliad*; 6X in the *Odyssey*; as opposed to the middle: 16X in the *Iliad*; 40X in the *Odyssey*. Further, this active present is isolated in Homer. In the aorist only middle and passive forms exist. Outside of Homer, an active future τέρψω and perfect forms, τέτροφα, τέτραμμαι are also attested. The overall Homeric paradigm is thus

	Active	*Middle*	*Passive*
prs.	τέρπει	τέρπομαι	
aor.		τετάρπετο, ταρπώμεθα τερψάμενος τέρψομαι (aor. subj.)	τάρπησαν, τάρφθα, +
fut.		τέρψομαι	

The variety of passive forms is interesting, but the chronology is clear. The passive in -φθ- occur only in the *Odyssey* and is certainly secondary: τάρφθη πολυδακρύτοιο γόοιο 'delighting in many-teared wailing' [φ 57 = τ 213 = 251, cf. ζ 99] (note the full grade ἐτέρφθητε [ρ 174, θ 131, cf. ε 47]). The unenlarged passive is

---

[31] But note that Latacz, *Zum Wortfeld* 58 f., argues that these are in fact true accusative complements. For a general discussion of Latacz' interpretation of the middle reduplicated aorist see §4.1.8.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

common in both the *Iliad* and the *Odyssey*, although it is restricted to the dual and plural: ταρπήτην, ταρπήμην, τάρπησαν. This η-aorist also means 'to enjoy, have one's fill of' usually with a genitive (ἐδωδῆς, εὐνῆς, ὕπνου, ἥβης), but other syntagmata occur, e.g., αὐταρ ἐπεὶ τάρπησαν ἐς ἀλλήλους ὁρόωντες 'but when they had had their fill, looking at one another' [Ω 633]; also with dative [Γ 441]. Despite the distribution, it is likely that this unenlarged passive in -η- is the more archaic form (cf. Prévot, *Aoriste* -θην 166; Chantraine, *Gr.Hom.* 399; Sihler, *NCGGL* 563 f.).

Even excluding the alternation τάρφθα : τέρφθα, there is still an excessive degree of polymorphy for this root. In addition to τάρπη*, there exist the three remaining aorist forms τετάρπετο, ταρπώμεθα, and τερψάμενος. Despite the formal variations, all of these aorist forms have essentially the same meaning. The *s*-aorist [μ 188 and π 26] is probably late, having been built to the isomorphic *s*-future τέρψομαι [Υ 23],[32] although this future may itself be a relatively late addition to the paradigm. This still leaves the three isofunctional forms τάρπη*, ταρπώμεθα and τετάρπετο.

Let us turn to the etymological details. The handbooks equate τέρπομαι with Sanskrit *tŕṃpati*, *tṛpṇóti* 'sich sättigen', Lith. *tarpti* 'thrive, prosper', Goth. *þaurban* 'have need of' (and other Germanic forms), and TA and TB *tsārw* 'to be comforted' (Pokorny, *IEW* 1077 f.; Frisk, *GEW* II 881 f.; Chantraine, *DELG* 1108; Mayrhofer, *EWAia* I 634 f.; Lehmann-Feist, *GED* 356 f.). Although the Tocharian form makes a good semantic match, there are several phonological difficulties here where Indo-European **terp-* should yield **cärp-*. The semantics of the remaining forms is as follows. The original sense was apparently 'take for oneself' as in Avestan *tarəp-* or the final of Sanskrit *paśutṛp-* 'cattle stealing' , but latter became 'use for oneself, enjoy' of the Greek lexeme and even 'seek to have use of -> need' in Germanic (cf. Lehmann-Feist, *loc.cit*; Mayrhofer, *loc.cit.*).

---

[32] There is no doubt that within the history of Greek, sigmatic aorists and futures influenced each other, e.g., ἐλεύσομαι < ἦλ(υ)θον, (Chantraine, *Morph.hist.* 247). For another example of the pattern see -κεύσω (§3.1.2) above.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

In Indic the active nasal present *tṛṇóti* was originally intransitive, but was replaced in later Sanskrit by the more marked intransitive *tṛpyati* [*AV*+]. Although *tarpáyati* dates from the Rig Veda, Jamison, *áya-Form*. 140 f., suggests that this form was built to this intransitive *tṛpyati* through a regular analogical process. This causative present shows a reduplicated aorist *atītṛpat* 'made to delight' [*AV*], but both the late attestation and the functional disagreement argue against any direct equation between this and the Greek form. Finally, a thematic aorist *átṛpat* occurs in the *Brāhmaṇas*, but this is also probably an innovation for an older root aorist, **térp-t* (or even *tṛp-tó*), which we expect in contrast to the archaic nasal present.

More importantly, the Greek τάρπη* may itself point to the athematic aorist we expect, since pairs of *-*ē*- and athematic aorists are not unknown in Homer, e.g., μίκτο and μίγη. Such pairs suggest that the aorist in *-*ē*- productively replaced older athematic forms not long before and even within the Homeric period (Chantraine, *Gr.Hom.* I 400; Jasanoff, *SMIE* 120). Assuming that our pre-Greek paradigm contained a root aorist middle **tṛptó* 'partook of, delighted in', both our reduplicated and non-reduplicated thematic aorists are presumably more recent innovations.

This original root aorist may have left other traces as well. As with several of the forms above (see esp. §3.1.1), the Greek thematic present may itself be the relic of an older root aorist subjunctive. The pre-Greek pattern may then have been identical to that proposed above for πεύθομαι, with a full grade middle present contrasting with an older zero-grade middle aorist. In addition to the thematic present and *ē*-aorist, other traces of the older root aorist may survive in the zero-grade thematic aorists; however, the chronology between reduplicated and non-reduplicated forms is not clear.

The thematic aorist ταρπώμεθα is restricted to the first person plural subjunctive and occurs only three times in identical lines: ὕπνῳ ὑπο γλυκερῷ ταρπώμεθα κοιμηθέντες# 'that we, lulled to sweet sleep, may rejoice...' [Ω 636 = δ 295 = ψ 255]. Cardona, *Them.Aor.* 125 n. 1, takes this as a de-reduplicated form of the more common

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

τετάρπετο where the reduplicated form here would produce an un-metrical line. This may well be correct. We would then have the converse of the pattern described above for πεπύθοιτο and πεπίθοιτο where the older τεταρπώμεσθα (beginning at the feminine caesura) has been altered to ταρπώμεθα (beginning at the masculine caesura). Now, this does not pose any real problem for our analysis since the same process is at work. Although here the process appears to function in reverse order (reduplicated τεταρπ-ώμεσθα > ταρπώμεθα), the basic fact remains: reduplicated and non-reduplicated verb forms function within *one and the same* verbal paradigm, the choice between forms being determined purely on the basis of metrical considerations.

Moreover, an alternative to viewing ταρπώμεθα as a de-reduplicated form exists. The reduplicated middle may have come about in several ways, but here again some metrical motivation is likely. Only two of the eight attestations are in the indicative (the remaining forms being three participles and three subjunctives). The two indicative forms both occur after a vowel before the feminine caesura. It is at least possible that an older *ἐτάρπετο was remade to τετάρπετο to avoid the hiatus. Thus, although the rare ταρπω-μεθα is not the best evidence for a thematic aorist to this root, it is still possible that an older thematic form underlies τετάρπετο. If we assume that the process of creating the reduplicated form occurred at a time when syllabic *ṛ still scanned as a single short vowel, ie., *tṛp-e/o-,[33] then a thematic *e-tṛ́ponto would be metrically equivalent to ἐ-πύθοντο. From this older thematic form the reduplicated *te-tṛ́poito was built exactly as πε-πύθοιτο was built from ἐ-πύθοντο. However, with the change of *ṛ to αρ the metrical use of both reduplicated and non-reduplicated aorists would be considerably restricted, giving us the distribution we find in the texts.[34] Again, a metrical solution for

---

[33] For a general discussion of the survival of syllabic ṛ in epic see R. Janko, *Il.Comm.* IV 11 (with bib. n. 11).

[34] For another possible root in ṛ, with a different restructuring of the paradigm see με-μαποῖεν below (§4.1.7).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

τετάρπετο cannot be given with any absolute certainty, but such an explanation remains possible, and the restricting of τετάρπετο to the position after the feminine caesura strongly supports this conclusion.

§3.4 *Ce-C̥C-e.o-*

§3.4.1 δέδαε 'taught'

The reduplicated aorist δέδαε 'taught' occurs only five times in Homer, twice in identical lines. All instances are in the *Odyssey*. Three times the form is active taking a double accusative[35] construction: ὃν ποτέ μιν δέδαε φρεσὶ πότνια Κίρκη '(a knot) which the mistress Circe taught him' [θ 448]; ὃν ῥ' Ἥφαιστος δέδαεν καὶ Παλλὰς Ἀθήνη / τέχνην παντοίην 'whom Hephaestus and Pallas Athene taught all manner of craft' [ζ 233 = ψ 160]. The forth active occurrence has a single expressed accusative: ἔργα δ' Ἀθηναίη δέδαε κλυτὰ 'and the Athenian-one taught famous crafts' [υ 72].

The single instance in the middle voice is also transitive but with a single object: ...σε γυναῖκας ἐγὼ δεδάασθαι ἄνωγα# '...I bid you to learn about the women (those which have dishonored me, those which are guiltless)' [π 316].[36] The ending in -ασθαι for the expected - εσθαι is peculiar, but unlike the alphathematism above (εἶπον : εἶπας §1.1), here there is a simple explanation where verbs in final -α often have infinitives in -ασθαι (cf. Chantraine, *Gr.Hom.* 82, δεδάασθαι "...semble reposer sur δεδάεσθαι, cf. δεδάε", parallel to φαάνθην [A 200 +] for *φαένθην).

---

[35] For a recent study of the usage is Homer and elsewhere see Jacquinod, *Double acc.* 186 f.

[36] Zenodotus offers one additional reading of δεδάασθαι as a replacement for δεδιδάχθαι in τά σε προτί φασιν Ἀχιλλῆος δεδιδάχθαι 'those things which they say you have been taught by Achilles' [Λ 831 & *app.crit.*]. Zenodotus may, however, have been motivated by other considerations. Thus, van der Valk, *Researches* II 49, argues that δεδιδάχθαι seemed out of place to the Alexandrians since it implies that the older Patroclus was the pupil of the younger Achilles.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

This reduplicated form contrasts with the so-called passive aorist (ἐ)δαη- 'came to know, learned'. But this latter form can also be transitive, e.g., ἐδάην φυὴν 'I knew the form...' [Γ 208], ἐδάην βουλήν τε νόον... 'I came to know the counsel and mind (of many men)' [δ 267]'; in the infinitive δαῆναι ἐμὸν νόον 'to learn my mind' [δ 493] and ταῦτα δαήμεναι 'to learn this' [Z 150 = Y 213]. Intransitive uses are also attested as in the infinitive: πολέμοιο δαήμεναι [Φ 487, cf. ν 335, Ψ 262]; and perhaps also in the participle: μή πῶς με προϊδὼν ἠὲ προδαεὶς ἀλέηται 'lest he seeing or knowing me should avoid me' [δ 396], although here the με can go with any or all of the three verbs. In addition, a subjunctive is attested, always intransitive, always line final, always in an ὄφρα clause and only in the first person δαείω : δαῶμεν < *δαήω : *δαήομεν [ι 280 +, plural at B 299].

In addition to the reduplicated and so-called passive forms, this verb attests a future δαήσεαι [γ 187, τ 325] and two perfect formations. Like the reduplicated aorist, both future and perfect forms are attested only in the *Odyssey*. The unextended perfect occurs only once in Homer, in one of the later books of the *Odyssey* ἀείδει δεδαὼς 'he, having been taught, sings (songs)' [ρ 519]. The other three instances of the perfect are built to the root δαη- of the aorist passive (and future), but with the perfect - κ- δεδάηκε 'has learned' [θ 134, cf. θ 146, β 61]. Thus, the Homeric paradigm is as follows

	Active 'teach X to Y'	*Middle* 'teach X'	*Middle/Passive/Perfect* 'know, learn'
aor.	δέδαε	δεδάασθαι	δαῆναι
pft.			δεδαώς, δεδάηκε
fut.			δαήσεαι

The hiatus in ἔδαην/δέδαε requires some original root final consonant. The comparative evidence suggests an original *-s-*. A preform **dn̥s-* allows us to equate this root with the nominal form δήνεα 'counsels' < **denses-* which matches Skt. *dáṃsas-*

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

'weises, wunderbares Können', Av. *daŋhah-* 'id.' (Pokorny, *IEW* 202; Frisk, *GEW* I 338; Chantraine, *DELG* 278; Mayrhofer, *EWAia* I 689), and probably also Hittite *daššu-* 'mighty' (Mayrhofer, *loc.cit.*; Melchert, *AHP* 163). Further, pre-Greek **dn̥s-* is also the root of the present διδασκ- ε/ο- (Homeric διδασκόμενος [Π 811 +]), which has generated its own paradigm within the history of Greek[37] (A. Debrunner, *Mélanges Boisacq* 252 f.; Frisk, *loc.cit*; Chantraine, *loc.cit*). However, although the comparative evidence suggest an original **dn̥s-*, the phonology is not entirely clear since there is some evidence that *C-n̥s-V* gives Greek *C-as-V-* as in δασύς 'thick' < **dn̥s-u-* (Sihler, *NCGGL* 171; Schwyzer, *Gr.Gr.* 307, notes δέδαε as an exception without explanation).[38]

Outside of Greek only Indo-Iranian offers evidence for a verbal formation to this root. Avestan attests a verb in two verses: *hyaṯ xšmā uxδāiš dīdai ŋ́hē paourvīm* 'when I was first instructed by your words' [*Y* 43.11c; Insler *GZ* 63]; *nōiṯ ərəš.vacå sarəm didąs drəgvātā* 'The truly speaking man has never expounded alliance with the deceitful one' [*Y* 49.9b; Insler *GZ* 96]. Insler, *GZ* 237, takes the former form as a redactional change for a perfect middle **dăđai ŋ́hē*. Hence, the Iranian evidence points to a reduplicated present

---

[37] An older etymology equates διδασκ- ε/ο- with Latin *discō* 'to learn' both from **di-d^{h}k-sḱ-e/o-*. Such a reconstruction, however, requires some theory of 'reduced vowel' which is unlikely here. Further, the semantics do not quite agree, since the Greek form is causative (taking a double accusative). E. Hamp, *Classical Philology* 63 (1968) 285 f., assumes a proto-form **di-dk-sḱō* agreeing with Latin, but which was analogized to διδασκ- ε/ο- in Greek due to the influence of the semantically similar root δα-. Most recently, Keller, *Verbes en -sc-* 114 f., is also suspicious of the etymology of διδασκ- ε/ο- from the root **dens-*. Such reservations are groundless (see Debrunner, *op.cit.*, and above).

Because of the semantics, it is, of course, tempting to imagine some sort of contamination between **di-dk-sḱe/o-* and **di-dn̥s-(s)ḱe/o-*. Still, at an Indo-European level the two roots have relatively little in common except for the initial consonant, and the difference in usage, as noted above, also poses problems.

[38] For a possible phonological explanation see H. Hoenigswald, *Laryngaltheorie* 208, although I don't see how his appeal to syllabicity can differentiate δέδαε from δασύς. For arguments against the change **n̥sV* > ασ*V* see D. Ringe, *MSS* 50 (1989) 141 f.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

active **di-d(e)ns-t* 'announced (?)' and middle perfect **ded(e)ns-e* 'I was instructed' (see Insler, *loc.cit.* for a note on the unexpected vowel gradation). The interpretation of *dīdai ŋ̊hē* as a perfect works well and clarifies the passage. The exact usage of the injunctive *didaṣ* is slightly less clear. Nevertheless, in view of the usage both here and at *Y* 43.11c, something akin to 'teach/be taught' is certain (cf. Kellens, *Verbe avestique* 67 'enseigne'; H. Humbach, *GZ* II 211 'teaches, approves of'; and Insler's reading above). In Sanskrit we find a single verbal form, *daṃsáyas* 'du machtest leistungsfähig' [*RV* X 138.1]. This form may well be denominative (Jamison, *áya-Form.* 83), but it is also possible that the *aya*-formation is a replacement for an older transitive reduplicated form matching the Avestan (cf. above §3.1.2 where *bodháyati* may have replaced an older nasal present; see Mayrhofer, *EWAia* I 688 f. with bib.).

The Avestan *didaṣ* < **did(e)ns-t* would then appear to be a reduplicated formation with especially transitive force built to the root **dens-* 'have knowledge/power'.[39] In Greek, however, the form was extended with the *-ske/o-* suffix yielding **di-dn̥s-ske/o-* > διδασκ- ε/ο- . Most likely the **-ske/o*-form was a Greek innovation. Since Greek preserves few traces of the Indo-European transitive formations, it is likely that the change from **di-d(e)n̥s-* to **di-dn̥s-ske/o-* would have been all but required to insure the usage 'teach' as opposed to 'have knowledge/power' (cf. Schwyzer, *Gr.Gr.* 710, who notes that the reduplicating -σκε/ο- verbs are "...oft transitiv, auch kausativ...").[40] However, even if διδασκ- ε/ο- is a Greek creation, it must be old enough to account for the deverbative διδάσκλος 'teacher' (Chantraine, *Formation* 245), a form which is already attested in

---

[39] For a general discussion of reduplication as a transitive formative see G. Klingenschmitt, *MSS* 28 (1970) 84 n. 5.

[40] The origins of the Greek reduplicating *sk*-presents are unclear. Note, however, that despite the view in the standard handbooks there is some evidence that this formation may be archaic. Outside of Greek only Latin *discō* is certain, but circumstantial evidence for such a formation occurs in both Hittite and Tocharian.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

Mycenaean *di-da-ka-re*, an adjective of children, perhaps *διδασκάλει 'at the school teacher's' or the like (Ventris-Chandwick, *Documents* 162 f.; despite the controversy over the exact meaning the general sense is clear, cf. Aura Jorro, *DMic.* I 171 for bib.). Hence, both comparative evidence, as well as the pattern of attestations in Greek, suggests that διδασκ- ε/ο- is the primary form of the verbal paradigm, and that the aorist stems have been built to a root δα- within the history of Greek.[41] Certainly, if the present διδασκ- ε/ο- (< **di-dn̥s-ske/o-*) is the more archaic form the double *-ss-* would simplify and re-segment as δι- δα- σκ- ε/ο-. From this present, the abstracted root -δα- could then serve as the building block for the newer aorist, although it is not clear what form would serve as an analogical pivot.[42]

---

[41] Jacquinod, *Double acc.* 264, assumes that the double accusative construction had its origin in the reduplicated aorist and then spread to the present. True, there are three clear double accusative formations of δέδαε in Homer, versus only one certain double accusative of a form of διδασκ-, and that of a derived aorist ἐδίδαξε [θ 480-1]. However, Jacquinod himself, *op.cit.* 185 f., notes that several passages containing διδασκ- are difficult to interpret and may contain double accusative constructions with elements gapped to avoid repetition. Further, the evidence above clearly suggests that δέδαε is the more recent form.

Such a chronology does not preclude the possibility that δέδαε was the first in its paradigm to take a double accusative construction. However, such a transfer of usage has little to recommend it. The example used by Jacquinod, *op.cit.* 190, as a parallel formation "λέλαθον 'faire oublier' est causatif par rapport à ἔλαθον" is quite a different matter (also *op.cit.* 264). Λέλαθον is the causative of the middle λελάθοντο 'they forgot' not the active ἔλαθον 'they escaped the notice of' (see below §4.1.4). Most likely the double accusative construction developed as an active counterpart to a transitive middle form, διδάσκομαι (as Keller, *Verbes en -sc-* 113 f.).

[42] Consider, e.g., 1) χάρ- η : κε- χαρ- ε/ο- :: δά- η : X, X = δέ- δα- ε; or 2) ἴζομαι : ἕζετο :: διδασκ- ε/ο- : X, X = δέδαε (?). But the first option is unlikely. Κε- χάροντο is itself a late analogical form (see below § 4.1.3). More importantly, the usage of δέδαε is unique among reduplicated aorists in that it is causative in the true sense of the word, that is, it turns a transitive verb (learn X) into a double accusative verb (make Y learn X). The second option would allow for an aorist with the same semantics as the present, although the middle voice of ἴζομαι : ἕζετο might make this a poor choice for an analogical model. Better in some ways is ἀραρίσκω : ἄραρε :: διδασκ- ε/ο- : X.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

Finally, it might be possible to take δέδαε not as an analogical creation built to some other reduplicated aorist, but instead from the perfect δεδαώς [ρ 519] just as τετύκ- οντο (trns.) was built to τέτυκ- το (intrns.). Although this approach suffers in that δεδαώς is both a hapax and late (occurring in the latter half of the *Odyssey*), this last possibility may be supported by comparative evidence. The pattern of reduplicated present : perfect here is matched by the Avestan data if we accept, with Insler, the reconstruction of the perfect **dădai ńhē*.

	**i*-reduplication	**e*-reduplication
Avestan	*didąs*	**dădai ńhē* (pft.mid./intrns.)
Greek	διδασκ- ε/ο- *	δέδαε (act.aor./trns.)

Thus, Greek not only extended the older present with a formant **-ske/o-*, but the older perfect was restructured into δέδαε taking on active endings and the semantics of the present just as in the pairing of πείθω : πεπιθεῖν.[43]

In conclusion, the idea that the present διδασκ- ε/ο- is built to the older reduplicated aorist is demonstrably false. Both the comparative evidence outside of Greek, as well as the inner-Greek history (Mycenaean *di-da-ka-re* versus δέδαε [ζ 233 +]) indicate the present is ancient. The late attestation combined with certain phonological problems make it appear that the reduplicated δέδαε is a Homeric development, a late addition to its paradigm, even though the exact analogical model is unknown.

Although this analogy requires an equation between *Ce-* and *aC-* reduplication, it better explains both the semantics and the dropping of the present -σκε/ο- in the aorist. The variation between - ίσκω and - σκω remains a separate problem (see below §6.6 f.).

[43] It remains, of course, possible that δέδαε represents the older form, with the Avestan as the innovation. This would indicate a pattern **di-dn̥s-e/o-* (prs.) : *de-dn̥s-e/o-* (aor.) equivalent to **si-sd-e/o-* : **se-sd-e/o-* (if this pattern is old, cf. above §1.4) and perhaps Greek πι- πτ- ε/ο- : Skt. *pá-pt-a-* (Klingenschmitt, *Verbum* 130 f.). Although possible, a number of factors argue against this interpretation here (see above and §7.2).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

§3.4.2 πέφραδε 'pointed out'

This root attests a number of forms: 1) a *ie/o*-present, φράζ-, which is always middle (e.g., φράζονται 'they consider' [δ 699]); 2) a future middle; 3) an *s*-aorist middle (e.g., -φράσσατο [A 537]); and 4) an *s*-aorist passive . The exact usage is difficult to isolate. Both intransitive and transitive forms occur (e.g., φράζετο θυμῷ 'he pondered at heart' [Π 646] versus, e.g., ὃ ἐσθλὴν φράζετο βουλήν 'he considered a better plan' [Σ 313]). Further, there is a distinction between the mere act of considering, and the reaching of a conclusion τὰ φράζεαι ὅσσα θέλῃσθα 'you devise what you wish' [A 554]. But the variation between 'ponder', 'consider', and 'devise' is not so great. All of these uses have the central etymological source 'to use one's φρένες'. The middle forms contrast with the reduplicated aorist, which is always active.

This reduplicated aorist is all but restricted to singular finite forms. The first person singular occurs only once: ἵνα γάρ σφιν ἐπέφραδον ἠγερέθεσθαι 'where I showed them to gather' [K 127]. The infinitive is almost as rare with two occurrences: ὅν με κελεύεις / πεφραδέμεν 'which you bid me to point out' [ζ 46-47]; and πεφραδέειν [τ 477]. A second person imperative πέφραδε and a single optative, πεφράδοι [Ξ 335], are the only other non-indicative forms. The remaining forms are all third person singular indicative πέφραδε(ν) [Λ 795 +].

Now the semantics of the reduplicated aorist are quite different from that of the present, *s*-aorist and future forms, and are also diverse, e.g., θεοῖσι δὲ πᾶσι μετελθὼν / πεφράδοι 'one going forth might tell the other gods' [Ξ 335]; τὴν πέφραδ' Ἀθήνη 'which (path) Athene pointed out' [α 444]; ἰὼν καὶ πέφραδε μῦθον 'going over (to him) make known (your) word' [θ 142]. However, again there is an underlying pattern which the English glosses obscure. 'Tell' and 'point out' are not so different from one another, and a strict translation of 'point out' would work passably well in most instances. In each instance, moreover, there is either a personal dative, μοί, τοί, οἱ, σφισί, Τρώεσσι [Ξ 500], κούρῃ [ζ 47], or an accusative [K 127 only], or an implied

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

dative as in ἡ δὲ μάλ' αὐτίκα πατρὸς ἐπέφραδεν ὑψερεφὲς δῶ 'she immediately pointed out (to him) the lofty house of her father' [κ 111 = ο 424, cf. α 444 (quoted above), Υ 340, η 49, θ 68, θ 142 (quoted above) +]. The semantics 'point out, make another see' probably have their origins in 'make another use their φρένες'. The function is a sort of quasi-causative here with the dative.[44]

Formally, φράζονται is certainly a *ie/o*-present to a root *φρ̥δ-, where the α-vocalism represents the outcome of the older syllabic nasal (cf. Chantraine, *Formation* 360; Frisk, *GEW* II 1038; Chantraine, *DELG* 1224 f.). It is tempting to see in this formation *ie/o*-present to the nominal φραδή 'plan', but this is unlikely. Formally, we would expect *φραδάω as the denominative to φραδή. Moreover, the semantics described above imply a connection directly to φρήν. The external history of this root is all but unknown outside of a small number of Germanic cognates, e.g., ON *grunr* 'Verdacht' < **g*ʰ*hren-* (cf. Pokorny, *IEW* 496). In this case we seem to be dealing with a verbal form in **-d-* further enlarged with the 'durative' *ie/o*-suffix (cf. Sihler, *NCGGL* 502 & 510). But in most verbs reconstructed with a **-d-*suffix, this suffix is itself used as a present formative (Sihler, *loc.cit.*). Hence, we might expect a present *φράδω from which the reduplicated aorist would be a late creation. But there is no evidence for such a present form.

Still, several scholars have voiced suspicions concerning this reduplicated aorist. Thus, Leumann, *HW* 60, notes the possibility that πέφραδε could be a reanalysis for an older *ἐπ(ι)- ἐ- φραδε, perhaps with the pivot form still preserved in διεπέφραδε [ρ 590] (cf. Veitch, *GV* 610 for similar objections and bib.). Although possible, such a scenario finds little support in the texts. If this were the case one would expect to find instances of ἐπι + φραδ- with the usage attributable to the reduplicated aorist. However, the few instances of ἐπι + φραδ- have essentially the same syntax and usage of the pres-

---

[44] As the contrast between such forms as δέδαε and πεπιθεῖν shows, the reduplicated aorist is not a transitive formation, but instead a semantic causative (see below, §6.1 & §6.2, for additional discussion).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

ent and *s*-aorist, not the usage of the reduplicated aorist, e.g., πᾶσαν ἐπιφρασσαίμεθα βουλήν# 'we shall consider all manner of counsel' [N 741, cf. B 282, σ 94]. Any attempt to derive πέφραδε from ἐπι + φραδ- suffers from an additional problem of explaining both the change in the final consonant and the change in voice. Although one could assume that the final stop is analogic, the change to active voice is difficult to motivate in a verb that is otherwise a *medium tantum* in Homer. Finally, the specific usage of πέφραδε implies that this form was created as a 'causative' to the present φράζονται. Probably the form was created in analogy to δέδαε or some other specifically causative reduplicated aorists.

§3.4.3 Conclusion on the *Ce-C̥C-e/o-* aorist

Both reduplicated aorists of the shape *Ce-C̥C-e/o-* lack comparative evidence suggesting a archaic formations. However, in neither case is there a clear analogic source.

§3.5 *Ce-C̥HC-e/o-*

Of all the reduplicated aorists built to roots in a resonant, only one shows any evidence for a resonant laryngeal cluster. Its designation as an aorist, however, is open to debate.

§3.5.1 πέπληγον 'they struck'

Considerable disagreement exits regarding the interpretation of this form. Both Brugmann, *Kurze* 565, and Schwyzer, *Gr.Gr.* 777, take πέπληγον as pluperfect. There are, however, three problems with this interpretation: 1) in most cases πεπληγ- ε/ο- is transitive, with resultative usage; 2) the form is thematic; 3) there can be no doubt that some forms of πέπληγον are synchronically aoristic. The first two points pose no real problem. Although rare, resultative perfect forms do occur: βεβίηκεν 'has afflicted' and

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

the pluperfects βεβλήκει 'struck', ἔστασαν 'they set up'.[45] As to the inflection, thematic perfect forms are rare, but are attested in Homer, particularly in the pluperfect (Schwyzer, *Gr.Gr.* 767 f.; Tichy, *Onomat.* 71 f.; Nussbaum, *GsCowgill* 234). The aorist forms cannot, however, be so easily discounted.

This is, of course, the most damaging point for any 'perfect' interpretation. In one passage πέπληγον is coordinated with two other aorist forms: οἱ δ' ἅμα πάντες ἐφ' ἵπποιιν μάστιγας ἄειραν (aor.) / πέπληγόν θ' ἱμᾶσιν, ὁμόκλησάν (aor.) τ' ἐπέεσσιν 'then they all lifted the whips above the horses, and they struck (them) with the reins, and they called together (to them) with words...' [Ψ 362-363]. These considerations above have led many scholars to view πέπληγον as a reduplicated aorist (Frisk, *GEW* II 561; Chantraine, *DELG* 917; Risch, *Wortbildung* 243).

There is, however, also a problem with the aoristic hypothesis: the existence of the perfect participles πεπληγώς and πεπλαγυῖα. An aoristic interpretation would seem to require both an aorist πεπληγ-*e/o*- and an isomorphic perfect formation πεπληγ- (Chantraine, *Parfait* 14 f.). One could, however, assume that the aorist form is older where the zero-grade root vocalism in -πληγ- (< **Cl̥h₂C*) was reinterpreted as a full grade. The parallels of such forms as πέπηγε or γέγηθε would then make the transfer possible (cf. below §4.1.4 and §4.2.3).

Most recently, Tichy, *Onomat.* 48 & 65 f., argues for a different analysis taking all instances of πεπληγ- as perfective, but considers this form 'ein präsentisches Perfekt'

---

[45] Compare: ὑπερθεν δὲ σκολόπεσσιν / ὀξέσιν ἠρήρει, τοὺς ἔστασαν υἱες Ἀχαιῶν 'and above were fixed sharp stakes, those which the sons of the Achaeans had set' [M 55-56]; or ὅτ' ἐν Ἄργεϊ νῆας ἐΐσας/ ...ἕταροι.../ ἔστασαν 'when the companions had set the curved ships in Argos...' [γ 180-82]. Both these passages (and others) are discussed by S. West, *Od.Comm.* 170 f. West favors the correction to the aorist ἔστασαν. But the need for a correction is based entirely on the presupposition that a transitive perfect is impossible. The reasoning is, of course, suspect: i.e., 1) there are no transitive perfects in Homer, therefore; 2) whenever we find a transitive perfect we emend it out of existence, therefore; 1) there are no transitive perfects; etc.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

with 'iterativer Funktion'. Tichy, *Onomat.* 65 f., correctly points out that the majority of cases of πέπληγον are iterative. This is clearest where the verb refers to feet pounding on the ground as in: #οὐρανόν ἐς πολύχαλκον ἐπέπληγον πόδες ἵππων 'the feet of the horses beat into all-bronze heaven' [E 504, cf. Π 728] or #πέπληγον δὲ χορὸν θεῖον ποσίν 'they beat the dance floor with their feet...' [θ 264]. Compare also the passage described above [Ψ 362-363] where 'they beat (the horses) with the reins' almost certainly refers to repeated action.

Slightly more interesting are those forms referring to striking the body. The infinitive πεπληγέμεν at Ψ 660 is intransitive, but the sense 'to box' requires an unexpressed reciprocal, and presumably is also iterative in origin, i.e., 'box' < 'beat over and over'. Twice, women in mourning beat their breasts #στήθεα πεπλήγοντο [Σ 31, 51]. In view of this mourning usage it is likely that such phrases as πε-πλήγετο μηρώ 'and he beat his thighs' [M 162 +], should also be seen as iterative (cf. Tichy, *Onomat.* 66, & see below n. 49).

Thus, not only are the forms of πεπληγ- ε/ο- specially marked as transitive, but they are also notable in being iterative. The most important contribution of Tichy's work is to show that πεπληγ- ε/ο- is not an isolated formation, but instead belongs to a small class of iterative thematic perfects including the formally similar κεκοπώς[46] 'striking' and perhaps a number of intransitive forms as well, e.g., βέβρυχεν, μέμηκον etc. (see Tichy, *op.cit.* 72 f.).

In addition to the reduplicated form two other aorists occur. Thus, in Homer we find the *s*-aorist πλῆξα 'strike' (14X *Iliad*, 9X *Odyssey*), the aorist passive πλήγη 'be

---

[46] Κεκοπώς [σ 335] is given in most manuscripts at B 60, although Allen prefers the Alexandrian correction to κεκοπών (see van der Valk, *Researches* II 5 f.). Chantraine, *Gr.Hom.* 397, accepts the latter reading and takes this as a reduplicated aorist. But this is unlikely. Formally, κεκοπών would be the only reduplicated aorist with *o*-grade vocalism. More importantly, the semantic use of this participle is not unexpected in a perfect of this kind as Tichy, *loc.cit.*, has shown.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

struck' (4X *Iliad*), and the passive participle πληγεὶς (5X *Iliad* 3X *Odyssey*). The semantic coincidence of the *s*-aorist and reduplicated πεπληγ- ε/ο- suggests that one or the other form is an innovation, and hence supports the theory that the reduplicated form is, in origin, a perfect.

In general, the standard handbooks (Frisk, *loc.cit.*; Chantraine, *DELG* 917) list the present as the formally similar -πλήσσεις (3X *Iliad* 1X *Odyssey*), but this present is always prefixed with a preposition. This patterning suggests that - πλήσσω is a late addition to the paradigm analogically built to the aorist -πλῆξα -> -πλησσ- ε/ο- (Fraenkel, *LEW* 603; Beekes, *Development* 200; Tichy, *Onomat.* 49).[47]

Greek πλήγη, πεπληγ- ε/ο- and the nominal πληγή 'blow' were probably originally associated with Greek πλάζω 'beat, drive off course, daze'. Judging from the non-present forms πλάγχασα [τ 187] and πλάγχθη the present here is in origin a nasal formation *πλαγγ-i̯ε/ο- < **plang-i̯e/o-* matching the Latin *plangō* : *plānxī* 'hit oneself, mourn' (Frisk, *GEW* II 548 f.; Chantraine, *DELG* 909; Walde-Hofmann, *LEW* II 315 f.; Ernout-Meillet, *DELL* 512; Schrijver, *Reflexes* 223).[48] The root **pleh₂g-* is also found in Germanic, OE *flōcan* 'strike' : Goth. *faiflokun* 'bewail', and possibly Irish *len* 'lamentations', and Irish *lésaid* 'heftig schlagen' (Pokorny, *IEW* 801 f.; Seebold, *EtWb* 205 f.; Lehmann-Feist, *GED* 119).[49]

---

[47] Alternatively πλήσσω < **pl(e)h₂k-i̯e/o-* may have been built to the by-form **pleh₂k-* which is found in Balto-Slavic, e.g., Lith. *plàkti* 'schlagen' (Fraenkel, *LEW* 602 f.; for similar by-forms see **pāk/ǵ-* §4.2.3 below). The OCS *plačǫ sę* also belongs here, but the metaphor of 'beating (the body)' -> 'weep' has here completely taken over the use of the verb (see n. 49 below).

[48] The short vowel is probably the result of Osthoff's Law **pl̥h₂ng* > **plāng-* > **plang-* (e.g., Sihler, *NCGGL* 58 f.), although it is also possible that we should posit a third by-form **plang-* to account for the presence of the nasal in the non-present forms.

[49] Regarding our root **pleh₂ǵ-* there is some evidence that the specialized usage of striking the body in moments of grief was already a 'proverbial' notion in Indo-European. This is particularly clear in those languages where the meaning of this root has passed en-

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

The Gothic *faiflokun* suggests a perfect formation in the parent language.[50] Hence, I am inclined to accept Tichy's analysis, taking πεπληγ- ε/ο- as perfective.[51] With the rise of the reduplicated aorist in epic diction, the iterative usage, the thematic inflection, and perhaps the Aeolic inflection of such forms as κεκληγο[ν]τες[52] may have created some doubt even in the mind of the epic poets as to the exact interpretation of these forms in Homeric tradition.[53]

---

tirely from 'beat, strike' to 'strike the body' -> 'mourn' as in Gothic and OCS. The key passage in Homer is πεπλήγετο μηρώ / χερσὶ καταπρηνέσσ' 'he struck his thighs with the hands flat' [O 397, cf. M 162, O 113, Π 791, *ν* 198]. For extensive study of these passages see Onians, *Origins* 184 "...in none of the instances recorded [of striking the thighs] is it a matter of surprise or any lighthearted occasion, but in all cases the person is in distress, in grief or anxiety"; cf. also Lowenstam, *Patroklos* 31 f. Note that the gesture remains significant even into the middle ages as in the XVIII[th] Canto of Dante's *Inferno* (A. Mandelbaum's translation):

> the farmer who is short of fodder rises
> and looks and sees the fields all white, at which
> he slaps his thigh (*si batte l'anca*), turns back into the house...

[50] One could, of course, argue that if Indo-European had a reduplicated aorist here, and such a form survived into proto-Germanic, it would simply have been reinterpreted as a perfect and hence be subsumed into the early Germanic preterite system.

[51] Tichy, *Onomat.* 71, denies the equation of πεπληγ- = Goth. *faiflok-*, instead tracing the origins of the iterative perfect to older iterative presents. Whatever the historical background, the significant point for our purposes is to show that πεπληγ- is in no way abnormal but synchronically functions as part of a small class of iterative perfects.

[52] As with κεκοπώς above and πεπληγώς [B 264] itself, there is some variation in the tradition regarding the exact form and accentuation of κεκληγο[ν]τες where the manuscripts show κεκλήγοντες and the unmetrical κεκληγότες. Aristarchus proposes the correction to κεκληγῶτες or κεκλήγοντες. The former is predictable, but the latter Aeolic form may well be original (see van der Valk, *loc.cit.*). Πεπληγώς [B 264] is the manuscript form, although many modern editions adopt the scholiastic "correction" to πεπλήγων. But this Aeolic form may well be based on κεκλήγοντες (again see van der Valk, *Researches* II 5, 6 n. 28).

[53] The confusion between aorist and perfect (and even present) continues in Classical poetry. J.E. Harry, *Classical Review* 24 (1910) 178 f., tries to sort out these forms, but it seems clear that Classical poets recognized both an aorist and perfect πεπληγ-.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

§3.6 Conclusions on the *Ce-CRC-e/o-* aorists

Of the forms above only πεπύθοιτο is matched by an analogous form in Sanskrit, but here the comparison is so weak as to be uninstructive. Further, there is reasonable evidence to suggest that this form as well as κεκύθωσι, τετάρπετο, and πεπιθεῖν are not old, but products of epic diction. For each of these, metrical expansion may have played a part in their creation. The aorists τετυκεῖν, δέδαε and πέφραδε are also unlikely to be old and may be explained as late analogical creations.

The metrical arguments above do not, in themselves, explain the origin of the forms so much as describe their use in the Homer texts. However, in a larger sense, the metrical variation is clearly responsible for the existence of some reduplicated aorists. Assuming that Greek inherited a small number of reduplicated aorists, say εἶπον, ἔπεφνον, εὗρον, ἐκέκλετο, ἐνεγκεῖν and ἄραρε, the analogical creation of more reduplicated aorists is difficult to motivate. The inherited forms should have been viewed as a small closed class of irregular verbs, and such closed classes typically resist analogical expansion. The awareness of this principle is no doubt responsible for the prejudice of seeing reduplicated aorists as archaisms. However, the requirements of epic diction offer a clear reason for the expansion in this case: to provide forms of varying metrical shapes.

Setting aside the vowel initial forms, eight reduplicated aorists show the distribution of πεπύθοιτο with no indicatives forms attested. The first four forms, πεπύθοιτο, κεκύθωσι, πεπιθεῖν, and πεφιδέσθαι are described above. The remaining four forms, λελαβέσθαι, τεταγών, - πεπαλών, λελάχωσι will be described in the following chapter. To this list of eight should probably be added three more forms τετάρπετο (with 2/8 indicative forms), κεκαδών (with 1/3) and λελάθοντο (2/9). The exact analogical processes are more difficult to locate, mostly because of the impossibility of clearly chronologizing the forms into strata of older and younger reduplicated aorists. The final stop of τετυκεῖν seems to require some sort of analogic expansion

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

from the perfect, while the semantics of δέδαε instead indicate a form modeled on the present. Positing an intermediate perfect, reflected in δεδαώς, might help connect the two processes, but some present influence would be required in any case.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

CHAPTER IV

§4.0 *Ce-CaC-e/o-*

In Greek α- vocalism has several possible origins: 1) a vocalized sonant as in τετάρπετο (< **te-trp̥-e/o-*); 2) from laryngeal origin; 3) from Indo-European **a*; 4) an analogic zero-grade (e.g., κε- χαρ- ε/ο- below).[1] Zero-grade types which are demonstrably derived from vocalized sonant have been described in the preceding Chapter. The forms below are, therefore, of the remaining three types. No absolute distinctions are possible between types two and three. Some scholars deny the existence of any Indo-European **a*, and reconstruct a number of words with an internal laryngeal (cf., e.g., R.S.P. Beekes, *Kratylos* 26 [1981] 107), while other scholars take a more traditional approach (e.g., Mayrhofer, *Idg.Gr. I/2* 169 f.). To a large extent such controversy need not concern us here, although I have reconstructed internal laryngeals where the evidence suggested them.

Those forms which belong to the final class are demonstrably innovations of the Greek language. Certainly, a number of *Ce-CaC-e/o-* aorists fall into this category. However, even those forms below which show evidence for "real etymological" *CaC* (or *CHC*) vocalism are unlikely to be old as the evidence will show. For ease of presentation I have divided the forms below into two groups, separating out four aorists, τεταγών, κεκαδών, - πεπαλών, πεπαγοίην, which are best discussed together and with special attention to possible Italic cognates. Seven other α- grade aorists, λελάκοντο, λελαβέσθαι, κεχάροντο, ἐκλέλαθον, λελάχωσι, δέδακε and μεμαποῖεν, will be discussed first.

---

[1] For the canonical discussion see Kurylowicz, *Apophonie* 174 f.; For the Greek material see also C.J. Ruijgh, *Fachtagung VIII* 83.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

§4.1.1 λελάκοντο 'they bark'

This form occurs in the *Hymn to Hermes*, #οὐδὲ κύνες λελάκοντο 'nor did the dogs bark' [145], and is listed as a reduplicated aorist in the standard handbooks (Curtius, *Grk.Verb* 293; Schwyzer, *Gr.Gr.* 748; Frisk, *GEW* II 88 f.; Chantraine, *DELG* 622). However, two recent works, Nussbaum, *GsCowgill* 234 f. and Tichy, *Onomat.* 63 n.1, have independently taken this form as a pluperfect. Nussbaum, *GsCowgill* 232 n.7, notes that the active λέλακε(ν) 'shouts, bellows' [*Cyn.* 2 141] is certainly a perfect. Further, the usage at *Hermes* 145 points to a pluperfect interpretation since the perfect λέληκ- typically takes an animate subject, while the aorist consistently takes an inanimate (similar comments in Tichy, *loc.cit.*). Since λελάκοντο is best taken as a pluperfect, it need not concern us further.

§4.1.2. λελαβέσθαι 'to grasp'

The present λαμβάνω 'I grasp, take' is not attested in Homer. Instead the *ie/o*-present λάζομαι is the present of the aorist ἔλαβε. This present is relatively rare, occurring only in the middle third singular (ἐ)- λάζετο (9X *Iliad*, 2X *Odyssey*) and one third plural optative λαζοίατο [B 418]. Contrasting with this middle present, the aorist in Homer is almost always active. There are only two exceptions, both in the early books of the *Odyssey*: 1)...ἐλλάβετ' αὐτῆς# 'he seized it' [ε 325]; 2) λελαβέσθαι in #τόν γ' εἴ πως σὺ δύναιο λοχησάμενος λελαβέσθαι# 'if you, lying in ambush, are able to catch him...' [δ 388].

The use of the middle voice at ε 325 is clearly a neologism for the more common ἔλλαβε, created to fill out the line.[2] In Homer, there are several instances of middle forms built to verbs which, in general, occur only in the active voice. The poets created

---

[2] In a slightly different position in the line the poet has taken the more obvious solution with ν-movable: ἔλλαβεν αὐταρ Ὀδυσσεὺς# [σ 394], and one wonders why this solution was not used at ε 325.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

these middle verbs so as to produce forms which were one syllable longer, e.g., διώκε-το|– ∪∪|– –|# [Φ 602] from the active δίωκε. Such forms are most common in the latter part of the line used to avoid the fourth foot caesura, the so-called Hermann's Bridge location (Meister, *Kunstsprache* 19 f.), and to a lesser extent in the fifth foot as in ἐξερεείνετο μύθῳ# [K 81] (Meister, *loc.cit.*; Hainsworth, *Comm.Il.* III 165). A similar fifth foot violation may underlie the creation of our reduplicated form. That is, λοχ-ησάμενος λελαβέσθαι was preferred to – ∪ λαβέσθαι# thus avoiding the fifth foot inter-dactyl caesura. But before accepting such a metrical solution for our lone reduplicated λελαβέσθαι, let us examine more closely the etymology and usage of this verb.

Already in Homer, the correspondence between aorist ἐλλάβε and present λάζομαι had become obscured. This led to the creation of the newer present λαμβαν-ε/ο- directly from the aorist stem, presumably on the model of ἔλαθον : λανθάνω. As a late creation, the present λαμβαν- ε/ο- can tell us nothing about the prehistory of this verb. Etymologically, the ζ : β alternation between aorist and the original present λάζομαι argues for a final labiovelar. The double λλ of the augmented forms points to an initial **sl-*, as does the Aeginetan spelling λ*h*αβών (cf. Schwyzer, *Gr.Gr.* 310; Lejeune, *Phonét. hist.* 119; Sihler, *NCGGL* 171 & 216). Hence, we are dealing with a root **slăĝ*$^{\mu}$- at the earliest level (cf. Frisk, *GEW* II 77-78; Chantraine, *DELG* 616).

However, the root **slăĝ*$^{\mu}$- is rather poorly attested in Indo-European. Outside of Greek, Pokorny, *IEW* 958, lists a single Germanic cognate in OE *læccan* 'fassen, ergreifen' < **(s)lakjan*. An alternative, older, and almost pre-Neogrammarian etymology instead connects Greek ἔλλαβε with a root **labh-* reflected in Sanskrit *lá(m)bhate* 'seize'. Although this etymology works well with the unoriginal λαμβαν- ε/ο-, it cannot explain the more archaic λάζομαι. However, this older etymology is still invoked by some scholars who assume a contamination in Greek of Indo-European **labh-* and **slăĝ*$^{\mu}$- once the latter had passed to **hlab-* (see most recently Sihler, *NCGGL* 575). This approach easily explains the noun λάφυρον 'booty', although this noun can also be ex-

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

plained as a derivative of the perfect ἔιληφα (Chantraine, *DELG* 616).[3] For the following discussion I will work with a single root **slăg$^{\mu}$-*. Finally, concerning the *a*-vocalism of the root it is not clear whether we have a root **slăg$^{\mu}$-* or **sl(e)h$_2$g$^{\mu}$-*. Some scholars have argued for the laryngeal root, although there is no outside confirmation (cf. C.J. Ruijgh, *Fachtagung VIII* 83 '**sleh$_2$g$^{\mu}$-*'; Schrijver, *Reflexes* 305 '**lh$_2$g-*').

Whatever the status of the final consonant of the Attic perfect ἔιληφα, this form shows the correct outcome of the reduplicated **se-slā-*. In contrast, the reduplicated λε-λαβ-ε/ο-, as if from a root **lab-* (or **lag$^{\mu}$-*), can only be the result of analogy. Further, if we accept the laryngalist solution, the vocalism would also need to be explained analogically, as *se-s̥lh$_2$g$^{\mu}$-e/o-* would yield Greek *ἔιλᾱβ- ε/ο- (cf. Ruijgh, *loc.cit.*, who takes ἔλλαβε, and hence also λελαβέσθαι, as having a 'nouveau degré zéro').

Regardless of the status of the root vocalism, the analogized reduplicating syllable is sufficient to guarantee that λελαβέσθαι is a late analogical creation, most likely after the formally similar λελαθέσθαι (cf. West, *Comm.Od.* I 219; see §4.1.4). Chantraine, *Gr. Hom.* 396, takes λελαβέσθαι as an analogic form but with special force, an "...hapax qui semble expressif au lieu de λαβέσθαι". However, I see nothing in the usage which requires any special force here. Given the conclusions above, the motivation for this reduplicated form was most likely metrical.

§4.1.3 κεχάροντο 'they rejoiced'

The reduplicated aorist κεχάροντο 'they rejoiced' occurs six times in Homer. Three instances are in the third plural indicative: ²|– μέγα δὲ Τρῶες κεχάροντο# 'the

---

[3] The status of the root **labh-* in Greek is ultimately connected with the origin of the aspirated Attic perfect. If we accept **labh-* in Greek prehistory, we can explain the perfect ἔιληφα as having been altered from an earlier *ἔιληβα 'I have taken' on the influence of the semantically similar λάφῡρον (Sihler, *loc.cit.*; D. Ringe, *Glotta* 62 [1984] 125 f.). From this we can now explain the whole class of aspirated perfects. For criticism and an alternative approach, see S.R. Slings, *Glotta* 64 (1986) 9 f.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

Trojans greatly rejoiced' [Π 600]; [2]|– κεχάροντο δὲ πάντες' Αχαιοί 'all the Achaeans rejoiced' [δ 344 = ρ 135]. The other three instances are all in the optative: 1) 3sg. #οὐ κέν οἱ κεχάροιτο γυνή 'the woman would not rejoice in this' [β 249]; 2) 3sg. [3]| ἴν' ἄγαλμα θεὰ κεχάροιτο[4] ἰδοῦσα# 'that the goddess, seeing the offerings, might rejoice' [γ 438]; and 3) 3pl. ἄλλοι τε Τρῶες μέγα κεν κεχαροίατο θυμῷ# 'and the other Trojans would greatly rejoice at heart' [A 256].

In addition to the reduplicated aorist, two other aorist forms occur in Homer: 1) a sigmatic (descriptively a liquid) aorist χήρατο [Ξ 270 only]; 2) the aorist passive χάρη (12X *Iliad*, 3X *Odyssey*). A single perfect participle, κεχαρηότα occurs at H 312 which matches the use of the aorist forms, all of which are intransitive with the sense 'to enjoy, rejoice' (but see below for further discussion). Strikingly, the active present χαίρω and the active future χαιρήσειν [Θ 363 only] are also intransitive meaning 'to enjoy, rejoice'. Finally, a reduplicated future occurs twice in Homer: 1) the middle #οὐ μέν τοι θυμὸς κεχαρήσεται 'your heart will not rejoice' [ψ 266]; 2) the active οὐδέ τί φημι / πᾶσιν ὁμῶς θυμὸν κεχαρησέμεν '...I do not think this will delight the spirit(s) of all in the same way' [O 97-98]. The usage is remarkable in that this appears to be the only transitive form of χα(ι)ρ- in Homer. Note, however, that some scholars take this as an intransitive form, presumably taking τί θυμὸν 'any spirit' as the subject of the infinitive (Chantraine, *Gr.Hom.* 448; Janko, *Comm.Il.* IV 239; for the transitive reading see Schwyzer, *Gr.Gr.* 783). Thus, the Homeric forms are

	Active	*Middle-Passive*
prs.	χαίρω	
aor.		κεχάροντο, χήρατο, χάρη
pft.	κεχαρηότα	
fut.	κεχαρησέμεν, χαιρήσειν	κεχαρήσεται

[4] Note that most manuscripts read a third plural κεχαροίατο here, although the singular is certainly called for (see R. Janko, *Comm.Il.* IV 351 for discussion).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

The Greek present χαίρω is a *ie/o*-derivative of the root **ǵher-* (Chantraine, *DELG* 1240 f.; Frisk, *GEW* II 1064). Outside of Greek, verbal forms are attested in Sanskrit *háryati*, and Italic: Latin *horitur*, Oscan *herest* 'volet', Umbrian *heri* 'volt' (Pokorny, *IEW* 440 f.; Walde-Hofmann, *LEW* I 657; Ernout-Meillet, *DELL* 299; Mayrhofer, *KEWA* III 583).[5] Although each of these forms is in origin a *ie/o*-present, there is some disagreement in stem vocalism. Both the Greek and Latin forms appear to show a zero-grade formation, **ǵhr̥-ie/o-* > Greek χαιρ- ε/ο-, Latin *horitur*, while the Sabellic and Indic forms indicate a full grade (Meiser, *Lautgeschichte* 78 "o[scan] *herest*... u[mbrian] *heriest* */herjest/* 'wird wünchen' < vorursab. **ǵher-ie-s-t*").

The variation in vocalism has caused some scholars to suspect parallel denominative formations. Certainly the Goth. *gairnjan* 'lust after' is likely to be a denominative formation (Pokorny, *loc.cit.*; Lehmann-Feist, *GED* 140; but note C. Barton, *IF* 84 [1989] 143 n. 26 [quoted below]). Sihler, *NCGGL* 517, takes χαίρω from < **kʰar-ie-ō* as if from χαρά (see also Walde-Hofmann, *loc.cit.*). Finally, Ernout-Meillet, *loc.cit.*, suggest that the Latin *horitur* may well represent a 'causative' *o*-grade formation.

Nonetheless, such parallel formations seem counterintuitive. To begin with the *o*-grade Latin form *horitur* is more likely to represent the phonological outcome of a zero-grade formation agreeing with the Greek (Frisk, *loc.cit.*). This still leaves both full and zero-grade forms, but this vowel gradation may have a simple solution. A. Nussbaum, *Glotta* 54 (1976) 248 f., argues that we have an ablauting formation **ǵher-ie/o-*/**ǵhr̥-ie/o-*. But because ablaut is unexpected in a thematic form, Nussbaum goes on to suggest

---

[5] Hittite *karii̯a-* 'comply with' does not belong here, despite several attempts to connect this verb with the root of Greek χα(ι)ρ- (see Lehmann-Feist, *GEW* 140; Tischler, *HEG* III 506 for bib.). There is universal agreement that this verb is the equivalent of the collocation *kari tii̯e-* 'id.' but the identity of the nominal is open to debate. Oettinger, *Stambildungen* 552 n. 5, takes this as a relic of an older locative *kár-i* 'heart' noting a second collocation *kar-di-aš-ta-aš i-ia-mi* [*KBo* III 7 I 26] 'ich willfahre dir' < 'tue das deines Herzens', which seems probative.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

that this thematic present replaced an athematic form which may still be reflected in Umbrian *pisher* < **k^{u}is=ǵher-e/o-*. Hence we have evidence for an archaic **ǵhér-t(o)-* (3sg.) : **ǵhr-ént(o)* (3pl.), later altered to **ǵhér-i̯e-t(o)-* : **ǵhr̥-i̯é-nt(o)*.[6]

Significantly, the Sanskrit verb is all but limited to the present system, with other tense forms existing only in grammatical writings (Whitney, *Roots* 203; Mayrhofer, *KEWA* III 583). Likewise the Italic forms offer no evidence of archaic aorist or perfect formations. Hence, the Greek non-present forms must be analyzed without comparative data. But here it is immediately clear that none of the non-present forms of χα(ι)ρ- can predate the Greek era. Nussbaum, *op.cit.* 249, notes that the aorist, future, and perfect forms are all built on the χαρ- form of the present. If these forms were old, they should be built to a root **ǵher-* and, thus, the expected form of the *s*-aorist should be *χειρατο < **ǵhĕrs-* while the reduplicated form would be *κέχροντο, parallel to πέφνον.

The origin of the *s*-aorist is clear. This form is simply the phonological result of adding an *-s-* on to the root of the present, i.e. *χαρ-σ-. In the same way the aorist passive was made by adding the intransitive *-ē-*, i.e. χαρ-η-. Further, the reduplicated future is clearly secondary to either the reduplicated aorist or perfect (Risch, *Wortbildung* 350 f.). The exact chronology between perfect and reduplicated aorist is, however, more difficult to sort out. The reduplicated aorist may have been built to the older perfect (as apparently with τετύκοντο above). But the single attestation of κεχαρηότα, added to the fact that perfects in *-ē-* are part of later strata of Homeric diction, makes it likely that the aorist is the older form.

But even assuming that this aorist predates the corresponding perfect and future

---

[6] The original voice of this present is also a point of controversy, but the active intransitive of Greek seems most likely, since both the transitive force of the Indic forms (cf. Jamison, *aya-Form* 146), and the middle inflection of the Italic can easily be explained as analogical. Contrariwise, the active intransitive usage in Greek is extremely difficult to motivate analogically. Note Barton, *loc.cit.*, "[i]n surface syntax *háryati* is (mostly) transitive... this state of affairs is less archaic than the intransitive usage of Gk. χαίρω...".

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

forms, the phonological evidence above shows that κεχάροντο is a late addition to its paradigm. Once, again, however, the exact analogical process is unrecoverable. Certainly an analogy from the formally and semantically similar τετάρπετο is possible, i.e., ταρπ-η : τε-ταρπ-ε/ο- :: χαρ-η : κε-χαρ-ε/ο-, and I have no doubt that these forms have influenced each other. Other analogical pivots might lie in ἐπλήγη : πέπληγον (§3.5.1), or even some slightly divergent formation, e.g., τύχθη : τετύκοντο.

Regardless of what form served as source for the analogic creation of κεχάροντο, it is still necessary to motivate its creation: i.e. why was a second middle aorist created alongside χάρη? The simplest explanation, again, is that κεχάροντο was produced so as to create a metrically distinct form. Prévot, however, *L'aor. grec en* -θην 166 f., argues that the different aorist stems have distinct usages: χάρη shows the 'état de joie', κεχάροντο expresses 'l'idée d'une explosion de joie'. Compare also the extended discussion of Latacz, *Zum Wortfeld* 58 f. (I will discuss Latacz's analysis in detail below §4.1.6). However, at this point I should say that I am extremely skeptical of such analyses. One can always argue that a given passage has some emphatic force. Given such parallelism as $^{2}|-\cup$χάρη δ' ἄρα οἱ προσιόντι# 'he (Sarpedon) rejoiced at his coming' [Ε 682] and #οὐ κέν οἱ κεχάροιτο γυνή 'the woman would not rejoice in this' [β 249], it is difficult to see any strong semantic variation. Note however that both forms in this example occur in the third foot with κεχάροιτο beginning at the masculine caesura, while χάρη begins at the feminine caesura, one syllable later. Hence, the placement strongly suggests a purely metrical distinction.

The two verbs above offer some of the clearest proof for reduplicated aorists whose existence in the Homeric texts is based on metrical motivation. Phonologically, neither form can be old. Nor is there is any compelling reason, aside from metrics, which can explain either λελαβέσθαι or κεχάροντο.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

§4.1.4 (ἐκ)λέλαθον 'caused to forget'

This reduplicated aorist is attested in both the active and middle voice. The middle reduplicated aorist occurs seven times: 1) one indicative form οὐδὲ σέθεν, Μενέλαε, θεοὶ μάκαρες λελάθοντο# 'the blessed gods did not forget thee' [Δ 127]; 2) one subjunctive ἐκλελάθωντ' Ἀφροδίτης# 'they should forget love' [χ 444]; 3) one optative ἐκλελάθοιτο γάμοιο# 'they would forget marriage' [γ 224]; 4) two imperatives Μυρμιδόνες, μή τίς μοι ἀπειλάων λελαθέσθω# 'Myrmidons, let none be forgetful of (your) threats!' [Π 200] and ἀπεκλελάθεσθε δὲ θάμβευς# 'forget your wonder!' [ω 394]; and finally 5) two infinitives #φαίην κε φρέν' ἀτέρπου ὀϊζύος ἐκλελαθέσθαι# 'I might say that my heart had forgotten its woe' [Z 285] and #οὐ δυνάμην λελαθέσθ' Ἄτης 'I was not able to forget *Ate*' [T 136]. The sense is consistently 'forget', always with the genitive. This sense of 'forget' agrees with the semantics of the non-reduplicated middle thematic aorist λαθέσθαι (10X *Iliad*, 5X *Odyssey*) and the perfect: τῶν δὲ λέλασται# '...he has forgotten them...' [E 834]; Τυδεϊδῃ τί παθόντε λελάσμεθα θούριδος ἀλκῆς# 'Son of Tydeus, what have we suffered that we have forgotten our furious valor?' [Λ 313]; and the middle perfect participle λελασμένος (4X *Iliad*, 2X *Odyssey*). Contrasting with the middle, there are two active forms of λελαθ- ε/ο- : αἰτὰρ ἀοιδὴν / θεσπεσίην ἀφέλοντο καὶ ἐκλέλαθον κιθαριστύν# '(the Muses)... took (his) wondrous song and made (him) forget (his) *kithar*-playing skill' [B 599-600]; Ἕκτορα δ' ὀτρύνῃσι μάχην ἐς Φοῖβος Ἀπόλλων / ...λελάθῃ δ' ὀδυνάων# 'let Phoebus Apollo rouse Hector and make (him) forget his pains' [O 60].[7].

---

[7] Note, however, that the syntax of these clauses does not agree: in the first instance we have no agent of 'forget' and an accusative of the 'thing forgotten' (κιθαριστύν); in the second instance the accusative is gapped, with the 'thing forgotten' in the genitive. It is possible that κιθαριστύν has been 'promoted' into the accusative from underlying oblique (i.e. genitive), and that at some level the two constructions are analogous. Nevertheless the lack of consistent usage is one more indication of the secondary nature of this

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

The usage of the active λελαθ- ε/ο- is remarkable in another respect in that it differs from the active non-reduplicated λαθ- ε/ο- which has the familiar Attic usage 'escape the notice of' with the 'effective' accusative #οὐδ' ἔλαθ' Ἀτρέος υἱόν... / Πάτροκλος 'Patroclus [the body!] did not escape the notice of the son of Atreus' [P 1-2]; δὲ Τρῳὰς λάθεν '...and she escaped the notice of the Trojan women...' [Γ 420]. Of the aorists described in the proceeding sections, pairs of reduplicated and non-reduplicated forms have been all but restricted to the middle voice and show no clear semantic variation (but see the *excursus* below). For the active forms we have only κύθε : κεκύθωσι both 'hide' and ἔτευξα : τετυκεῖν both 'make' (although the reduplicated form does have the specialized sense of 'prepare' used only with food—see above §3.1.3). A few other isofunctional reduplicated aorists are described below: ἄλαλκε : ἀλεξήσειε both 'ward off' (§5.3); ἦραρε : ἄρσε both 'fit' (§6.1); ὦρορε : ὦρσε both 'stir up' (§6.2). None of these forms show anything like the semantic variation described for (λε)λαθ- ε/ο-.

The usage of the remaining forms of *λᾰ̈θ- is also complex. The nasal present λάνθανον in Homer usually does not show the 'escape the notice' usage of the Attic present or Homeric thematic aorist, but instead has an agent subject: οἱ δ' ὄπιθεν βάλλοντες ἐλάνθανον 'but the others, casting from behind, hid themselves (intransitive)' [N 721]; ἄλλους μὲν πάντας ἐλάνθανε δάκρυα λείβων 'he hid (transitive) from all the others, shedding tears' [θ 93 = 532]. There is also a future λήσομαι (4X *Iliad*, 7X *Odyssey*) which is closer in usage to the thematic aorist with the effective accusative, e.g., οὐδέ σε λήσει '...you will not miss it' [Ψ 326, cf. 413, λ 126 +]; and an *s*-aorist with the causative usage of the reduplicated aorist ὃ γάρ τ' ἐπέλησεν ἁπάντων 'for it (sleep) makes one forget all things' [υ 85 only]. Finally, a by-form of the present exists in the long vowel present λήθειν and in the hybrid present

---

form.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

ἐκ δέ με πάντων / ληθάνει 'it (my stomach) makes me forget all...' [η 220-21 only]. The semantics of each form in Homer can be tabulated as:

	ACTIVE			MIDDLE
	'elude'	*'escape the notice of*	*'make forget'*	*'forget'*
prs.	λάνθανον λήθειν		ληθάνει	λανθανόμην λήθετο
aor.		λάθε	ἐκ-λέλαθον ἐπ-έλησε(ν)	λελαθέσαι λαθέσθαι
pft.				λέλασται
fut.		λήσει		

In contrast to the middle 'forget', three different semantic notions are attached to the active forms. In order to sort out the divergence in meaning and explain the history of these forms let us turn to the etymology. Hesychius gives a form of this root without the final stop λῆτο· ἐπελάθετο. Since other such pairs of unenlarged : enlarged roots are attested in Greek, e.g., πλήθω : πίμ-πλη-μι, ἀλήθω : ἀλέω, it is likely that the final -θ- is a late addition to the root (cf. Benveniste, *Origines* 187 f.; Chantraine, *Morph.hist.* 228 f.). By analyzing our root as a pre-Greek **lă̆-(i)/t^h-* it becomes possible to equate this form with Latin *lateō* 'be hidden', OCS *lajati* 'nachstellen' ON *lōmr* 'Verrat, Betrug' (Pokorny, *IEW* 649; Frisk, *GEW* II 80 f.; Chantraine, *DELG* 618 f.; Walde-Hofmann, *LEW* I 768 f.; Ernout-Meillet, *DELL* 343). Now both the long vowel in some nominal forms (e.g., ON *lōmr*) and the 'verba pura' alternations, i.e. **lă̆-i-*, point to a laryngeal root **leh$_2$-*. Thus, we seem to have a root **leh$_2$-* which is enlarged with either **-i-* or one of two dental enlargements. Such alternating forms are not unparalleled. Note the similar variation in Latin *gaudeō*, Greek γηθέω, Tocharian *kātk-* 'rejoice' (< **gā-dh[t?]-sḱ-e/o-* cf. H.C. Melchert, *KZ* 91 [1977] 128), but also Tocharian A *kāw-* 'begehren' which point respectively to the stems **geh$_2$-u-dh-e/o-*, **geh$_2$-dh-e/o-*, and **geh$_2$-u-* (Pokorny, *IEW* 353).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

Positing an internal laryngeal poses additional problems where we might expect the aorist to have developed with a long vowel **lh₂dh-e/o-* > **lādh-e/o-* > **ʔληθ- ε/ο-*. However, it is also possible that initial *RhC* developed into *RaC* (R.S.P. Beekes, *IF* 93 [1988] 22 f.; Schrijver, *Reflexes* 166 f.). Such a formulation is probably necessary to account for the Latin *lateō*, while the Greek aorist would probably replace a long vowel with an analogical short ᾰ in the aorist in any case.

Sorting out the chronology of the Greek forms is difficult. Frisk, *loc.cit.*, takes λήθω as the starting point for the Greek verb. From this form, the thematic aorist is built as a morphological zero-grade which then serves as foundation for a new nasal present λανθάνω. Sihler, *NCGGL* 559, also begins with a form of λᾱθ-, but assumes this must have been an aorist in the parent language which was later transferred to the present. But both of these views may have been colored by the expectation of a zero grade **ʔληθ- ε/ο-*, which is by no means certain.

Any attempt to understand this verb in Greek should begin with the especially Greek element: the final dental enlargement. Certainly, in standard examples, e.g., πλήθω, -θω began as a present formative (cf. Schwyzer, *Gr.Gr.* 703 f.; Chantraine, *Morph.hist.* 228 f.; Sihler, *NCGGL* 510; although Risch, *Wortbildungen* 278, takes this as an aorist or present formative). This would imply some present form, presumably λήθω 'hide, elude', as the oldest form. The meaning 'hide' is supported by the Greek adverb λάθρη 'secretly' and perhaps the Hesychian λαθροῦν· κλέπτειν. Nevertheless, Greek has a number of nominal forms which show that the semantic notion 'forget' was attached to this root at an early point in time. The nominal forms show the long vowel, e.g., λήθη 'forgetfulness, oblivion' [B 33], ἔκλησις 'a forgetting', ἀληθής 'true <- unforgettable'. Presumably, the middle usage must have developed from a sense of 'hide for/from oneself' -> 'suppress' -> 'forget'. Further, this must have happened early enough in the history of Greek for the nominal forms to have developed.

Now, the earliest aorist to this root would presumably have been λαθεῖν, created

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

as a zero-grade to the present λήθω or perhaps even as a real zero-grade **lh₂dh-e/o-* (see above). This aorist in turn engendered the new present λανθάνω, perhaps on the model of πυθόμην : πυνθάνομαι.[8] Now we come to the question of the reduplicated aorist. Certainly if the final **-dh-* is in origin a present formative this would argue that all the aorist forms (with the exception of the obscure λῆτο) are purely Greek developments. But there are more obvious reasons for viewing λε-λαθ- ε/ο- as an innovation.

To begin with, we must note that - λέλαθον 'they made forget' is not the causative of ἔλαθε 'it escaped the notice of' but of the middle (λε)λάθοντο 'they forgot'. Only two other forms in this paradigm have causative semantics: 1) the *s*-aorist ἐπ-έλησεν 'it makes (one) forget' [υ 85 only]; 2) (ἐκ...)ληθάνει 'it makes (me) forget' [η 220-21 only]. But both of these forms are best taken as late additions to the paradigm and are unlikely to have served as the source for our reduplicated λέλαθον.[9] Hence, this active causative form can only have been created in opposition to the middle. Once again, the starting place is the middle non-reduplicated aorist, where such forms as οὐδὲ λάθοντο / ἀλκῆς 'and they did not forget their valor' [N 834] or ὃς κέλεαι Ζηνὸς μὲν ἐριγδούποιο λαθέσθαι# 'you who bid me forget the counsels of loud-thundering Zeus' [M 235] show no functional difference from the reduplicated forms above. As noted above, the reduplicated middle forms of this aorist occur seven times, six of which are non-indicative forms. Even in the one indicative form, a metrical variation is possible

---

[8] The exact origin of the Greek - ανω presents is unknown. The closest formal match to λάνθανον is λάγχανον, and this form apparently descends from a root with internal nasal **leng-*, cf. e.g., λόγχη 'part'. However, this latter present does not occur in the *Iliad*, and may not be the best choice for the pivot form in the analogy (see below). For general discussion see, e.g., Sihler, *NCGGL* 519 f.

[9] The present λήθανε is one of several late formations in - αν- which appear to be built as polymorphic variants in Homer. Other such forms are ἀλύσκανε [χ 330 only] to ἀλύσκω or κεύθανον [Γ 459 only] to κεύθω. Note, however, that these forms are not specifically transitive or causative, and thus λήθανε may have been built from the aorist.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

since reduplication is required to produce the necessary long syllable before the verb: θεοὶ μάκαρες λελάθοντο# [Δ 127], where the augmented *μάκαρες ἐλάθοντο# would give an unmetrical line.

All in all, the simplest solution is to assume that the middle λελάθοιτο was created first, to fill a metrical slot in the non-indicative modes matching the augmented indicative form. Thus, the following Iliadic examples are completely isofunctional, except in regards to meter:

∪ λάθοντο# 'they forgot' [N 834]
∪ λαθέσθαι# 'to forget' [M 235]
ἐκλελαθέσθαι# 'to forget' [Z 285]

Because λελαθέσθαι was, however, a late addition to the paradigm, its usage was not bound to the more idiomatic usage of the older thematic aorist. Hence, when the active -λέλαθον was created to match λελαθέσθαι it acquired meaning in opposition to that newer aorist form, i.e. 'forget' -> 'make forget', and not the usage of the older non-reduplicated thematic form 'escape the notice'.

Thus we have the following contrast: isofunctional reduplicated and non-reduplicated middle aorists, distributed according to metrics, contrasting with an active transitive reduplicated aorist. Note that this distribution is identical to that proposed above for πεπιθεῖν. Although the former example relied on a single middle as a pivot form, the evidence for metrical origins is much stronger here for λελαθ- ε/ο-, and the parallelism between the two forms can only strengthen our supposition.

§4.1.5 λελάχωσι 'they may give (one) his due'

This reduplicated aorist occurs only in the *Iliad*, always in the formulaic collocation with funeral rites. Three times this form comes from the mouth of Hector: ...ὄφρα πυρός με / Τρῶες καὶ Τρώων ἄλοχοι λελάχωσι θανόντα# 'so that the Trojans

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

and the wives of the Trojans may grant me the fire when I am dead' [H 79-80 = X 342-43]; οὐδέ νυ τόν γε / γνωτοὶ τε γνωταὶ τε πυρὸς λελάχωσι θανόντα# 'nor shall his kinsman and kinswomen grant the dead the fire' [O 349-350]. In a dream the shade of Patroclus tells Achilles he will not come back: ἐπήν με πυρὸς λελάχητε# 'when you (pl.) have granted me the fire' [Ψ 76]. Once again, as in the case of τετύκοντο (§3.1.3), the formulaic consistency is noteworthy.

Only the thematic aorist ἔλαχε 'win ones share' is well attested in Homer (17X *Iliad*, 4X *Odyssey*), e.g., τὸ γὰρ λάχομεν γέρας ἡμεῖς 'for we are allotted this share' [Δ 49]. The existence of Mycenaean *ra-ke* may further support an archaic thematic aorist here (Chantraine, *DELG* 612; Ventris-Chadwick, *Doc.* 261 f., *ra-ke* = λάχε 'was allotted a portion' ; although see Aura Jorro, *DMic.* II 216, for other possible interpretations). Outside of the aorist system a single perfect is attested in Homer τιμὴν δε λελόγχᾰσιν 'they won honor' [λ 304].[10]

The present is also restricted to the *Odyssey* with three occurrences: ἐς δὲ ἑκάστην / ἐννέα λάγχανον αἶγες 'into each (ship) nine goats were allotted' [ι 159-160]; πολλά δ' ὀπίσσω / λάγχανον 'afterwards I obtained much by lot' [ξ 232-233], ὡς αὐτοὶ περ ἐλάγχανον 'as much as they themselves received' [υ 282]. The single instance in the first person narrative of Odysseus is unique in that its subject is the 'thing acquired' and not the person who obtains it. Of the remaining two forms, one is intransitive, the other transitive. Both this lack of syntactic unity and absence of this present in the *Iliad* suggest that the present λάγχανον was a relatively late addition to its paradigm.

---

[10] The 3pl. - ᾰσι with its short vowel is restricted to this form and πεφύκᾰσι [η 114], although the latter form is an emendation for the singular πέφυκει (subject δένδρεα), presumably based on λελόγχᾰσιν itself (see J.B. Hainsworth, *Comm.Od.* I 329). The short vowel may indicate an archaism, or simply a metrical shortening in the Homeric text (Schwyzer, *Gr.Gr.* 664 lists this and other later forms).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

The etymological evidence for this verb is slim indeed. Greek has several nominal forms, e.g., λόγχη = λάχος 'Anteil' (Frisk, *GEW* II 69 f.). The nasal of the nominal and perfect forms implies a root ***lengh-***; however, there are no clear extra-Greek cognates (Frisk, *loc.cit.*; Chantraine, *DELG* 611 f.). Formally it is certain that the presents λαν-θαν- ε/ο- and λαγχαν- ε/ο- have influenced each other, although it less clear which form is more archaic. The nasal root λαγχαν- ε/ο- < ***l̥ngh-*** could serve as a convenient starting place for these double nasal verbs, but the late attestation and functional inconsistency speak towards an analogical formation.

This is not the place to solve the history of the Greek double nasal presents, and even if the present λανθαν- ε/ο- has been built analogically to λαγχαν- ε/ο- this tells us nothing about the chronology between the formally similar (ἐκ)λε- λάθωσι and λελάχωσι. Since a clear motivation for the former aorist has been given above, the simplest solution is to take the latter form as an analogical creation: λαθ- ε/ο- : (ἐκ)λε- λάθωσι :: λαχ- ε/ο- : X, X= λελάχωσι.[11]

§4.1.6 δέδακε 'bit, knawed' [*AP* XII 15]

Although listed as an aorist by Curtius, *Grk.Verb* 292, this form may just as well be a perfect formation (Frisk, *GEW* I 343; Chantraine, *DELG* 249). The aorist to δάκνω 'bite' is otherwise the simple thematic ἔδακε, which is likely to be old (= Skt. *daśati* ?, cf. Mayrhofer, *EWAia* I 688). Although Schwyzer, *Gr.Gr.* 748, accepts the form as an aorist, he considers it, as with τέθιγε above, "spät (künstlich)". Whatever its synchronic usage it is certainly a late form, and I will not discuss it further.

---

[11] Λελαχ- ε/ο-, is also restricted to non-indicative modes, and a metrical solution may also be possible here. However, the situation is complicated by the usage of this reduplicated aorist which is functional distinct from the simple ἔλαχε.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

§4.1.7 μεμαποῖεν 'they should catch'

In describing the fates (Κῆρες), the poet of the Hesiodic *Aspis* tells us that they were longing to drink blood and that when they first caught a man, ὃν δὲ πρῶτον μεμάποιεν# [252, app.crit. με-μάρπ-], one would throw her claws about him and cast him into Tartarus. The non-reduplicated form of the same verb occurs twice in the same text #ἱέμενοι μαπέειν 'desiring to snatch' [304, 231]. There are no other forms of this verb in Greek, although Frisk, *GEW* II 173, sees further evidence for an old thematic aorist in the adverb ἐμμαπέως 'hastily' (< *ἐμ-μαπ-εῖν). But the difference in usage between μαπ- 'grab' and μαπ- 'be swift (?)' is notable. In view of the textual confusion between με-μαπ- and με-μαρπ-, Schwyzer, *Gr.Gr.* 747 n. 7, sees a contamination of μαρπ- 'seize, grab' and the root of ἐμμαπέως or μάψ 'in vain'. A contamination theory is likely here, although perhaps not quite in the manner envisioned by Schwyzer. Let us begin with the verb μαρπ-.

A present (μαρπτῃσι 'shall strike' [Θ 405 +]) and *s*-aorist (γῆρας ἔμαρψεν# 'old age seized (him)' [ω 390 +]) both occur in Homer. The thematic aorist and perfect are attested only from Hesiod on. But the latter form μέμαρπε(ν) occurs alongside the reduplicated μεμαποῖεν in the *Aspis* in an echo of the Homeric aorist cited above: γῆράς τε μέμαρπεν# [245]. The broader etymology is less clear, although a connection the Sanskrit *mṛśáti* < **merḱ-* is possible, either involving a root with multiple enlargments **mer-ḱ-/kᵘ-* (Mayrhofer, *EWAia* II 331 f.), or perhaps as the result of labial assimilation *μ ... κ > μ ... π (Frisk, *GEW* II 178; also Schwyzer, *Gr.Gr.* 302, with other possible examples).

Regardless of the status of the broader etymology, a connection between μαπεῖν and μαρπ- is likely. The formal disagreement is puzzling, but I believe a simple solution exists if we begin with a zero-grade thematic aorist **mr̥p-e/o-* with syllabic **r̥*.[12] In an ar-

---

[12] See τετάρπετο above (§3.3.1) for a similar zero-grade formation. In that case a dif-

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

chaic formula, the phrase #–⏑⏑|– *mr̥̆p-ĕ-* could not develop into the expected *– ⏑⏑| – *marp-ĕ-* without producing an unmetrical half-line. Nevertheless we know that such unmetrical formulae do survive into the epic period. Thus we find ἀνδροτῆτα [Π 857, X 363, Ω 6] scanned as if **ănr̥̆tāta* or ἀμφιβρότης = **amp^{h}ĭmr̥̆tās* (R. Janko, *Comm.Il.* IV 11 and 421). Unlike such forms as ἀνδροτῆτα,[13] with *marp-* as a rare and otherwise isolated formation, there was no paradigmatic model to preserve the etymologically correct *-ar-*,[14] so the vowel alone won out. It is possible that at this final stage, such forms as ἐμμαπέως did contribute to the formation of μαπέειν, although the disagreement in the textual tradition at *Aspis* 252 indicates that the root μαπ- was never well established in Greek.

Given the above we might argue that if the α- vocalism of μεμάποιεν is also archaic pointing to a preform **me-mr̥p-e/o-*. But this hardly follows. Given solid evidence for an isofunctional thematic aorist we must ask if an analogical origin for the reduplicated form is possible. Certainly the simplest solution is to suppose that this form has simply been created from its non-reduplicated counterpart: πυθέσθαι : πεπύθοιτο :: μαπέειν : X, X = μεμάποιεν. Once again the motivation here was metrical.

* * *

---

ferent solution was proposed where the verbal root had strong analogical support.

[13] Additional reasons for the survival ἀνδροτῆτα may lie in a metrical ambiguity where the first syllables of ἀνδροτῆτα may form an old aeolic base. Compare E. Tichy, *Glotta* 59 (1981) 55 f. (with bib.), who uses this form as evidence for viewing a single hexameter as a relic of an older multi-line verse.

[14] Although the spelling -ορ- /-ρο- is typical in the relics of older syllabics (e.g., ἀνδροτῆτα), in this case we would expect -αρ- (:-α-) where the poets did not explicitly recognize the older syllabic as such.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

§4.1.8 *Excursus*: On the usage of the middle reduplicated aorist

In the majority of cases above the reduplicated aorist serves as a functionally redundant by-form of a non-reduplicated aorist. This is particularly clear in the middle where several pairs of reduplicated and non-reduplicated thematic middle aorist occur in the Homeric texts.

κέκλετο : καλέσσατο both 'call to'
πεπύθοιτο : πύθοντο both 'learn of'
πιθέσθαι : πεπίθοιτο both 'trust in'
τετάρπετο : ταρπώμεσθα, τάρπη* all 'enjoy'
κεχάροντο : χάρη, χήρατο all 'rejoice'
λελαβέσθαι : ἐλλάβετο both 'seize'
λελαθέσθαι : λαθέσθαι both 'forget'

Above I have argued that the reduplicated forms are distinguished from the non-reduplicated solely on the basis of metrical variation. Nevertheless several scholars have attributed some special force to the middle reduplicated aorist. Thus, note Curtius' assumption, *Grk.Verb*, 294 (quoted extensively above §0.3 n. 8), that the reduplicated aorist had a special 'intensive force'. Again, Prévot, however, *L'aor. grec en* -θην 166 f., argues for a distinction where the aorist in -η- indicates 'state' while the reduplicated aorist indicates the entrance into a state, that is 'l'idée d'une explosion de (joie)' (see above §4.1.3). Now certainly, no one would argue with his claim that the aorist in -*ē*- is essentially 'stative'. But the contrast he makes (using χάρη : κεχάροντο) is indefensible. Any aorist in Homer can have an 'inceptive' force, but such a function is simply one function of the aorist and cannot be tied to certain stems as opposed to others .

Finally, the most extensive discussion of the use of the middle reduplicated aorist is that of Latacz, *Zum Wortfeld* 59 f., who notes the use of reduplication in the imperative, such as λε-λάθ-εσθε (etc.), πε-πάλ-ασθε. Since the middle reduplicated λελαθ-έσθε and third singular λελαθ-έσθω are the only imperatives to λαθ- attested in

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

Homer, Latacz argues that a phrase such as μή τις μοι ἀπειλάων λελαθέσθω# 'let none forget the threats' [Π 200] has a "...subjektiv-tätiges, absichtsvolles Verdrängen...", that is, an 'active subject'. This usage stands in contrast to the use of the non-reduplicated middle, 'to forget', that is, 'slip ones mind', with a "...nichts absichtsvolle *Re*aktion..." (*op.cit.* 58 and other examples). Thus, although in English we say 'don't forget', there is a problem with such a statement, requiring a sense of 'take care not to forget' or the like. Reduplication would then add this subjective force to imperative. This analysis is essentially equivalent to Risch's hesitant interpretation of the reduplicated aorist as "λελάθοντο eigentlich wohl 'sie machten sich vergessen' (?)" (*Wortbildung* 243; quoted by Latacz, *loc. cit.* 59 n. 61).[15]

Although this analysis is striking in the case of λαθ- where the only two imperative forms of this root in Homer are reduplicated, Latacz is unable to find other verbs with the same distribution. Moreover, this distribution is not so much a restriction to the imperative, but is part of a pattern of restriction in the reduplicated aorist to non-indicative modes, a distribution demonstrated by a number of other reduplicated aorists. Hence, Latacz's theory explains only a small part of the data, leaving the majority of non-indicative forms aside. The metrical theory sketched above, however, will easily account not only for the imperative forms but also the other non-indicative forms and even the sin-

---

[15] The distribution of imperative forms is reminiscent of the irregular imperative κέκλυθι, κέκλυτε (see above §3.1.2 n. 12). This form had been classed a reduplicated aorist in some earlier treatments (cf. Curtius, *Grk.Verb* 295; Birwé, *Gr.-Ar.Sprach.* 29 "...*aśuśruvat* "hörte" (cf. hom. κέκλυτε)..."), although it is now usually considered either a 'mixed form' (cf. Chantraine, *Gr.Hom.* 379, who notes that κέκλυτε "...semble présenter un redoublement déconcertant..."; see also *DELG* 541; see also Risch, *Wortbildung* 233) or as a reanalysed collocation of two words (*κε κλυ-, cf. Schulze, *Quaest. epicae* 391; Schwyzer, *Gr.Gr.* 804). Latacz's explanation could explain the co-occurrence of reduplication in this imperative. However, Mackie, *Talking Trojan* 25 f., notes that κέκλυτε is used in the *Iliad* only to address the Trojan army (or a group which includes the Trojan army). Although not incompatible with some 'intensive' reading, this observation may well indicate some other significance.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

gle λελάθοντο. More importantly, as a number of modern studies of Homer have shown, it is not possible to ascribe significance to every Homeric formal nuance without a careful review of the metrical distribution. Perhaps the most important contribution of oral-poetry theory has been its caution always to subject usage to a careful review of the metrical distribution.[16]

* * *

[16] Parry, *MHV* 114 f. and *passim*, argues that such forms as μερόπων ἀνθρώπων 'of mortal men' (metrically ∪∪ –) and θνητῶν ἀνθρώπων 'id' (– –) are semantically identical. There has, of course, been some attempt to weaken Parry's original emphatic statements.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

§4.2.0 Reduplicated aorist with possible Italic cognates

The Italic perfect system contains both relics of older perfect stems, e.g., Lat. *stetī* < **ste-(s)tā-* (cf. Greek ἔστα- μεν) as well as old aorists, e.g., Lat. *dīxī* = Greek ἔδειξα, or *fidī* ≡ Skt. *abhet* (Sihler, *NCGGL* 580 f.; above §3.2.2). In general, most reduplicated perfects in Italic are assumed to descend from Indo-European perfect forms (Leumann, *LLuF* 586 f.). However, if Indo-European had some number of reduplicated aorists, it is at least possible that some Italic reduplicated perfects descend from old aorist forms.

In 1904, H. Hirt, *IF* 17 (1904/1905) 278 f., cited possible evidence for relics of the reduplicated aorist in Latin. Hirt notes that in Indic there is a productive pairing of *o*-grade-*e̯je/o*-presents and reduplicated aorists, e.g., *bodháyati* 'causes to wake' : *abūbudhat*. Hence, both these Indic forms and the Italic type *mordeō* : *momordī* would seem to show the same pattern: *CoRC-e̯je/o-* : *CV-C(V)RC-e/o-*.[17] Nevertheless there are serious problems with this attempted equation.

In addition to the disagreement in the reduplicating vowel (Italic *-o-*,[18] Greek *-e-*, Indic *-ī-*), the perfects *momordī*, *spopondī* and *totondeō* differ from the Greek zero grade reduplicated aorists showing *o*-grade root vocalism, representing either an older perfect or vocalism copied from the present. There are also functional problems. The Indic formation is used to create transitive-causative forms while the Italic formation is apparently iterative in origin. This is clearest for *tondeō* 'to cut' but usually 'to cut (hair), mow, reap, etc.' where the reference is to repeated action (Ernout-Meillet, *DELL* 695; Sihler, *NCGGL* 532).

---

[17] F. Specht, in his article on the Indo-European stative suffix *-ē-*, *KZ* 62 (1935) 56 f., echoes Hirt's equation.

[18] The *o*-reduplication is likely to represent a purely Latin development since *spepondit* is attested (*CIL* VI 10241), and Gellius lists both *memordī* and *spepondīt* (Leumann, *LLuF* 586). Recently, however, M. Niepokuj, *Explanation* 193 f., notes that the *o*-reduplication (as well as the Indic *i-* and *u*-reduplication) may represent the oldest state of affairs. Overall, I am not persuaded by her arguments, which are largely typological.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

The etymological details are complex. There is spotty evidence for an Indo-European root **tend-* 'cut' not only in Italic but also in M.Ir. *tennaid* (**-nn-* < **-ndn-*) and possibly in Greek τένδω[19] (Ernout-Meillet, *DELL* 694 f.; Walde-Hofmann, *LEW* II 689 f.). The general consensus is that this form is a by-form to the more common *set* root **tem-h$_1$-* in Greek τέμνω and τάμνω and proto-Slavic **tĭnō* (Pokorny, *IEW* 1062; Beekes, *Development* 222; Vine, *Diss.* 112). But in this case we might expect the **-de/o-* suffix to be a present formative (Sihler, *NCGGL* 510), perhaps requiring a perfect **teton-*.

The Latin *spondeō* presents better cognates with Greek σπένδω and Hittite *išpand-* (Walde-Hofmann, *LEW* II 578 f.; Pokorny, *IEW* 989; Ernout-Meillet, *DELL* 643 f.; Tischler, *HEG* III 411; Puhvel, *HED* II 441), and possibly Sanskrit *pánate* 'admire' (Mayrhofer, *EWAia* II 80 f.; Goto, *I Prs* 26 f.). The Umbrian *spefa* < **spend-to-*, whatever its exact meaning, also belongs here (Walde-Hofmann, *LEW* II 579; Ernout-Meillet, *DELL* 644; cf. Meiser, *Lautgeschichte* 70). Once again, however, there is little evidence for any non-present form. Homer attests only the present and an *s*-aorist σπείσας [Π 253] and future σπεῖσον [Ω 287]. The formation here may also suggest a segmentation of **spen-de/o-* with a present **-de/o-*. If Sanskrit *pánate* is also cognate this would further point in that direction.[20]

Finally, *mordeō* 'bite' is also barely attested outside Latin, the best possible cognate would seem to be Sanskrit *mardáyati* : *mamṛdé* 'wipe, rub, destroy' (Walde-Hofmann, *LEW* II 111 f.; Pokorny, *IEW* 736 f.; Ernout-Meillet, *DELL* 414). The semantics are clearer once it is realized that the Sanskrit form is almost certainly a conflation of

---

[19] The Greek is obscure. It occurs only in the Hesiod ὅτε ἀνόστεος ὸν πόδα τένδει 'when the boneless one knaws (?) his foot' [*Erga* 524]. For recent studies see C. Watkins, *FsLejeune* 231 f. = *Selected Wr.* II 588 f.; E. Campanile, *FsRisch* 355 f.).

[20] The alternation between **spen(de/o)-* and **spon(de/o)-* may also point to an older athematic formation, but with '*molō*'-inflection as described in a number of articles by J. Jasanoff (e.g., *FsPolomé* 299).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

two separate verbs: 1) **mrad-* 'sanft machen' (< **(h₂)mled-*) and **mard'-* 'zerdrücken, zerreiben' (Mayrhofer, *EWAia* II 386; Goto, *I Prs* 248 f.). Although the *seṭ* quality of the latter may be analogical, cf. the Avestan *mōrəndaṭ* 'ruined' [*Y* 32.9a, 10a], *mōrəndən* [*Y* 32.11a, 12b] (Insler, *GZ* 47; Kellens, *Verbe avestique* 169). But here again it is possible that this form is an older **-de/o-* present. This point is explicit in Walde-Hofmann, *LEW* II 112, and may be supported by the parallelism in the two Avestan verbs *duš.sastiš sravå mōrəndaṭ* 'the evil doctrine has ruined the (true) words' (Insler, *GZ* 47 [*Y* 32.9a]) and *nōiṭ ... dūš.sastiš ahūm mərąšyat* 'may not the deceitful one... destroy the world' (Insler, *GZ* 75 [*Y* 45.1]) pointing to **m(e)r-d-* and **m(e)r-ḱ-* respectively (cf. Humbach, *GZ* II 83; also Pokorny, *IEW* 735 f. for different enlarged forms of **mer-*).

Hence, although there are no Indo-European perfects reconstructible for *momordī*, *spopondī* or *totondī*, there is also no evidence to suggest that these form are aoristic. Further, if the final **-de/o-* was a present suffix in the proto-language, it would be necessary to assume that, whatever the exact source, these three Italic reduplicated perfect forms were products of the Italic period. The status of these forms remains unclear, but they are unlikely to represent relic reduplicated aorists.[21] In addition to these three *o*-grade forms, however, there are four other Latin perfects which are more significant for the present study. Each of these forms has, at one time or another, been considered as a cognate to a reduplicated aorist in Greek.

Greek	*Latin*
τεταγών 'grasping'	*tetigī* : *tangō* 'touch'
κεκαδών 'robbing'	*cecidī* : *cadō* 'fall'
πεπαγοίην 'I would fix'	*pepigī* : *pangō* 'fix'
(ἀμ)πεπαλών 'brandishing'	*pepulī* : *pellō* 'drive'

[21] Not all Italic reduplicated perfects represent Indo-European archaisms as is clear from such forms as Oscan *fefacid* 'fecerit' and Latin *vhe vhaked* 'fecit' [*CIL* I² 3 (if not a forgery)], which has certainly been built to the root of the older aorist, i.e., **dheh₁-k-* (θῆκε) -> **d(h)e-dheh₁-k-*.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

In order to determine the validity of this claim, both the Greek and Latin forms must be examined in detail.

§4.2.1 τεταγών 'grasping'

The participle τεταγών 'grasping, seizing' is a virtual hapax which occurs twice in Homer in passages with similar diction. At the end of the first book of the *Iliad* Hephaestus tells of his fall from Olympus: με... / ῥῖψε ποδὸς τεταγὼν ἀπὸ βηλοῦ θεσπεσίοιο 'grabbing (my) foot he hurled me from the divine threshold' [A 590-1]. The second passage containing τεταγών occurs in the Fifteenth Book. Zeus speaking to Hera says: ὃν δὲ λάβοιμι / ῥίπτασκον τεταγὼν ἀπὸ βηλοῦ 'whom I caught, I, seizing, would hurl from the threshold' [O 22-23]. There are no other forms of this verb attested in Greek.

The standard handbooks give this form as a word equation to Latin *tetigī* < **te-tag-ai* (Pokorny, *IEW* 1054; Frisk *GEW* II 884; Chantraine, *DELG* 1109; Walde-Hofmann, *LEW* II 647 f.; Ernout-Meillet, *DELL* 676; Schrijver, *Reflexes* 98; some doubt in Sihler, *NCGGL* 581). Three questions arise out of this comparison. First, is the root etymology sound? Second, if the root etymology is sound, do we have a formal word equation? Third, if the word equation is solid, does this indicate an archaic reduplicated aorist, or is τεταγών simply an odd perfect formation? I will begin with the root etymology.

Outside of the classical languages other cognates are more difficult to judge. O.Ir. *tong-* 'swears' is possible (Pokorny, *loc.cit.*), but this Irish form is more likely a match to Lat. *tongēre* 'know', Goth *þagkjan* 'think, consider' (Vendryes, *LEIA* T-106 f.; Lehmann-Feist, *GEW* 353). Pokorny, *loc. cit.*, also refers to the more dubious Phrygian form *eti-te-*

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

tik-menos 'verflucht', but this form is in such dispute it is in no way instructive.[22] Finally, both Tocharian and Germanic have words meaning 'touch' which seem related, but are difficult to equate phonologically. Ultimately the interpretation of Tocharain B *tek-* 'touch' (prs. II *ceśäṃ*, subj. I *tekäṃ*) is intertwined with the Germanic verbs, Goth *tēkan** 'touch', ON *taka* etc., which seem incompatible with the Greek and Latin forms (cf. Lehmann-Feist, *GEW* 342 f.). Now in principle Tocharian *tek-* can descend either from a root **tVg-* or **d$^{h)}$Vg$^{(h)}$-* (let us set the vocalism aside for the moment), since Tocharian has apparently merged all three manners of articulation from Indo-European (Adams, *THMP* 36).[23] However, Ringe, *Sprache* 34 (1988-1990) 106, notes that the palatalization of the present *ceśäṃ* requires a vocalism in **ē*. Since Gothic *tēkan** also implies a lengthened grade, this tips the scales for Ringe in favor of connecting Tocharian B *tek-* with the Germanic, and, hence, separating this form from the Greek and Latin **tăg-*.

An alternative to positing two roots **tăg-* and **dĕg-* (or even **tĕh$_2$g-* and **dĕh$_2$g-*), both meaning 'touch', may exist. Recently, K. Mottausch, *HS* 106 (1993) 148 f., has proposed an ingenious solution to this problem. He argues for an Indo-European **teh$_2$g-*, with an original Narten present, **tég-*,[24] reflected in Tocharian B *ceś-*, Germanic **[t]ækan*. Already in Indo-European a reduplicated aorist (or perfect?), **te-th$_2$g-*, gave **tetag-* in Greek and Latin but Germanic (pre-sound shift) **te-tg-* (with the laryngeal loss comparable to Gothic *dauhtar* < **dugh$_2$ter-*). In turn this assimilated to **te-dg-* which in turn was analogically altered to **de-dg-* and then underwent the proper shift to **tetk-*.

---

[22] Haas, Haas, *Phrygische* 88, gives it as a perfect participle **stetigmenos* < **ste-stig-menos* : Gk. στίζω. A third etymology explains it as a perfect participle of **deik-* 'show'-> 'accuse' -> 'condemn' (Diakonoff-Neroznak, *Phrygian* 137 f.). A.M. Lubotsky, *Kadmos* 28.1 (1989) 79 f., briefly discusses this participle.

[23] Note, however, that the status of initial **#d-* may well be exceptional and is still a matter of some dispute (W. Winter, *IF* 67 [1962] 24 f., who takes **#d-* > **#ts-*).

[24] Where *Cēh$_2$C* > *CēC* per Eichner's Law, see *MSS* 31 (1973) 53 f.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

Finally, this preterite was then remade to fit the standard ablaut pattern **tetōk-/*tetk-*. At some point the older present **tēg-* was also remade to **dēg-* in analogy to the preterite, giving the **tēkan* of the attested forms, e.g., Goth. *tekan**. Significantly, the argument shows that what we have is not so much an Indo-European problem, but a Germanic one. In what follows I assume that the Greek, Latin, and Tocharian verbs all descend from the same proto-form. I further find Mottausch's explanation possible, but since the Germanic forms will play little part in this examination, I do not require this analysis.

Several scholars, Schrijver, *loc.cit.*; Ringe, *loc.cit.*; Mottausch *loc.cit.*, have assumed an internal laryngeal for this root. Certainly the alternation of **ē* : *a* is easily explained by appealing to a proto-form **teh₂g-* (although Ringe assumes **-h₁-*). Only the Tocharian subjunctive *tekāṃ* cannot be explained by this preform, since the Tocharian *e*-grade without initial palatalization implies **tok-*, but this form is certainly analogical (Ringe, *op.cit.* 106 f.). Further evidence for a reconstruction **t(e)h₂g-* lies in the Avestan *θang-* 'ziehen', e.g., *vāšəm θanjayånte* 'sie sollen den Wagen ziehen' (H. Humbach, *MSS* 9 [1956] 68 f. [*Yt* 19.44]), where the initial aspirate implies a consonant cluster. Of the standard handbooks only Walde-Hofmann, *LEW* II 648, list Avestan *θang-* with the Greek and Latin **tăg-*. But both the agreement of usage and the nasal present make a comparison with Latin *tangō* likely.[25] Hence, for this root we have fairly solid evidence in favor of a root **t(e)h₂g-* with a nasal present *th₂-n-g-* (for the vocalization of the laryngeal in this environment see Schrijver, *Reflexes* 114; Sihler, *NCGGL* 498)

Given the validity of our root equation, Greek τεταγών, Latin *tetigī*, and perhaps (pre-)Germanic **tetk-*, would seem to form a solid word equation.[26] Further, there is

---

[25] This verb is lacking in the oldest texts and but occurs several times in the *Yašts* [*Yt.* 10.136 +]. Despite some philological problems the meaning is clear. For the apparent intransitive use at *Yt* 10.136 see Gershevitch, *Hymn* 281 f.; Kellens, *Verbe avestique* 26. For the apparent scribal alternation of *θangasa-* and *θangaya-* see Humbach, *loc.cit.*

[26] Even the ever cautious Leumann, *LLuF*² 588, separates τεταγών and ἀμπεπαλών

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

good evidence that this *CăC* root should be reconstructed with an internal laryngeal. However, the final question remains: does the Greek form argue that we have a relic reduplicated aorist in Latin or, conversely, does the Latin form argue for a Greek perfect participle with Aeolic inflection? On the one hand, certain facts suggest an aoristic reading. The Greek form is synchronically an aorist. Moreover, the semantics of the proto-form are somewhat inappropriate for perfect being both transitive and telic (Sihler, *NCGGL* 581). On the other hand, there is still the Old Latin subjunctive *tagam* which may offer evidence for an archaic aorist as does the existence of the nasal present **th₂-n-g-*.[27] Thus, the shift of **teth₂g-* to the aorist could be a purely Greek affair. Such an approach does require us to posit a perfect with odd semantics in the proto-language, but that is probably no more radical than positing a reduplicated aorist.

§4.2.2 κεκαδών 'robbing'

The finite middle κεκάδοντο occurs twice in Homer: ὑπὸ δε Τρῶες κεκάδοντο# 'the Trojans shrank back' [Δ 497 = Ο 574]. The proceeding half line and following lines are also identical suggesting a small type-scene which may well indicate an archaism. The active transitive participle κεκαδών 'separating' appears only once: τοὺς... Διομήδης / θυμοῦ καὶ ψυχῆς κεκαδὼν 'Diomedes separating them from their *thumos* and soul' [Λ 333-4]. Finally, a reduplicated future also occurs: πολλοὺς γὰρ τόδε τόξον ἀριστῆας κεκαδήσει / θυμοῦ καὶ ψυχῆς 'this bow will separate many of the best from their *thumos* and soul' [φ 153-4 ≅ 170]. This future functions in the same metrical slot as κεκάδοντο, but with the formular construction of the active κεκαδών. The late attestation agrees with the general view that reduplicated futures are

---

from the other reduplicated Latin perfects as possible relics of archaic aorist formations.

[27] It is also possible that *tagam* is a precious relic of the oldest layer of Indo-European morphology where the modal forms were built directly to the root and not to the present stem (for a recent discussion see McCone, *IE Origins* 85 f.).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

late derivatives of older reduplicated aorists (see above §3.2.1 n. 26; Risch, *Wortbildung* 350 f.). A pluperfect form is also given by Hesychius: ἐκεκήδει· ὑπε<κε>χωρήκει.

An older etymology connects this aorist (and future and pluperfect) to Greek κήδω 'vex', -ομαι 'be distressed, care for'. But such an approach requires a considerable semantic shift. Hence, Schwyzer, *Gr.Gr.* 748, assumes that this root is the aorist to χάζομαι 'withdraw, give ground', but now we are faced with an unexplained phonological change *κεχαδ- > κεκαδ- ε/ο-. Chantraine, *DELG* 510 f., is unsure and gives both χάζομαι and κήδω as possible related verb forms. Neither derivation, however, is possible. Although κήδω and κεκαδ- ε/ο- have superficial formal similarity, Schwyzer is certainly right to object to this pairing on semantic grounds. At the same time, the phonological problems of equating χάζομαι and κεκαδ- ε/ο- are insurmountable.

Ignoring the inner-Greek problems, Pokorny, *IEW* 516-17, takes κεκαδών with Sanskrit *kadanam* 'destruction' (also Mayrhofer, *KEWA* 149 *kadanam* "...vielleicht zu gr. hom. κεκαδών..."). There is also a Sanskrit verbal root attested in the perfect *cakāda*, but Mayrhofer, *loc. cit.*, is certainly correct in seeing this as a misreading where *cakāda kadanam* should be emended to *cakāra kadanam*. The connection with the Toch. A noun *kat* "Zerstörung", Pokorny, *loc. cit.*, is too weak to be instructive. Both the Sanskrit and Tocharian etymons further fail in that they abstract a rather specific usage out of the formular context. In Greek the contrast between middle and active forms implies that κεκαδ- ε/ο- does not mean 'robbing' so much as 'make separate'. That is, if we start from an active 'robbing', it is difficult to see how we could develop the middle 'separate' from 'rob oneself (?)'. Alternatively, if we begin with the middle 'separate (oneself)' the active usage poses no real problem.

In this way, the older etymology with Latin *cecidī* : *cadō* 'fall' (cf. English 'fall away, fall apart') is superior (Walde-Hofmann, *LEW* I 128; Ernout-Meillet, *DELL* 81 f.).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

The Latin root is connected to Modern Irish *casair* 'Hagel', Sanskrit *śad-* 'fall'[28] [*AV śatsyati*] and perhaps Armenian *c'amum* 'fall' (Walde-Hofmann, *loc.cit.*; Pokorny, *loc.cit.*; Mayrhofer, *KEWA* III 294 f.; Schrijver, *Reflexes* 100; some doubt regarding the Armenian form in Klingenschmitt, *Verbum* 258). In theory, this approach would offer Greek κεκαδών as a possible word equation for the Latin *cecīdī*. Nevertheless, although this approach is possible, a better etymology lies in a connection with καίνυτο [γ 282] : κέκαστο 'to surpass, excel'.[29]

A semantic change of 'to excel' -> 'to separate oneself' (mid.) and, hence, -> 'make separate' (act.) does not seem particularly unlikely. Formally the perfect κέκαστο implies a dental final stop, which appears in the participle κεκαδμένος [Pindar, *Olympian* I 27]. The present ἐκαίνυτο [γ 282, θ 127 and 219] is probably a late addition to the paradigm, either an analogical creation on the model of δέδασμαι : δαίνυμαι :: κέκασμαι : X (Frisk, *GEW* I 811 f.; Chantraine, *DELG* 480), or perhaps is even a suppletive form (D.A. Hester, *Lingua* 13 [1965] 373; Mayrhofer, *loc.cit.* and *EWAia* I 296).

Outside of the rare present form all other attestations of this verb in Homer are of the perfect/pluperfect κεκασ- (10X *Iliad*, 8X *Odyssey*). The usage is clear, e.g., #ἐγχείῃ δ' ἐκέκαστο 'he excelled (over the Achaeans) with the spear' [B 530], πᾶσαν γὰρ ὁμηλικίην ἐκέκαστο / κάλλεϊ 'she excelled over all her age group in beauty' [N 431-2]. The existence of the Greek form and the comparable Sanskrit *śāśad-ur* 'to excel, to distinguish one's self' requires us to posit a second root ^{2}**ḱăd-* 'excel', formally equivalent to ^{1}**ḱăd-* 'fall' (Pokorny, *IEW* 516 f.; Frisk, *GEW* I 811 f.; Chantraine,

---

[28] Sanskrit ^{1}*śad-* does not occur in the oldest texts, but this hardly constitutes grounds for doubting the Indic : Italic correspondence (as in Ernout-Meillet, *loc.cit.*).

[29] Most Hellenists apparently have given up on the attempt to equate κεκαδών with ^{1}**ḱăd-* 'fall'. Compare, Frisk, *GEW* I 811, "[f]ern bleiben lat. *cadō*". Chantraine, *loc.cit.*, does not even mention it.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

DELG 511; Mayrhofer, *KEWA* III 294 f.).[30] Since only the perfect ***ke-kăd-*** 'excel' is reconstructible in Indo-European the original use of this perfect may well reflect an older 'one had separated him/herself from'.[31]

Our aorist κεκάδοντο is then built to the older κέκαστο in the same manner as τέτυκτο -> τετύκοντο. The difference in usage would then, to a large extent, be attributed to aspect, although this form, like τετύκοντο, may have quickly developed a specialized meaning once its association with the corresponding perfect was forgotten. However, since κεκάδοντο preserves the older sense of 'separate', the root may reflect an archaic formation, although a development within the epic period is also possible.

§4.2.3 πεπαγοίην

In considering the phrase #τῷ κεν ἐπισχοίης λιπαροὺς πόδας 'you may stretch out your sleek feet upon it (i.e. the footstool)' [Ξ 241], the scholiast notes some confusion where both ἐπισχοίης and ἐπισχοίες are suggested in the tradition.[32] Arguing for the former the scholiast points to other such optatives noting 'ιοίην, ἀγαγοίην (παρὰ Σαπφοῖ) and πεπαγοίην παρ' Εὐπόλιδι. This final scholiastic form is known as Eupolis fragment 472 (*Poetae Comici Graeci* V 533 [= Kock 435]). Some scholars have taken this form as a reduplicated aorist (Curtius, *Grk. Verb* 293; Brugmann, *Grund-*

---

[30] The Irish *cáid* 'saint, pur, noble', cited in the sources above, does not belong here (Vendryes, *LEIA* C 9 f.).

[31] I remain agnostic as to the further claim tracing the root of ἐκέκαστο = *śāśadur* to the root of Latin *censeō* 'estimate' (Chantraine, *DELG* 511). Although possible, there are a number of problems with this supposed connection (for a general discussion see most recently J.L. García-Ramón, *Sprache* 34 [1988-1990] 27 f. and *FsRix* 95 f.). Further, whatever the broader etymology, the agreement between Greek and Sanskrit indicates that the perfect forms were already disassociated from any present in the parent language with the sense of 'excel'.

[32] For the authorship of this entry and general discussion see Van der Valk, *Researches* I 113.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

riss II.3 143; Schwyzer, *Gr.Gr.* 748 f.), although it is equally possible that πεπαγοίην is a perfect optative. Unlike some forms discussed above, for **πᾰ̆γ-* both the forms in Greek and the etymology are well understood. In Homer the following forms occur

	active	*middle*	*passive*
prs.	---	πήγνυται	
aor.	-πῆξε	κατέπηκτο	ἐπάγη, πῆχθε
pft.	πέπηγε		
fut.	πηξεις		

The Greek verb is matched by the Latin *pangō* : *pepigī* 'fasten, fix' (Pokorny, *IEW* 787 f.; Chantraine, *DELG* 894 f.; Frisk, *GEW* II 525; Walde-Hofmann, *LEW* II 245 f.; Ernout-Meillet, *DELL* 479). Both languages have a reduplicated perfect, but because πέπηγε shows a 'long-vowel' formation, the Latin *pepigī* (< **pe-păg-*) might seem a better match to the Greek πεπαγοίην which has caused some scholars to suggest that *pepigī* is a relic of an older aorist form (e.g., Brugmann, *loc.cit.*). Beyond the Greek and Latin forms, the root **păg-* (**peh₂g-*?) is also found in Indic in the adjective *pajra-* 'fest' and perhaps the verbal root underlying the hapax *pāpaje* 'bleibt zurück' (Mayrhofer, *EWAia* II 65 [*RV* X 105.3]).[33]

In addition to the question of the vocalism (i.e., Indo-European **ă̆* versus a laryngeal solution) matters are complicated by the existence of a common by-form **păk-* to the Indo-European **păǵ-*. Thus, we also find Latin *pacīscō*, *pacīscī*, *pactus sum*, where *pactus* = Greek *πακτος (< *παγτος ?) > πακτόω 'mache fest' (Walde-Hofmann, *LEW* II 231 f.; Ernout-Meillet, *DELL* 473 f.). The root in final **-ḱ-* also occurs in Germanic, Goth. *fāhan* 'capture' (Pokorny, *loc.cit*; Seebold, *EtWb* 185 f.; Lehmann-Feist, *GEW* 102), as well as in Avestan *ava.pašat* 'one should chain together', and *fšāb̄īš*

---

[33] Tocharian B *pyāk-* 'einschlagen' may also belong here (J. Duchesne-Guillemin, *BSL* 41 [1940] 159). The initial palatalization, however, is unexpected.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

'fetters' [both in *Vend.* 4.50] (Pokorny, *loc.cit.*; Kellens, *Verbe avestique* 107, with *-pašat* < **pas-i̯a-*).[34] Finally, some forms are ambiguous. Thus, the Greek aorist κατέπηκτο has probably simply assimilated its final consonant, i.e., **-pāg-to*, yet Ernout-Meillet, *DELL* 473, seem inclined to connect this form to Latin *pacīscor* et cetera. There is also the Old Latin *pacit, pacunt* [*XII* 8.2, 1.6], but here the old epigraphic usage, i.e., <c> = /g/, is possible (Walde-Hofmann, *LEW* II 231; Ernout-Meillet, *DELL* 473 f.; see also the extended philological discussion in Keller, *Verbes en-sc-* 235 f.).[35]

The two by-forms, **păk-* and **păǵ-*, may have been distinguished in the parent speech (Walde-Hofmann, *LEW* II 323), but the standard view takes them as interchangeable. Thus, it is common practice to equate the two nasal present forms represented by Latin *pangō* and Goth *fāhan* 'capture' < **faŋxana-* both from **pang/k-* or **ph₂ng/k-* (Frisk, *GEW* II 526). The Greek present with lengthened grade and the *nu*-suffix probably also belongs here, although it has undergone some modification. Still, the replacement of an older **πάγγω with πήγνυμι compares well with the similar contrast in ζεύγνυμι < **ζύγγμι = Lat. *iungō* (Frisk, *loc.cit.*).

For the following we will assume a single root, with two by-forms. In principle, such alternations may well imply a laryngeal root where our root **păk/ǵ* is likely to reflect an enlarged root, **peh₂-k/ǵ-* (compare above §4.1.4). Recently Weiss, *Studies* 32 f., has pointed out that the evidence seems to point in both directions. The zero-grade in the Avestan instrumental plural *fšəbīš* 'with fetters' seems to reflect an original **ph₂k-*, while *zaranya-paxšta-pad-* 'whose feet are fixed with gold' [*Yt* 127.9] and the verbal *paiti.ava.pašāt* 'let them be fastened' apparently from **păk-*. Both the *-tó* participle and pas-

---

[34] Lithuanian *puõšti*, Latvian *pùost* 'schmükken, putzen' (Lehmann-Feist, *loc.cit.*) does not belong here —see Fraenkel, *LEW* 669.

[35] The status of Hittite *pašk-* 'plant, fasten' is unclear. This may be built directly to the unenlarged stem, **peh₂-sḱ-*, or perhaps to the enlarged form with consonant simplification, **păg/k-ske/o-* (Melchert, *AHP* 167).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

sive in *-ya-* require a zero grade. But a root with an internal laryngeal would give **ps* > *fs-* as in *fšəbīš* above. Still it is more likely that the full grade *pas-* would be created analogically as opposed to the zero-grade in *fšəbīš*. The laryngeal solution poses some dificult for the *a*-vocalism of the Indic *pajra-* 'fest' and *pāpaja*, but this may be explained if we accept A.M. Lubotsky's claim, *MSS* 40 (1981) 133, that tautosyllabic *-h_2- is lost before a voiced stop plus one other consonant in Indic (cf. also Mayrhofer, *loc.cit.*).

Returning to our verbal forms, the root aorist -πηκτο may be an archaism, since to some extent we expect a root aorist in contrast to the nasal present. But, here the full grade is unexpected and this hapax may have been created to the more common aorist ἐπάγη.[36] However, it is also possible that Greek inherited a root aorist *-πακτο but that this form was altered to the metrically equivalent -πηκτο in analogy to the present.[37] There is also ample evidence for a perfect formation, but as noted above the vocalism is unclear. Greek shows a full grade form πέπηγε[38] while the Latin *pepigī*, and the Greek hapax πεπαγοίην show a zero-grade. The Gothic *faifahun* is ambiguous, but this third plural would probably imply a third singular *faifāh** (Hirt, *Hb.Urg.* II 147).

As for the Greek perfect, however, this is itself analogized from an older **pepōg-* < **pepoh$_2$g-* (cf. S. Kimball, *Laryngaltheorie* 249; Chantraine, *Morph.hist.* 193). In Latin, the vowel grade of the perfect is also unlikely to represent anything archaic since Latin has consistently matched the vocalism of the perfect to that of the present stem (Sihler, *NCGGL* 580). Ultimately, there is no form which clearly illuminates the original proto-

---

[36] The Indic *pajra-* may be a relic of an older 'Caland system' where the *ē*-aorist would then be entirely expected. Under this analysis, however, the nasal present is harder to account for.

[37] Such a change could happen either within the history of Greek, or even have been a redactional change in the transmission of the text (cf. Wackernagel, *Spr.Unt.z.Hom.* 11 for a similar comment on πηκτός).

[38] The Sanskrit *pāpaja* may also descend from a full grade formation if we accept Lubotsky's Law (see above).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

form, yet something like **pepōg-* (< **pepoh₂g-*) is most expected. Hence, our reduplicated πεπαγοίην could be an aorist. However, since the status of the non-indicative forms of the perfect are both unclear from a comparative point of view,[39] and subject to analogical reworking, it is not impossible that πεπαγοίην is a perfect optative. More importantly, even if πεπαγοίην is synchronically an aorist, this non-indicative form may be another example of a metrically expanded form.

§4.2.4 (ἀμ)πεπαλών 'brandishing'

The participle ἀμπεπαλών occurs eight times in the *Iliad*, always in the same formula: ἦ ῥα, καὶ ἀμπεπαλὼν προΐει# 'and thus brandishing (it) he cast (his spear)' [Γ 355, +]. In the *Odyssey* the participle occurs only twice in a different but similar formula: αἶψα μάλ' ἀμπεπαλὼν προΐει# 'straightaway brandishing (the spear) he cast...' [ω 519, 522]. The exact semantics are not clear, but this participle must mean 'brandishing', 'shaking' or the like. The reduplicated aorist πεπαλ- ε/ο- exists in two other instances, both referring to the shaking of lots: κλήρῳ νῦν πεπάλασθε διαμπερὲς ὃς κε λάχῃσιν 'cast now lots for that one who will be chosen!' [H 171]; αὐτὰρ τοὺς ἄλλους κλήρῳ πεπαλάσθαι ἄνωγον 'I bid them to cast lots' [ι 331]. The reduplicated form is matched by both an *s*- and root aorist. A present also occurs in both active and middle forms, giving the following Homeric paradigm:

	active	*middle*
prs.	πάλλειν	πάλλεται
aor.	πῆλε, ἀμπεπαλών	πάλτο

The perfect forms of Attic tragedy are late and need not concern us here. The *s*-aorist πῆλ- < **pals-* (4X *Iliad*) in general functions in similar formulaic contexts as the

---

[39] For a brief discussion of the vocalism of the non-indicative modes of the perfect see above §3.1.2 n. 6.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

reduplicated form, e.g., —ἀλλά μιν οἷος ἐπίστατο πῆλαι Ἀχιλλεύς 'but only Achilles knew how to handle it (the spear)' [Π 142 = T 389, cf. Π 117]. Note, however, the phrase ὅ γ' ὃν φίλον υἱὸν ἐπεὶ κύσε πῆλέ τε χερσίν 'he kissed his dear son and dangled (?) him in his hands' [Z 474].

The active present πάλλω (15X *Iliad* 1X *Odyssey*) in some instances matches the active aorist meaning 'brandish' with spears or stones: πάλλων δ' ὀξέα δοῦρα 'brandishing the sharp spears...' [E 495 = Z 104 = Λ 212, cf. Γ 19, X 320]; ὁ δέ μιν ῥέα πάλλε καὶ οἷος 'easily he alone wielded it (the stone)' [E 495 = M 449 = Y 287]. In several instances, however, the active present is used with the same semantics as the reduplicated middle aorist to describe the shaking of lots in a helmet: #κλήρους ἐν κυνέῃ χαλκήρεϊ πάλλον 'they shook the lots in the dog-skin helmet' [Γ 316, 324, H 181, +].

In the middle, things become more complicated. The root aorist to πάλλ- ε/ο- is πάλτο 'leapt'. But the usage here is notably different from the other forms of this verb. In general, this form is a late addition to the paradigm, having been abstracted from an older ἀν- ἐπ- αλτο > ἀν- ἐ- παλτο (Veitch, *GV* 451; H. Fränkel, *FsWackernagel* 278 f.; Leumann, *HW* 60 f.; H. Geiss, *MSS* 11 [1957] 62 f.; Harðarson, *Studien* 196 f.).[40] The question remains: was this root aorist created in this way, or did an older πάλτο 'shake' have its usage altered? After discarding all instances where the meaning is clearly 'leap', only a single form remains: μετόπισθεν ἐν ἀσπίδος ἄντυγι πάλτο# 'and he tripped on the rim of his shield' [O 645]. But even here, this usage can more easily be derived from a sense of 'jump back' (with μετόπισθεν) than from some sense of 'shaking'.[41]

---

[40] Gehring lists six occurrences of the aorist πάλτο; however, four of these, ἀνέπαλτο [Θ 85, Y 424, Ψ 694], ἐκκατέπαλτο [T 351], could just as easily be synchronically traced to ἄλλομαι (see Fränkel, *loc.cit.*; Geiss, *loc.cit.* for the philological details).

[41] R. Janko, *Comm.Il.* IV 299, suggests instead that this aorist is from *παλέω 'be wrecked', cf. Hesychius πεπαλμένος· βεβλαμμένος.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

Finally, the middle present πάλλεται 'shakes, jumps' (5X *Iliad*) has had its usage altered in response to the innovated aorist πάλτο: ὡς ... ἀναπάλλεται ἰχθὺς 'as a fish leaps' [Ψ 692];. Two passage refer to the movement of the heart: ἐν δ' ἐμοὶ αὐτῇ / στήθεσι πάλλεται ἦτορ ἀνὰ στόμα 'my heart jumped up into my mouth' [X 451-2]; μεγάροιο διέσσυτο μαινάδι ἴση / παλλομένη κραδίην 'she hastened from the hall like to one mad, quivering at heart' [X 460-1]. Here, the usage 'quivering' or 'trembling' might seem closer to 'shaking', but there is evidence for exactly this usage of **sal-* 'leap' (> ἅλτο) in Indo-European (cf. Plautus *iam horret corpsus cor salit* [*Cist.* 551]). In two other instances, however, this present has the usage we would expect from the middle reduplicated aorist, i.e., παλλομένων [O 191] and παλλόμενος [Ω 400], both 'casting lots'.

Hence, our paradigm is pruned down to a present πάλλω - ομαι and the two aorist forms πεπαλ- ε/ο- and πηλ-. But Watkins, *Celtic Verb* 53, is certainly right in his suggestion that πῆλε is "...chronologically late, a purely Greek development, since it is simply built of the Greek stem παλ-, not on any Indo-European full grade". Thus, only the present and reduplicated aorist are original. Further, we have only a narrow range of formulaic phrases on which to base our understanding of the original semantics of this root. For the aorist we have ultimately three separate attestations, since all instances of the active ἀμπεπαλών 'brandishing' are essentially identical. The two middle forms present more interesting cases. The imperative at H 171 and the infinitive at ι 331 are peculiar in that they show the alphathematic forms παπαλάσθε -ασθαι instead of the expected πεπάλεσθε* - εσθαι*. There is some variation in the manuscripts here where πεπάλαχθε -αχθαι occur as alternate readings (Chantraine, *Gr.Hom.* 396). But πεπάλαχθε 'besprinkle!' is nonsensical, nor is a perfect likely here (G.S. Kirk, *Comm.Il.* II 259; A. Heubeck, *Comm.Od.* II 32). An alternative is the emended πεπάλεσθε -

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

ἔσθαι found in some versions,[42] but since παπαλάσθε -ασθαι is the *lectio difficilior* it cannot be discarded so easily.[43] Dybo, *Slavjanskaja* 250, argues that the alphathematism is result of the *set* quality of the root. But this can hardly be possible. A reduplicated **pe-plh₂-e(s)dhe* (however one reconstructs the ending) should have given *πεπλεσθε with the laryngeal loss of εὗρον or κέκλετο (above §2.1 f.).[44, 45]

Within Greek, a small number of nominal forms are also attested, but such forms as ἐγχέσπαλος or σακέσπαλος are largely uninformative both formally and semantically, and may well have been built in analogy to older verbal phrases. Outside of Greek, both Frisk, *GEW* II 469, and Chantraine, *DELG* 854, accept the connection with Latin *pellō*.[46] Within Italic, Umbrian *(am)pendu* 'offer up' may also be included here if we accept Meiser's analysis, *Lautgeschichte* 165, *-pendu* < **pel-n-ə-*. Further connections may

---

[42] Compare Heubeck, *loc.cit.* Note that the computer *TLG* text keeps πεπάλασθε at H 171, but has πεπαλέσθαι at ι 331.

[43] A. and O. Strachov, *Studia-Slavico-Byz.* 252 f., argue that this aorist πεπαλάσθε -ασθαι belongs not with the present πάλλω but instead with παλάσσω '(be)-sprinkle' (< *παλακ-*i̯*ε/ο-, plpft. πεπάλακτο). Their reasoning, however, is suspect. The problem here is that παλάσσω means '(be)-sprinkle'. Now it is true that most dictionaries gloss the verb as '(be)-sprinkle, shake'; however, the reading 'shake' is based on *Iliad* H 171 when it is emended to read πεπάλαχθε. But in this way the argument has now emended out of existence the very form it tried to explain. Hence, there is no way to construct an argument that will derive πεπαλάσθε/αι from παλάσσω.

[44] Even without laryngeal loss, **pe-plh₂-e(s)dhe* couldn't really give πεπάλασθε since the thematic vowel would certainly be restored, or not colored at the morpheme boundary, e.g., **e-kṃh₂-e* > κάμε not **κάμα.

[45] For alphathematism in the reduplicated aorist see §2.1 and §5.1.

[46] There, are, however several older etymologies. Thus, in 1936 L.J.D. Richardson, *TPS* 101 f., argued that a single lot would be shaken from the helmet only if πεπαλάσθε (&c.) described a circular motion. Hence, he derives πεπαλέσθε from the root **kʷel-* 'to turn around in a circular motion'. One year later this etymology was challenged by L.R. Palmer, *Glotta* 27 (1939) 134 f., both on philological grounds and on the assumption that **kʷel-* is always intransitive. Pokorny, *IEW* 801, derives πάλλω &c. from the root [1]**pel* 'fliessen, schwimmen'. None of these approaches is very promising.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

lie with the Slavic **pőlti* 'flap, wave' < **palə-ti* and perhaps also with O.Ir. *-ella* < **-pelnā-*, fut. *eblaid* 'will drive' < **pi-plāse-ti* (Dybo, *loc.cit.*; McCone, *IE Origins* 34 f.).

The four forms listed above may well be cognate, but the semantic match could be better. Certainly the Greek and Slavic forms, which both point to something like 'shake', fit well together. And the difference between the Italic 'push' and Celtic 'drive' is well within the bounds of what we normally expect. However, the broader connection between 'shake', on the one hand, and 'push/drive' on the other, is more difficult to motivate. Further, there may be a formal difference as well. The Slavic **pőlti* (e.g., Slovenian *pláti* 'wave'; Russian (dialectal) *polótĭ* 'winnow' —Dybo, *loc.cit.* n. 80 for additional forms) would seem to point to a *seṭ* root *pelh$_x$-*, while the Greek forms are more difficult to pin down.

The present πάλλω could point to either a nasal or *i̯e/o*-present, although the latter form would require an *aniṭ* formation, since **pl̥h$_x$-i̯e/o-* would yield ***plāi̯e/o-* (or ***plēi̯e/o-* or ***plōi̯e/o-*). The nasal present is ambiguous where either a *seṭ* or *aniṭ* form is possible. Here **pl̥-n-eh$_x$-* would give **παλνᾱ-* (or *-νη-* or *-νω-*), but this in turn would certainly become πάλλω (cf. βάλλω < **balnē-* < *g^{u}ḷ-n-eh$_l$-*). The *aniṭ* form would presumably give the same thing, i.e., **pl̥-ne/o-* > **πάλνω* > πάλλω. But this latter formation can only be a late thematic formation since *CeC* roots did not build nasal presents in Indo-European (Sihler, *NCGGL* 499). Overall, the simplest explanation is to assume a nasal formative to a root *pelh$_x$-*, which allows us to equate πάλλω with the Slavic **pőlti* above (Harðarson, *Studien* 198). Note that this analysis will not allow the root aorist πάλτο, which can only be an *aniṭ* form.

For the Italo-Celtic 'push/drive' the evidence seems to point in two different directions. Thus, the O.Ir. *ella* : *eblaid* clearly points to a root in *ā*, e.g., *eblaid* from **pi-plā-ti* < **pi-pleh$_2$-*. However, the Latin *pellō* 'push, drive' may indicate an *aniṭ* root. Ernout-Meillet, *DELL* 494, take this present as a root enlarged in **-d-*, *pellō* < **peldō* which nicely explains the past participle *pulsus*. An alternative explanation takes *pellō*

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

from **pelnō* which then forms a better match to the Irish present *(ad)-ella* < *pelnā-* (Walde-Hofmann, *GEW* II 277). Although this approach requires an analogical source for *pulsus*, this is not such a great problem (Walde-Hofmann, *loc.cit.*, where *pulsus* "...ist nicht altes **peld-tos*... sondern jüngere Bildung gegenüber *pult-āre*"; Schrijver, *Reflexes* 406; Sihler, *NCGGL* 506). Further, the O.Ir. present *·ella* (found only in the compound *ad·ella* 'visits', but clearly from the same root as the future *eblaid* above[47]) and the Latin *pellō* and perhaps Umbrian *-pendu*, each show evidence for a full grade formation. Since a full grade is anomalous in a nasal present formation, it is likely that each of these descends from a single akrostatic proto-form.[48] Hence, both our Italo-Celtic root 'drive/push' and our Greco-Slavic root 'shake' were *set* forms in the parent language. Still, this is not sufficient to guarantee an equation between these two roots and ultimately between our reduplicated *pepulī* and -πεπαλών.

In point of fact, the semantic difference between these two roots is insurmountable (cf. Schrijver, *loc.cit.*). Although broader connections cannot be made with certainty, it is likely that the Italo-Celtic **pelh$_x$-* belongs with Greek πελάζω 'bring near', which may well point to a root **pelh$_2$-* (Walde-Hofmann, *loc.cit.*). The Greek πάλλω : -πεπαλών is best removed from the Italo-Celtic **pelh$_2$-* and connected to the Slavic forms. Since the central notion of both Greek πάλλω and Slavic **pőlti* is brief repetitive motion, Greek πελεμίζω 'quiver' also probably belongs here, which would point to a root **pelh$_1$-*.[49] The Greek and Latin forms then have no relation to one another. Greek πεπαλ- ε/ο- may well be archaic, but there is no evidence which might suggest a possible source for

---

[47] Compare, McCone, *IE Origins* 32 "[i]ndeed, OIr. pres. *-ella*, M[iddle] W[elsh] subj. *el(h)-* and O.Ir. fut. *-ebla* obviously once belonged to a single paradigm..." (with bib.).

[48] For a fourth possible cognate, cf. Avestan *parəṇti* 'attacks' if this is really from **pel-n-h$_2$-ti* (McCone, *IE Origins* 32, K. Strunk, *MSS* Beiheft 13 [1986] 445 f.).

[49] Perhaps also the Germanic adjective *us-filma* 'frightened' and Armenian *atmuk* 'agitation' (Lehmann-Feist, *GEW* 381)

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

this form.

§4.2.5 Conclusion

Concerning our four sets of correspondences τεταγών : *tetigī*, κεκαδών : *cecidī*, πεπαγοίην : *pepigī* and (ἀμ)πεπαλών : *pepulī*, only the comparison of τεταγών = *tetigī* is possible. But even here it is not certain whether or not the proto-form was descriptively an aorist or a perfect. Hence, as with the *C(e)RC* roots, there is no comparative evidence that any *CaC* reduplicated aorist predated the Greek era.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

CHAPTER V

§5.0 *HRe-HRC-e/o-*

In addition to the standard *Ce*-reduplication, Greek presents us with a vowel-initial, so-called 'Attic reduplication'. Thus, the perfect to ὄλλυμι is ὄλωλε, to ἔδω, ἐδηδώς [P 542]. The origins of this formation almost certainly lie in roots beginning with a laryngeal sonant cluster, e.g., ἐλήλουθα < **h₁le-h₁loudh-* (Beekes, *Development* 115; cf. also Rix, *Hist.Gr.Gr.* 204 f. and C.J. Ruijgh, *FsChantraine* 211 f. = *Script.Min.* 349 f.). Thus, the initial laryngeal reduplicates in the same manner as an initial *s-* where such contrasts as Latin *sistō* = Sanskrit *tiṣṭhati* imply an original **sti-sth₂-*. The reduplicated aorist has a similar formation,[1] except the root vowel is not lengthened, e.g., aorist ἄρᾰρε 'fit together' : perfect ἀρηρώς 'be fitting'. However, the historical phonology is not clear (see below §6.1).

Now, in principle, it is possible that the *Ce*-reduplicated aorist of such forms as εἶπον represents an archaic formation, but these vowel-initial aorists are an innovation. This is the position of Ruijgh, *FsChantraine* 228 f. = *Script.Min.* 365 f. Ruijgh notes that already in the language of Homer, one finds reduplicated aorists which resemble perfects except for the grade of the root. Hence, the production of new aorists such as ἄρᾰρε from ἀρηρώς would seem to be a simple matter of analogy. Unfortunately, this analogical process cannot be correct in all instances. Greek ἄραρε must date from Indo-European times (or at least late Indo-European, see below §6.1) since it forms a word equation with Armenian *arari*. Further, a second archaic formation may exist in the aorist ἐνεγκεῖν. Even so, Ruijgh's central insight is valid. The vast majority of vowel-initial reduplicated aorists, as with the *Ce-* aorists above, are creations of epic diction. These analogical developments will be explored more closely in the next chapter.

---

[1] Note that there is some variation in terminology, thus Schywzer, *Gr.Gr.* 766, reserves the term 'Attic reduplication' only for the perfect type.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

§5.1 ἐνεγκεῖν 'to bear, carry'

Ἐνεγκεῖν is the standard Attic suppletive aorist to φέρω 'carry' (also once in Doric ἀνενέγκαι cf. M. Meier-Brügger, *KZ* 100 [1987] 316) The form is lacking in epic where the alphathematized ἔνεικα occurs instead [N 213, 453, cf. Ξ 255 +], although a thematic form, ἐνεικέμεν, occurs at T 194. The Homeric ἐνεικ- α- is more common in the *Odyssey*, but an all but complete paradigm exists in the *Iliad*: -ένεικα [T 261], -ένεικας [Ξ 255, O 28], ἔνεικε [Ψ 564], and the third plural, e.g., #ἀλλά μ' ὑπήνεικαν ταχέες πόδες 'but my swift feet bore me away' [E 885, cf. I 306 +].

The relationship between ἐνεγκεῖν and ἔνεικα is the subject of considerable controversy. Both Chantraine, *DELG* 346 f., and Frisk, *GEW* I 512 f. & 720, follow Brugmann, *IF* 1 (1892) 174, in the belief that there is no direct historical relationship between the two forms: ἐνεγκεῖν is a reduplicated formation while ἔνεικα is a compound of ἐν- plus -εἶκαι, a form of ἵκω (< **se(i)k-*?). The semantics of ἵκω 'reach, arrive' are not incompatible with that of ἔνεικα 'bear', and if the formal comparison were certain we could at least consider it. Nevertheless, this analysis has little to recommend it. The formal correspondence is weak, both concerning the root vocalism and the verbal ending where the present ἵκω has a common by-form ἥκω, which may well imply a root **sĕk-* (with a present **sisk-e/o-*?).[2] As for the ending, there is no morphological reason why ἐν- εἶκαι, apparently a root aorist, should have alphathematic inflection.

Before attempting other extravagant etymologies for ἔνεικα, we should consider the simplest solution: ἔνεικα is related to the formally similar ἐνεγκεῖν. That ἐνεγκεῖν and ἔνεικα should be equated is hardly a new idea (cf. Curtius, *Grk.Verb* 291 with reference to J. Schmidt). Further, this is still the position of some modern scholars (e.g., Rix, *Hist.Gr.Gr.* 215). Attempts to separate the two verbs point to two distinctive

---

[2] For a possible reduplicated formation here, cf., G. Klingenschmitt, *MSS* 33 (1975) 75.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

features: vocalism (ε̄ versus ει) and the thematic versus alphathematic endings. However, the latter distinction is artificial. As noted above one instance of ἐνεικ- is thematic. More importantly, the inflection of ἐνεγκεῖν is not simply thematic but is mixed. Even in Attic, alphathematic forms are attested frequently, especially in the second person (Veitch, *GV* 593 f.). The use of the alphathematic endings primarily in the second person is, of course, not unparalleled, cf. εἶπον : εἶπας above (§ 1.1).

Hence, the distinction between ἐνεγκεῖν and ἔνεικα rests almost exclusively on the contrast of vocalism and nasalization. But the question here is whether - ει- can be the result of dissimilation of an original ε̄-. Most recently, M. Meier-Brügger, *op.cit.*, and K. Strunk, *Laryngaltheorie* 571 f., have re-examined this question. Neither Meier-Brügger nor Strunk is able to find any exact parallel for a change of -ει- < *- ε̄-.[3] Nevertheless, as was the case with εἶπον above, the lack of examples or counterexamples does not solve the question one way or another. The root vocalism remains a problem, but one that is in no way insurmountable. More importantly, both Meier-Brügger, *op.cit.* 318 and Strunk, *op.cit.* 576, note that alphathematic endings are more easily explained by *assuming* an equation with the reduplicated ἐνεγκεῖν, where alphathematic inflection has parallels, than by the *ad hoc* assumption of an otherwise unattested *-εῖκαι. Finally, as Meier-Brügger points out, *op.cit.* 318, the semantic contrasts sometimes cited for a distinction between ἐνεγκεῖν and ἔνεικα (e.g., in Frisk, *loc.cit.*) are not valid. All in all taking ἐνεικ- as a dissimilated form is the simplest solution. Since I take ἔνεικα as secondary, the remaining discussion will focus on ἐνεγκεῖν.

Like εἶπον above (§1.1) the apparent full grade is anomalous in a thematic aorist and, hence, many scholars argue for a reduplicated formation, *ἐν- ε̄- κ- ε/ο- < *$h_x$ne-

---

[3] Meier-Brügger, *op.cit.* 320, compares ἰγνύη, apparently from **en-ǵnu̯-ā-* as perhaps indicating the same dissimilation. Strunk, *op.cit.* 579, points to the Lesbian change of secondary *-ns* > *-is*, but such a change is slightly different both in environment and chronology.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

(*h*$_x$)*nḱ-e/o-* (e.g., Schwyzer, *Gr.Gr.* 748; Frisk, *GEW* I 513; Chantraine, *DELG* 346 -see below for additional sources). If we assume that the loss of laryngeal between vowels precedes the change of *n̥* > α, then a preform **h*$_1$*ne-h*$_1$*n̥ḱ-* with the loss of laryngeal and resyllabification would yield ἐνεγκεῖν directly (W. Cowgill, *Evidence* 151). Alternatively, we could assume **h*$_1$*ne-h*$_1$*nḱ-* first yielded *ἐνηγκ- which was shortened to ἐνεγκ- by Osthoff's Law (C.J. Ruijgh, *Lingua* 26 [1971] 197 = *Script.Min.* 346; Strunk, *op.cit.* 574). But neither the assumption that this root began with **h*$_1$- nor the subsequent phonology is clear. In fact, not even the underlying reduplicated formation is certain. Other options are also possible, e.g., Cowgill, *op.cit.* 151, "[h]ence I prefer to reconstruct **Hnenḱ-* as a root aorist... with double nasal..."; or Rix, *Hist.Gr.Gr.* 215, who argues for a root with nasal introduced from the third plural **ēnken* < **e-h*$_1$*-n̥ḱ-ént* (cf. also Sihler, *NCGGL* 563). In order to evaluate the different possibilities let us move on to the comparative data.

Pokorny, *IEW* 316 f., gives a root **(e)neḱ-* as the basis for ἐνεγκεῖν; Sanskrit *aśnoti* (aor. *āṇat*, aor. subj. mid. *náśate*) 'erreichen, erlangen'; Iranian *-nasaiti* (*s*-aor. subj. *nāš-ā-mā* [< **nāḳṣ-*], thematic [reduplicated?] *nąsaṭ* [see below n. 14]) Lith. *nèšti* 'tragen, bringen' and OCS *nesǫ* 'id.'; Armenian *hasi* 'bin angekommen'; Old Irish *ro'icc* 'erreicht'; Goth. *genah* 'es reicht = genügt'; Tocharian A *eṃts-*, B *enk-* 'nehmen, fassen'; both Hittite *ninikzi* 'hebt' and *henkzi* 'teilt zu' and other forms not listed here (cf. also Fraenkel, *LEW* I 497 f.; Lehmann-Feist, *GEW* 70 f.; Mayrhofer, *EWAia* II 27 f.). Both the prothetic vowel in Greek and the lengthened augment in Indic guarantee an initial laryngeal. However, the two Hittite forms point in two very different directions: the latter points to an initial **h*$_2$-, while the former and the Greek ε- would seem to indicate **h*$_1$-. Further, the semantics are not quite as simple as one might wish.

Kuiper, *Nasalpräsentia* 50 f., argues that the semantics are too diverse and suggests that there were originally two homophonous roots, [1]**enek-* 'erreichen' and [2]**enek-* 'tragen'. More recently, R.S.P. Beekes, *MSS* 38 (1979) 18 f., has resurrected Kuiper's

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

semantic contrast and further correlated this contrast with a formal distinction indicating two roots, **h₁nek-* 'to carry' and **h₂nek-* 'to reach'. Not all scholars are convinced, and many continue to work with either a proto-form in **h₁-* or in **h₂-*. There are, then, essentially three competing reconstructions: 1) a single root **h₁nek-* (e.g., Chantraine, *DELG* 346; Klingenschmitt, *Verbum* 212 f.); 2) a single root **h₂nek-* (e.g., Mayrhofer, *loc.cit*, *FsNeumann* 191 n. 51, *Idg.Gr. I/2* 132 n. 143); or 3) both.

Of the forms cited by Pokorny, most can tell us nothing about the color of the original laryngeal. Balto-Slavic and Indo-Iranian have presumably merged the three laryngeals. More interesting, however, is the evidence from Hittite, Old Irish, and Latin. Pokorny gives two possible Hittite cognates for **(e)nek*. The first, Hittite *ninink-*, is morphologically interesting and the double nasal formation is suggestive. However, the usage 'to mobilize, set (people) in motion, loosen, break' (other uses as well, cf. *CHD* L-N 441) is incompatible with the root under examination (cf. Fraenkel, *LEW* 503, who takes Hittite *ninink-* with Lith. *nìkti* 'über einen herfallen' pointing to a root **neiK-*). This brings us to Hittite *henk-* 'bestow, consign'. Here Hittite *h* (< **h₂-* or **h₃-*) with *e*-vocalism implies 'Eichner's Law' (cf. H. Eichner, *MSS* 31 [1973] 53 f.; Mayrhofer, *Idg.Gr. I/2* 132), which supposed that *ē* was not colored by an adjacent *h₂*.[4] Hence, we suppose a root **h₂ēnḱ-*[5]

---

[4] Puhvel, *HED* III 289, explains the Greek form ἐνεγκ- < *E₂*ne*-E₂*nḱ*- where his E₂ is a non-coloring laryngeal with a reflex of *h* in Hittite. Such a solution is unlikely, and requires seriously re-evaluating our understanding of Hittite phonology.

[5] Note, however, the Old Hittite spellings *ha-in-kán-ta* and *ha-ik-ta-ri* which cause Neu, *Anitta* 43 n. 30, to read the form as *haink-*. Melchert, *SHHP* 23 and n. 46, noting that the nominal *hengur* 'offering' has a genitive *henkuwaš* not **henkunaš* (which we expect based on the parallel *mēhur* : *mehhunaš*) concludes that *henk-/hink-*should be analyzed as a compound, **h₃e-h₁enḱ-*, with the prefix of ὀ- κέλλω and other forms. In his most recent work, *AHP* 143 f., however, Melchert retracts this view and favors the analysis of *henk-* as an acrostatic present. Hittite also attests a middle form *hink-* 'bow', which may be related.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

with an acrostatic (i.e., so-called Narten) present.[6]

Further evidence for an acrostatic present may occur in the Irish *ro'icc* [*-ig'*] 'erreicht' which the older handbooks take as a zero-grade nasal formation **(h$_x$)n̥ḱ-(i̯e/o)-* (Thurneysen, *GOI* 126, 130).[7] But, M. Mayrhofer, *FsNeumann* 191 n. 51, notes that *'icc* may also be traced to the same acrostatic root proposed for Hittite *hénk-*. McCone, *IE Origins* 50 f.; and *Eriu* 42 (1991) 1 f., has recently re-examined the whole matter. He notes that both this newer analysis and the older explanations have problems. In the end, however, McCone opts for the **h$_2$ēnḱ-ti* paradigm assuming an analogical replacement of the third singular by the thematic stem of the third plural[8] as follows:

3rd sg. **h$_2$énḱ-ti* > **īnx-ti* ——> **ink-et(i)* > *'icc*
3rd pl. **h$_2$énḱ-n̥ti* > **ank-anti* -> **ink-o-nt(i)* > *'ecat*

Such a paradigm does not in itself point to *h$_2$-*, since a preform *h$_1$ēnḱ-*would have exactly the same outcome, but the comparison with Hittite is suggestive.[9]

---

[6] Compare also M. Peters, *Sprache* 21 (1975) 51, who see this form as a relic of an older system where the present and aorist are distinguished by different ablaut patterns. He sets up pairs of proterodynamic presents contrasting with hysterodynamic aorists: PD **h$_2$éḱ-ti* -> Hittite *henkzi* : HD **(e)h$_2$néḱ-t -> RV ánaṭ*. See also Oettinger, *Stammbildung* 173 f.

[7] Compare also Klingenschmitt, *Verbum* 212 f.; F.O. Lindeman, *FsSchmidt* 114 n. 18. The zero grade **h$_x$n̥ḱ-* finds some support in Tocharian, TA *eṃts-* TB *enk-* 'seize' pointing to Common Tocharian **enk-* (the TA form is lautgesetzlich, J. Hilmarsson, *TIES* 1 [1987] 49 f., **ank-s- > *ăns > emts-*). Common Tocharian **enks-* reflects either **(h$_x$)onḱ-(s)-* or a zero-grade formation **h$_x$nḱ-tó* (see D. Ringe, *Sprache* 34 [1988-1990] 68 f. for discussion). A preform **(h$_x$)ēnḱ-(s)-* would give the correct vocalism, but we would expect the initial palatalization to be preserved as **i̯enks*, cf. TA *ysār* TB *yasar* < **yäsār* < **h$_1$esh$_2$ōr*.

[8] One of the theses of McCone's book is that thematization spread from the third plural, so that this form fits into his general framework.

[9] Alternatively J. Jasanoff, *FachtagungIX* 161, takes (*ro*)*'icc* as a reduplicated thematic present < **h$_x$i-h$_x$nḱ-e/o-*.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

The real evidence for *h_2-* lies in the Old Irish preterite. Cowgill, *op.cit.* 154 n. 22, notes "...OIr. *:ánaic*, not **·inaic*, suggests that the laryngeal here was not [*h_1*]". Clearly this is the persuasive fact for Mayrhofer, *Idg.Gr. I/2* 132 n.143, "...**h₂e-h₂no(n)ḱ-e* als einzig mögliche Vorstufe von altir. *·ánaic* = ved *ānaṁśa*". Now there are other solutions, but all are lacking in some respects. G. Schmidt's reconstruction, *Orbis* 26.1 (1977) 114 f., **h_1ōn-h_1ōnḱ-* is phonologically adequate, but is little improvement on Brugmann's **ōn-onke*, *Grundriss* II^2 3.431, or for that matter on E. Windisch's **ā-na-n-ka-*, *KZ* 21 (1873) 411 f. An initial *ō-* would give the correct form in Irish, but is unmotivated.[10]

One final piece of evidence in favor of assuming a root **h₂enk-* lies in the Latin *nancior*, *nanciscor* 'obtain'. The root is often taken to be a nasal infix verb with secondary *-ie/o-* comparable to Latin *uinciō* (cf. Ernout-Miellet, *DELL* 429; Walde-Hofmann, *LEW* II 143 f.; Schrijver, *Reflexes* 491). It is also possible, however, that we have a newer present built to an older reduplicated aorist stem **nank-* (see Weiss, *Studies* 18 f.). But this analysis also requires a root **h₂ĕnḱ-*, since a reduplicated **$h_1$ne-h_1nḱ-e/o-* should give ***nēncior* > ***nencior*.[11]

The forms above would seem to provide a formidable case for **h₂enḱ-*. However, as the Latin example above shows, a preform **h₂ne-h₂nḱ-e/o-* should yield Greek **anŋk-e/o-* > ***ἀν-αγκεῖν*. For ἐνεγκεῖν a pre-form **$h_1$enḱ-* would seem certain. Three possibilities remain: 1) the root in question is **$h_1$enk-* and the Hittite, Irish, and Italic forms require some other explanation; 2) the Greek form is from **h₂enḱ-*, but needs some phonological or analogical restructuring; 3) we have, as per Beekes, two roots **h₂enḱ-* and **$h_1$enḱ-*. The first possibility can be discounted. Even if we explain away the Hittite *henk-* (< **h₂e-$h_1$enk-*?) and *ro·icc* (< **$h_x$nk-e/o-* or **h_xi-h_xnḱ-e/o-*), the Irish preterite *·anaic* is

---

[10] F.O. Lindeman's objection, *Eriu* 35 (1982) 184, that *·anecc-*, attested once in the glosses, requires a zero grade formation is not probative (see McCone, *IE Origins* 50).

[11] Schmidt, *op.cit.* 117, assumes an archaic **nenk-* was colored in analogy to the past participle *nactus*).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

probative.

The second possibility is more controversial. Certainly, the initial ε- of ἐνεγκεῖν would seem to imply the first laryngeal, although it is also possible that the ε- is the result of assimilation, i.e., ἐνεγκ- < *ἀνεγκ- as ὄνομα 'name' < *enoma* (cf. Doric ἐνυμα- [in a personal name]) < **h₁n(e)h₃men-* (Mayrhofer, *EWAia* II 36).[12] This approach makes more sense if we assume a root aorist with secondary nasal as per Rix and Mayrhofer (above). But there are a number of problems with such an assumption. First, this double nasal form is chronologically peculiar in that it is both an archaism, showing an anomalous full grade against the standard zero-grade thematic aorist of Greek, and an innovation, showing the secondary nasal. Second, the nasal is somewhat difficult to motivate since it would seem to be a sporadic change, i.e., if **e-h₁nek-* became **e-h₁nenḱ-* why not **μένδω for μέδω or any other Greek word beginning with a nasal? Finally, such a formation cannot explain the formally similar **nank-* underlying the Latin *nancior*.[13]

It is, in fact, the contrast between Greek ἐνεγκεῖν and Latin *nancior* which defies analysis. Both forms would seem to imply a reduplicated formation, but while the former can only come from **h₁ne-h₁nḱ-e/o-* the latter requires **h₂ne-h₂nḱ-e/o-*. In a recent analy-

---

[12] R. Anttila, *Lingua* 18 (1972) 34 f., argues for such an assimilatory change where a root *ἀνεγκ- gives, both a verbal ἐνεγκ- and a nominal ἀναγκ- ἡ 'necessity'.

[13] Greek offers two forms which may be related to the root of ἐνεγκεῖν. 1) διηνεκής (Doric διανεκής) 'continuous'. M. Mayrhofer, *FsNeumann* 191 n. 51, derives this directly from **dia-h₂nek-*. Chantraine, however, *DELG* 282, explains the Doric form as a backformation for an older **diēnek-* (cf. also M. Meier-Brügger, *Sprache* 33 [1987] 102 f. for extensive discussion with similar conclusions). 2) ἕνεκα 'on account of'. The older etymology of ἕνεκα : εἵνεκα : ἕννεκα from *ἐν- Fεκα (Frisk, *GEW* I 513) is doubtful in view of Myc. *e-ne-ka* (see Frisk, *GEW* III 88). In this case, this form may belong with the root of ἐνεγκεῖν (P. Chantraine, *RPh* 34 [1962] 193; A. Heubeck, *Sprache* 9 [1963] 193 = *Kl.Schr.* 396; Chantraine, *DELG* 347). Note, however, that the usage of the Mycenaean form is not entirely clear (Aura Jorro, *DMic* 217). Since both forms are subject to different possible interpretations, they cannot decide the issue here one way or another.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

sis Weiss, *Studies* 18 f., has argued for Beekes' two root hypothesis with an important addition. Weiss argues that Indo-European had two roots: 1) **h₁nek-* which built a reduplicated aorist preserved in Greek ἐνεγκεῖν; and 2) **h₂nek-* with a root aorist reflected in Sanskrit *ānaṭ*. However, he goes on to note: "considering the well-known tendency for rhyming form to influence one another, it would not be surprising if **h₂nek-* also had or acquired a reduplicated aorist..." (*op.cit.* 20).

The solution has two fundamental advantages. First it allows us to explain both Greek ἐνεγκεῖν and Italic **nank-*, without requiring any exotic phonological rules. Second, the apparent Indo-Iranian polymorphy reflected in Avestan *nąsaṭ*[14] and Sanskrit *ānaṭ*

---

[14] K. Strunk, *Laryngaltheorie* 571 f., argues that the Avestan *nąsaṭ*, implying an Indo-Iranian **na-nś-at*, forms a word equation to ἐνεγκεῖν (cf. E. Pirart, *MSS* 47 [1986] 177 for an earlier analysis). The existence of a third example would give additional support for a preform **hₓne-hₓnḱ-e/o-*, although this form has its own phonological problems. The form occurs only in the following verse

> *tąm kavā vīštāspō magahyā xšaθrā nąsaṭ / vaŋhāuš padəbīš manaŋhō . . .* [*Y.* 51.16a-c]
> Through his rulership (following) along the paths of good thinking, Kavi Vishtaspa reached this understanding of our task...' (Insler, *GZ* 107); 'Kavi Vištāspa, by (his) power over the offering, presents that (insight)...' (Humabach, *GZ* I.190)

Kellens, *Verbe avestique* 355, argues that *nąsaṭ* is a corrupted form of *nasaṭ* 'will reach' with the nasal vowel intruding here under the influence of *nąsvå* above [*Y.* 51.13]. This assumption is both unlikely and unnecessary. Copying and transcription errors are common in any text transmission. One factor in such errors is similar words which occur in close proximity. Such corruptions, however, become less likely when two stanzas separate the words in question. Further, the claim is unnecessary, as Humbach, *GZ* II 232, points out: "[t]he meaning 'presents' [or: 'offers'] of our *¹nąsaṭ* is close to the meaning of 'to bring, carry' found with the *s*-aor. *nāš* of the root *¹aṣ/nas*".

The Avestan form suffers from an additional difficulty in that the phonology is unclear. If Sanskrit *vāta-* (trisyllabic) descends from **h₂u̯eh₁n̥to-*, as is generally admitted (B. Vine, *IIJ* 33 [1990] 267 f.), a preform **hₓne-hₓnḱ-e/o-* might be expected to yield Avestan ***nāśa-* not *nąśa-*. The formations, however, are not exactly parallel where the morpheme boundary in **hₓne-hₓnḱ-e/o-* might alter the outcome (See also Strunk, *op.cit.* 574, for additional discussion of the phonological problems.).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

is the result of the merger of two near homophonous forms. Thus, an Indo-European reduplicated aorist is directly attested by the Greek ἐνεγκεῖν and (and perhaps Avestan *nąsaṭ* —see note), while the Italic ***nank-*** indirectly reflects the same formation.

§5.2 ἄλαλκε 'defended'

The indicative ἄλαλκε is found only at Ψ 185 and ἀπ- άλαλκε [δ 766]. The third person optative, ἄλαλκοι# occurs seven times: Τρώεσσί δὲ λοιγὸν ἀλάλκοι# '(so that) he might ward off destruction from the Trojans' [Φ 138 = 250, cf. 539, Φ 548], ἀπαλάλκοι# [X 348] and the plural ἀλάλκοιεν βελέσσι# '(if somehow) they might defend (him) from the darts' [X 196 cf. δ 167]. The second person ἀλάλκοις [ν 319] occurs once. There is a single instance of the aorist subjunctive ἀλαλκῃσιν [κ 288]; a single instance of the participle ἀλαλκών [I 605]; and three infinitive forms ἀλαλκεῖν [T 30], ἀλαλκέμεναι [P 153], and ἀλαλκέμεν [γ 237].

The heavy distribution of modal forms is reminiscent of the metrically generated reduplicated aorists above (see esp. §3.1.1 f. and §4.1.2 f.); however, no such solution will apply here. To begin with, there is no non-reduplicated **ἄλκοντο. More importantly, the diction surrounding ἄλαλκε heavily favors modal uses, not only in the reduplicated aorist but in all forms of this verb. Thus, the two other aorist stems are also restricted to non-finite forms: the middle ἀλεξασθαι 'ward off (transitive)' (*Iliad* 3X, *Odyssey* 1X); and the active optative ἀλεξήσ- αιμι [Ω 371] - ειε [η 346]. The same lack of indicative forms is found in the present: 1) the infinite ἀλεξέμεν(αι) [A 590 + (8X all in the *Iliad*, all at the feminine caesura)]; 2) the middle present subjunctive ἀλεξώμεσθα 'that we might ward (him) off' [Λ 348 & X 231]; 3) a single participle ἀλεξόμενοι (the only attested form of this present in the *Odyssey*). Only in the future do we find finite forms ἀλεξήσεις, - ει, but again mostly in the *Iliad* (3X, once in the *Odyssey*). And there are future participles (3X all in the *Iliad*). The Homeric occurrences can be tabulated as:

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

	s-forms	*s-aorist*	*reduplicated*
prs.	ἀλεξ-ε/ο-		
aor.	ἀλεξήσ-	ἀλέξασθαι	ἄλαλκ-ε
fut.	ἀλεξήσ-		

In addition to the verbal forms above, Greek attests a number of nominal forms, e.g., a feminine ἀλκή 'strength' and an apparent '*r/n*-stem' ἄλκαρ 'defense' (only in nominative and accusative). A root noun is also attested in the formula ἀλκὶ πε- ποιθώς [O 490] (Chantraine, *Form.* 3; Risch, *Wortbildung* 6). The nominal forms in *-s-*, e.g., Ἀλέξανδρος Myc. *a-re-ka-sa-da-ra*, may have been derived from the *s*-present or may show a formation in *-ti-* < **alekt̂i-anr-* (cf. Aura Jorro, *DMic* 98).

The thematic *s*-present underlying ἀλεξ-ε/ο- is equivalent to Sanskrit *rakṣati* where both descend from a preform **h₂leks-e/o-* (Pokorny, *IEW* 32; Frisk, *GEW* I 68 f.; Chantraine, *DELG* 58; Mayrhofer, *EWAia* II 422; Goto, *I Prs* 257).[15.16]. Beyond the Greek and Indic material, the root is rather poorly represented with forms only in the Northermost languages: Goth. *alhs* 'temple (i.e., protected place)', OE *ealgian* 'protect' < **algō-̂jan* (Pokorny, *loc.cit.*; Lehmann-Feist, *GED* 27), and the Lith. *ālkas* 'Hain' (Frisk, *loc.cit*; Fraenkel, *LEW* 7). Outside of the present there is little comparative evidence for any other verbal formation. The Germanic *̂je/o*-formation is probably an innovation. In

---

[15] The Tocharian *alāsk-* cited by Pokorny, *loc.cit*, does not belong here. Nor does Armenian *aracel* (V. Pisani, *KZ* 68 [1944] 157 f.) which is probably from < **ar-hat̂i-* < **pṛ-peh₂-d-̂je/o-* (Klingenschmitt, *Verbum* 153). For Middle Iranian *ark* 'Arbeit' see below.

[16] The *s*-present may be a relic of an older *s*-desiderative (Chantraine, *Gr.Hom.* 440; J. Jasanoff, *Laryngaltheorie* 232 f.; Kuipers, *Nasal-präsentia* 36 f. The formular context of the Iliad is compatible with a desiderative analysis. Throughout the *Iliad*, the Achaeans try, and fail to ward off the attacking Trojans. Hence, the usage of ἀλεξέμεναι μεμαῶτα '...him, desiring to ward off' [A 590, cf. E 799, Γ 9]. On the other hand, if this usage dates from Indo-European, as required by the Sanskrit cognate, we might not expect such phrases to survive into the epic period, and perhaps the Iliadic usage is accidental, pointing to a non-desiderative formative (cf. Sihler, *NCGGL* 509 f.).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

Sanskrit all forms of this root have been built from the present, e.g *rarakṣa*, aor. *arakṣīt*, and are likely to be later additions to the paradigm. In the same way, both the *s*-aorist and the enlarged *ēs*- of Greek are similarly late analogical formations.

The 'prothetic' α- of Greek in contrast with the initial Sanskrit *r*- points to an initial laryngeal (as does the rare Indic perfect *rārakṣāṇá* -), but a number of small details are less clear. In order to incorporate the Germanic forms a schwebeablaut alternate is required **h₂elk-* > **algō̯-ian* > OE *ealgian*. The Greek ἀλκ- is ambiguous, perhaps indicating a zero-grade from an older ablauting formation, **h₂lélk-* : **h₂̯lk-* (Beekes, *Development* 89), or perhaps indicating a schwebeablaut form as well. Finally the *seṭ* quality of some Sanskrit forms, e.g., *arakṣīt*, is certainly secondary (Narten, *Sigm.Aor.* 216 f.).[17] Hence, we have a proto-form **h₂lek-s-*, which if it reduplicated as **hₓneḱ-* above would give **h₂le-h₂lk-e/o-*, from which our ἄλαλκε falls out directly.

There is no comparative evidence suggesting that ἄλαλκε is old, yet no analogical solution suggests itself. Thus, while ἐνεγκεῖν is a parallel formation, there is no clear analogical pivot.[18] Further, since all known analogic forms of this verb show the *-s-* of the present stem, e.g., ἀλεξησ- , ἀλεξασ- (cf. the Indic pft. *rarakṣa*, aor. *arakṣīt*) an analogic ἄλαλκε is particularly difficult to motivate. These facts suggests that, like ἐνεγκεῖν, ἄλαλκε may be a relic dating from the Indo-European era.

---

[17] The velar should probably be reconstructed as a palatal (i.e., **h₂leḱ-s-*) since the Indic cluster *-kṣ-* can be resolved either as **ks* or **ḱs*. This excludes the Middle Iranian *ark*, but the semantic connection is hardly certain. The Baltic forms also point to true velar, but the Baltic languages are unreliable for such determinations, e.g., Lith. *akmuõ* versus Indic *aśmā*. In what follows, however, I will follow convention and reconstruct **h₂lek-*.

[18] We could posit a present **ἐνεκσ- ε/ο- (cf. Sanskrit *nakṣati*), or perhaps a root form *ἐγκ- such that *ἐγκ- : ἐνεγκεῖν :: ἀλκ- : ἀλαλκεῖν. But despite the existence of Sanskrit *nakṣati* the possibility of a Greek present **ἐνεκσ- ε/ο- is virtually nil. The Greek aorist is in a suppletive relationship with φέρω and the evidence of Germanic **bring-*, Goth. *briggan*, Modern English *bring*, all from **bhr-(hₓ)enḱ-* indicates that this relationship is ancient. Nor is a root *ἐγκ- likely where all forms of **h₁enḱ-* (and *h₂enḱ-*) in Greek are reduplicated.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

§5.3 ἀείδω 'I sing'

The relationship between Greek αὐδή 'voice' and ἀείδω 'sing', although universally accepted (Chantraine, *DELG* 22; Frisk, *GEW* I 22) is puzzling. Assuming a root **h₂ud-eh₂* for αὐδή, we expect the denominative present αὐδάω 'speak', which we have, or any number of possible root or derived presents. A full grade present, however, should have the form *αFεδ-, leaving the diphthong - ει- unexplained. J. Wackernagel, *KZ* 29 (1888) 151 f., suggests that ἀείδω is backformed from an unattested aorist *α-Fειδ- which is, like εἶπον, dissimilated from *α- Fε- Fδ- and hence, in modern terms a reduplicated **h₂u̯e-(h₂)u̯d-*. He went on to suggest a connection with the root of Sanskrit *vad-* 'say' (more recently see Mayrhofer, *KEWA* III 134).[19]

There are, however, both formal and functional problems with such a preform. Why should this aorist have been transferred to the present while leaving no trace of its origin? One might argue that the full grade was out of place for a thematic aorist so that an older aorist ἄειδον was reinterpreted as an imperfect. However, although apparent full grade thematic aorists are peculiar, they do occur (cf. above §1.1 n. 3). Furthermore, when such forms are restructured the usual pattern is to convert such forms to alphathematic inflection, hence we might expect **ἄειδας parallel to εἶπας.

More difficult are the phonological problems where Wackernagel's reconstruction is possible only with an inexact notion of prothesis, but the initial alpha of ἀείδω and αὐδή would seem to imply a laryngeal root **h₂u̯ed-*. A reduplicated form of this root should yield *ἀ-Fα-Fδ- parallel to ἄλαλκε (Frisk, *GEW* I 22 f.). Hence, we find the

---

[19] The semantic mismatch between Sanskrit *vad-* and Greek ἀείδω is not a problem, since even within Greek the relationship to αὐδή 'voice' shows that the specialized meaning 'sing' was a late addition to this root. Presumably, the contrast with Greek φωνή which can refer to the noise of animals and αὐδή, always a human 'voice', gave rise to the notion of 'sound only humans make' -> 'sing'.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

same problem we encountered above trying to derive ἐνεγκεῖν from *h_2nek-. It might be possible to imagine some phonological rule that would allow us to get from h_2Re-h_2RC- to a pre-Greek *aR-eRC*- although such a rule would seem to contradict what we know of laryngeal coloring. Perhaps the morpheme boundary blocked the coloring here, or better perhaps the *e*-quality was analogically restored in the reduplicating syllable, but Latin *nancior* would seem to require that coloring took place as expected in *h_2nek-. It is unlikely that a single solution will apply to ἐνεγκεῖν and ἀείδω.

W. Winter, *Language* 26 (1950) 366, noting that *h_2 did not color contiguous *-*o*- argues for a perfect to this root with 'saṃprasāraṇa' reduplication, i.e., *h_2u-h_2oud- which would give *ἀFoFδ- > *ἄοιδ-α from which a present could then be created analogically. But this theory faces its own difficulty, namely, why do we find no trace of the older perfect which served as the pivot form? Further, the existence of 'saṃprasāraṇa' reduplication is unknown outside of Indic.

F. Specht, *KZ* 59 (1932) 119 f., argues against the reduplicated analysis. Noting the existence of the noun ἀηδών 'nightingale' and the verbal αὖε 'shout' [Λ 461 +], he sees the origin of Greek ἀείδω in the root *$au\bar{e}$-*i*-*d*-. This approach separates the Greek forms from Sanskrit *vad*-, but has other advantages. Most importantly, although it is generally admitted that ἀηδών belongs with the family of αὐδή (Frisk, *loc.cit.*; Chantraine, *DELG* 26), the lengthened grade is unexplained.[20] Working with a root *$au\bar{e}i$- we can now posit an internal laryngeal (cf. Peters, *Untersuchungen* 14 f.). Given such contrasts as λαθ- : λῆ-το (§4.1.4), a root alternately extended with a dental in some forms, with an *-i*- in others may underlie the Greek forms.[21]

---

[20] Beekes, *Development* 89, simply posits a lengthened grade *$h_2u\bar{e}d$- here, but he offers no reason for the lengthening in this form (cf. also Solmsen, *Untersuch.* 238; Wackernagel, *loc.cit.*).

[21] Additional evidence for the enlarged stem is found in Celtic (cf. Joseph, *Problems* 47 f., who explains the O.Ir. *ai* 'inspiration' as equivalent to Welsh *awydd* 'desire', Old Cornish

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

*$h_2u̯(h_1)$-[22]	>	αὖε
*$h_2u̯eh_1$-d-	>	ἀηδών
*$h_2u̯eh_1$-i-d-	>	ἀFείδω

Finally, our root *$h_2u̯eh_1$-i- may be related to the homophonous *$h_2u̯eh_1$-i- 'blow' (Pokorny, *IEW* 81 f.; see Vine, *Diss.* 142 f., Joseph, *loc.cit.*). All in all, although a reduplicated aorist is possible here, other options are more likely.

§5.4 ἐνένῑπε and ἠνίπαπε 'I rebuked'

The reduplicated ἐνένῑπε 'rebuked' superficially bears some resemblance to the forms above, although the distribution of forms in Homer (4X *Iliad*, 11 *Odyssey* [all in the second half]) suggests that this aorist is secondarily built to an older ἠνῑ́παπε (4X *Iliad*, 2X *Odyssey*). The priority of ἠνῑ́παπε is further supported by the analogical developments, where a formally opaque ἠνῑ́παπε could be easily restructured to a reduplicated ἐν- ἐν-ῑπε after *ἐν- εῃ- κε. Contrariwise, given a reduplicated ἐνένῑπε, only ἐρῠ́κακε[23] could serve as a building block for a newer ἠνῑ́παπε, but this change is both unmotivated and unexpected. In addition to the aorists, two present formation also exist in Homer, ἐνίπτε [Γ 438], -οι [Ω 768] and ἐνισσέμεν (3X *Iliad*, 2X *Odyssey*), as well as the nominal ἐνῑπή 'rebuke' (3X *Iliad*, 3X *Odyssey*). The contrast between final -σσ- and -π- implies a labial-velar final root (Chantraine, *DELG* 349). Two possibilities have been proposed.

Most recently, J.A. Harðarson, *FsRix* 163 f. (after J. Bendahman), has suggested a

---

awit < *$h_2u̯eh_1$-id-).

[22] The root final laryngeal is necessary to explain the enlarged ἀῡτέω 'shout' as well as the aorist ἀῦσαι.

[23] The aorist ἐρῠ́κακε 'held back', sometimes cited as a parallel has most likely been built in analogy to ἠνῑ́παπε (Chantraine, *Gr.Hom.* 398; Beekes, *loc.cit.*).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

connection to the root **u̯ek$^{u̯}$-* 'speak'. Harðarson and Bendahman assume a reduplicated present **u̯i-u̯k$^{u̯}$-e/o-* formally equivalent to Sanskrit *vívakti* which when prefixed will give more or less the correct form, i.e., **eni-u̯i-u̯k$^{u̯}$-e/o-* '(heftig) auf jm. einreden' > **eni-u̯i-ik$^{u̯}$-e/o-* (as in εἶπον) > **enīk$^{u̯}$-e/o-* > *ἐνῑπ-. Nevertheless, in addition to the obvious semantic problem,[24] there are a number of formal problems as well.

First, an equation between Sanskrit *vívakti* and a preform **u̯i-u̯k$^{u̯}$-e/o-* is weak. The Indic from is probably a innovation (cf. §1.1), but even if it were old, there is disagreement both in root vocalism and thematic versus athematic endings. Second we would expect the preform **eni-u̯i-u̯k$^{u̯}$-e/o-* to become **eni-u̯i-ik-e/o-* with the de-labialization of βουκόλος < **g$^{u̯}$ou̯-k$^{u̯}$ol-os* (Weiss, *Studies* 153 f. and above §1.1). In the case of εἶπον there is strong analogical support to block the change of *u̯k$^{u̯}$* > *u̯k*, but with the semantic change required for **eni-u̯i-u̯k$^{u̯}$-e/o-*, that analogical support is much harder to motivate. Finally, even if we accept both the preform **eni-u̯i-u̯k$^{u̯}$-e/o-* and the phonological development, the analysis fails in that it explains neither the present ἐνίπτε nor the peculiar aorists ἐνένῑπε and ἠνῑπαπε. That is, if ἐνῑπ- was no longer recognizable as a present and a newer ἐνίπτε was required, why didn't the poets simply use ἐνῑπ- as the aorist form instead of generating the truly bizarre ἠνῑπαπε? To put it another way, any account of ἠνῑπαπε (and ἐνένῑπε) should generate the irregular forms first, then apply analogical restructuring to give the more regular forms.

A older approach derives ἠνῑπαπε from the root of **ok$^{u̯}$-* (i.e. **h$_3$ek$^{u̯}$-* 'see') and further suggests a relation to the present ὀπιπεύω 'stare at' (K. Brugmann, *IF* 12 [1901] 31). Certainly this present is a reduplicated formation, reflecting **h$_3$(e)k$^{u̯}$i-h$_3$k$^{u̯}$-eu̯-*, but how this relates to ἠνῑπαπε is less clear. No obvious preform will work here, since ἠνῑπαπε seems to imply a zero-grade root with zero-grade reduplication. Beekes, *De-*

---

[24] Certainly a change from 'speak forcefully' to 'rebuke' is possible, but is unmotivated here.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

velopment 129 f. n. 59, in fact, suggest **eni-h_3k^{μ}-h_3k^{μ}-e/o-* as a possibility, but he is bothered by the phonology *Ch_3C > CoC*. This is the least of our problems, since the formation is all but impossible, and bears no resemblance to the other reduplicated aorists above. We might expect, **eni-h_3k^{μ}e-h_3k^{μ}-e/o-* > **ἐνῑπωπε, and perhaps this form would be analogized in some way to ἠνῑπαπε. I see no way to salvage this approach either. The form remains without etymology or analysis.

§5.5 Conclusion on *HRe-HRC-e/o-*

Both Greek ἐνεγκεῖν and ἄλαλκε offer evidence for an Indo-European reduplicated formation *h_xRe-h_xRC-e/o-*. As with εἶπον above, ἐνεγκεῖν seems to have isolated from any paradigm already in the Indo-European era, while ἄλαλκε is paired with an *s*-present. Although no absolute conclusions are possible for either ἀείδω or ἐνέν̄ιπε (and ἠνῑπαπε), it is unlikely that either form represents an Indo-European reduplicated aorist.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

CHAPTER VI

§6.0 *aC-aC-e/o-*

In addition to ἐνεγκεῖν and ἄλαλκε, five other vowel initial reduplicated aorists occur in Homer: ἤραρε, ὤρορε, ἥπαφε, ἤκαχε, and ἤγαγον. Of these only the last is common outside epic poetry (see §6.5). Like ἐνεγκεῖν and ἄλαλκε, these vowel initial reduplicated aorists show a short vowel in the second syllable; however, a preform **h₂re-h₂r-e/o-* analogous to **hₓne-hₓnḱ-e/o-* would yield not the expected ἀρᾰρε but *ἀρᾱρε. This consideration led G. Schmidt, *Orbis* 26 (1977) 114, to rule out **hₓne-hₓnḱ-e/o-* as a preform for ἐνεγκεῖν. But there is no *a priori* reason to assume that ἄραρε should descend from exactly the same formation as ἐνεγκεῖν and ἄλαλκε.[1] While the existence of the *HRe-HRC-e/o-* aorist has been described above, this chapter will focus on the remaining vowel initial forms.

For ἄραρε and ὤρορε, there exist perfect stems ἀρηρ- and ὤρωρε, the former of which alternates with a weak stem ἀρᾰρ- (ἀραρυῖα and the pluperfect [?] ἄραρον). Hence, there is some ambiguity concerning the designation of ἀραρ- as an aorist. This formal ambiguity is matched by a functional variation in that both ἄραρε and ὤρορε are used both intransitively and transitively in Homer. However, no simple division equating intransitive forms and perfects, or transitive forms and aorists, is possible. The analyses below will begin with a philological discussion of these intransitive uses.

§6.1 ἤραρε 'fixed'

Of the ten uses of ἤραρε, four refer to craftsmanship: τούς τε κλυτὸς ἤραρε τέκτων 'those which the famous carpenter had fixed' [Ψ 712]; τὰ μὲν ἀσκήσας

---

[1] Compare Meier-Brügger, *Gr.Spr.* II 58: "[d]iese sg. attische Reduplikationweise basiert auf dem Prinzip **HK-e-HK-*... [d]er Typ ἀλαλκεῖν, ἀραρεῖν bildet eine Nebengruppe...".

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

κεραοξόος ἥραρε τέκτων 'these the horn-worker having wrought had fit together' [Δ 110]; ἴκρια δὲ στήσας, ἀραρὼν θαμέσι σταμίνεσσι 'he set the beams, fitting (them) to dense ribs' [ε 252]. In the last instance, although the accusative object has been gapped, the verb is clearly transitive. The fourth 'technologic' use will be described below.

Two uses of ἥραρε seem to be metaphoric, describing things fit together as a craftsman would fit together wood: οἱ δ' ἐπεὶ ἀλλήλους ἄραρον 'when they had fit one another together' = 'hemmed each other in' [M 105]; ὡς ἂν μνηστῆρσιν θάνατον καὶ κῆρ' ἀραρόντε 'thus (when) you two have fit death and fate upon the suitors' [π 169]. The metaphor between this 'technologic use' and the other usage of ἥραρε is explicit at Π 212-214 where there is a simile comparing the battle array to the stones of a house:

> ὡς δ' ὅτε τοῖχον ἀνὴρ ἁράρῃ πυκινοῖσι λίθοισι
> δώματος ὑψηλοῖο, βίας ἀνέμων ἀλεείνων
> ὡς ἄραρον κόρυθές τε καὶ ἀσπίδες ὀμφαλόεσσαι.
>
> As when a man would fit together a wall of a high house with tight stones,
> avoiding the force of the winds,
> thus the helms and embossed shields were fit together.

Here the transitive subjunctive ἁράρῃ 'fit together (a wall)' contrasts with the intransitive ἄραρον '(helms and shields) were fit' two lines later. The former aorist requires no comment, but the latter usage is both intransitive and unaccusative. M. Leumann, *Celtica* 3 (1955) 244 n. 1 = *Kl. Sch.* 254 n. 1, sees this as a contrast between transitive aorist and intransitive pluperfect: "Neben trs. Aor. Konj. ἁράρῃ 'er fügt' kann intr. ἄραρον 'waren gefügt' nur Plqpf. sein..." (see also Chantraine, *Parfait* 27). Alternatively, it is possible that the transitive aorist was influenced by the existing finite pluperfect participle

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

ἀραρυῖα (cf. R. Janko, *Comm.Il.* IV 346).[2] Whether ἄραρον reflects an archaic pluperfect or a secondary contaminated form, it is unlikely to indicate any archaic intransitive usage in the reduplicated aorist. There remains, however, a second intransitive usage of ἤραρε.

This second intransitive cannot be viewed as a pluperfect. The phrase ὃ δὴ καὶ πᾶσιν ἐνὶ φρεσὶν ἤραρεν ἡμῖν [δ 777] seems to mean 'that which is fitting to all of us at heart'. However, this passage must be viewed in the light of the two uses of ἤραρε yet to be discussed. In these two instances the verb is used idiomatically in the phrase ἤραρε θυμὸν ἐδωδῇ 'to fit the spirit with food' = 'to be satisfied, pleased' [ε 95; ξ 111]. Hence, this intransitive use of ἤραρε is best interpreted as 'which pleases all of us', where the idiom has been altered by dropping the accusative θυμός in favor of the phrase ἐνὶ φρεσὶν. Although this passage (and perhaps also ἄραρον above) does represent a later development in the diction of ἤραρε, the basic meaning of ἤραρε was 'fitted X together'.[3]

In Homer, the reduplicated aorist occurs alongside an *s*-aorist ἄρσε (3X *Iliad*, 4X *Odyssey*), and a middle ἄρμενον (1X *Iliad*, 2X *Odyssey*). There is also a single instance of the passive ἄρθεν [Π 211], and a single instance of the present ἀράρισκε [ξ 23]. A large number of perfect forms also occur, although curiously only one of these is a finite form, ἀρήρῃ [ε 361]. All other forms of the perfect system in Homer are either pluperfect forms (6X *Iliad*, 1X *Odyssey*) or participles (28X *Iliad*, 18X *Odyssey*), e.g., ἐν δ' ἔπεσε ζωστῆρι ἀρηρότι πικρὸς ὀϊστός 'and the bitter dart fell against the well-

---

[2] Janko, *op.cit.*, notes that the reverse process is also attested, where the perfect is used transitively at *Erga* 431 and in the variant ἀρηρεν (for sigmatic ἄρασσεν) [ε 248].

[3] S. West, *Comm.Od.*. I 241, considers the possibility that ἤραρε at δ 777 is the aorist to ἀρέσκω 'satisfy' not ἀραρίσκω. Compare also B. Vine, *HS* 106 (1993) 56 n. 39, for possible relations between ἀρέσκω and ἀραρίσκω. All in all, I do not think such measures are necessary.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

fitted belt' [Δ 134]. The adjectival perfect participle is already attested in Mycenaean: *a-ra-ru-wo-a* = *ἀραρϜόϝα and *a-ra-ru-ja* = ἀραρυῖα (Aura Jorro, *DMic* I 95), and may predate the Greek era.[4]

	act.	*mid.*
prs.	ἀράρισκε	—
aor.	ἦραρε, ἄρσε	ἄρμενον, ἄρθεν
pft.	ἀρήρῃ	—

Pokorny, *IEW* 55 f., gives a root **ar-* attested in several branches of Indo-European. Although nominal forms of this root abound, e.g., Greek ἀρθμός 'Verbindung', Latin *artus* 'limb', Sanskrit *ṛta-* 'proper, right, fit', Avestan *aša-* 'truth', verbal forms occur only in Greek, Armenian, and Avestan *ārāntē* 'sie setzen sich fest'.[5] The most significant of these forms for our present purposes is the Armenian *ārnem* : aor. *arari* 'I made', which gives a word equation for Greek ἦραρε (Pokorny, *IEW* 56; Frisk, *GEW* I 128 f.; Chantraine, *DELG* 102; Schmitt, *Gr.Kl.Arm.* 153; Klingenschmitt, *Verbum* 284; Clackson, *Arm.-Grk.*).

Although Greek and Armenian do share certain features not present in other Indo-European languages, even if this form dates from some Greco-Armenian epoch, it still

---

[4] Note Meier-Brügger, *Gr.Spr.* II 58: "Der Typ ἀλαλκεῖν, ἀραρεῖν... is beriets myk. (vgl. neben *a-ra-ru-wo-a araruoho* auch *a-ra-ro-mo-te-me-na ararmotmena*) und hat Parallelen im Armen.". Exactly how *ararmotmena* reflects an aorist in not entirely clear to me (cf. also the objections in Clackson, *Arm.-Grk.* 232 n. 78). The details of the formation are obscure (Ventris-Chadwick, *Doc.* 533), but it appears to be a perfect formation (cf. Aura Jorro, *DMic* I 97 f.).

[5] Aside from quoting the form, Kellens, *Verbe avestique* 378, gives no details. For a discussion see Harðarson, *Studien* 199 f. Harðarson, *op.cit.* & n. 205, also argues for a possible Indic cognate at *RV* IV 19.9 *sám aranta párva* 'die Gelenke heilten'. This may, however, be a form of *[1]ar-* 'in Bewegung setzen' with the specialized meaning attributable to the preverb (cf. also Mayrhofer, *EWAia* I 107, who notes both options with literature).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

must pre-date Greek history.[6] This contrasts with the other portions of the paradigm. The lack of agreement between the present forms in Armenian, Avestan and Greek implies that these forms were innovated in each of the separate languages (for additional discussion of the present see §6.6.1 below). Nor is the *s*-aorist likely to reflect an archaism as we have only the evidence of the rare Greek forms. A root aorist middle as reflected in Greek ἄρμενον has a slightly better chance of being old since we expect a root formation in contrast to the derived presents of Greek and Armenian.[7] Hence, only the reduplicated active aorist and perhaps also the middle root aorist have any chance of being archaic.

Now the ablaut between full grade **ar-* and zero-grade **r̥-* is most easily explained in terms of a laryngeal root **h₂er-* (Beekes, *Development* 139; Klingenschmitt, *Verbum* 284; Mayrhofer, *EWAia* I 254 "*r̥ta-*"; E. Tichy, *Sprache* 32 [1986] 91 f.; Harðarson, *Studien* 199 f.).[8,9] As noted above, however, a laryngeal preform **h₂re-h₂r-e/o-* should give **arār-e/o-* instead of the required short vowel form. Still, this is not the only possible reconstruction. A formation showing what is traditionally called intensive reduplication, **h₂ér-h₂r-e/o-*, would give the correct ἀραρ- ε/ο- in Greek. However, this reconstruc-

---

[6] In this regard, note also Clackson's central conclusion, *Arm.-Grk.* 199 f., that the common features of Armenian and Greek represent preserved archaisms not innovations.

[7] Arguments, e.g., Schwyzer, *Gr.Gr.* 751, in favor of seeing ἄρμενον as an *s*-aorist, < **ars-men-*, are not probative (cf. Watkins, *Celtic Verb* 52).

[8] Further evidence for initial **h₂-* is sometimes cited in the Luvian *hīrūn* 'curse' < **h₂ēr-u-* (Mayrhofer, *Idg.Gr. I/2* 133), but this is hardly probative. C. Watkins, *FsRix* 469 f., connects *hīrūn* with Greek ἀρά 'prayer', which is certainly a better semantic match.

[9] Note in particular the comparison of Skt. *r̥tu-* 'period of time' = Lat. *artus* 'straight, narrow, strict' = Greek ἀρτύς (·σύνταξις Hesychius) = TA *ord* 'friend' (see below n. 11). The evidence for 'Lex Rix', i.e., $h_xR > e_xR$ in anlaut (where $e_x = e$, *a*, or *o*), is certain for Greek and has also been proposed for Italic, which would allow a proto-form **h₂r-tu-* to explain all three forms (see most recently Schrijver, *Reflexes* 68 for discussion). Nevertheless we expect an ablauting proto-form **h₂ér-tu-s* : genitive sg. **h₂r-téu̯-s* (Mayrhofer, *EWAia* I 257) so perhaps an **ár-tus* : **r̥-téu̯-s* is not impossible. If the Armenian *ard* 'Ordnung' belongs here an ablauting form would be required.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

tion is not without problems. In Armenian we expect a laryngeal in a non-initial syllable to be reduced (e.g., *dowstr* 'daughter' < **dhugh₂ter*) and hence, **h₂er-h₂r-e/o-* should give **arr-e/o-* > ***arī*.[10] Since the reconstruction **arar-e/o-* will give both Greek and Armenian forms without problem, there is circumstantial evidence that we may be dealing with a root with a 'real' Indo-European **a-*. Further, Hittite significantly shows no trace of a laryngeal in the noun *arāš* 'companion', that is 'one who fits' (cf. Puhvel, *HED* I 120).[11]

Working with a non-laryngeal root **ar-* is not without its own problems. First, the alternation between **ar-* and **r̥-* requires **a* : *Ø* ablaut, evidence for which is rare in Indo-European. Second, we now have no phonological source for our perfect ἀρηρ-. However, this latter problem is easily dealt with. M. Leumann, *Celtica* 3 [1955] 243 f. (= *Kl.Sch.* 253 f.), argues that the short vowel forms ἄραρον [Π 214] and ἀραρυῖα are the more archaic while the long vowel has been introduced from the masculine participle *ἀραρFώς > *ἀρᾱρώς. Note, however, that Szemerényi, *Syncope* 209 f. n. 5, argues for the reverse process, an older ἀρᾱρFώς shortened by Osthoff. But this is unlikely. Both the antiquity of the aorist and the defective paradigm of the perfect suggest that perfect forms have been built to the older aorist **arăr-*. The simplest solution both phonologically and chronologically is to assume an Indo-European root **ar-* (with **a* vocalism) which formed a reduplicated **ar-ar-e/o-* already in the Indo-European era.

---

[10] Klingenschmitt, *Verbum* 284, argues for a laryngeal root, but generates the reduplicated formation in the "nachgrundsprachlicher Zeit" so as to work with a root **ar-ar-*. This chronology seems to me unworkable: **ar-ar-e/o-* must predate the Greek era, while a variety of factors (most notably the triple representation of h_1, h_2, and h_3) require that the laryngeals survived at least into very early Greek.

[11] Note the semantically similar Tocharian *ort* 'friend' (J. Hilmarsson, *MSS* 43 [1984] 107 f.) which apparently requires a zero-grade formation **r̥t-u-* (with *u*-umlaut) formally equivalent to Sanskrit *ṛtu-* (see above n. 9). The form is ambiguous as to the initial laryngeal where both **R* and **HR* develop as **æR* in anlaut, cf. TA *arkaś* TB *erkenta* (Nom.pl.f.) 'black' < **h₁r̥g^{w}ónt-es*, but Greek ἔρεβος 'hell' < **h₁rég^{w}os* (cf. D. Ringe, *MSS* 52 [1991] 142 f. for this and other examples).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

The formation of **arar-e/o-* is distinct from that of ἐνεγκεῖν. Therefore, the vowel initial reduplicated aorists have (at least) two possible sources: 1) $C_1Re\text{-}C_1RC_2\text{-}e/o$- where C_1 is [+ continuant], that is *s-* or *R-*; and 2) **aC-aC-e/o-*. While this first formation resembles the canonical reduplication in *Ce-*, this second formation, like the hypothesized **h₂ér-h₂r-e/o-* above, shows a formation more akin to root or intensive reduplication. While both Greek and Armenian agree in showing **ar-ar-e/o-* as an aorist formation, both the peculiarity of the formation and the formal equivalence of aorist and perfect forms in Greek leave some doubt as to the position of this formation in the Indo-European verbal system.

§6.2 ὄρορε 'roused'

As with ἦραρε both transitive and intransitive uses of the aorist ὄρορε exist in the Homeric corpus. Three out of seven times ὄρορε(ν) is clearly transitive and requires little comment: ...τὰ μέν τ' Εὖρός τε Νότος τε / ὄρορ[ε] 'those (waves) which Euros and Notos roused' [B 145-6]; ...ἤ τίς μιν θεὸς ὄρορε 'or some god roused him' [δ 712]; τὴν... θεὸς ὄρορεν... 'a god roused her (to do an unseemly deed)' [ψ 222]. The remaining four passages have at one time or another been described as intransitive.

At τ 201, χαλεπὸς δέ τις ὄρορε δαίμων 'some difficult Daimon roused [something] up', there is no syntactic object in the accusative, but the clause requires something to complete the thought. The overall sense implies that this δαίμων has stirred up something, presumably the winds. The previous line εἶλει γὰρ βορέης ἄνεμος μέγας 'for a great north wind held (them)' supplies the context. Ὄρορε is causative because of the semantic relation of this stem 'made rise, stirred up' to the root aorist ὦρτο 'arose', cf. ὦρτο πολὺ πρῶτος ...'Αγαμέμνων 'and first sprang up Agamemnon' [H 162]. The stem ὄρορε, used without an accusative complement, is still causative as is the functionally similar *s*-aorist ὦρσε in τότε δὲ Ζεὺς ὦρσε μάχεσθαι 'Zeus then urged (them) to fight' [N 794].

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

The next instance is more complex. The phrase καί μοι μένος ὄρορε (N 78) seems to mean 'my might was roused', cf. Chantraine, *Gr.Hom.* 398, who notes that this usage "...semble équivaloir sensiblement au parfait ὄρωρε" (cf. also R. Janko, *Comm.Il.* IV 53). Here the question of transitive/causative usage rests on the interpretation of μένος. Since μένος is neuter there is nothing which prevents a reading of '(someone) roused my might'. But without any overt subject such a reading sounds unlikely to our ears. Nevertheless, A. Heubeck, *KZ* 97 (1984) 91 f., has recently studied the diction of this passage and shows that a reading of μένος as an accusative is quite possible. Heubeck notes that only a few lines before, the poet describes Poseidon filling the heros with might [N 59-61]:

> ἦ, καὶ σκηπανίῳ γαιήοχος ἐννοσίγαιος
> ἀμφοτέρω κεκοπὼν πλῆσεν μένεος κρατεροῖο
> γυῖα δὲ θῆκε ἐλαφρά, πόδας καὶ χεῖρας ὕπερθεν.
>
> The Enfolder and Earthshaker spoke, and having struck
> the two with his staff filled them with strong might
> and made their limbs light, their feet, and hands above.

Heubeck goes on to note certain similarities in the diction of this passage to phrases surrounding καί μοι μένος ὄρορε [N 78]. In the next line Aias describes his feet below ...νέρθε δὲ ποσσὶν... [N 79], and one line later he says that he is μεμαῶτι μάχεσθαι 'eager to fight' [N 80]. Since both of the phrases echo the diction of Poseidon's inspiration at N 59-61, it is probable that μένος is the object and not the subject of ὄρορε.

Additional support for Heubeck's claim lies in the formulaic consistency of ὄρορε. Each of the passages noted above, B 145-6; δ 712; ψ 222; τ 201, describes divine inspiration. Similar formulaic consistency was noted above concerning τετύκοντο 'they fashioned, they prepared', which is always used to describe the preparation of food (see above §3.1.3, cf. also §4.1.5). For ὄρορε, however, a second formulaic context exists.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

The last two instances to be discussed both describe the effect of music on an audience.

Heubeck explains the next example, θ 540, ἐξ οὗ δορπέομέν τε καὶ ὄρορε θεῖος ἀοιδός 'since we began to eat, and the divine singer roused/was roused (?)' as another absolute usage, "[s]eitdem nach dem Nachtmahl der göttliche Sänger uns in heftige (innere!) Bewegung versetzte..." (*op.cit.* 93). The last example is ω 62, τοῖον γὰρ ὑπώρορε Μοῦσα λίγεια, which must mean 'such did the clear-toned Muses rouse (the hearts)' (again see Heubeck, *op.cit.* 94 f.). This usage is similar to the intransitive use of ἦραρε 'fit (the heart)' described above, and poses no special problem.

The major point to be gleaned from the data above is that despite some syntactic problems, the aorist ὦρορε is clearly a causative in the sense outlined above. The interpretation of this usage appears complicated only because semantically transitive verbs can, in Greek, appear without an external argument, even though the sense requires it (cf. English 'eat'). A non-causative meaning of 'arose, be stirred' can apply only for the phrases καί μοι μένος ὦρορε [N 78] or ὦρορε θεοῖς ἀοιδός [θ 540]. Heubeck's analysis explains these apparent exceptions, and in view of the diction of the other five passages, his line of reasoning will be followed here.

Unlike ἦραρε, ὄρορε does not build a present in -ίσκω; however, the two paradigms have other similarities. Both verbs build, in addition to the reduplicated aorist, an *s*-aorist active, a root aorist middle and a perfect.

	Active	*Middle*
prs.	ὄρνυε	ὄρνυται
aor.	ὦρσε, ὄρορε	ὦρτο, ὄρετο[12]
pft.	ὄρωρε	
fut.	---	

[12] The thematic aorist ὄρετο is certainly a late addition to the paradigm (cf. S. Insler, *IF* 72 [1967] 250 n. 1)

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

The nasal present and root aorist are also found in Sanskrit *ṛṇoti* : *ārta* (Pokorny, *IEW* 326 f.; Frisk. *GEW* II 422; Chantraine, *DELG* 823; Mayrhofer, *EWAia* I 105 f.). The same pairing may underlie the Tocharian subjunctive forms TA *arāñtär* 'they will call forth', certainly a relic of the older present, and TB *ertär* which may well descend from the older aorist.[13] The contrast of TA *ar-* TB *er-* could point to either an Indo-European zero-grade **r̥-*, or more likely an *o*-grade agreeing with the Greek. Finally, a nasal present also occurs in Armenian *yarne-*.[14]

Although the Sanskrit *ārṣīt* is not attested in the *RV* and is probably a later innovation (see Whitney, *Roots* 14), an *s*-aorist is assured by the correspondence of Greek ὦρσε 'stirred, roused up' and the Tocharian class III preterite, TA *arsāt* TB *ersate*. Further, both Tocharian A and B agree in having a class VIII present, TB *erṣäṃ*, TA *aräṣ*, e.g., *orṣi pekeṣi pemu arämpāt wrasaśśi tunk kāpñune aräṣ* 'a wooden or painted beauty arouses love and affection' [*Puṇyavantajātaka* 9b1]. In a series of articles J. Jasanoff, *GdCowgill* 101; *TIES* 2 (1986) 58, has argued that the class VIII presents of Tocharian are reflexes of old *s*-aorist subjunctives. Even if the class III preterites were extended to roots which did not originally have an *s*-aorist in Indo-European, the coincidence of both a class III preterite and a class VIII present provides strong evidence for assuming an *s*-aorist in Indo-European.

Thus, the forms above point to a nasal present active contrasting with a root aorist middle and *s*-aorist active: **h₃r̥néu̯-ti* : **é-h₃rto* : **e-h₃ér̥s-*. The initial laryngeal is justified both on theoretical grounds and because of the expected zero-grade in nasal present. The intransitive Latin *orior* 'arise', although an Italic *ie/o*-formation, also points to an initial **h₃-* (Schrijver, *Reflexes* 69 f.; cf. also Walde-Hofmann, *LEW* I 722; Ernout-Meillet,

---

[13] J. Jasanoff, *FsLehmann* 145 n. 30, compares *ertär* with the Hittite *āri* and notes the possibility of a direct comparison of both with ὦρτο.

[14] Klingenschmitt, *Verbum* 161 f. derives this from **h₃r̥-nhₓe/o-*, an innovated form built to **h₃r̥-mu̯e/o-* (but see below [n. 16] for a possible *set* formation).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

DELL 468 f.). Beyond this the etymological details become complicated. In addition to the nasal present, Sanskrit also shows a reduplicated present *íyarti* : *írte* < **h₃i-h₃er-* : *h₃i-h₃r-to-i* (cf. H. Rix, *MSS* 27 [1970] 93; S. Insler, *KZ* 86 [1972] 98 f.) and a causative *īráyati*, although this latter form is certainly an innovation (Jamison, *aya-Form*. 124).

Older handbooks also include the *ske/o*-present *ṛcchati* 'erreicht' (e.g., Whitney, *Roots* 14); however, this form is probably best taken with a second root *²ar-* 'hinkommen, erreichen' (Mayrhofer, *EWAia* I 106) perhaps cognate with Greek ἔρχομαι, both from **h₁ṛ-ske/o-* (Rix, *op.cit*. 98). This simple division into two roots **h₃er-* 'arise/stir up' and *h₁er-* 'come to, arrive' is, however, complicated by the existence of the Hittite *armu-* 'stir up, move along', which is at first glance both formally and semantically a match to Greek ὄρνυε = TA *arāñtār* (Mayrhofer, *EWAia* I 105 f.; Puhvel *HED* I 162). Nevertheless, there is clear evidence showing that **h₃-* should be preserved in anlaut in that language[15] and, hence, ὄρνυε = Hittite *armuzi* would seem to require **h₁or-*, a form perhaps justified by the Hesychian ἔρετο· ὡρμήθη.

Still, both the expected zero-grade of the root in a nasal present, as well as over-riding evidence for **h₃-*, make it likely that the Hittite form (and ἔρετο?) is to be explained in another way. Hittite *armu-* is only one of no less than three Hittite verbs which have been compared to Greek ὄρνυε at one time or another. But, Hittite *armu-* is simply the causative to the root *ar-* 'come' (prs. *ārhi*), a root best compared with the Sanskrit *²ar-* 'arrive, reach', with the productive nasal suffix (Puhvel, *op.cit* 111). Both its formal and semantic match to ὄρνυε is thus accidental. The second possible match, *arāi-* 'arise' (J. Jasanoff, *AArmL* 2 [1981] 15 f.) works semantically, but this form is clearly *seṭ* and cannot be directly equated.[16] The final possible cognate to ὄρνυε is the Hittite deponent

---

[15] Compare, e.g., *harganāu* 'palm, sole' < (virtual) **h₃reǵnṓw* : Greek ὀρέγω; *happariye-* 'sell' < **h₃ep-* (H.C. Melchert, *Sprache* 33 [1987] 22 n. 10; *AHP* 72),

[16] H. Rix, *IF* 70 (1965) 25 f., raised the possibility of a *seṭ* variant of the root in order to explain the irregular thematic perfect ὀρώρεται. Rix then tried to equate this *seṭ* perfect

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

ar- 'stand (by)' (prs. *arhari*), 3sg. *arta* 'he stands' which Puhvel, *op.cit.* 107, takes as a word equation to Greek ὦρτο = Sanskrit *ārta* (also Mayrhofer, *loc.cit.*).[17] Despite the problems of broader connections, a root ***h₃er-*** 'arise' is securely attested for Greek, Sanskrit and Tocharian.

Returning to the aorist ὤρορε, we now see that the apparent synchronic similarities between ἦραρε and ὤρορε mask distinct formations in the parent language, where the latter formation certainly began with a laryngeal. Again the same problems raised above apply here (§6.1). Assuming a proto-form ***h₃er-***, only an intensive ***h₃ér-h₃r-e/o-*** would give the required ὤρορε in Greek. In fact, attempts have been made to equate ὤρορε with the Sanskrit intensive *alarti* (Brugmann, *Kurze* 500; Schwyzer, *Gr.Gr.* 749). They are not convincing. First, the same phonological problems concerning reduplication with a laryngeal arise here: ***h₃re-h₃r-*** would presumably give ****lār-***, while ***h₃er-h₃r-*** would give ****alir-***. Second, *alarti* is athematic in contrast to the thematic ὤρορε. Third, the use of Sanskrit *alarti* is clearly intransitive/non-causative: *álarṣi yudhma bhajakṛt purandara* 'Regst du dich, Kämpfer, Streiterreger, Burgenbrecher?' [*RV* VIII 1.7]; *...alarti dákṣa utá manyur indo...* 'Es regt sich der gute Wille und der Eifer, o saft' [*RV* VIII 48.8]. Since neither the meaning nor the formal characteristics coincide, there is no reason to perceive ὤρορε and *alarti* as descending from the same proto-form.

With clear evidence for both a root and *s*-aorist, the reduplicated ὤρορε is redundant in the proto-paradigm. This fact, combined with the phonological difficulty, makes it likely that ὤρορε is a late analogical creation. The source for the analogical ὤρορε poses no real problem, particularly with the formal agreement between the two perfect stems

---

formation with the present ὀρίνω. The major problem with this analysis is how to explain ὀρίνω for an expected *ορένω. A possibility not discussed by Rix might see this form as another example of Strunk's zero-grade *CRh₁* sequence (see *Glotta* 47 [1970] 109 f.; also B. Vine, *HS* 106 [1993] 49 f.)

[17] Assuming a stative use here, a connection between *ar-* (act.) 'come' and *ar-* (mid.) 'stand (< is arisen)' is possible.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

ἀρήρῃ and ὄρωρε. Thus the aorist ἦραρε served as a model for the creation of the transitive ὤρορε (already in Buttmann, *Aus.Gr.Spr.* 259): *ἀρᾱρε (intrans. pft.) : ἦραρε (trans.aor.) :: ὤρωρε (intrans.pft.) : X, where X = ὤρορε (trans.aor.).

§6.3 ἤκαχε '(it) grieved (them)'

The active aorist ἤκαχε occurs once in the *Iliad*, twice in the *Odyssey*: μέγα δ' ἤκαχε λαὸν Ἀχαιῶν '(it) greatly grieved the host of the Achaeans' [Π 822, cf. ο 357, π 427]. The middle forms are more common, although the indicative occurs only once μνηστῆρες δ' ἀκάχοντο 'the suitors grieved' [π 342]. The remaining middle reduplicated forms are all in the optative οὐδ' ἀκάχοιτο# 'nor did he grieve' [Ν 344, cf. Π 16, Θ 207, α 236]. A single *s*-aorist is also found ἀκαχήσε [Ψ 223 only].

Outside the aorist, one finds a perfect, ἀκάχημαι (always middle), and a reduplicated present ἀκαχίζεις : ἀκαχίζεο. To these forms we should add the non-reduplicated present ἄχομαι, the participle ἀχεύων (or ἀχέFων —see below), and the nasal present ἄχνυμαι. The following forms then occur

	red. mid.	*red. act.*	*non-reduplicated*
prs.	ἀκαχίζεο	ἀκαχίζεις	ἄχνυμαι, ἄχομαι, ἀχεύων
aor.	ἀκάχοντο	ἤκαχε, ἀκαχήσε	
pft	ἀκάχημαι		

A few other forms exist in later Greek but none of these is likely to be old (Frisk, *GEW* I 202).

Both the distribution of forms [Z 486, λ 486, π 432 only], and the general status of the ιζω-present as a deverbative formation suggest that the reduplicated present, ἀκαχίζεο : ἀκαχίζεις, is a late addition to the paradigm built to the aorist stem. The *s*- aorist ἀκαχήσε probably owes its creation to the same era where the language of

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

Homer shows a large number of ιζω-presents paired with presents in -εω, e.g., αἰνίζομαι : αἰνέομεν (ἤνησ[ε]); αἰτίζεις : αἴτει (αἰτήσασα); -καλίζεο : καλέω (καλέσσατο); κομίζω : κομέουσι (prs. only); χατίζει : χατέουσι (prs. only). The same pairing is also found between *ē*-aorists and -ιζω verbs with no present attested, κανάχιζε : κανάχησε; κονάβιζε : κονάβησε : μοχθίζοντα : μοχθήσειν (Risch, *Wortbildung* 299). The lack of present forms may well be an accident of attestation, but it is worth noting that while the former class of ιζω-presents are mostly deverbative, the latter forms are denominative built to καναχή 'sound'; κόναβος 'noise' and μόγος 'toil' (or μόχθος in Hesiod).

Strunk, *Nasalpräsentien* 105 f., takes ἀχεύων as a relic aorist formation reflecting an archaic present : aorist contrast, ἀχνυ- : ἀχευ-, parallel to κινυ- : σευ- (but see Beekes, *Development* 279). The status of the thematic ἄχομαι [σ 256, τ 129] is less clear. Although it is possible that ἄχομαι is a relic of the same polymorphic pattern set out above (nasal prs. act. : thematic mid., cf. κεύθω §3.1.1 or πεύθομαι §3.1.2), the attestation in the latter half of the *Odyssey* may well indicate a late form (Chantraine, *DELG* 151). These considerations leave us with a core of four forms: a nasal present ἄχνυμαι, an aorist (?) ἀχεύων, a perfect[18] ἀκάχημαι, and finally the aorist ἤκαχε.

In addition to the verbal forms above, Greek shows a neuter *s*-stem ἄχος 'grief, pain, sorrow' (≅ ἄχθος) which forms a word equation with the Germanic **ag-iz*, OE *ege* 'Furcht', Gothic (remade to *a*-stem) *agis* 'id.' (Pokorny, *IEW* 7 f.; Frisk, *GEW* I 202 f.; Chantraine, *DELG* 151; Seebold, *EtWb* 362 with other Germanic forms; Lehmann-Feist, *GED* 10). Germanic also shows a pretero-present **ōg-* in the Gothic 2sg. in *ni ogs þus* 'fear not', probably a thematic subjunctive **āgh-e-s(i)* (W. Cowgill, *Language* 3 [1959] 10 f.; A. Bammesberger, *FsRisch* 673 f.). In addition to the Greek and Germanic forms,

---

[18] For the perfect forms ἀκαχέδαται [P 637] < *ἀκαχήδαται see Beekes, *Development* 119; also Chantraine, *Gr.Hom.* 435.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

only Old Irish attests a verbal form to this root in *ad-ágathar* 'fears'. This Irish form, although showing some traces of *a*-inflexion, is shown to be in origin an *i*-present by such forms as *'áigder* (Thurneysen, *GOI* 358). This verb is best analyzed as *āg-i̯o/ī*, pointing to a formation identical to that of the Gothic *ōg-* (Watkins, *Celtic Verb* 122). Hence, both the Germanic and Celtic evidence points to a perfect **āg-*, which assuming a laryngeal root (Beekes, *Development* 113) implies a preform **h₂e-h₂gh-* to **h₂égh-* 'suffer in spirit' (Pokorny, *loc.cit.*).

Although the comparative evidence is scanty, a root showing an *s*-stem neuter, an enlarged root in *-dh-*,[19] the *u*-extension in the verbal forms and the enlarged stative formation (reflected in the perfect) imply a Caland system root in the proto-language. It is likely that a stative present/aorist form **agh-eh₁-* (or **h₂(e)gh-eh₁-*) served as the basis for a new perfect ἀκάχημαι, remade with Attic reduplication according to the standard Greek pattern, e.g., ὄρ- ωρε < *ωρε = Skt. *ā́ra* both from **h₃e-h₃r-* (Sihler, *NCGGL* 489). Although no firm conclusion regarding the reduplicated ἤκαχε is possible, this form was probably created analogically on the model of ἤραρε, even though formally *ἄρᾱρε : ἤραρε does not exactly parallel ἀκάχημαι : X, X = ἤκαχε.

Further evidence for a late formation here is suggested by the preponderance of middle optative forms in a metrical pattern ⏑⏑ — ⏑ (especially with the third singular -οιτο), a feature of many of the secondary reduplicated aorists described above (πεπύθοιτο, πεπίθοιτο, πεφίδοιτο, κεχάροιτο, λελάθοιτο). That is to say, as *CV-* and VC- reduplication became increasingly associated with one another, influence from the CV-reduplicated forms became increasingly possible. In this way, the creation of ἀκάχ- οιτο from ἀκάχημαι (itself secondary to an original *ἄχη- μαι) may have received further initiative from the consonant initial reduplicated forms of epic diction.

---

[19] For the Caland implications of such forms as ἄχος : ἄχθος, or μέγας : μέγεθος see G. Pinault, *MSS* 38 (1979) 165 f.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

§6.4 ἤπαφε 'deceived'

This reduplicated aorist usually occurs with a preposition, either παρά- or ἐξ-. There are three examples in the active indicative: παρά μ' ἤπαφε δαίμων# 'some god has deceived me' [ξ 488]; ἐξ οὗ δή μ' Αἰτωλὸς ἀνὴρ ἐξήπαφε μύθῳ# 'from that time that one of Aitolios deceived me in word' [ξ 379]; #Ἥρη δ' ἐν φιλότητι παρήπαφεν εὐνηθῆναι# 'Hera deceived (him) to lie in love' [Ξ 360]. A first person subjunctive also occurs: αἴ κέν σ' ἐξαπάφω, κτεῖναί μ' οἰκτίστῳ ὀλέθρῳ 'if I deceive you, then kill me with pitiless death' [ψ 79]. Finally, there are three occurrences in optative middle, all with the -οιτο ending typically associated with the reduplicated aorist: μή τίς με βροτῶν ἀπάφοιτο ἔπεσσιν# 'lest some of men deceive me with words' [ψ 216]; ὅππως ἐξαπάφοιτο Διὸς νόον αἰγιόχοιο·# '...how she might deceive the mind of aegis bearing Zeus' [Ξ 160]; οὐδ' ἂν ἔτ' αὖτις / ἐξαπάφοιτ' ἐπέεσσιν· 'never again will he deceive me with words' [I 375-76]. In addition to this aorist only the present ἀπαφίσκει occurs in Homer (once in Hesiod). There is also an aorist ἐξαπάφησα at *Hymn to Apollo* 376. Attestations in later Greek appear to be Homeric (and Hesiodic) imitations (cf. Veitch *GV* 76).

The verb lacks a clear etymology (Frisk, *GEW* I 119; Chantraine, *DELG* 96). Chantraine, *loc cit.*, refers to the noun ἀποφώλιος 'idle, fruitless (?)' (4X *Odyssey*), although the details of this equation do not seem to have been worked out. More recently, G. Dunkel, KZ 95 (1981) 228 f., uses this verb as an example of the so-called *āmreḍita* preverb, hence *ἀπ- απ-ἰσχω 'keep off repeatedly' with aspiration metathesis. Dunkel bases his interpretation on the phrase, οὔ τί σε Περσεφόνεια Διὸς θυγάτηρ ἀπαφίσκει# 'in no way does Persephone, daughter of Zeus, deceive you' [λ 217-18]. Here, he argues that the diction supports a reading of 'keeps off repeatedly' (Dunkel does give other examples but none is persuasive). At any rate this is mere speculation. The reading 'deceive' works well here. Moreover, if this meaning is derived from an earlier

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

'keeps off repeatedly', then this must be the oldest form with ἧπαφε derived form this present. Unfortunately, the distributional evidence, not only from ἧπαφε (3X *Iliad*) : ἀπαφίσκει (1X *Odyssey*), but of the other Homeric verbs in -ίσκω clearly shows that these present in -ίσκω are late additions to their paradigms with thematic aorists functioning as the 'foundation forms' (cf. B. Vine, *HS* 106 [1993] 53 n.22 & below §6.7).

Vine, *op.cit.* 58 f., noting the aorist ἐξαπάφησα and the Hesychian gloss ἀποφεῖν· ἀπατῆσαι, argues that this form may originally have been a compound *απ-οφέω (or perhaps *απο-φέω, *α-ποφέω?), but that this aorist was reinterpreted as a reduplicated form and ἀπαφίσκω was then backformed on the model of ἧραρε : ἀραρίσκω.[20] There is no evidence that either the present or aorist are archaic. The paucity of extra-Greek cognates, as well as the isolation of the root in Greek speak to an innovation, as perhaps also does the **a*-vocalism. Once again, no absolute conclusion is possible here. Nevertheless, although it is impossible to state what analogical steps led up to the creation of ἀπαφ- ε/ο-, the formal agreement between this form and ἀκαχ- ε/ο-, imply some similar origin. Further the occurrence of both these reduplicated forms with the optative ending -οιτο may further indicate an analogical source.

§6.5 ἤγαγον 'I led'

Like εἶπον, ἤγαγον is extremely common in Homer (48X *Iliad*, 56X *Odyssey*), e.g., #τοῦτον δ' ἄλλον ἐγὼ νέον ἤγαγον ἐκ πολέμοιο# 'just now I led the other from battle' [Λ 663]. Also like εἶπον, it survives into the Classical period, although it is relatively rare in prose (Veitch, *GV* 12 f.). Finally, like εἶπον, the dialectal evidence assures that this aorist dates at least as far back as common Greek: cf. Cretan αγαγε

---

[20] Puhvel, *HED* III 118 f., offers a similar chain of events, ἀπαφ- < *ἀπ(ο)- αφ-, but from the Indo-European root ***h₂ebh-* seen in the Hittite *hapanzuwai-* 'be trusted, be dependable', "...perhaps ἀπαφίσκω, ἀπαφεῖν 'deceive' contains the reversive ἀπο- (like e.g. ἀπο- κοσμέω 'dis-array')...".

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

(*Gortyn*) and Phocrian εαγαγον (Buck, *Greek Dialects* 315 & 238 respectively [although the latter form is not certain]). Unlike εἶπον, however, this form does not complete its paradigm through suppletion. The following forms of ἄγω occur in Homer.

	act.	*mid.*
prs.	ἄγω	ἄγεται
aor.	ἤγαγον, ἄξετε	ἠγαγόμην, ἄξοντο
fut.	ἄξω	

The thematic present to this root is certain: ἄγω (Myc. *a-ke*) = Latin *agō* = Skt. *ájati* = Av. *azaiti* = ON *aka* = Arm. *acem* = O.Ir. *·aig* = TA *āk-*, TB *āk-* all from ***h*$_2$*ég-e/o-* (Pokorny, *IEW* 4 f.; Frisk, *GEW* I 18; Chantraine, *DELG* 17 f.; Seebold, *EtWb* 74 f.; Kellens, *Verbe avestique* 102; Mayrhofer, *EWAia* I 50; Aura Jorro, *DMic* 35 f.).

In contrast to the present, however, evidence for any aorist stem is sparse. Thus, Sanskrit supplies an aorist through suppletion. Likewise the Tocharian preterite to *āk-* is TA *wānt* (3pl.), TB *wāya*. The Armenian *aci* might argue for a root formation, but this form could just as easily be the result of analogy, e.g., *berem* : *beri* 'bear' :: *acem* : X. Further, we again face the phonological problem described above where a laryngeal root ***h*$_2$*ge-h*$_2$*g-e/o-* should yield **αγηγε, and not the short vowel form ἤγαγον. As above we could posit some kind of intensive reduplication, i.e. ***h*$_2$*eg-h*$_2$*g-e/o-*, but an analogical solution is preferable. In this case there is circumstantial evidence for just such an analogical formation. There is no perfect to this verb in Homer,[21] but later Greek supplies a perfect ἀγάγοχα on the model of ἐνήνοχα. Since the perfect of ἄγω has been built in analogy to the older perfect of ἐνεγκεῖν (§5.1), it is likely that the aorist has also been influenced by the same root (cf. F. Specht, *KZ* 63 [1936] 225).

* * *

[21] A relic of perfect may exist in the Greek noun ἄγυια 'way, path' < *'driven' (Chantraine, *Parfait* 45; *DELG* 15 f.).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

§6.6 *Excursus*: Greek presents in -ίσκω

Of our five *aC-aC-e/o-* (and *oC-oC-e/o-*) aorists, two, ἦραρε and ἧπαφε, build presents in -ίσκω. The origins of this formation remain a mystery, although B. Vine, *HS* 106 (1993) 49 f., has recently given these forms a thourough treatment. Vine notes, *op.cit* 53 n. 22, that only four such present are found in Homer[22]

ἧπαφε (7X) : ἀπαφίσκει (1X *Od.*) 'deceive'
ἄραρον (10X) : ἀράρισκε (1X *Od.*) 'fit'
εὗρον (90X) : εὑρίσκω (1X *Od.*) 'find'
ἔπαυρε (10X) : ἐπαυρίσκονται (1X *Il.*) 'reach'

Since, in addition to ἦραρε and ἧπαφε, εὗρον also represents a reduplicated formation at the oldest level, a few words regarding this formation are here in order.

Vine, *op.cit.* 54, notes that three of these forms end in a final -ρ-, and that in later Greek verbs in -ίσκω show a "similar predilection for liquids". From this he offers a phonological source such that *CR̥h₁-ské/o-* would yield *CaRiské/o-* directly with a disyllablic outcome of such forms as δολιχός = Skt. *dīrghá-* (both from **dl̥h₁gh-*?).[23] There are, however, problems with this analysis. To begin with, evidence for disyllabic laryngeal developments is scanty at best (see most recently, Sihler, *NCGGL* 104). More importantly, the chronology is difficult to defend since the textual evidence suggests that these presents were created during, or not long before, the Homeric period, long after the period of laryngeal developments. Finally, as I have argued above (§2.1 f.), there is strong evidence to suggest that a laryngeal would be lost in a reduplicated formation. Even if

---

[22] For arguments against taking Mycenaen *te-ra-pi-ke* as a verb in ίσκω see Vine, *op.cit.* 56 f.

[23] For this development see K. Strunk, *Glotta* 47 (1970) 1 f. For additional bibliography see Vine, *op.cit.* 55 ns. 25, 26, and 27.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

εὗρον is built to a *set* root **u̯reh₁-*, the reduplicated form reflected in the aorist was *aniṭ* already in Indo-European (**u̯e-u̯rh₁-e/o-* > **u̯e-u̯r-e/o-* > εὑρ- ε/ο- = O.Ir. *·fuar*). Hence, of our four oldest forms only the simple thematic ἔπαυρε would have any chance of giving the correct environment. To be sure, it remains possible that ἔπαυρε (or some other unattested form) served as the 'foundation form' for the other ἰσκω-presents, and that the thematic nature of ἔπαυρε (or another form) was the basis for newer ἰσκω-presents. Nevertheless, although any account must consider the correlation of ἰσκω-presents to thematic aorists, the fact that three of our four oldest examples are reduplicated also deserves consideration.

§6.6.1 ἀράρισκε

As with all four -ἰσκω presents, ἀράρισκε occurs only once, in the second half of the *Odyssey*: αὐτὸς δ' ἀμφὶ πόδεσσιν ἑοῖς ἀράρισκε πέδιλα 'he himself was fitting shoes upon his feet' [ξ 23]. Unlike ἀπαφίσκει, εὑρίσκω and ἐπαυρίσκονται, however, a possible word equation exists for ἀράρισκε in the Armenian subjunctive *arar‘ic‘* (Godel, *Introduction* 117; E. Hamp, *KZ* 89 [1975] 100 f.; Schmitt, *Gr.Kl.Arm.* 153). This 'equation' is formally impeccable,[24] and the semantic change of an old present to newer subjunctive is possible (cf. the Tocharian subjunctive *arāntär* above §6.2).

However, such an 'equation' is little more than a desperate attempt to explain the origins of the Armenian subjunctive morpheme *-ic‘-*. All Armenian weak verbs built their aorists in *-Vc‘*, e.g., *sirem* 'I love' : (aor.) *sirec‘i*, *bnakem* 'I dwell' : *bnakec‘i*, presumably based on such forms as *harc‘anem* : *harc‘(e)-* where the present has been secondarily en-

---

[24] Indo-European **-ske/o-* yields Armenian *-c‘e-* as in **prḱ-ske/o-* > **hartsʰe-* > *harc‘anem* 'I ask'; *h₂(e)iske/o-* > *hayc‘e-* 'suchen' (cf. Klingenschmitt, *Verbum* 60 f.; Godel, *Introduction* 113 f.). The alternate present formant *-č‘i-* is also a descendant of **-ske/o-*; however, this is not the place for further discussion. For the phonological problems see Klingenschitt, *Verbum* 67 f. & esp. 84, also *MSS* 28 (1970) 75 f.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

larged with the productive nasal present. These weak verbs contrast with strong aorists which lack the suffix -Vc'-, e.g., *berem* 'I carry' : *beri*. However, all subjunctives[25] in Armenian contain the morpheme *-ic'-*. Thus, the subjunctive to *sirem* : *sirec'i* is *sir-ic'-em* : (aor.sub.) *sir-ec'-ic'*, while the aorist subjunctive of the strong verbs are such forms as *beric'*, *harc'ic'*.

Viewing the aorist and subjunctive forms together we see the problem with the attempt to date **-iśke/o-* to the Indo-European era. The standard explanation for the aorist marker *-c'-* in Armenian derives this formative from the iterative **-śke/o-* (the Greek φιλέσκον type) in late Indo-European (Schmitt, *Gr.Kl.Arm.* 134; Klingenschmitt, *Verbum* 286). But projecting both **-iśke/o-* and **-śke/o-* to the parent language does not explain how **-iśke/o-* ends up as a model for the subjunctive, while the simple **-śke/o-* ends up as a subjunctive marker. Stated another way, the formally significant element of the morpheme *-ic'-* would seem to be *-i-*, not *-c'-*. This being the case, it is reasonable to suppose that *-ic'-* is, in origin, a collocation of two separate parts. It is further reasonable to suppose that this compound morpheme is derived from an earlier **-ih$_1$-śke/o-* containg the optative suffix before **-śke/o-*, just as H. Pedersen, *KZ* 38 (1905) 207, explains it (cf. also Meillet, *Esquisse* 122; J. Jasanoff, *KZ* 93 [1979] 135 n. 5 "[i]n my view, however, Godel's rejection [of the reconstruction **-ih$_1$śke/o-*]... is premature").

But even granting the functional development of an Indo-European **-iskō*, there remain certain formal problems the most telling of which is the scanty evidence for **-iskō* in Indo-European. Godel's appeal, *op.cit.* 118, to such preforms as **bheriskō* is unlikely. As Vine has shown in great detail, *op.cit.* 49 f., most of the evidence in favor of an Indo-European **-iskō* is illusory. Now the argument can still be saved if we accept the possibility that **arariskō* was a rare formation in Indo-European but survived in both Greek and pre-Armenian. This argument, however, requires that the displaced **arariskō* is now

---

[25] The status of the second plural, which lacks *-ic'-*, need not concern us here.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

the source for the entire Armenian aorist subjunctive system, with the present subjunctive built to the older aorist subjunctive. But this theory is also unlikely. First it has the odd characteristic of separating the origins of the subjunctive (< **ararisko̅*) from the aorist (< 'iterative' **-ske/o-*). Further, such forms as the third singular present of the aorist subjunctive, *siresc'ē* < **sirec'-c'-ē*, seem to have been created in analogy to the present, *sir-ē* < **-e-y* < **-eti*, not the other way around.

All in all, I see no way to salvage this argument. As noted above, there is simply no evidence outside of Greek and Armenian that **ar-* 'fit' built any present in Indo-European (cf. Pokorny, *IEW* 55 f.). The comparative evidence supports only the irregular aorist **arar-*, but this form is odd enough without imagining a hypothetical **ararīskō*. Hence, Greek ἀράρισκε is a late development with no relation to the Armenian *araric'*.

§6.6.2 εὑρίσκω

Like ἀράρισκε, εὑρίσκω is also attested in the latter poriton of the *Odyssey*: νῦν δ' οὔτ' ἐκφυγέειν δύναμαι γάμον οὔτε τιν' ἄλλην / μῆτιν ἔθ' εὑρίσκω 'now I am unable to flee marriage, nor find some other counsel' [τ 157 f.]. As noted above (§2.1.1), although there is strong evidence for a *seṭ* root here, the aorist is *aniṭ* and no purely phonological explanation for the present is likely. Not only is there no evidence that this present is old, there is no evidence for any present to this root. The O.Ir. *'fuar* is in a suppletive relationship with the present *fo'gaib* 'finds'. A nasal present is attested in Balto-Slavic, Lit. *randù* 'ich finde' < **u̯rh$_1$-n-dh-* (Pokorny, *IEW* 1160), but this form is almost certainly an innovation.

§6.6.3 ἐπαυρίσκοντ[αι]

Of the four -ίσκω verbs, only ἐπαυρίσκοντ[αι] occurs in the *Iliad*, in the famous maxim at N 730 f.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

ἄλλῳ μὲν γὰρ δῶκε θεὸς πολεμήϊα ἔργα
ἄλλῳ δ' ὀρχηστύν, ἑτέρῳ κίθαριν καὶ ἀοιδήν
ἄλλῳ δ' ἐν στήθεσσι τιθεῖ νόον εὐρύοπα Ζεὺς
ἐσθλόν, τοῦ δέ τε πολλοὶ ἐπαυρίσκοντ' ἄνθρωποι

to one man god has given warlike deeds
to one the art of dance, to another the *kithar* and song
to one, into his breast, wide seeing Zeus has set useful *noos*
of which many men profit

Although this is the only Iliadic instance of an -ίσκω verb, we cannot simply assume that this is the most archaic form. In his book on Homeric language, Shipp, *Studies passim*, has argued that similes, theological maxims and other digressions are late additions to the poem. Although his work has not met with universal approval, we must be somewhat circumspect in dating this passage.[26]

As with the other ἰσκω- verbs, the aorist ἐπαυρεῖν is much more common (7X *Iliad*, 2X *Odyssey*), although often with a slightly different usage μή τις χρόα χαλκῷ ἐπαύρῃ 'lest someone touch his flesh with bronze' [N 649, cf. Λ 573 +]. A future ἐπαυρήσεσθαι 'will profit' [Z 353] also occurs. Both the distribution of forms and the semantics, where a change of 'touch' to 'profit' is more likely, indicate the aorist as the 'foundation form'.

The aorist ἐπαυρεῖν 'touch' also lacks any clear etymology, although a connection with εὑρίσκω has been suggested by a number of scholars (e.g., Schywzer, *Gr.Gr.* 709 n. 3; Chantraine *DELG* 99 & 356; Peters, *Untersuchungen* 22 f). Peters, *loc.cit.*, has explored this relationship between εὑρίσκω and -αυρίσκονται at length and offers a tentative solution with the preform *$\underset{\sim}{u}erh_1$- as the root of both εὗρον and -αυρον. Both

---

[26] Shipp's main criterion, *Studies* 87, for dating this passage is the verb ἐπαυρίσκονται itself. For additional comments on the diction of the surrounding passage see *Studies* 283. For a more recent analysis see Lardinois, *Diss.* 53 f., who argues that the gnomic material is often archaic, but also notes that definite conclusions are not possible.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

forms are the result of metatheses: 1) zero-grade *$\underset{\smile}{u}\underset{\bullet}{r}h_1$-e/o- > *$\underset{\smile}{u}ar$-e/o- > *$aur$-e/o- (where $\underset{\smile}{u}ar$- > $au\underset{\smile}{r}$); 2) full grade *$\underset{\smile}{u}erh_1$-t > *$\underset{\smile}{u}ere$-t > *eure-t (where $\underset{\smile}{u}eC$ > $eu\underset{\smile}{C}$). A third possibility is given later, *op.cit.* 31, where εὗρον is argued to be the result of a laryngeal metathesis *$\underset{\smile}{u}\underset{\bullet}{r}h_1$- > *$h_1\underset{\smile}{u}r$- with -αυρον < *$h_2\underset{\smile}{u}er$-, but as Peters himself notes this invalidates the connection with εὑρίσκω .

I have given my own analysis of εὗρον above (§2.1.1). Although a derivation of ἐπ-αυρ-ε/ο- from *$\underset{\smile}{u}\underset{\bullet}{r}h_1$- is in principle possible, the root meaning 'touch' makes a derivation from 'find' unlikely. Alternatively, some scholars take -αυρεῖν 'touch' as a derivative of ἁπηύρα, -ούρας 'take away'. The semantic change is notable, and there is little value in the comparison excepting that both verbs are always prefixed (cf. T.A. Sinclair, *Classical Review* 39 [1925] 99 f. for bib.). Now Frisk, *GEW* I 125, takes εὑρίσκω and ἁπηύρα, ἀπούρας together while excluding ἐπ-αυρίσκονται. The participle ἀπούρας is apparently *απο-Fρᾱ-, while the indicative form must come from απο-ε̄-Fρᾱ-, where the digamma induces both lengthening and vocalization (cf. J. Wackernagel, *Die epische Zerdehnung* 270 = *Kl.Schr.* III 1523).[27] The internal problems of this form make any comparison uninstructive.

Finally, the extra-Greek cognates are also unhelpful. Armenian *gerem* 'take' < *(h_x)$\underset{\smile}{u}er(h_x)$-$e\underset{\smile}{i}e/o$- is a fair match for ἀπούρας, ἁπηύρα < *$h_x\underset{\smile}{u}erh_2$-. Further, it is not unthinkable that these forms should be connected with ἐπ-αυρ-. However, the further connection with εὑρίσκω seems to me unlikely, both on semantic and phonological grounds (for my phonological analysis of εὑρίσκω : εὗρον see above). Hence, as the etymology of ἐπαυρίσκομαι is a dead end.

---

[27] Perhaps we are dealing with a root *απ(ο)-εFFρ-α- , *$ap(o)$-e-$h_x\underset{\smile}{u}r$-h_x-, with a treatment similar to Lindeman's Vh_xiV > $V\underset{\frown}{i}\underset{\frown}{i}V$ (*Introduction* 68 —for evidence for this treatment with $\underset{\smile}{u}$ see 69 f.).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

§6.6.4 ἀπαφίσκει

This last - ῐ́σκω present to be discussed occurs in the Nekuia section of the *Odyssey*: οὔ τί σε Περσεφόνεια Διὸς θυγάτηρ ἀπαφίσκει# 'in no way does Persephone, daughter of Zeus, deceive you' [λ 217-18]. The verb is without etymology (§6.4). Hence, not only is there no evidence for a **-ske/o-* present which could serve as foundation form for the other Greek verbs in - ίσκω, there is no evidence these four roots had *any* present formation in Indo-European. It is, in fact, this very lack of a present formation which makes an analogical solution to the origin of the Greek verbs in - ίσκω all the more likely. Contra Vine, *op.cit.* 54, I find such reanalysis as ἐΐσκω < *Fε- Fικ- σκ- fairly appealing, although I doubt that ἐΐσκω was the pivot form.

My own belief is that ἀπαφίσκει should probably be taken as the starting point in the history of these forms. Now this solution would require that this is the oldest of the four ίσκω-verbs, but since our four passages are more or less contemporary, this is not such a great leap. Further, if one believes that some degree of analytic divisions can be made in Odyssey, the first-person narration of Odysseus is more likely to reflect an older strata of diction than any other discrete section (A. Heubeck, *Comm.Od.* I 5 f.; II 3 f.; for the antiquity of the Nekuia passage see esp. Frame, *Myth of Return* 34 f.).

I want to begin by pointing out certain parallelisms between the verbal paradigms ἧκαχε and ἧπαφε.

prs.	ἀκαχίζεο	ἀπαφίσκεις
aor.	ἧκαχε	ἧπαφε
s-aor.	ἀκάχησε	- απάφησα*

This parallism was noted long ago by W. Schulze, *KZ* 43 (1910) 185, who argued that we expect a present *ἀκαχίσκω (cf. also Risch, *Wortbildung* 299). However, rather than hypothesize an unattested *ἀκαχίσκω, if we want to connect these two paradigms, we are better off to posit *ἀπαφίζω, which was later replaced by ἀπαφίσκεις. First, the

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

origin of the present ἀκαχίζεο is clear. The present forms in -ίζω have been generalized from older stems in -ιδ plus -*ie/o*, e.g., ἐλπίς (-ιδ) -> ἐλπίζω (Chantraine, *Morph.hist.* 235). After its creation the morpheme -ίζω became productive, yielding both denominative and deverbal present forms. As noted above, the suffix -ίζω was especially used in early Greek in a polymorphic opposition to *e*-contract presents or in some cases directly to an *ē*-aorist. It is this productive pattern which not only guarantees that ἀκαχίζεο is not a recent reformulation of an older *ἀκαχίσκω, but also makes *ἀπαφίζω likely in contrast to the aorist -απάφησα.[28]

Since ἤραρε lacked a present formation, the parallel structure of ἤραρε to ἤπαφε allowed for the analogic creation of ἀραρίσκω. At this point however the sequence -ρ-ίσκω was exploitable for the creation of newer present forms such as ἐπαυρίσκονται and εὑρίσκω (which was probably no longer recognizable as a reduplicated form).

---

[28] The reason for the replacement of older -(ι)ζω with the newer -(ι)σκω can only be guessed at. G. Dunkel, *KZ* 95 (1981) 228, may be correct in seeing an iterative formation in ἀπαφῖσκε/ο- [λ 217]; although not because of preverb reduplication (see above §6.4). Perhaps *ἀπαφ-ιζε/ο- was remade to ἀπαφ-ισκε/ο- on the influence of the iterative imperfect γενέσκετο in the same passage (although this is some lines above). Note such pairs as ῥιπτά-ζων [Ξ 257] : ῥίπτα-σκον [O 23]; however, this approach fails to explain why a present backformed to an older iterative imperfect was preferred over the regular *ἀπαφ-ιζε/ο-. Alternatively, note that all four -ίσκω verbs are transitive in Homer. Perhaps the reduplicated transitive forms had some influence here, cf. e.g., διδάσκω, βιβάσκω, γιγνώσκω (Schwyzer, *Gr.Gr.* 710).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

CHAPTER VII: Conclusions

§7.1 *The Reduplicated Aorist in Greek*

This dissertation has shown that Greek inherited a small number of reduplicated aorists: 1) εἶπον 'I spoke' (= Skt. *ávoc-a-* 'id.') < **u̯e-u̯ĺ̥ᵘ-e/o-* (§1.1); 2) εὗρον 'I found' (= O.Ir. *·fuar* 'id.') < **u̯e-u̯r(h₁)-e/o-* (§2.1.1); 3) ἐνεγκεῖν 'carry' (= Av. *nąsaṭ* 'reaches, presents'[?] ≅ Italic **nank-*) < **hₓne-hₓnḱ-e/o-* (§5.1); 4) ἄραρε 'fit' (= Arm. *arari* 'I made') < **ar-ar-e/o-* (§6.1). Two other forms should probably be added to the list of archaic reduplicated aorists, although the comparative evidence for these additional examples is much weaker: 5) ἔπεφνε 'killed' (=Av. *ava-jaγnaṭ* 'struck' [?]) < **gᵘhe-gᵘhn-e/o-* (§1.2); 6) κέκλετο 'called' < **ke-kl(h₁)-e/o-* (§2.1.2).

In addition to knowing that a small number of forms are inherited, we can also be certain that some forms are innovations of the Greek era. This is clearest for those forms whose formal appearance is incompatible with the regular phonological developments of the Greek language: 1) λελαβέσθαι 'grasp' (not *ἐλᾱβ- ε/ο- < **se-slăgᵘ-e/o-* [§4.1.2]); 2) κεχάροντο 'they delighted' (not *κέχροντο < **ǵhe-ǵhr-e/o-* [§4.1.3]); 3) τετύκοντο 'they prepared' (not *τετύχοντο < **dhe-dhugh-e/o-* [§3.1.3]); 4) ὦρορε 'inspired' (not *ὤρωρε [as aorist] < **h₃re-h₃r-e/o-* [§6.2]); 5) ἤγαγον 'led' (not *ἤγηγον < **h₂ǵe-h₂ǵ-e/o-* [§6.5]); and perhaps 6) δέδαε 'taught' (not *δέδασε < **de-dn̥s-e/o-* [§3.4.1]).

Unfortunately, the etymological analysis above is helpful for less than half of the attested reduplicated aorists in Greek. We can, of course, further separate out those forms which are demonstrably late: δέδακε (§4.1.6); παφών (§1.2 [n. 10]); πεπορεῖν (§2.2.1); τεθιγεν (§3.2.3); τεθορεῖν (§2.2); τετορεῖν (§2.2); τετύποντες (§3.1.4). A few other forms can also be set aside.[1]

---

[1] Most obvious are those forms which do not exist, e.g., *δεδεχθαι (§0.1 n. 1); those forms which are not reduplicated, ἔγρετο (§0.1 n. 1); ἔσπετε (§1.3.1), ἔσπετο

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

This still leaves about twenty forms without explanation. Nevertheless, although etymological analysis is unhelpful here, the metrical distribution of some of these forms suggests a possible source for these more ambiguous cases. Thus, as noted in the proceeding pages, a number of reduplicated aorists occur only (or almost only) in modal forms. These modal reduplicated aorists typically correspond to non-reduplicated forms in the indicative. Note the following examples (forms are listed with the ratio of indicative to non-indicative forms, the corresponding non-reduplicated aorist and the location of the discussion of the forms above).[2]

reduplicated aorist	ratio of non-modal : modal forms	corresponding simple thematic aorist	section where discussed
κέκυθωσι	0:1	κυθ- ε/ο-	§3.1.2
-λέλαθον	2:7	λαθ- ε/ο-	§4.1.4
λελάχωσι	0:4	λαχ- ε/ο-	§4.1.5
πεπιθεῖν	0:12	πιθ- ε/ο-	§3.2.1
πεπύθοιτο	0:3	πυθ- ε/ο-	§3.1.1
τετάρπετο	2:8	ταρπ- ε/ο-	§3.3.1

Further, this distribution is not typical of those reduplicated aorists which lack an attested simple thematic form (e.g., ἔτετμεν [7:1]; ἐπέφραδον [20:5]).[3]

The evidence of the six forms above can be strengthened with the addition of λελαβέσθαι, which has been clearly built to the older λαβ- ε/ο-, and κεχάροντο whose

---

(§1.3.2), ἐρύκακε (§5.4 (n. 21), and those forms which are synchronic perfects, πέπληγον (§3.5.1), λελάκοντο (§4.1.1), and also τεταγών (§4.2.1) which, if not synchronically perfect, is a perfect in origin.

[2] This short list could be expanded with the inclusion of μεμαποεῖν : μαπ- ε/ο- in the *Aspis* (§4.1.7) and πεπαγοίεν (contrasting with a root aorist) in Eupolis (§4.2.3).

[3] Πεπαλών and ἄλαλκε, and πεφιδέσθαι also show a strong preference for non-indicative forms, but as noted above, in the case of ἄλεξω non-indicative usage is a feature of this verb in all tense forms. The distribution of πεπαλών and πεφιδέσθαι is probably accidental.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

history is slightly more complex, but is also a late reduplicated form (see above). This distribution of forms does not in itself prove that these reduplicated aorists are innovations, since complementary distribution of forms in epic poetry can be the result of mergers based on the principle of economy. However, given that we know that λελαβέσθαι and κεχάροντο cannot be old, and that they have been created for metrical reasons, there is strong circumstantial evidence that our other modal reduplicated aorists were secondary developments of Greek poetic diction. Finally, once it is granted that some reduplicated aorists are the products of epic diction it become increasingly possible to view such forms as πέφραδον, or ἔτετμεν as products of the Greek era and not Indo-European archaisms, even though their origins are unknown.

Although the inherited and the innovated reduplicated aorists have different histories, the end result for both classes was the same: marginalization and elimination. The first class was once part of the spoken language which in the course of time became increasing associated with poetic diction. The latter class, however, was never part of the spoken language and existed only in the language of poetry. A few inherited forms would survive into later Greek, e.g. εἶπον and ἐνεγκεῖν, but only because they had ceased to be recognizable as reduplicated formations.

§7.2 *The Reduplicated Aorist in Indo-European*

The status of those forms which are old (εἶπον, εὗρον, ἐνεγκεῖν, ἄραρε, ἔπεφνε, κέκλετο) is unclear. To begin with, we may be dealing with two unrelated formations: 1) *C_1Re-C_1RC_2-e/o-* (where C_1 is [+ continuant]); and 2) **aC-aC-e/o-* for ἄραρε. A number of questions remain, both regarding the function of this category in the parent tongue and its position in the verbal system.

A possible paradigmatic link between reduplicated aorist and reduplicated present

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

has been suggested above. There are four examples,[4] all lacking in some respects: 1) Greek πι- πτ- ε/ο- : Skt. *paptat* (Mayrhofer, *EWAia* II 71 f., Klingenschmitt, *Verbum* 130); 2) διδάσκω : δέδαε < **di-dn̥s-e/o-* : **de-dn̥s-e/o-* (if the latter is old); 3) ἵζει : ἔζετο < **si-sd-e/o-* : **se-sd-e/o-* (if the latter is old); 4) OIr. *·icc* : ἐνεγκεῖν < **h₂i-h₂nḱ-e/o-* : **h₁ne-h₁nḱ-e/o-* (if the former is reduplicated and assuming a merger of **h₁neḱ-* and **h₂neḱ-*). None of the examples inspires confidence and so this remains merely one possibility.[5]

The function of this formation (or formations) in Indo-European is also difficult to judge. Although the numbers are small, the reduplicated aorists above show a restriction to *CeC* roots (or *CeRH* [which reduce *CR*] or *HReC* [in the case of ἐνεγκεῖν]). That is to say, roots of the shape *CeRC* are excluded, unless one of the consonants is a laryngeal. Since the two thematic aorists traceable to Indo-European are built to roots of the ex-

---

[4] A fifth possible example (Greek ἔνῑπτε ≅ (?) Skt. *vivakti* : εἶπον = Skt. *ávocat*) is based on a false etymology (see above §1.1 and §5.4).

[5] An older analysis of the Tocharian causative present (e.g., TA *śärsäṣt* [2sg.], TB *śarsäṣṣäṃ* 'teaches' from the root *kärs-* 'know') derives these forms from an older reduplicated formation **k'ə-k'ə́rsəsk-* > **śäśärsäsk-* with syncope of the first vowel (Krause-Thomas *Toch. El.* 211). Although this view has been generally discarded it would give us another example of the same contrast proposed for διδάσκω : δέδαε. However, even if we accept this analysis for the Tocharian causative present, an analogical model is again possible here. Note the contrast:

	non-Caus.	Caus.
Ps.	**kərs-nā-*	**k'ə-k'ə́rs-əsk-*
Pt.	**k'ərs-*	X, X = **k'ə-k'ə́rs-*

Since the non-causative class I preterite is made from the corresponding present by the elimination of the present suffix *-nā-* we would expect a causative preterite of the shape **k'ə-k'ə́rs-*. Since we know that the reduplicating vowel (CT **æ*) of the past participle was remade from an older **e* (above §0.1 n. 6) it is likely that the CT **ə* would likely be remade to **æ*. Further, if we suppose the replacement of **æ* happened only in TA, it becomes easier to explain the corresponding TB *cāla* type (although again some analogical restructuring is needed). Just as in the case of ἔζετο above it is impossible to say whether or not this is an Indo-European archaism, or simply a late analogical development.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

cluded *(H)CeRC* shape (**h₁ludh-e/o-* 'came' and **u̯id-e/o-* 'saw'), it is possible that these forms were in complementary distribution.[6] That is to say, perhaps reduplicated aorists were built to roots which did not easily build zero-grade thematic aorists.

It is also possible that the reduplicated aorist may have had some special semantic force in the parent language. However, as the study above has shown, aside from a tendency for active inflection, Greek offers no evidence for a causative reduplicated aorist formally equivalent to the Sanskrit *ajījanat* formation (see above §2.4.1). Only ἤραρε clearly represents some kind of derived causative where **ar-ar-e/o-* 'fit together (trns.)' is built to a root **ar-* 'fit together (intrns.)'. But an Indo-European **ar-ar-e/o-* is formally quite different from the *Ce-CC-e/o-* implied by the other Greek reduplicated aorists, and this may well imply a different verbal category. A second example of the same *VC-* reduplicated formation may underlie the Hittite causative *ašeš-* 'make sit' < *aš-* 'sit'. Nevertheless, this *VC-VC-(e/o-)* formation has left us so few clues that its status in the parent language remains a mystery.

* * *

[6] In this way the reduplicated aorist might parallel the sigmatic aorist, where roots of the form *CeRC* are a later development than those of the shape *CeC* (cf. Narten, *Sigm.Aor.* 80; Hollifield, *System* 47 f.).

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.

INDEX of Principal Greek Forms

ἀγ- αγ- ε/ο-	§6.5	λε- λαθ- ε/ο-	§4.1.4
ἀειδ- ε/ο-	§5.3	λε- λακ- ε/ο-	§4.1.1
ἀκ- αχ- ε/ο-	§6.3	λε- λαχ- ε/ο-	§4.1.5
ἀλ- αλκ- ε/ο-	§5.2	με- μαπ- ε/ο-	§4.1.7
ἀπ- αφ- ε/ο-	§6.4	ὀρ- ορ- ε/ο-	§6.2
ἀρ- αρ- ε/ο-	§6.1	πα- φ- ε/ο-	§1.2 (n. 11)
δε- δα- ε/ο-	§3.4.1	πε- παγ- ε/ο-	§4.2.3
δε- δακ- ε/ο-	§4.1.6	πε- παλ- ε/ο-	§4.2.4
ἐζ- ε/ο-	§1.4	πε- παρ- ε/ο-	§2.2
εἰπ- ε/ο-	§1.1	πε- πιθ- ε/ο-	§3.2.1
ἐν- εγκ- ε/ο-	§5.1	πε- πληγ- ε/ο-	§3.5.1
ἐν- ενιπ- ε/ο-	§5.4	πε- πορ- ε/ο-	§2.2
ἐρυκακ- ε/ο-	§5.4 (n. 21)	πε- πυθ- ε/ο-	§3.1.1
ἐσπ- ε/ο-	§1.3.1	πε- φιδ- ε/ο-	§3.2.2
ἑσπ- ε/ο-	§1.3.2	πε- φν- ε/ο-	§1.2
ἐνίπαπε/ο-	§5.4	πε- φραδ- ε/ο-	§3.4.2
εὑρ- ε/ο-	§2.1.1	τε- θιγ- ε/ο-	§3.2.3
κε- καδ- ε/ο-	§4.2.2	τε- θορ- ε/ο-	§2.2
κε- κλ- ε/ο-	§2.1.2	τε- ταγ- ε/ο-	§4.2.1
κε- κλυ- τε	§4.1.8 (n. 15)	τε- ταρπ- ε/ο-	§3.3.1
κε- κοπ- ε/ο-	§3.5.1 (n. 46 & 52)	τε- τμ- ε/ο-	§2.1.3
κε- κυθ- ε/ο-	§3.1.2	τε- τορ- ε/ο-	§2.2
κε- χαρ- ε/ο-	§4.1.3	τε- τυκ- ε/ο-	§3.1.3
κι- χ- ε/ο-	§0.1 (n. 1)	τε- τυπ- ε/ο-	§3.1.4
λε- λαβ- ε/ο-	§4.1.2	τε- τυχ- ε/ο-	§3.1.3 (n.16)

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.